GREEN
CLIMATE
FUND

Meeting of the Board
17 – 20 February 2025
Songdo, Incheon, Republic of Korea
Provisional agenda item 11

GCF/B.41/02/Add.07

27 January 2025

Consideration of funding proposals – Addendum VII

Funding proposal package for FP259

Summary

This addendum contains the following six parts:

a) A funding proposal titled "Adapting tuna-dependent Pacific Island communities and economies to climate change";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Secretariat’s assessment;
d) Independent Technical Advisory Panel’s assessment;
e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
f) Gender documentation.

Table of Content

Funding Proposal

Programme title:	**Adapting tuna-dependent Pacific Island communities and economies to climate change**
Countries:	**Cook Islands** (Climate Change, Cook Islands Division of the Office of the Prime Minister); **Federated States of Micronesia** (Department of Finance and Administration); **Fiji** (Ministry of Economy), **Kiribati** (Ministry of Finance and Economic Development); **Marshall Islands** (RMI Climate Change Directorate - NDA office); **Nauru** (Department of Foreign Affairs and Trade); **Niue** (Ministry of Finance); **Palau** (Office of the President); **Papua New Guinea** (Climate Change and Development Authority); **Samoa** (Ministry of Finance); **Solomon Islands** (Ministry of Environment, Climate Change, Disaster Management and Meteorology); **Tonga** (Ministry for Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications); **Tuvalu** (Government of Tuvalu); **Vanuatu** (Ministry of Climate Change, Change Adaptation, Meteorology, Geo-Hazards, Environment, Energy and Disaster Management)
Accredited Entity:	Conservation International Foundation (CI)
Date of first submission:	2024/04/06
Date of current submission	2025/01.20
Version number	V.8.1

GREEN CLIMATE FUND

Contents

Disclaimer

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	**A.2. Public or private sector**	Public
	Not applicable		

A.4. Result area(s)		**GCF contribution**	**Co-financers' contribution**
	Mitigation total		
	☐ Energy generation and access		
	☐ Low-emission transport		
	☐ Buildings, cities, industries and appliances		
	☐ Forestry and land use		
	Adaptation total	100 %	100 %
	X Most vulnerable people and communities	47%	97%
	X Health and well-being, and food and water security	53%	3%
	☐ Infrastructure and built environment	0%	0%
	☐ Ecosystems and ecosystem services[1]	0 %	0 %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	Not applicable	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	790,000 direct beneficiaries (50% male, 50% female)	2.5 million indirect beneficiaries (50% male, 50% female)
			5.8% of the region's projected total population in 2030 (31.9% of population within 1 km of the coast)	18.3% of the region's total projected population in 2030
A.7. Total financing (GCF + co-finance)	156.8 million USD	**A.9. Project size**	Medium (Upto USD 250 million)	
A.8. Total GCF funding requested	107.4 million USD			
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 107.4 million USD ☐ Loan ☐ Guarantee	☐ Equity ☐ Results-based payment		
A.11. Implementation period	7 years	**A.12. Total lifespan**	20 years	

[1] Ecosystems and ecosystem services are targeted as a co-benefit.

A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.* Click or tap to enter a date.	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* C
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes[2] ☒ No ☐
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☒ No ☐		
A.20. Executing Entity information	The Pacific Community (SPC). SPC is a regional entity serving as the principal scientific and technical organisation supporting development in the Pacific Island region. It was established by the Agreement Establishing the South Pacific Commission in 1947 and is owned and governed by its 26 members, including all 22 Pacific Island countries and territories.		

[2] Included in GCF Country Programme for some participating countries

A.21. Executive summary

Climate change problem:

1. This GCF Regional Tuna Programme (RTP) is designed to work with and support the governments of some of the most climate vulnerable countries in the world to build resilience and adapt to two major threats posed by anthropogenic climate change to Pacific Island communities and economies. These are the continued degradation of coral reef ecosystems and the redistribution of Pacific tuna resources due to the impacts of ocean warming.

2. These climate change impacts are of grave concern to the 14 Pacific Island Countries (PICs) participating in the Programme for two reasons. First, coral reefs have traditionally supplied much of the fish needed to provide the recommended protein intake required for domestic nutritional security (hereafter 'food availability'), given the ease of fishing and limited number of alternative options for obtaining sufficient protein for a healthy diet from agriculture and animal husbandry throughout the region. A gap between how much fish should be consumed by coastal and urban communities for a healthy diet and how much can be harvested sustainably from coral reefs is already emerging in several PICs, driven mainly by population growth. The projected future degradation of coral reef ecosystems due to ocean warming and acidification will exacerbate this problem.

3. Second, 9 of the 14 participating countries have significant dependence on access fees from industrial fleets fishing for tuna in countries' exclusive economic zones (EEZs). Participating countries receive 4-70% of all their (non-grant) government revenue from these fees, which contribute to national programmes for health, education, disaster preparedness and post-disaster recovery. Current modelling indicates that under continued high GHG emissions, 10-30% of the access fees, currently totalling ~$500 million per year across the region, could be lost by 2050 as tuna biomass is redistributed to the east (and poleward to some extent) due to ocean warming. These losses will result from reduced fishing effort in the EEZs of PICs as tuna becomes more available in the high seas.

Theory of Change and Key Programme Interventions:

4. The Programme's Theory of Change describes how the Programme will reduce impacts of climate change on the Pacific's most important resource by applying the priority interventions identified by the participating countries.

5. By strengthening national programmes to install anchored Fish Aggregating Devices (FADs) in nearshore waters (A1.1), training fishers to safely access FADs (A1.2), and improving post-harvest practices for FAD-caught fish (A1.3), national capacities to access tuna for coastal communities will be increased (Output 1). By delivering more bycatch from fishing fleets during transshipping operations (A2.1) and improving post-harvest practices and market opportunities for bycatch (A2.2) in urban communities, there will be an increased supply of fish from industrial tuna fishing for these communities (Output 2). By achieving these two Outputs, the availability of nutritious food for vulnerable communities will be improved (Outcome 1) and livelihoods will be positively impacted (co-benefit 1). To address the impacts of climate-driven tuna redistribution, the Programme will deliver an Advanced Warning System (AWS) that will serve as an economic planning tool for participating countries to understand and adapt to the shifting distribution of fish stocks (A3.1). The Programme will support adaptive planning by assessing the impacts of redistribution (A3.2) and provide training to national institutions to better negotiate for retention of fishing license revenue (A3.3). These activities will facilitate more effective adaptations to the redistribution of tuna stocks (Output 3) and lead to strengthened capacity of participating countries to retain essential benefits from tuna (Outcome 2), along with strengthened management of industrial fisheries (co-benefit 2). Key interventions are summarized below:

6. <u>Increasing access to tuna for coastal communities</u> to help fill the gaps in fish supply by expanding the use of anchored FADs to make it easier for small-scale fishers to catch tuna and other pelagic species. Although FADs have been installed by PICs for many years, the use of these tools for improving tuna catches in coastal waters has not scaled sufficiently because national fisheries agencies have only had *ad hoc* support to deploy FADs. The Programme will shift the paradigm so that FADs become part of countries' national infrastructure for food availability, installed and maintained through strengthened national FAD programmes and progressively expanded by recurrent government expenditure. The activities to be implemented during the Programme to scale-up FAD infrastructure in all 14 Participating Countries will be 'best practice' based on methods developed by SPC to ensure that the benefits to coastal communities are maximised, while ensuring that any impacts on non-target species are minimised. Shifts in the distribution of tuna biomass are not expected to reduce the effectiveness of this intervention.

7. Harnessing the potential of bycatch available from transhipping and unloading operations by industrial tuna-fishing fleets in regional ports to supply more fish for urban and peri-urban communities. Small enterprises are already capitalising on regulations requiring purse-seine fishing vessels to tranship their catch in port, which creates opportunities for selling low-cost fish unsuitable for canning (bycatch) to urban communities. However, there is considerable scope for increasing the supply and distribution of this bycatch. The Programme will assist national fisheries agencies to i) assess the potential supply of bycatch and the best ways to handle fish so that it feeds a larger number of people in urban and peri-urban areas; and ii) develop policies based on the information from the Advanced Warning System (AWS) described below to ensure that this potential is maximised under climate change.

8. Develop an Advanced Warning System (AWS) to serve as an economic planning tool that allows countries to plan adaptations to the redistribution of tuna and other species from the EEZs of many Pacific Island countries to the high seas with confidence. The AWS will reduce uncertainty in modelling by improving the spatial resolution of current models from 2 x 2° to 1 x 1°. It will produce shorter-term forecasts that facilitate identification of impacts on tuna fisheries at the EEZ scale as well as longer-term projections of the impacts of climate change on tuna fisheries and on national economies. The AWS will employ innovative methods for measurement of changes in tuna abundance and distribution among and within EEZs, and between EEZs and the high seas, providing actionable information to countries on future tuna catches and fishing license revenue.

Key benefits:

9. The deployment of 333 FADs across the region, and the training and equipping of small-scale fishers to fish safely and effectively around these FADs, in conjunction with strengthening national FAD programmes is expected to deliver additional tuna meals for a total of 560,000 men, women and children from coastal communities in the 14 participating countries. In the case of the smaller countries, 50-100% of the entire population will have access to additional tuna for local consumption. In the medium to large Pacific Island countries the most vulnerable communities are targeted, ranging in size from 30,000-90,000 people. Strengthened national FAD programmes are expected to deliver between 9 and 18 million fish meals per year. In the five smallest participating countries, the Programme will deliver up to 4-10 additional fish meals per person per month.

10. Following improvements to the supply and distribution of bycatch and tuna from transhipping and unloading operations at regional ports, a total of 290,000 men, women, and children from the five countries where transhipping operations will have improved food availability, and additional countries where longline catches are unloaded are expected to have increased access to tuna. It is expected that this intervention will supply ~13 million additional fish meals per year, an average of 3-4 meals per month for target beneficiaries.

11. The AWS will enable the nine tuna-dependent Pacific Island economies to understand the future timing and extent of tuna redistribution with much greater confidence. This information will be used as an economic planning tool by participating countries to better manage tuna fisheries within their EEZs, plan future fisheries infrastructure, and to negotiate through the Western and Central Pacific Fisheries Commission (WCPFC) and/or the United Nations Framework Convention on Climate Change (UNFCCC) to retain the socio-economic benefits they have historically received from tuna resources, regardless of the redistribution of the fish.

12. The five non-tuna-dependent participating countries, located in subtropical areas outside the prime equatorial tuna-fishing areas, will also benefit from the AWS. The AWS will enable these countries to determine whether projections from preliminary modelling, which indicate that the biomass of tuna could increase in their EEZs due to poleward redistribution, are likely to eventuate. If so, it will enable these countries to evaluate and select options to capitalise on opportunities including increased industrial fishing and/or tuna processing.

13. The information produced by the AWS will also improve the management decisions and strategies required to sustain the productivity of the industrial fisheries across the Western and Central Pacific Ocean (WCPO), and the ecosystem that underpins these fisheries. A key indicator for the effectiveness of the AWS will be the extent to which the WCPFC's Scientific Committee (SC) incorporates information from the AWS into its recommendations to the Commission for the conservation and management measures (CMMs) required to sustain the production of these globally significant tuna resources under warming ocean conditions.

14. The Programme will be implemented in close coordination with the 14 participating countries by the Pacific Community (SPC) as the Executing Entity, with the Pacific Islands Forum Fisheries Agency (FFA) and the Australian Commonwealth Scientific and Industrial Research Organisation (CSIRO) as Technical Partner grantees.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Programme Region

15. The Oceania region in the WCPO includes some of the world's smallest countries surrounded by an area of ocean that is significantly larger than their land surface area (**Figure 1**). Contemporary Oceania is generally grouped along ethno-geographic and cultural lines as “Melanesia, Micronesia, and Polynesia”. The 14 countries participating in the GCF Regional Tuna Programme (RTP) are spread among these three sub-regions. Given cultural ties, geographic proximity, and relatively small economies / populations, these countries have an established track record of cooperation and coordination through government and member-state organizations to achieve common objectives. Key information for the 14 countries is provided in **Table 1** and further details are in **Annex 02, Chapter 1** and in appendices for each country. These profiles give a brief introduction to each country, with a summary of their history, politics and governance, maritime regions, biogeography, population (including ethnic composition), agriculture and fisheries sector, and economics.

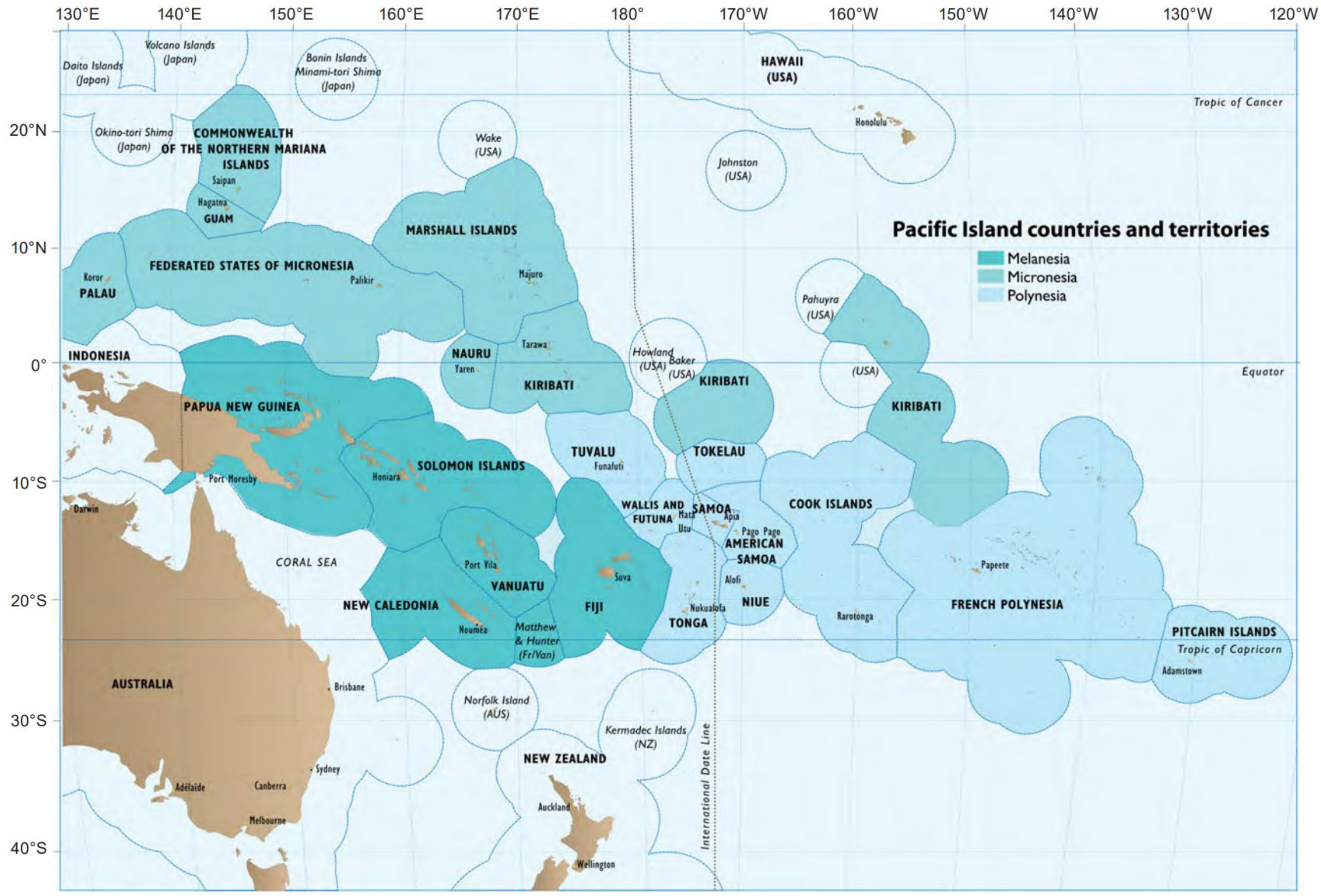


Figure 1.The Oceania region of the Western and Central Pacific Ocean with the ethno-geographic and cultural sub-regions of Melanesia, Micronesia, and Polynesia.[3]

[3] This map is indicative only of agreed and potential maritime boundaries between Pacific Island countries and territories (PICTs). It does not reflect the claims of PICTs to offshore areas.

B

16. Approximately 4.1 million people are projected to live within 5 km of the coast in Melanesia by 2030, with 1.9 million residing within one km of the coast (**Table 1**). The entire population of Micronesia, estimated to be 330,000 by 2030 km are expected to continue living within 1 km of the coast, with the exception 15,000 people who will be within 5 km of the coast. In Polynesia, 91% of the projected population of 335,000 in 2030 will live within 1 km of the coast.

Country and region	Population (latest census)*	Population (2030 estimate)	Land area (km^2)	EEZ area (km^2)	Coastline length (km)	% of popn. within 1 km of coast	No. of people within 1 km of coast (2030)	% of popn. within 5 km of coast	No. of people within 5 km of coast (2030)
Melanesia	**11,244,956**	**13,000,869**	**521,684**	**4,956,561**	**14,122**		**1,930,038**		**4,145,489**
Fiji	901,603	920,980	18,333	1,255,290	1,129	27.2	250,507	76.4	703,629
PNG	9,311,874	10,824,596	462,840	1,558,660	5,152	8	865,968	21.1	2,283,990
Solomon Is	744,407	892,093	28,230	1,547,600	5,313	65.1	580,753	91.4	815,373
Vanuatu	307,941	363,200	12,281	595,011	2,528	64.1	232,811	94.3	342,498
Micronesia	**296,241**	**329,943**	**2,158**	**8,906,382**	**9,174**		**315,598**		**329,943**
FSM	105,987	106,507	701	2,907,950	6,112	88.5	94,259	100	106,507
Kiribati	122,735	138,935	811	3,333,170	1,143	100	138,935	100	138,935
Marshall Is	42,418	53,983	181	1,774,280	370	100	53,983	100	53,983
Nauru	11,928	12,588	21	309,044	30	92.6	11,656	100	12,588
Palau	17,976	17,930	444	581,938	1,519	93.5	16,765	100	17,930
Polynesia	**333,441**	**335,158**	**4,205**	**3,765,421**	**1,030**		**235,918**		**329,059**
Cook Is	15,406	15,889	237	1,969,960	120	90.7	14,411	100	15,889
Niue	1,719	1,393	259	317,787	64	24.7	344.1	83	1,156
Samoa	200,999	209,369	2,934	123,278	403	61.1	127,924.5	97.2	203,507
Tonga	99,283	97,257	749	628,614	419	84.3	81,987.7	100	97,257
Tuvalu	10,778	11,250	26	725,782	24	100	11,250.0	100	11,250
Total	**11,874,638**	**13,665,970**	**528,047**	**17,628,364**	**24,326**		**2,481,553**		**4,804,491**

*Latest censuses were conducted between 2017 and 2022, depending on the country (**Annex 23**)

Table 1. Key geographical and population information for the 14 GCF participating Pacific Island countries.[4]

17. Two and a half million people, and the economies of nine countries, in the Pacific Islands region face severe threats due to the impacts of climate change on fisheries. This Programme will focus on two primary threats. The first is the degradation of coral reefs, which provide much of the animal-based protein required for appropriate nutrition across the region.[5] The second major threat is climate-driven redistribution of tuna which many PICs depend on heavily for revenue (from tuna-fishing access fees) to fund basic services for their citizens (See **Annex 02, Chapter 1**). The substantial effects that global greenhouse gas (GHG) emissions are having on coral reef fish and tuna in the Pacific Island region are described in the IPCC Sixth Assessment Report,[6] IPCC Special Report on 1.5°C and the IPCC Special Report on the Oceans and Cryosphere, and in even more detail in the

[4] Data sources: Phil Bright (*pers. comm.*) SPC's Statistics for Development Division and website - https://sdd.spc.int/ - population, land areas, population 1 and 5km from coast. EEZ areas drawn from SPC's Geoscience Maritime Boundaries dashboard (https://pacificdata.org/dashboard/maritime-boundaries). Coastline length was sourced from website: https://www.citypopulation.de/en/world/bymap/coastlines/. Also Figure 1 of the CI-SPC GCF RTP Concept Note.

[5] Except in some inland areas remote from the coast in Papua New Guinea, Solomon Islands and Vanuatu.

[6] IPCC, 2023: Sections. In: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team: H. Lee and J. Romero (Eds.)]. IPCC, Geneva, Switzerland, pp. 35-115, doi: 10.59327/IPCC/AR6-9789291691647

comprehensive assessments of the vulnerability of Pacific Island fisheries to climate change by the Pacific Community (SPC)[7] and FAO.[8]

18. For coral reefs, current and projected degradation is driven largely by increasing sea surface temperature (SST) and declines in pH (**see Annex 02, Chapter 2, Section 2.3**), with well-documented examples of thermal bleaching and acidification undermining the physical structure of reef habitats.[9, 10, 11] In turn, degraded coral reefs are projected to support fewer fish due the progressive loss of habitat and food associated with reductions in the structural complexity of reefs.[12] In addition, the growth, reproduction, and survival of coral reef fish and other coastal fish species are also projected to be directly affected by ocean warming, and in some cases by ocean acidification.[13] Taken together, the direct and indirect negative effects of continued GHG emissions on coral reef fish and other coastal fish species are highly likely to decrease the production of coastal fisheries in the Pacific Island region by 10-73% by 2050 under RCP8.5 and 7-65% under RCP4.5 (**see Annex 02, Chapter 2, Section 2.7.2**). For all 14 PICs, coral reef and seagrass habitats are assessed to have very high or high vulnerability to climate change[12].

19. Ocean warming is affecting the distribution of the region's rich tuna resources.[14] In the WCPO, the abundant skipjack tuna is caught most easily at the convergence of the two tropical ecological provinces − the western Pacific warm pool and the Pacific equatorial divergence (**See Annex 02, Chapter 1, Section 3**). This convergence zone is already known to shift by up to 4,000 km due to El Niño Southern Oscillation (ENSO) events and is projected to move further to the east as the warm pool expands with increasing SST caused by anthropogenic climate change.

20. Preliminary modelling indicates that skipjack tuna and the other tropical tuna species in equatorial areas are highly likely to shift progressively to the east as the ocean continues to warm and, to a lesser extent, into subtropical waters.[15] The resulting redistribution of tuna resources is projected to reduce the average annual combined tuna catch from the exclusive economic zones (EEZs) of tuna-dependent PICs in equatorial waters by 20% (range 10%−30%) by 2050 under RCP 8.5 compared to the annual average catch from 2009−2018. Under RCP 4.5, projected losses average 3% (**see Annex 02, Chapter 2, Section 2.9.4-2.9.7**).[16]

Food availability impacts

21. The implications of climate change for the availability of nutritious food for Pacific Island people are profound. Across the region, annual national fish consumption per capita ranges from approximately 20–100 kg (**Figure 2**), up to five times the global average, and fish traditionally caught from coral reefs and other coastal habits in small-

[7] Bell, J.D., Johnson, J.E. and Hobday, A.H. 2011. Vulnerability of tropical Pacific fisheries and aquaculture to climate change. Secretary of the Pacific Community, Noumea, 927 pp. https://www.spc.int/cces/climate-book/spc-publicationson-climate-change#tab-682-2

[8] Barange, M., Bahri, T., Beveridge, M.C.M., Cochrane, K.L., Funge-Smith, S. and Poulain, F. 2018. Impacts of climate change on fisheries and aquaculture Synthesis of current knowledge, adaptation and mitigation options. FAO Fisheries and Aquaculture Technical Paper 627, Rome. http://www.fao.org/3/i9705en/i9705en.pdf

[9] Hoegh-Guldberg, O., Poloczanska, E.S., Skirving, W., Dove, S. 2017. Coral reef ecosystems under climate change and ocean acidification. *Frontiers in Marine Science* 4: 158.

[10] IPCC, 2018: Global Warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty [Masson-Delmotte, V., P. Zhai, H.-O. Pörtner, D. Roberts, J. Skea, P.R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J.B.R. Matthews, Y. Chen, X. Zhou, M.I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, and T. Waterfield (eds.)] https://www.ipcc.ch/site/assets/uploads/sites/2/2019/06/SR15_Full_Report_Low_Res.pdf

[11] IPCC, 2019: IPCC Special Report on the Ocean and Cryosphere in a Changing Climate [H.-O. Pörtner, D.C. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, J. Petzold, B. Rama, N.M. Weyer (eds.)] https://www.ipcc.ch/site/assets/uploads/sites/3/2019/12/SROCC_FullReport_FINAL.pdf

[12] Hoegh-Guldberg, O., Poloczanska, E.S., Skirving, W. and Dove, S. 2017. Coral reef ecosystems under climate change and ocean acidification. Frontiers in Marine Science 4: 158

[13] Pratchett, M.S. et al. (2011). Vulnerability of coastal fisheries in the tropical fisheries to climate change. In: Bell, J.D., Johnson, J.E. and Hobday, A.H. (eds.. Vulnerability of tropical Pacific fisheries and aquaculture to climate change. Secretary of the Pacific Community, Noumea, 927 pp. https://www.spc.int/cces/climate-book/spc-publicationson-climate-change#tab-682-2

[14] Bell, J.D., Senina, I., Adams, T. et al. (2021). Pathways to sustaining tuna-dependent Pacific Island economies during climate change. Nature Sustainability 4, 900-910. https://doi.org/10.1038/s41893-021-00745-z

[15] Senina, I., Lehodey, P., Calmettes, B., Dessert, M., Hampton, J., Smith, N., Gorgues, T., Aumont, O., Lengaigne, M., Menkes, C. and Nicol, S., 2018. Impact of climate change on tropical Pacific tuna and their fisheries in Pacific Islands waters and high seas areas. *14th Regular Session of the Scientific Committee of the Western and Central Pacific Fisheries Commission, WCPFC-SC14-EB-WP-01.* Busan, Republic of Korea, 8-16 August 2018. 44 pages.

[16] Bell, J.D., Senina, I., Adams, T. et al. (2021). Pathways to sustaining tuna-dependent economies during climate change. Nature Sustainability 4, 900-910. https://www.nature.com/articles/s41893-021-00745-z

scale fisheries provide 50–90% of dietary animal protein for coastal communities.[17] By 2035, population growth and the negative effects of climate change on coral reef fish production will create demand for an additional 75,000 tonnes of fish per year for good nutrition of coastal and urban communities, relative to 2020.[18]

22. The rich tuna resources of the region are the most practical choice for filling this gap in fish supply.[19] Importantly, climate-driven shifts in the distribution of tuna biomass are not expected to alter this conclusion, the tuna remaining in each EEZ is projected to far exceed the quantity of fish needed for appropriate domestic nutrition.[20] Furthermore, apart from imports, there are few alternative sources of high-quality animal protein due to the limited potential for agriculture, horticulture, animal husbandry and aquaculture in many PICs.[21]

23. The region's tuna resources are not overfished nor subject to overfishing[22] due to exemplary management by PICs, and no problems are anticipated in using tuna to provide most of the additional 75,000 tonnes of fish needed. Just 6% of the total annual tuna catch from the Pacific region is needed to fill the gap in fish supply by 2035.[23]

24. Communities need additional support to catch and distribute tuna because these fish are usually found a significant distance from the coast. To make tuna a cornerstone of national food systems and increase the food availability and resilience of PIC populations, urgent action is required to i) empower coastal communities to catch more tuna efficiently and safely; and ii) ensure that the supply of tuna bycatch for urban communities, delivered by industrial fishing vessels, is increased and not disrupted as fleets catch and tranship or unload their fish further to the east.

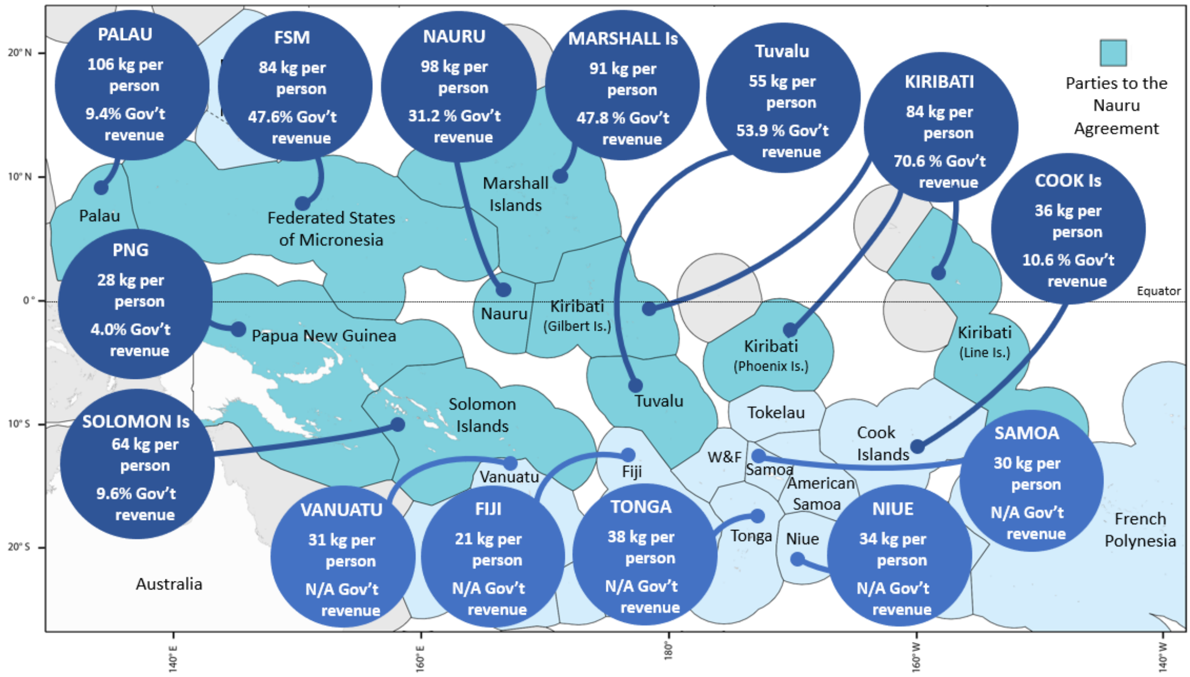


Figure 2. Annual average national fish consumption per capita (kg whole fish weight)[24], and the average percentage contribution of tuna-fishing access fees to (non-aid) government revenue (2015−2018),[25] in the 14 Pacific Island countries participating in the Regional Tuna Programme.

Economic impacts

25. PICs have already done much to sustain the economic benefits they receive from tuna. Regrettably, climate change poses a serious threat to this notable achievement. Tuna-dependent PICs have established a fisheries management system to deal with the effects of ENSO and climate change on the distribution of tuna within their combined EEZs. This mechanism – the Parties to the Nauru Agreement (PNA) 'Vessel Day Scheme' (VDS) – enables the eight participating countries[26] to share the economic benefits from tuna resources equitably, regardless of where the tuna are caught within their combined jurisdictions.[27] The VDS is widely recognised as a world-leading, climate-smart fisheries management framework.[28] However, the VDS is currently unable to secure all the present-day benefits that PNA members receive from tuna as the fish move progressively from their combined EEZs into high-seas areas, leaving these countries highly vulnerable.[29,30]

26. Preliminary modelling indicates that climate-driven redistribution of tuna threatens to undermine the economies of the PNA member countries and the Cook Islands, which obtain an average of 32% (range = 4–70%) of their total (non-aid) government revenue from tuna-fishing access fees **(Figure 2)**.[31] This modelling projects that by 2050, under RCP8.5, significant redistribution of tuna from the EEZs of these countries to the high seas is likely to occur, reducing the total fishing access fees for these nine countries by an average of ~$90 million (range $40−$140 million) per year at current prices.[32] For several of these countries, the projected loss of fishing access fees could reduce total (non-aid) government revenue by 6−13% per year (range 2−9% to 11−18%) (**see Annex 02, Chapter 2, Section 2.9**). Under RCP 4.5, losses are reduced, but still significant at ~$12M/yr.[33]

27. Such a significant reduction in government funding would have direct impacts on vulnerable populations in these countries, with fewer resources available for health, education, disaster preparedness and post-disaster recovery. Tuna redistribution may also affect employment in the region, where tuna fishing and processing provide ~25,000 jobs.[34]

28. The priority adaptation for the tuna-dependent economies involves ensuring they have improved data, forecasts, and projections on climate-driven tuna redistribution. This will allow countries to negotiate to retain access to the tuna resources that have historically occurred within their EEZs as tuna are redistributed to the high seas.[35]

29. For the 'subtropical' countries (Fiji, Niue, Samoa, Tonga, and Vanuatu), preliminary modelling indicates that the more modest tuna resources that occur in their EEZs could increase.[36] For these economies, adaptations centre

[24] Annex 26, Technical Study 2 (Table 17), and Bell, J.D. et al. 2009. Planning the use of fish for food security in the Pacific. Marine Policy 33:64-76. https://www.sciencedirect.com/science/article/abs/pii/S0308597X08000778
[25] Bell, J.D. et al. 2021. Pathways to sustaining tuna-dependent economies during climate change. Nature Sustainability 4, 900-910. https://www.nature.com/articles/s41893-021-00745-z
[26] Federated States of Micronesia, Kiribati, Marshall Islands, Nauru, Palau, Papua New Guinea, Solomon Islands and Tuvalu, plus Tokelau.
[27] Clark, S., Bell, J., Adams, T. et al. .2021. The Parties to the Nauru Agreement (PNA) 'Vessel Day Scheme': A cooperative fishery management mechanism assisting member countries to adapt to climate variability and change. In: Bahri, T., Vasconcellos, M., Welch, D.J., Johnson, J., Perry, R.I., Ma, X. and Sharma, R. (Eds.) *Adaptive management of fisheries in response to climate change*. Pages 209-224. FAO Fisheries and Aquaculture Technical Paper No. 667. Rome, FAO. https://doi.org/10.4060/cb3095en
[28] Havice, E. 2013. Rights-based management in the Western and Central Pacific Ocean tuna fishery: Economic and environmental change under the Vessel Day Scheme. *Marine Policy*, *42*, pp.259-267.
[29] Lam, V.W.Y. et al. 2020. Climate change, tropical fisheries and prospects for sustainable development. *Nature Reviews: Earth and Environment.* https://doi.org/10.1038/ s43017-020-0071-9
[30] Bell, J.D., Senina, I., Adams, T. et al. (2021). Pathways to sustaining tuna-dependent Pacific Island economies during climate change. Nature Sustainability 4, 900-910. https://doi.org/10.1038/s41893-021-00745-z
[31] Pacific Islands Forum Fisheries Agency, fisheries development data for the 4-year period 2015-2018.
[32] Based on the most recent, but as yet unpublished, modelling and economic analyses by the project partners. These analyses update the information in SPC Policy Brief 32.
[33] Pacific Islands Forum Fisheries Agency, fisheries development data for the 4-year period 2015-2018.
[34] Ruaia, T., Gu'urau, S. and Wheatley, L. 2023. Economic and Development Indicators and Statistics: Tuna Fisheries of the Western and Central Pacific Ocean 2022. Pacific Islands Forum Fisheries Agency, Honiara, Solomon Islands. 74 pages.

[35] SPC 2019. Implications of climate-driven distribution of tuna on Pacific Island economies. SPC Policy Brief 32/2019. https://www.spc.int/DigitalLibrary/Doc/FAME/Brochures/Anon_19_PolicyBrief32_TunaClimate.html
[36] Senina, I. et al. 2018. Impact of climate change on tropical Pacific tuna and their fisheries in Pacific Islands waters and high seas areas. Western and Central Pacific Fisheries Commission 14th Scientific Committee, Working Paper SC14-EB-WP-01 https://www.wcpfc.int/node/30981

on capitalising on any significant opportunities. To implement effective adaptations for both categories of Pacific Island economies, more reliable information on the timing and extent of tuna redistribution is essential.

30. The 14 PICs participating in the Programme have an extraordinary dependence on coral reef fish for food and on tuna for economic development. The vital socio-economic benefits derived from these natural resources are at severe risk from climate change. Given the extensive social and economic dependence of PICs on tuna, the GCF Regional Tuna Programme (RTP) is designed to be both regional and national in scope. The Programme is of regional significance because tuna are migratory fish species shared by PICs. Therefore, reliable information is needed on how tuna resources will respond to climate change and how tuna biomass will shift among national jurisdictions, and from EEZs to high-sea areas. The GCF RTP is also relevant at the national level because adaptations to increase access to tuna for food availability need to be customised for each country, due to varying contexts, and the wide differences in population size and location of Pacific Island nations. These adaptations centre around increasing access to the tuna and associated species that occur in nearshore waters through the deployment of artisanal fish aggregating devices (FADs)[37] and leveraging increased utilisation of tuna and bycatch transhipped or unloaded in Pacific Island ports by commercial tuna fleets.

31. The Programme supports the following goals of the Framework for Resilient Development in the Pacific: An Integrated Approach to Address Climate Change and Disaster Risk Management (FRDP),[38] endorsed by Pacific Island Leaders: i) strengthened adaptation and risk reduction to enhance resilience to climate change, including managing risks caused by climate change within social and economic development planning processes; and ii) strengthened preparedness, response, and recovery to natural disasters caused by climate change. The Programme will also build resilience to climate change by supporting the following important management frameworks, strategies, and resolutions for the fisheries sector in the region (see **Annex 24** for additional national climate plans and interventions**)**:

- Western and Central Pacific Fisheries Commission (WCPFC) Resolution on Aspirations of Small Island Developing States and Territories (Resolution 2008-01)[39]
- The Commission's Resolution 2019-01 on Climate Change as it Relates to the WCPFC.
- Regional Roadmap for Sustainable Pacific Fisheries[40] designed to improve the sustainability of tuna resources, add value to tuna catches, increase employment, and provide better access to tuna for food availability, as well as building resilience of coastal habitats (by progressively shifting fishing effort from coral reefs to tuna).
- A New Song for Coastal Fisheries – Pathways for Change,[41] an innovative regional approach to maintaining the benefits of small-scale fisheries in the face of declining coastal ecosystems and associated fish stocks.
- The Parties to the Nauru Agreement (PNA) Vessel Day Scheme that supplies 95% of tuna caught in the region.
- Nationally Determined Contributions: 10 of the 14 participating countries have included the need for adaptations to the effects of climate change on the ocean and coastal marine habitats in their NDCs.[42]
- The Forum Fisheries Agency's Climate Change Strategy adopted by the Forum Fisheries Committee in August 2023 to bring additional focus to the implications of climate change to the industrial tuna fisheries administered by FFA member countries.

Pacific Island country	Vulnerability to shortages of	Vulnerability to loss of government	Comments on vulnerability of national economies to climate-driven redistribution of tuna by 2050

[37] Fish aggregating device (FAD) is a generic term that applies to a range of floating objects that aggregate fish. The industrial tuna surface fishery (purse seine and pole and line vessels) take advantage of fish aggregations beneath naturally occurring floating objects such as logs that are disgorged from rivers. The industrial purse seine fishery also deploys floating rafts that drift with the currents to aggregate tuna. Anchored FADs are also used by the industrial fleets. Coastal artisanal and commercial fishers almost always use anchored FADs to aggregate tunas and other neritic species. The subject of discussion here is anchored FADs used by artisanal fishers. This is the definition used in Technical Study 9.
[38] https://pacificclimatechange.net/document/frdp_2016
[39] https://www.wcpfc.int/doc/resolution-2008-01/resolution-aspirations-small-island-developing-states-and-territories
[40] https://ffa.int/system/files/Roadmap_web_0.pdf
[41] https://www.spc.int/DigitalLibrary/Doc/FAME/Reports/Anon_2015_New_song_for_coastal_fisheries.html
[42] Gallo, N. et al. 2017. Ocean commitments under the Paris Agreement. Nature Climate Change 10.1038/nclimate3422

	coral reef fish per capita	revenue	
Cook Is	**Low-moderate**	**Low**	Preliminary modelling indicates that annual losses of total non-aid government revenue due to climate-driven redistribution of tuna under continued high GHG emissions could be limited to **<1%** in Cook Islands in 2050.
Fiji	**Low-moderate**	N/A	There could be an increase in tuna biomass in Fiji's EEZ due to climate-driven tuna redistribution under continued high GHG emissions by 2050 according to preliminary modelling.
FSM	**Low-moderate**	**Moderate**	Preliminary modelling indicates that FSM could lose **~6%** of total non-aid government revenue per year by 2050 due to climate-driven redistribution of tuna under continued high GHG emissions.
Kiribati	**Moderate**	**Moderate**	Preliminary modelling indicates that Kiribati could lose **~6%** of total non-aid government revenue per year by 2050 due to climate-driven redistribution of tuna under continued high GHG emissions.
Marshall Is	**Low-moderate**	**Low**	Preliminary modelling indicates that annual losses of total non-aid government revenue due to climate-driven redistribution of tuna under continued high GHG emissions could be limited to **<1%** in Marshall Islands in 2050.
Nauru	**Very High**	**Moderate-high**	Preliminary modelling indicates that Nauru could lose **~7%** of total non-aid government revenue per year by 2050 due to climate-driven redistribution of tuna under continued high GHG emissions.
Niue	**Low-moderate**	N/A	There could be an increase in tuna biomass in Niue's EEZ due to climate-driven tuna redistribution under continued high GHG emissions by 2050 according to preliminary modelling.
Palau	**Low-moderate**	**Low**	Preliminary modelling indicates that annual losses of total non-aid government revenue due to climate-driven redistribution of tuna under continued high GHG emissions could be limited to **<1%** in Palau in 2050.
PNG	**High**	**Low**	Preliminary modelling indicates that PNG could lose **>1%** of total non-aid government revenue per year by 2050 due to climate-driven redistribution of tuna under continued high GHG emissions.
Samoa	**Moderate-high**	N/A	There could be an increase in tuna biomass in Samoa's EEZ due to climate-driven tuna redistribution under continued high GHG emissions by 2050 according to preliminary modelling.
Solomon Is	**Moderate-high**	**Low-moderate**	Preliminary modelling indicates that Solomon Islands could lose **~3%** of total non-aid government revenue per year by 2050 due to climate-driven redistribution of tuna under continued high GHG emissions.
Tonga	**Low-moderate**	N/A	There could be an increase in tuna biomass in Tonga's EEZ due to climate-driven tuna redistribution under continued high GHG emissions by 2050 according to preliminary modelling.
Tuvalu	**Low-moderate**	**High**	Preliminary modelling indicates that Tuvalu could lose **~13%** of total non-aid government revenue per year by 2050 due to climate-driven redistribution of tuna under continued high GHG emissions.
Vanuatu	**Very high**	N/A	There could be an increase in tuna biomass in Vanuatu's EEZ due to climate-driven tuna redistribution under continued high GHG emissions by 2050 according to preliminary modelling.

Table 2. Summary of the vulnerabilities of the 14 Pacific Island countries eligible for support from the Green Climate Fund with respect to shortages of coral reef fish and continued contribution of access fees from industrial tuna fishing to government revenue. For all countries, improved modelling is needed to reduce uncertainty in projections of the economic impacts indicated above.

The following section provides background information on coastal and tuna fisheries, their management, and the importance of fisheries as a basis for good nutrition in the region. See **Annex 02, Chapter 1** for comprehensive background information.

National and regional fisheries policies and institutional arrangements

32. Three organisations serve in pivotal roles in coordination of policy relating to coastal and oceanic fisheries and the provision of technical advice to support decision-making.

33. **The FFA** is a multilateral organisation composed of 15 Pacific Island countries and a territory (it includes Australia, New Zealand, and Tokelau). The purpose of the FFA is to strengthen national capacity and regional cooperation for the sustainable management and development of their offshore fisheries resources. The most prominent of the groups within FFA is the **Parties to the Nauru Agreement (PNA)**, comprised of Federated States of Micronesia, Kiribati, Marshall Islands, Nauru, Palau, Papua New Guinea, Solomon Islands and Tuvalu plus Tokelau. The EEZs of PNA Members produce around 50% of the tropical tuna caught in the WCPO and 30% of the world's skipjack tuna.[43] PNA countries work together to cooperatively manage tuna fisheries within their EEZs, with the primary tools being the purse-seine and longline Vessel Day Schemes (VDS). Since its establishment in 1947, the **Pacific Community (SPC)** has provided scientific and technical development support to its 27 country and territory members.[44] SPC's vision for the region is a secure and prosperous Pacific Community whose people are educated and healthy and manage their resources in a sustainable way. The organisation's priorities are elaborated in its 2022-2031 Strategic Plan. See **Annex 02, Chapter 1, Section 4,** for additional information on FFA, PNA, and SPC.

34. The SPC convenes a Heads of Fisheries (HoF) Meeting for its 27 island and metropolitan Members annually. The HoF is the only regional forum for discussion and priority-setting for coastal fisheries and aquaculture, and for the joint consideration of oceanic and coastal fisheries, where all SPC Members participate. HoF reports to SPC's governing body, the CRGA. In collaboration with FFA and other regional agencies that participate in CROP, SPC also provides strategic and policy support to the Regional Fisheries Minister's Meeting.

Coastal fisheries

35. Coastal fish stocks are resident within EEZs and are managed nationally without formal multinational collaborative management. Regional cooperation in relation to coastal fisheries focuses on sharing information and lessons learned at the national level for application in fishery development and management situations in neighbouring countries supporting similar fisheries. This particularly relates to community-based fisheries management.

36. Community-based resource management (CBRM) approaches are receiving increasing attention for the management and conservation of inshore fisheries resources in PICs. CBRM is actively promoted by non-governmental organisations (NGOs) across the region, largely because of the predominance of traditional management systems, poor formal governance structures for coastal fisheries management and under-resourced central agencies. The CBRM model focusses on strengthening the capacity and responsibilities of local communities, supported through governance reforms undertaken by governments to regulate and manage resources by utilising local institutions, such as Local Government Councils or village institutions. These institutions are more effective when resource users are themselves actively engaged in decision-making.

37. Historical estimates of harvests of coastal pelagic species in the WCPO are approximately 43,000 t annually. These fisheries are managed at national levels for local and national consumption, generating significant benefits to national economies which is largely undocumented.[45] Efforts to estimate the value of small-scale fisheries in

[43] Clark, S., Bell, J., Adams, T., Allain, V., Aqorau, T., Hanich, Q., Jaiteh, V., Lehodey, P., Pilling, G., Senina, I., Smith, N., Williams, P. and Yeeting, A.2021. The Parties to the Nauru Agreement (PNA) 'Vessel Day Scheme': A cooperative fishery management mechanism assisting member countries to adapt to climate variability and change. In: Bahri, T., Vasconcellos, M., Welch, D.J., Johnson, J., Perry, R.I., Ma, X. and Sharma, R. (Eds.) *Adaptive management of fisheries in response to climate change*. Pages 209-224. FAO Fisheries and Aquaculture Technical Paper No. 667. Rome, FAO. https://doi.org/10.4060/cb3095en.

[44] SPC membership comprises American Samoa, Australia, Cook Islands, Federated States of Micronesia, Fiji, France, French Polynesia, Guam, Kiribati, Marshall Islands, Nauru, New Caledonia, New Zealand, Niue, Northern Mariana Islands, Palau, Papua New Guinea, Pitcairn Islands, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, United States of America, Vanuatu, and Wallis and Futuna.

[45] Dalzell, P.J. 1993. Small pelagic fishes. In: Wright, A. and Hill, L. (Eds.). *The nearshore marine resources of the South Pacific.* Pacific Islands Forum Fisheries Agency, Honiara, Solomon Islands and Institute of Pacific Studies, Suva, Fiji. pp.97-134; Dalzell, P., Lewis, A.D., St, S., Village, L. and Makati, M.M.S. 1988, Fisheries for small pelagics in the Pacific Islands and their potential yields. In: *Workshop on Pacific Inshore Fishery Resources, Working Paper No* Vol. 9, pp. 14-25.

the WCPO indicate that they may be undervalued by a factor of five.[46] Historical catch and effort data are sparse, but in some regions, catches may have been larger in the past than they are today, emphasizing the historical importance of proximal fisheries for nutritious food availability.[47]

38. Most coastal fisheries are characterised by small-scale, relatively low cost, fishing methods.[48] A considerable amount of fishing takes place from the shore or in shallow waters without the use of fishing vessels. Where vessels are used, these are generally small canoes and dinghies powered by outboard motors or sails. Larger (8-20m) vessels powered by outboard motors or inboard diesel engines are used for commercial purposes (fishing for demersal species beyond the reef slope and for catching tuna on the open ocean).[49] Gears include hook and line, traps, nets, and spears. Hook and line fishing may include drop lines, bottom, and surface long lines, or towed or trolled baits and lures.

Tuna fisheries

39. The migratory nature of tuna in the WCPO means that fishing occurs across many jurisdictions (national and high seas) and consequently, the regulatory framework is multi-layered. The Convention for the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (WCPF Convention) is the international agreement supporting decision-making; which is adopted by consensus in the form of Conservation and Management Measures (CMMs).[50] All members of the WCPF Convention are obligated to implement all CMMs formally adopted by the WCPFC throughout the Convention Area.[51] The Commission also adopts Resolutions, which are voluntary. The WCPF Convention provides that Members endeavour to achieve compatibility between measures adopted by the Commission and the national rules and regulations which apply in countries' EEZs.

40. The WCPFC ensures international cooperation on tuna fisheries. Since entering into force in 2004, acting on the advice of its subsidiary bodies[52], the WCPFC has adopted numerous CMMs. The primary CMM that regulates the management of fisheries for tropical tuna species (skipjack, bigeye, and yellowfin) is CMM 2023-01 and its predecessors. CMM 2023-01 describes permitted activities in the purse-seine and longline fisheries through provisions such as catch and effort limits, retention of catch requirements, and seasonal closures in relation to fishing with FADs.

41. In relation to climate change, the provisions of CMM 2023-01 note the SEAPODYM[53] analyses presented to the 11th Regular Session of the WCPFC Scientific Committee (SC11), SC12, and SC13 on the projected impacts of climate change on tuna distribution, larval numbers and stock biomass, indicate that the WCPFC needs to build resilience into the medium and long-term planning and manage WCPO fish stocks in a precautionary manner, and Article 30(2)(c) of the Convention requires the Commission to ensure there is no disproportionate burden of conservation action on developing States and Territories.

---

[46] Zeller, D., Booth, S. and Pauly, D. 2006. Fisheries Contributions to GDP: Underestimating Small-scale Fisheries in the Pacific. *Marine Resource Economics* 35. pp. 1-38.

[47] Weng, K.C., Glazier, E., Nicol, S.J. and Hobday, A.J., 2015. Fishery management, development and food security in the Western and Central Pacific in the context of climate change. *Deep Sea Research Part II: Topical Studies in Oceanography*, *113*, pp.301-311.

[48] Wright, A. 1993. Shallow water reef-associated finfish. In: Wright, A. and Hill, L. (Eds.). *Nearshore Marine Resources of the South Pacific*. Institute of Pacific Studies, University of the South Pacific and Pacific Islands Forum Fisheries Agency, Honiara, Solomon Islands. 710 pages; Wright, A. and Richards, A. 1983. A multispecies fishery associated with coral reefs in the Tigak Islands, Papua New Guinea. *Asian Marine Biology* 2, 69-84; Dalzell, P.J. 1996. Selectivity and yields of reef fishing. In: Polunin, N.V.C. and Roberts, C.M. (Eds.), *Reef Fisheries*. Chapman & Hall (Chapman & Hall Fish and Fisheries Series 20), London. 477 pages.

[49] Lee R. *In prep*. Needs analysis for safe small vessel designs for operating around nearshore FADs (GCF Study 12). A report prepared for the Pacific Community and Conservation International as a contribution to a funding proposal to the Green Climate Fund. 106 pages. https://purl.org/spc/digilib/doc/i66qp

[50] Decision making is recorded in the form of Conservation and Management Measures (CMMs) which are adopted at the annual sessions of the Commission. CMMs are binding on Members, Cooperating Non-members and Participating Territories (CCMs).

[51] .Except for archipelagic waters which are subject to national sovereignty.

[52] A Scientific Committee (SC), a Technical and Compliance Committee (TCC) and a Northern Committee (NC). A Finance and Administration Committee (FAC) provides the Commission with advice in relation to financial matters, the budget, staff establishment and headquarters affairs among other administrative subjects.

[53] SEAPODYM is a numerical model initially developed for investigating physical-biological interaction between tuna populations and the pelagic ecosystem of the Pacific Ocean. Using predicted environment from ocean-biogeochemical models, SEAPODYM integrates spatio-temporal and multi-population dynamics and considers interactions among populations of different species and between populations and their physical and biological environment (including intermediate trophic levels). The model also includes a description of multiple fisheries and then predicts spatio-temporal distribution of catch rates, and length-frequencies of catch based either on observed or simulated fishing effort, allowing respectively to evaluate the model or to test management options (e.g., changing the fishing effort, implementing marine reserves).

42. Fishing for tuna plays a vital role in the economic development and/or food availability for the 14 countries participating in the Regional Tuna Programme (RTP). However, in the case of ten of these Pacific Small Island Developing States (SIDS), the contributions of tuna to the economy are so substantial that these SIDS are considered 'tuna-dependent'.[54] Four species of tuna are the primary targets for WCPO fisheries: skipjack (*Katsuwonus pelamis*), yellowfin tuna (*Thunnus albacares*), bigeye (*T. obesus*) and South Pacific albacore (*T. alalunga*). These four species represent > 90% of the total catch taken by industrial fleets. The remainder of the catch is predominately billfish (marlin & swordfish), oceanic sharks, and the principally N. Pacific catch of Pacific bluefin tuna (*T. orientalis*).

43. The WCPFC Convention Area (WCP–CA) tuna catch (2,701,239 t) for 2022 represented 80% of the total Pacific Ocean tuna catch of 3,370,920 t, and 54% of the global tuna catch (the provisional estimate for 2022 is 4,962,310 t), noting that unlike other oceans, over 85% of the WCP–CA tuna catch occurs in the waters of coastal States.

Contribution of nearshore caught fish to local diets

44. The availability of high-quality, subsistence production data across the region is patchy and as a range of methodologies are used, analysis is challenging. However, estimates of annual average consumption by coastal rural populations range from 30–118 kg per person in Melanesia, 62–115 kg in Micronesia, and 50–146 kg in Polynesia with aquatic foods providing 50–90% of the dietary animal protein in rural areas. While consumption in urban centres is often lower than in rural areas, it usually still exceeds the global average of ~20 kg per person per year.[55] Despite the rate of highly nutritious aquatic food consumption, many PICs experience diet-related poor health outcomes due to the prevalence of low nutritional value, high sugar, imported substitutes in their diets.[56,57]

45. In 2023, estimates for the coastal artisanal and subsistence fisheries production were reviewed.[58] On the basis that the coastal subsistence harvest was 123,961 t in 2021, and the artisanal coastal harvest was 50,123 t, a per capita production of 13.8 kg annually was estimated for coastal fisheries across 22 Pacific Island Counties and Territories (PICTs).[59,60] This represents a declining trend (14% over 21 years) in the contribution of coastal fisheries to PIC economies, livelihoods and dietary welfare which is of significant concern to PICs (**Table 3**).

Study year	Per capita coastal fisheries production (kg/person/year)
2007	16.1
2014	14.9
2021	13.9

Table 3. Per capita coastal fisheries production estimates from reviews published in 2007, 2014 and 2021.[61]

[54] Bell, J.D., Senina, I., Adams, T., Aumont, O., Calmettes, B., Clark, S., Dessert, M., Gehlen, M., Gorgues, T., Hampton, J. and Hanich, Q., 2021. Pathways to sustaining tuna-dependent Pacific Island economies during climate change. *Nature sustainability*, *4*(10), pp.900-910.

[55] FAO. 2014. Global Blue Growth Initiative and Small Island Developing States; Food and Agriculture Organization of the United Nations: Rome, Italy.

[56] Brewer T., Kottage H., Andrew N. *in prep*. Options for supplying dietary protein for growing Pacific Island populations. Technical Study 2 prepared for the Pacific Community in support of the preparation of a Funding Proposal for the Green Climate Fund. 93 pages. https://purl.org/spc/digilib/doc/4aszp.

[57] Andrew, N. L., Allison, E.H. Brewer, T., Connell, J., Eriksson, H., Eurich, J.G. Farmery, A., Gephart, J.A., Golden, C.D. and Herrero, M. 2022. Continuity and change in the contemporary Pacific food system. *Global Food Security* 32: 100608.

[58] Gillett R. and Fong, M. *in press*. Fisheries in the economies of Pacific Island countries and territories (Benefish Study 4). Noumea, New Caledonia: Pacific Community.

[59] The Benefish4 review used a total population for the region of 12.5 million with PNG accounting for 78.3% of that population (*Op. cit.* Footnote 116). In PNG, 21% of the country's total population of 9,311,874 was estimated to live within 20 km of the coast (1.95 million residents). The contribution of the large inland population to coastal fisheries production is probably insignificant with the result significant caution is necessary in interpreting these estimates. If the large and rapidly-growing inland population of PNG influences analysis it is possible that the strong effect of this large population hides a reasonably healthy supply of coastal fish per capita for many of the smaller countries, and for coastal communities in larger countries. It underscores the critical need for improved methodology and standards for collecting this type of basic information that should underpin a range of services and planning across the region.

[60] While it could be argued that the non-coastal population of some of the larger countries in Melanesia could distort this finding, the populations were treated consistently in the 2007, 2016 and present study – so the trend is valid.

[61] Gillett R. and Fong, M. *in press*. Fisheries in the economies of Pacific Island countries and territories (Benefish Study 4). Noumea, New Caledonia: Pacific Community.

46. Analysis of Household Income and Expenditure Survey (HIES) data also shows that fish provides 51–94% of animal protein in the diet in rural areas, and 27–83% in urban areas, across the region.[62,63] PNG is the exception, where the large inland population generally has much less access to fish, except for communities living near rivers or who grow tilapia in ponds.[64] Importantly, the great majority of fish is produced via coastal subsistence fishing. In 14 PICs, 52–91% of fish eaten in rural areas is caught by households from coral reefs and adjacent coastal habitats.

47. In six of the 14 participating countries (Kiribati, Nauru, PNG, Samoa, Solomon Islands and Vanuatu) coastal fisheries are not expected to be able to meet the future demand for fish and the gap between the fish required and the fish expected to be available from coastal fisheries will be substantial. In another five of the participating countries (Federated States of Micronesia, Fiji, Niue, Tonga, and Tuvalu), it may not be economical to transport fish to urban centres from remote, productive coral reefs. If so, future demand for fish in the rapidly growing urban centres in these PICs may not be met.[65]

Present and emerging gaps in fish supply and the potential of tuna to fill this gap.

48. It is essential that new sources of fish are utilised to meet the growing need for high-quality dietary protein in most PICs.[66] An important opportunity is the allocation of a relatively small proportion of the rich tuna resources of the tropical Pacific to improve national food availability. This option promises to make a significant contribution to filling the gaps between the amount of fish required for nutritious food availability and the fish available from coastal fisheries in the many countries where these gaps exist.[67]

49. In relative terms, the percentage of the region's tuna catch needed by 2035 to fill the gap in domestic fish supply is small, i.e., ~6%. The greatest quantities of tuna will be required in PNG, Solomon Islands, and Kiribati (**See Annex 02, Chapter 1, Section 9**).[68]

50. The RTP will implement two approaches to utilize tuna to fill this gap in fish supply. The first is to increase opportunities for coastal fishers to access oceanic pelagic fish resources through the deployment and management of nearshore artisanal Fish Aggregating Devices (FADs).[69] The second is improving access to tuna and bycatch transhipped or unloaded by industrial fishing vessels in Pacific ports.[70]

Related Initiatives in the region

51. In preparation for the Programme, regional and national initiatives were surveyed, and projects related to fisheries and climate change that were being implemented or were in planning stages of relevance to the GCF Programme. The projects reviewed were a balance of single-country (bilateral), groupings of between 2 – 7 PICs, and regional arrangements, projects, and initiatives. Regional arrangements included participation by the majority of the membership of SPC and FFA.

---

[62] Bell, J.D., Kronen, M., Vunisea, A., Nash, W.J., Keeble, G., Demmke, A., Pontifex, S., Andréfouët, S. 2009. Planning the use of fish for food security in the Pacific. *Mar. Pol*. 33 (1), 64–76.
[63] Bell, J.D., Allain, A., Allison, E.H., Andréfouët, S., Andrew, N.L. *et al*. 2015 Diversifying the use of tuna to improve food security and public health in Pacific Island countries and territories, *Marine Policy* 51 (2015) 584–591.
[64] Bell, J.D., Kronen, M., Vunisea, A., Nash, W.J., Keeble, G., Demmke, A., Pontifex, S., Andréfouët, S. 2009. Planning the use of fish for food security in the Pacific. *Mar. Pol*. 33 (1), 64–76.
[65] Bell, J.D., Allain, A., Allison, E.H., Andréfouët, S., Andrew, N.L. *et al*. 2015. Diversifying the use of tuna to improve food security and public health in Pacific Island countries and territories, *Marine Policy* 51 (2015) 584–591.
[66] SPC. 2008. Status Report: Nearshore and Reef Fisheries and Aquaculture. Secretariat of the Pacific Community, Noumea, New Caledonia. www.spc.int/mrd/ministers/2008/ MIN4WP03-coastal-isheries-status-annex-a.pdf.
[67] Bell, J.D., Allain, A., Allison, E.H., Andréfouët, S., Andrew, N.L. *et al*. 2015. Diversifying the use of tuna to improve food security and public health in Pacific Island countries and territories, *Marine Policy* 51 (2015) 584–591.
[68] Bell, J.D., Allain, A., Allison, E.H., Andréfouët, S., Andrew, N.L. *et al*. 2015. Diversifying the use of tuna to improve food security and public health in Pacific Island countries and territories, *Marine Policy* 51 (2015) 584–591.
[69] See: Chapman L. *in prep*. Regional Report: Feasibility of scaling-up National Fish Aggregating Device (FAD) Programmes in all 14 participating countries. Technical Study prepared for the Pacific Community as a contribution to a funding proposal being prepared for submission to the Green Climate Fund (GCF). October 2023. Lindsay Chapman Consulting Pty Ltd, Brisbane, Australia. 92 p. https://purl.org/spc/digilib/doc/au6vm
[70] See: MRAG Asia Pacific. *in prep*. Adapting tuna dependent Pacific Island communities and economies to climate change: Existing and future needs and conditions for distributing tuna bycatch to urban and peri-urban areas (GCF Study 5). Report prepared for the Pacific Community. 44 p. https://purl.org/spc/digilib/doc/5t6n9

52. A system of ranking projects/initiatives relevant to the GCF Regional Tuna Programme proposal was applied.[71] This was based on information available for each project, which was then used to rank the projects in terms of relevance. A total of nine projects were determined to be directly relevant to the Programme, nine were somewhat relevant, 6 slightly relevant, and 10 marginally relevant. Details of the projects reviewed can be found in **Annex 02, Chapter 4, Section 4.1.2.**

53. The RTP is designed to avoid replication and add value to existing projects/programmes to the maximum extent feasible. Although climate interventions are features across most of these projects, only a limited number support activities associated with FADs or utilisation of tuna to support national food availability. However, the projects with FAD components, apart from the World Bank PROPER project and the GEF-supported Oceanic Fisheries Management Project III (OFMP III), will have concluded by the time the GCF Programme commences in late 2025/early 2026.[72] The Programme PMU will reanalyse regional and national initiatives during the first year of implementation to ensure alignment.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Goal statement

54. At its highest level, the ***Theory of Change (ToC)*** presents a statement of the overall goal and intended impact of the Regional Tuna Programme, as follows:

> ***IF Pacific Island governments are supported to implement effective programmes for assisting coastal and urban communities to obtain and utilise more tuna, and are provided with improved information on climate change-driven redistribution of tuna;***
>
> ***THEN Pacific Island nations will be transformed to become more resilient to key climate change threats facing the fisheries sector;***
>
> ***BECAUSE communities and governments will be better informed and equipped to make optimal use of the fisheries resources on which they depend for food, livelihoods, and economic development.***

55. The Programme is designed to overcome the barriers identified below to bring about a paradigm shift in the way that tuna stocks, the most valuable food resource across the vast Pacific region, are assessed, managed, and utilised. The Programme will ensure that countries whose economies are tuna-dependent will be capable of adapting to climate change-affected relocation of tuna stocks, thereby enabling them to continue to access the resource and the accompanying benefits for improved food availability, health, livelihoods, and inclusive economic development.

Structure of the Programme

56. Programme activities are grouped within two broad components:

Component A – Adaptation to harness tuna to improve food availability of Pacific Island communities as coral reefs are degraded by climate change: This component contains sets of activities designed to (i) strengthen and invest at scale in national programmes through grants and technical assistance to support deployment and use of nearshore, anchored FADs to enable fishers in coastal communities to catch and process tuna and other pelagic species to improve access to dietary protein as the supply of fish from degraded coral reef ecosystems declines and human populations grow; and (ii) to design and implement tuna transhipment mechanisms that will provide urban and peri-urban communities with greater access to bycatch from industrial fishing fleets for domestic food availability.

Component B – Adaptations to reduce risks to Pacific Island economies from climate-driven tuna redistribution: This component includes a set of activities to establish a region-wide Advanced Warning System (AWS) to gather

---

[71] Detailed information was not always available for all the projects which made it difficult to determine relevance. By its nature, climate change adaptation or resilience is invariably interlinked. This is generally reflected at the broad project description or objective level.

[72] Relevant projects that will have concluded by 2026 include the Palau Northern Reefs Project, the food security project in Kiribati, and the following current regional projects: PEUMP and the FAO FishFAD project. The World Bank PROPER initiative will have at least 3 years remaining in some of the countries where activities have already started, with activities in several other countries yet to be launched and continue for ca. 6 years.

and analyse the data[73] needed to determine future abundance and distribution of tuna more accurately as stocks respond to ocean warming due to climate change. The evidence generated will support the preparation and negotiation of multi-lateral agreements, designed to enable tuna-dependent PICs to maintain their access rights to shifting tuna stocks and the associated economic benefits regardless of the redistribution of the fish. The data from the AWS will also be used to inform management decisions (harvest strategies, CMMs) made by the WCPFC to ensure continued effective management of tuna stocks and industrial fishing under impacts of climate change.

57. While Components A and B address different critical impacts of climate change on fisheries in the Pacific, they are interlinked. For example, Component B's AWS will allow countries to strategically plan the location and utilization of transshipment operations to provide additional tuna to urban communities under Component A, and the AWS will support decision making on the placement of FADs to maximize their utility, and national policies on minimum distances from the coast where commercial fishing can take place.

58. The ToC (**Figure 3)** illustrates the logical flow between programme activities and outputs, and how these will lead to achieving the desired outcomes, co-benefits, and ultimately, the transformational paradigm shift that will bring about greater resiliency to climate change threats, which is the overarching goal of the Programme.

Risks, barriers, and assumptions

59. Before examining the cause-and-effect flow of the Programme actions in the results chain, it is necessary to consider the *risks*, *barriers* and *assumptions* that have been identified, and the implications for Programme design and implementation. Risks and Assumptions are external factors over which the Programme has little control, but which may impact programme activities and results. Barriers are the obstacles the GCF RTP is designed to address to achieve its objectives and targets.

60. A *risk* is an uncertain threat that could have a negative impact in the delay and completion of the Programme. The Programme has identified the following primary risks to successful implementation (please see Section F for additional identified risks):

- Natural disasters or health crises disrupt regular economic, societal, or institutional functions (R1).
- Global or national economic or political situation becomes unstable (R2).
- Large-scale climate driven natural disasters significantly disrupt tuna fisheries (R3).
- Ineffective coordination between Regional Fisheries Management Organizations leads to increased uncertainties in managing redistributed tuna (R4).

61. The actions to be undertaken as part of the Programme are designed to overcome identified *barriers*. The barriers to be addressed by this RTP fall into two categories: (i) those that limit increased access to and acceptance of tuna to increase food availability in coastal and urban communities; and (ii) those that prevent governments from better understanding and responding to the implications of climate-driven tuna redistribution.

1. ***Barriers to use of tuna to improve food availability***: Coastal communities have not been able to access sufficient tuna to offset the effects of climate change-induced coral reef degradation and population growth on per capita fish supply for a variety of reasons. First, many Pacific Island governments do not have sufficient resources and capacity to scale-up the use of the nearshore FADs that are designed to help coastal communities catch tuna and other types of pelagic fish. Specific barriers include lack of resources to: institutionalise 'National FAD Programmes'; provide training in safe and effective FAD-fishing methods and post-harvest processing ; ensure that vessels used by small-scale fishers to fish around FADs are fit for purpose; mainstream disaster preparedness and recovery plans for small-scale fishing communities (**Annex 26, Technical Study 4**); cyclone-proof national FAD systems; and train communities without refrigeration in simple post-harvest methods (e.g., drying and smoking) to increase the storage life of FAD caught tuna and pelagic fish. Women have a primary role in the post-harvest part of the value chain, which creates opportunities for the Programme to address capacity building on leadership and management skills for women and contribute to empowerment of women in the local fishery sector (see **Annex 08**).

62. Second, despite existing regulations mandating that transhipment of tuna catches caught by industrial purse-seine fleets must occur in Pacific Island ports, many governments have not been able to invest in supply chains to deliver tuna from transhipping operations to rapidly growing urban communities, due to uncertainty about the locations of

[73] All fisheries data administered by SPC is subject to SPC's rules on data access and use available here: SPC FAME data policy ._ Data collected by a SPC member or by FAME on behalf of the member are, by default, owned by the member unless otherwise agreed.

future hubs of fishing activity. Patterns of transhipment in regional ports are already affected by the El Niño-Southern Oscillation (ENSO) and will be influenced further by climate change. Unless reliable information on the locations of future fishing hubs is available it will not be possible to identify the investments in market and supply-chain facilities needed to distribute tuna to urban communities effectively. In addition, local utilisation of tuna and bycatch unloaded in Pacific ports is constrained by limited local capacity to process and distribute these fish for urban and peri-urban markets. As for FAD-caught fish, women play a prominent role in processing and distribution of tuna in existing transhipping operations and there is a need for gender specific capacity and leadership gaps to be addressed as transhipping operations are expanded.

63. Third, there is a lack of awareness among both coastal and urban communities about the imminent climate-change induced shortage of coral reef fish, and the need to diversify fish consumption to include much more tuna in their diets to ensure nutritious food availability.

64. ***Barriers to national economic security***: The governments of tuna-dependent Pacific Island countries are acutely aware of the implications of climate-driven redistribution of tuna from their EEZs to high-seas areas for their economies. However, inadequate information for governments about the timing and extent of tuna movements is a fundamental barrier preventing adaptation to the economic impacts of tuna redistribution. This lack of information limits the ability of regional management bodies (e.g. WCPFC) to continue to sustainably manage the catch of industrial tuna fishing fleets and the ability of PICs to maintain their jurisdiction over the historical levels of tuna caught in their waters. For example, the existing fishing restrictions designed to constrain operation of purse-seine vessels in high-seas areas could be challenged by nations from outside the region once a greater proportion of tuna occur in international waters. To ensure that the effects of climate change are increasingly integrated into regional tuna management arrangements, the Pacific Islands Forum Fisheries Agency (FFA) succeeded in having a Climate Change Resolution adopted by the WCPFC in 2019 (**Annex 02, Chapter 1**). However, measures to secure the benefits of tuna for PICs under a changing climate remain to be identified and implemented. For example, obtaining the necessary international agreements in the relevant fora, including through the Convention on the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean, will only be possible with improved modelling and monitoring of tuna stocks. Negotiating for continued rights to the tuna that have been driven from PICs' EEZs to the high seas by ocean warming will require more accurate information on the timing and extent of tuna redistribution. Without such agreements, tuna-dependent PICs will lose vital government revenue, resulting in reduced ability to provide the basic services needed by their citizens.

65. The subtropical PICs are also expected to benefit from development of the information system described above. It will enable these countries where preliminary modelling indicates that abundance of tuna could increase to evaluate adaptations to capitalise on opportunities, e.g., increased industrial fishing and/or tuna processing. To provide the reliable information needed to predict the redistribution of tuna with confidence for use in management decisions, the key weaknesses in existing modelling approaches need to be addressed. Essential improvements to models include describing responses of tuna resources to climate change at higher resolutions (this will require integration of information on tuna connectivity, tuna stock structures and meso-scale oceanography), improved assessment of the effects of climate change on the food webs that support tuna resources and incorporating ocean forcings for all GHG emission scenarios. Further details of barriers preventing adaptations to secure the necessary contributions of tuna to food availability and economic development are included in **Annex 02, Chapter 4.**

66. These barriers are summarized and included in the ToC, as follows:

- Inadequate financial resources for National FAD programmes (B1)
- Mismatch of industrial tuna fishing operations with the food availability needs of urban and peri-urban communities (B2)
- Poor supply chain infrastructure and knowledge about the benefits of transhipped fish (B3)
- Lack of resources by regional agencies to develop AWS (B4)
- Limited contribution by fishing industry to the collection of climate data (B5)
- Poorly defined pathway for optimising national benefits as tuna biomass redistributes (B6)

67. The <u>*assumptions*</u> considered in developing the ToC include the following external factors:

- Large-scale climate events, natural disasters (cyclones, tsunamis, etc.), or public health calamities (e.g., COVID-19) do not significantly disrupt Programme activities for extended periods.
- Global economy and political situation in participating countries remain stable.

- Governments/communities continue their willingness to support and participate in the Programme.
- Data generated by AWS will be effectively utilised and applied by governments and regional bodies.
- Eastern Pacific Ocean (EPO) stakeholders engage with WCPO counterparts to strengthen mechanisms for collaboration.
- The 30x30 initiative[74] does not adversely effect transhipment and unloading operations by the commercial tuna fleet.
- FAD-sourced, transhipped, and unloaded tuna and other pelagic species are priced competitively with imported alternatives (affected by exchange rates and market forces)
- The cost of FAD acquisition and deployment remain economically viable.
- Fishers utilise FADs and become less dependent on reef fishing.
- With improved data, countries will be able to better manage tuna stocks and negotiate for retention of fishing license revenue.

68. Of the assumptions listed above, the first two reflect large-scale events which could potentially affect the implementation of the Programme. The latter eight are more narrowly focused and Programme-specific, with more direct effect on the transition from Programme outputs to outcomes and impact. To ensure effectiveness, the Programme will monitor these assumptions and implement adaptive management as necessary. Programme activities will be gender responsive, defined as: "identifying and understanding gender gaps and biases, and then acting on them, developing and implementing actions to overcome challenges and barriers toward improving and achieving gender equality (see **Annex 08**).

Results Chain: activities to deliver programme outputs and outcomes:

Component A: Adaptations to harness tuna to improve food availability of Pacific Island communities as coral reefs are degraded by climate change.

69. The purpose of Component A is to implement adaptative measures to better harness the potential of tuna to improve food availability for communities across the Pacific Island region.

The first set of activities under Component A is aimed at strengthening national programmes to expand the use of FADs to assist small-scale fishers catch tuna and associated pelagic fish species safely and effectively and build the capacity of women to extend the shelf-life of catches through training in post-harvest methods. The beneficiaries for this set of activities are residents of coastal communities across all 14 participating PICs. These activities are:

- Provide technical and logistical support to strengthen National FAD programmes (Activity 1.1).
- Augment national safety-at-sea initiatives (Activity 1.2).
- Strengthen post-harvest practices and improve market opportunities for FAD-caught fish (Activity 1.3).

2. Successful implementation of this set of activities leads to achievement of Programme Output 1: *Increased national capacity to access tuna and other pelagic fish for coastal communities*. Implementation will also lead to increased empowerment and awareness of women in the FAD fishery value chain through dedicated training and capacity building for women in FAD-caught fish post-harvest processing and marketing, as well as supporting the integration of Gender Equality and Social Inclusion approaches into national fishery ministries.

70. A second set of activities under Component A is also designed to provide greater access to tuna and other pelagic fish, but in this case, the source of the fish is off-loading of bycatch and tuna from industrial fishing vessels at ports during transhipping and unloading operations. Because transhipment and unloading ports are in larger population centres, this activity set will increase the availability of fish to urban and peri-urban populations. This set of activities include:

- Implement strategies to deliver more transhipped and unloaded bycatch and tuna to urban/peri-urban communities (Activity 2.1); and

[74] In December 2022, 30x30 was agreed at the COP15 meeting of the Convention on Biological Diversity and became a target of the Kunming-Montreal Global Biodiversity Framework. The 30x30 concept refers to a target for at least 30 per cent globally of land areas and of sea areas, especially areas of particular importance for biodiversity and its contributions to people, to be conserved through effectively and equitably managed, ecologically representative and well-connected systems of protected areas and other effective area-based conservation measures and integrated into the wider landscapes and seascapes.

- Strengthen and develop post-harvest practices and improve market opportunities to distribute bycatch and tuna from transhipping and unloading operations to urban/peri-urban communities (Activity 2.2).

71. This set of activities will result in achievement of Output 2: Increased supply of bycatch and tuna from industrial fishing operations for urban and peri-urban communities.

72. Taken collectively, the two sets of activities implemented as part of Component A are designed to ensure that coastal communities and urban / peri-urban communities have better access to tuna and other pelagic fish to help meet their nutritious food availability needs.

73. Achievement of Outputs 1 and 2 will lead to the realisation of ***Outcome 1: improved food availability of vulnerable communities in participating countries.***

Component B: Adaptations to reduce risks to Pacific Island economies from climate-driven tuna redistribution.

74. Current modelling and projections indicate that the effects of climate change on sea surface temperatures will cause an eastward redistribution of major tuna stocks across the Pacific. As a result, these stocks will be found less frequently, and in reduced numbers, within the EEZs of the Western Pacific nations where they currently occur. In the future, tuna will be more commonly found in central and eastern high seas areas of the Pacific. Negative impacts will be felt by Western Pacific nations whose government revenues are dependent on tuna being caught within their EEZs.

75. Recent advances in close-kin mark-recapture fisheries assessment tools[75] are expected to deliver a more accurate understanding of the status and spatial structure of tuna stocks in the WCPO. When this information is combined with improvements to SEAPODYM[76] to increase the spatial resolution and reduce uncertainties in the modelling, it will provide the foundation for creating an Advanced Warning System (AWS) to forecast climate-induced changes in the distribution of tuna biomass among the EEZs of PICs, and between EEZs and high-seas areas.

76. Creation of the AWS will depend in part upon genetic analysis of tuna tissue samples collected across the WCPO and Eastern Pacific Ocean (EPO). By cooperating with industrial fishing companies, major cost savings in conducting the sampling to gather the necessary scientific information will be realised **(Annex 26, Technical Study 7)**.

77. The data gathered through the AWS will be disseminated through reports, notifications, and advisories, which will be issued to the target countries at regular intervals. Scientific rigor in the design and implementation of the AWS will be annually reviewed by the WCPFC Scientific Committee, the results of which will be published in public reports of the Committee. AWS data will also allow for continued strong management of tuna fisheries by the WCPFC and countries under the impacts of climate change.

78. The AWS will provide better information on the extent of negative impacts to tuna-dependent Pacific Island economies from tuna redistribution and the need to develop international legal arrangements to retain the historical level of socio-economic benefits received from tuna in countries' territorial waters.

79. Successful negotiations based on the AWS will depend upon high-level country and regional representatives having a clear understanding of how the AWS can be used to develop legal arrangements to retain historical socio-economic benefits **(Annex 26, Technical Study 8)**. This will be achieved through design and implementation of several targeted training modules, plus provision of technical and legal support for the drafting of multilateral agreements.

80. The activities included within Component B are:

- Develop and deliver an Advanced Warning System (AWS) for tuna redistribution (Activity 3.1).
- Assess the impact of tuna biomass redistribution identified by the AWS on national economies at all levels (Activity 3.2).
- Provide AWS-related training to national institutions to equip them to engage in regional and international negotiations relating to impacts of climate change on the tuna resources historically caught in their EEZs and

[75] https://www.frontiersin.org/articles/10.3389/fmars.2023.1087027/full
[76] https://www.spc.int/ofp/seapodym/

the mechanisms needed to enable the tuna-dependent economies to the socio-economic benefits previously derived from their tuna resources, regardless of climate-driven redistribution of the fish (Activity 3.3).

81. Successful implementation of these activities will lead to achievement of Output 3: *science-based forecasts and projections that reduce uncertainty in climate change-driven tuna redistribution are available to all stakeholders*. Achievement of this Output will lead to **Outcome 2: strengthened capacity of tuna-dependent Pacific Island nations to negotiate for benefits from tuna stocks which are redistributed as a result of climate change.**

Results Chain: co-benefits

82. Co-benefits are important additional or ancillary benefits that are expected to result from successful implementation of the Programme's activities. In addition to the two outcomes described above, it is expected that two co-benefits will be realised through the Programme. These are:

- ***Co-benefit 1: 'Improved livelihoods of vulnerable communities in participating countries.'*** Beneficiaries of improved livelihoods will be fishers, processors, and transhipment vendors. This co-benefit is aligned with GCF Adaptation Results Area **ARA1: Most vulnerable people and communities.**

- ***Co-benefit 2: 'Strengthened management of industrial tuna fisheries by regional and national institutions.'*** The data generated by the AWS will inform the development of future WCPFC CMMs and Harvest Strategies to sustainably manage the stocks of tropical tuna species in the WCPO under the impacts of climate change.[77] This co-benefit is aligned with GCF Adaptation Results Area **ARA 4: Ecosystems and Ecosystem Services.**

Achieving paradigm shift

83. As illustrated in the ToC diagram (**Figure 3**, below), at the highest level, all outcomes and co-benefits will contribute to the overarching goal of the Programme, supporting the achievement of a paradigm shift where PICs are transformed to become more resilient to key climate change threats facing the fisheries sector.

84. In climate adaptation programmes, the overarching impact and paradigm shift may not be realised during the life of the programme. This is because climate impacts and responsive government policies may be realized / deployed over a longer timeframe than the Programme implementation period. To achieve paradigm shift, the Programme must establish the enabling environment that is needed to achieve the transformation to improved climate resilience.

[77] https://www.wcpfc.int/harvest-strategy

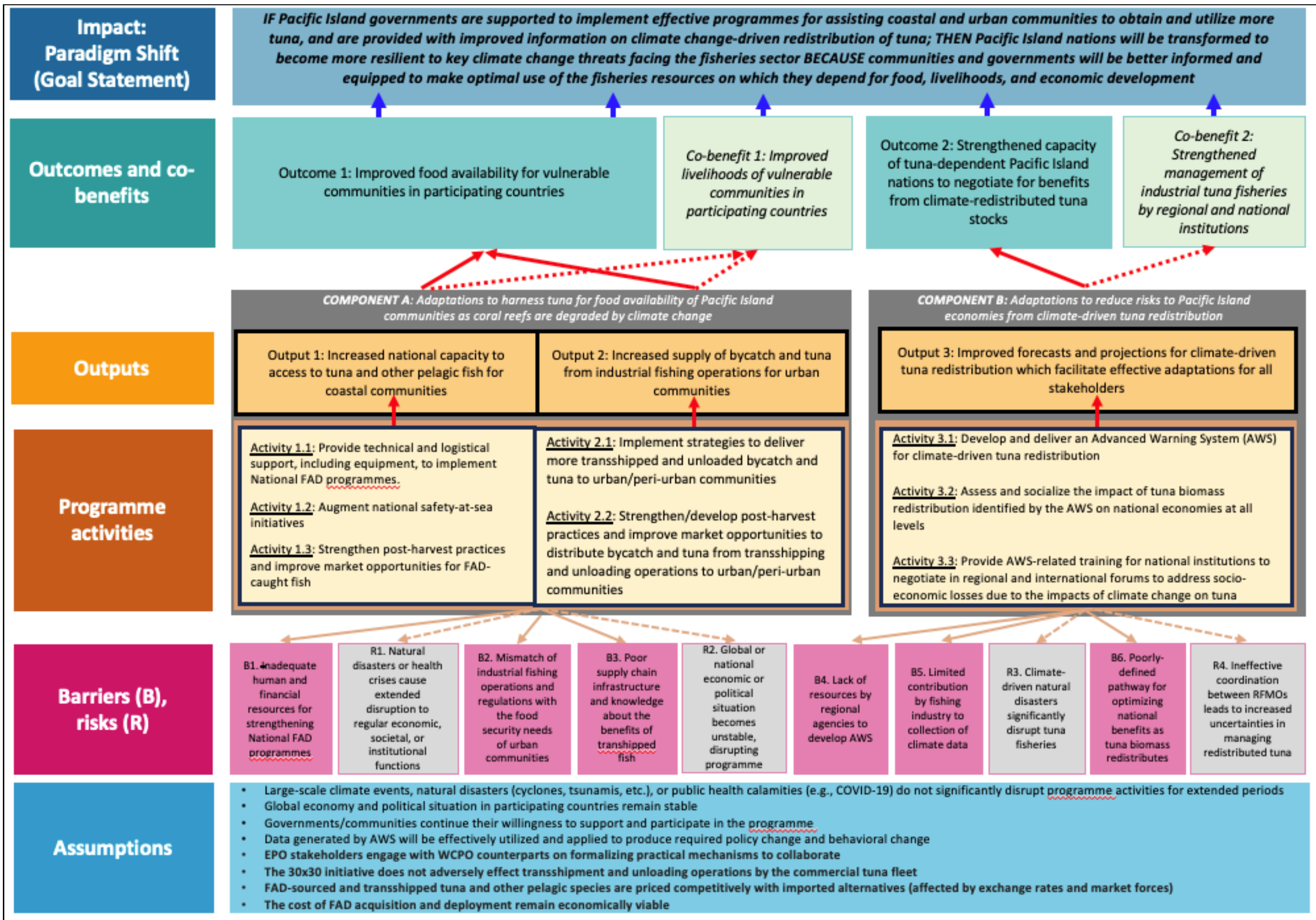


Figure 3: Programme Theory of Change diagram

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Adaptation Results Area (ARA 1-4)			
	ARA 1 Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1: Improved food availability of vulnerable communities in participating countries.	☒	☒	☐	☐
Outcome 2: Strengthened capacity of tuna-dependent Pacific Island nations to negotiate for benefits from climate-redistributed tuna stocks.	☒	☐	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1: Improved livelihoods of vulnerable communities in participating countries.	☐	☒	☒	☐	☐	☐
Co-benefit 2: Strengthened management of industrial tuna fisheries by regional and national institutions.	☒	☐	☒	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Programme Activities for each of the Programme Outputs are presented in this section. To find additional details on sub-activities and deliverables, please refer to the Feasibility Study (**Annex 02, Chapter 4, section 4.2**). **Table 4.5 in Section 4.2.3.3 of Annex 02** provides a summary of the participation of each of the 14 GCF countries in each Regional Tuna Programme activities.

Component A:

Output 1: Increased national capacity to access tuna and other pelagic fish for coastal communities.

Activity 1.1: Provide technical and logistical support, including equipment, to implement National FAD programmes.

85. Expanding the use of anchored, artisanal fish aggregating devices (FADs[78]) will provide the infrastructure needed to increase access to fish required for food availability (access to high quality food / nutrition security) of vulnerable coastal communities (**Annex 26, Technical Study 2**)[79], due to degradation of coral reefs and population growth (**Annex 2, Chapter 2** and **Annex 26, Technical Study 1**). Increasing the number of FADs in Pacific Island countries will provide a greater number of small-scale fishers with the opportunity to catch tuna and other pelagic fish species (hereafter 'tuna') efficiently and safely.[80] This Activity will assess the gaps to be filled to increase access to tuna for the target beneficiaries in each country documented in **Annex 26, Technical Study 3, Table 13**. National FAD programmes will then be strengthened to fill these gaps in ways that meet global standards, as described in **Annex 26, Technical Study 3**. This will be done through provision of materials, equipment, training, and capacity-building to ensure that FADs are appropriately constructed, deployed, utilised, and maintained to maximise the benefits of the investments for the target communities, and minimise any effects on non-target species, by applying the 'best practice' methods developed by SPC.[81] Small-scale fishers are equipped to fish safely around FADs to harness their full potential. Under this Activity, enumerators will also be hired to measure catch rates around a subset of FADs deployed in each country to determine the 'level of benefit' (number of additional fish meals) for beneficiaries (see **Annex 23**). In the smaller participating countries, this Activity will supply additional tuna to the entire population. In the larger countries, where the target beneficiaries are a subset of the national population, it will create a platform for further expansion of FAD infrastructure, supported by allocations from the national budget and/or development partners.

86. Across the 14 participating countries, more than 300 FADs will be deployed and routinely replaced as needed under this Activity. Train-the-trainer sessions will be conducted with more than 200 staff from national fisheries agencies in FAD rigging and deployment, and FAD-fishing methods. In turn, these trainers will train at least 800 small-scale fishers/community members across the region. In addition to measuring the contribution of FADs to food availability mentioned above, regular monitoring of the performance of FAD infrastructure and possible interactions between non-target species and FADs/FAD fishing will also be included in this Activity. Consideration will be given to the different roles played by men and women in increasing access to tuna for domestic food availability so that national FAD programmes are augmented in a gender-responsive manner.

87. The sub-activities of Activity 1.1 are detailed in **Annex 02, Chapter 4.** SPC will conduct these sub-activities with support from subcontracted consultants or services providers, and SPC will provide sub-grants to national governments to support the participation of national fisheries agencies in this Activity.

The sub-activities under Activity 1.1 are:

Sub-activity 1.1a: Audit progress towards requirements for scaling-up National FAD Programmes in the 14 participating countries.

[78] Fish aggregating device (FAD) is a generic term that applies to a range of floating objects that aggregate fish. The industrial tuna surface fishery (purse-seine and pole and line vessels) take advantage of fish aggregations beneath naturally occurring floating objects such as logs that are disgorged from rivers. The industrial purse-seine fishery also deploys floating rafts that drift with the currents to aggregate tuna. Anchored FADs are also used by the industrial fleets. Coastal small-scale fishers use anchored FADs to increase their catch rates for tuna and other large pelagic fish in nearshore waters, see https://www.spc.int/DigitalLibrary/Doc/FAME/Brochures/Anon_12_PolicyBrief19_FADs.html for further details about the differences between industrial and artisanal FADs.

[79] Technical Studies for the Programme are available at: https://fame1.spc.int/technical-studies-support-funding-proposal-green-climate-fund-regional-tuna-programme

[80] Bell et al. 2015. Optimising the use of nearshore fish aggregating devices for food security in the Pacific Islands. Marine Policy 56, 98-105. https://www.sciencedirect.com/science/article/pii/S0308597X15000457

https://www.spc.int/DigitalLibrary/Doc/FAME/Brochures/Anon_17_PolicyBrief31_FAD_Programmes.html

Manual on anchored fish aggregating devices (FADs): An update on FAD gear technology, designs and deployment methods for the Pacific Island region https://www.spc.int/resource-centre/publications/manual-on-anchored-fish-aggregating-devices-fads-an-update-on-fad-gear

Sub-activity 1.1b: Develop workplans for scaling-up National FAD Programmes based on the audit.

Sub-activity 1.1c: Review national policies and regulations to identify barriers to the equitable and sustainable use of FADs.

Sub-activity 1.1d: Design and implement capacity development activities to augment the skills of national staff to implement FAD programmes.

Sub-activity 1.1e: Design and implement a gender-responsive consultative stakeholder engagement strategy for national fisheries agencies to utilise in consultations with communities to identify suitable FAD deployment sites.

Sub-activity 1.1f: Procure materials to expand and maintain national FAD programmes.

Sub-activity 1.1g: Enable small-scale fishers to catch tuna and other pelagic fish around FADs that integrate traditional knowledge.

Sub-activity 1.1h: Improve the capacity of national FAD programmes to effectively collect and utilize catch data.

Sub-activity 1.1i: Establish/strengthen national response mechanisms to natural disasters affecting small-scale fishers using FADs.

Activity 1.2: Augment national safety-at-sea initiatives

88. The Programme will emphasize safety at sea, because some FADs will be deployed a considerable distance (up to 7 km) offshore, a different fishing environment for small-scale fishers accustomed to fishing within lagoons and relatively sheltered inshore coastal waters. This Activity will ensure that the risks for small-scale fishers operating around FADs are reduced by augmenting national safety-at-sea initiatives: First, by evaluating the suitability of small vessels currently used for fishing around FADs and providing governments with improved vessel designs, and practical options for regulating and promoting construction/purchasing of such vessels **(Annex 26, Technical Study 12)**. Second, by integrating sea conditions into national meteorological agencies' forecasts and identifying options for dissemination of these improved forecasts to small-scale fishers and coastal communities via mobile phone and other methods. Third, by providing training for small-scale fishers in the use of the boating safety equipment and associated regulatory reform to mandate its use. SPC will implement this activity with support from sub-contracted service providers, likely including the Food and Agriculture Organization of the United Nations (FAO).

The sub-activities included in Activity 1.2 are:

Sub-activity 1.2a: Conduct a needs analysis for improved vessel safety for small-scale fishers using FADs.

Sub-activity 1.2b: Establish the enabling conditions for national meteorological services to provide improved weather and extreme event warnings to small scale fishers.

Sub-activity 1.2c: Support procurement and delivery of boating safety equipment, linked to vessel registration where practical.

Sub-activity 1.2d: Increase national capacity to regulate and provide training in the use of the boating safety equipment.

Activity 1.3: Strengthen post-harvest practices and improve market opportunities for FAD-caught fish.

89. This Activity will provide training and support to coastal community representatives in post-harvest methods for tuna and associated species caught around FADs, and to develop supply chains and market opportunities for these fish. Improving the handling of tuna catches from FADs will extend the shelf life of the fish for communities catching tuna and other pelagic fish for subsistence consumption. Design of the training programme will commence with a needs assessment to determine the needs of the subsistence households and MSMEs receiving tuna from FADs. SPC, with support from subcontracted consultants or services providers, will conduct training on a variety of processing methods, including traditional knowledge where

appropriate, (for example, increasing the supply of ice, drying, smoking, salting, pickling, and canning) and be tailored to meet the specific needs and conditions prevalent in the target communities. The training will be developed and conducted in a transparent and gender-responsive manner that acknowledges the differing roles of men and women in beneficiary communities. Officials from the National Fisheries Agencies will be familiarised with post-harvest methods to provide further support to beneficiary households and MSMEs in post-harvest processing. National Fisheries Agencies will provide GCF-funded post-harvest equipment from SPC to priority communities via Programme subgrants, in compliance with GCF requirements.

The four sub-activities under Activity 1.3 are:

Sub-activity 1.3a: Provide training courses for subsistence households and MSMEs to improve preservation of FAD-caught fish using post-harvest methods, e.g., drying, smoking, and bottling, incorporating traditional knowledge.

Sub-activity 1.3b: Provide coastal communities with basic equipment to apply post-harvest methods, including practical options for cold storage where appropriate.

Sub-activity 1.3c: Identify and promote market opportunities for FAD-caught fish.

Sub-activity 1.3d: Conduct communication campaigns to raise awareness of coastal communities about the impacts of climate change on coral reef fish and the need to consume more FAD-caught tuna for community food availability.

Output 2: Increased supply of bycatch and tuna from industrial fishing operations for urban and peri-urban communities.

Activity 2.1: Implement strategies to deliver more transhipped and unloaded bycatch and tuna to urban/peri-urban communities.

90. This activity is designed to identify and address weaknesses in the supply chain for fish that come ashore from industrial tuna-fishing vessels while they are in port. These fish can provide urban and peri-urban communities with a nutritious source of protein at low cost, provided the quality of the fish is maintained throughout delivery and sale. These fish include bycatch caught by purse-seine fishing which are not accepted by tuna canneries because they are too small, damaged, or not the desired species. These fish are separated from the high-quality tuna when these vessels tranship their catch to carrier vessels in regional ports and lower-value fish from longline vessels that are also offloaded in port. The main transhipping ports where this Activity will be implemented are in FSM, Kiribati, Marshall Islands, Solomon Islands, and Tuvalu (and PNG, see below) (**Annex 26**, **Technical Study 5**).

91. The opportunity for trade in bycatch from purse-seine vessels has arisen from WCPFC's requirement that transhipping must occur within regional ports. The current availability of bycatch is insufficient to meet the fish consumption needs of rapidly growing urban populations in some countries, e.g., Solomon Islands (**Annex 26**, **Technical Study 5**). In other countries, the volume of bycatch exceeds local demand for food but provides the opportunity to improve food availability indirectly through use as a major ingredient in feeds for chickens and pigs and in fertilisers to improve the quality of soil for market gardening in atoll nations. In all cases, improvements to the volume of bycatch available and targeting its use will deliver important socio-economic benefits to urban communities.

92. The sub-activities in Activity 2.1 will identify and remove barriers to the supply of bycatch and tuna, thereby increasing food availability. Purse-seine bycatch is also offloaded in PNG when tuna is delivered for canning in Lae, Madang, and Wewak. However, it is already distributed effectively to nearby areas and the volume of bycatch exceeds the basic needs of the local population (**Annex 26**, **Technical Study 5**). Therefore, the Programme is not planning to include PNG in Activity 2.1. It will, however, be included in Activity 2.2. FFA will support the implementation of this Activity. SPC will enter into a grant agreement with FFA and CSIRO to support implementation of this activity, and sub-contracted service provider(s) specialising in marine food

systems will provide additional technical support. Nationally and regionally administered fisheries observer programmes (including PNAO or its third-party provider) will measure the increase in volume of bycatch delivered, and enumerators engaged under Activity 1.1 will collect and report data on the number of individuals involved in the trade of bycatch. National Health, Food Safety, Bio-Security and Maritime authorities from the participating countries will participate in the regulatory reform needed to facilitate improved access to bycatch and tuna for domestic consumption.

The sub-activities included in Activity 2.1 are:

Sub-activity 2.1a: Assess the supply of bycatch and tuna available for offloading at each transhipping and unloading port.

Sub-activity 2.1b: Evaluate the projected shortfalls in the supply of fish needed for nutritious food availability in urban and peri-urban communities by 2030 and in following decades.

Sub-activity 2.1c: Use the Advanced Warning System (AWS) (see Activity 3.1.1 below) to assess the implications of tuna biomass redistribution for transhipping and unloading activities across the region.

Sub-activity 2.1d: Build national capacity to conduct policy analysis on current and future transhipment and unloading of bycatch and tuna.

Sub-activity 2.1e: Develop procedures and regulations to increase availability of transhipped and unloaded bycatch and tuna where needed to fill the gap in fish supply.

Activity 2.2: Strengthen/develop post-harvest practices and improve market opportunities to distribute bycatch and tuna from transhipping and unloading operations to urban/peri-urban communities.

93. This Activity will provide training and support in post-harvest methods, and expanding markets, for bycatch and tuna landed in transhipping and unloading operations to urban/peri urban MSMEs. The activity will both extend the shelf life of the fish, and add value to bycatch and tuna sold by MSMEs. This activity differs from Activity 1.3 in the scale of MSME supported (due to the much larger volume of fish available for post-harvest processing) and access to infrastructure and essential services available to MSME in urban areas.

94. Training design will commence with a needs analysis to determine the needs of the MSMEs within the improved supply chains for bycatch and tuna resulting from Activity 2.1. The training will incorporate the processing methods requested by MSMEs and provide some of the equipment they need to process fish. The different roles played by men and women in post-harvest processing activities will be taken into consideration and the training will be developed and conducted in a transparent and gender-responsive manner The range of post-harvest methods available for bycatch offloaded from purse-seine vessels during transhipping operations are expected to be more limited than those for fresh or frozen tuna unloaded from longline vessels, because the fish from purse-seine vessels will have been stored in brine.

95. Activity 2.2 includes four sub-activities expected to increase demand for bycatch and tuna from transhipping and unloading activities: developing pilot marketing mechanisms, conducting communication campaigns about the benefits of bycatch and tuna (**Annex 26, Technical Study 6)**, and providing governments with fish market outlet designs at various scales to seek financial support to improve the sanitary conditions under which the fish are sold.[82]. FFA's Fisheries Development Division, under technical direction of SPC, will support the implementation of this Activity. National governments of ten countries will facilitate access and introductions to local networks and responsible government agencies, including market authorities.

The sub-activities included in Activity 2.2 are:

Sub-activity 2.2a: Provide training for urban communities to improve/develop post-harvest processing techniques for bycatch and tuna from transhipping and unloading operations.

[82] Solo, M.K. et al. 2023. [Assessment of Postharvest Practices of Tuna Sold at the Honiara Fish Market in the Solomon Islands (hindawi.com)]

Sub-activity 2.2b: Pilot alternative marketing mechanisms to support increased trade in bycatch and tuna in urban areas.

Sub-activity 2.2c: Conduct communication campaigns to raise awareness of urban/peri urban communities about the impacts of climate change on coral reef ecosystems and the need to consume more bycatch and tuna to meet future nutrition requirements.

Sub-activity 2.2d: Provide fish market outlet designs at various scales for countries where transhipping and unloading occurs.

Component B:

Output 3: Improved forecasts and projections for climate -driven tuna redistribution which facilitate effective adaptations for all stakeholders.

Activity 3.1: Develop and deliver an Advanced Warning System (AWS) for climate-driven tuna redistribution.

96. This Activity will deliver an Advance Warning System (AWS) on tuna redistribution using the latest in data analyses, communications, and climate & fisheries sciences. The AWS will provide decision-ready technical tools for countries to develop practical adaptations that protect their economies from disruptions to income streams and oceanic food systems caused by the redistribution of tuna biomass away from their EEZs due to climate change (see Activities 3.2 and 3.3).

97. Significant co-benefits will also be realized through the AWS, including long-term sustainability of Pacific tuna stocks and oceanic biodiversity through improved stock assessments and CMMs adopted by the WCPFC and the IATTC. SPC will lead the implementation of this activity with support from Technical Partner Grantees CSIRO and FFA and sub-contracted consultants and service providers (PNAO and other relevant institutions). SPC will provide the technical capacity to provide high resolution tuna-climate models (SEAPODYM) needed for developing and testing adaptations that minimise impacts of climate change on tuna dependent economies. CSIRO will support the technical transition of current fisheries monitoring activities to include genome-based approaches that are necessary for the AWS to provide absolute measures of change in tuna biomass at the EEZ scale to establish baselines for attributing climate change impacts. FFA will provide the technical capacity to modify SEAPODYM to support the coupling of a fleet dynamics model. CSIRO and the sub-contracted service providers will support the preparation of validated CMIP ensembles for forecasting and projecting the impacts of climate change.

The sub-activities included in Activity 3.1 are:

Sub-activity 3.1a. Transition existing fisheries and ocean monitoring systems to produce higher-resolution forecasts and projections of tuna biomass redistribution.

Sub-activity 3.1b. Establish baselines and indicators for quantification of climate change impacts on distribution of tuna biomass.

Sub-activity 3.1.c Enhance collection and curation of physical oceanography and micronekton data to inform the modelling of climate-driven redistribution of tuna biomass.

Activity 3.2: Assess and socialise the impact of tuna biomass redistribution identified by the AWS on national economies at all levels.

98. To improve adaptations and Conservation and Management Measures (CMMs) that secure national and regional benefits derived from the tuna industry, this Activity will develop and couple a fleet dynamics model to SEAPODYM, providing the AWS with the capability to perform bio-economic analyses. This will allow the AWS to simulate short-term and long-term changes in the productivity and locations of Pacific tuna fisheries under future climate variability and climate change scenarios and model associated economic impacts. This information will be used develop Regional and National forecasts and outlooks that provide advanced warning on the implications of impending climate extreme events and long-term guidance on the impacts of climate-driven tuna redistribution for Pacific Island economies. The economic and fleet dynamics modelling,

and capacity building of national agencies will be supported by technical experts from FFA staff or consultants engaged by FFA. Training of national counterparts to use the AWS to prepare national forecasting and resource outlooks will be provided to increase national and regional capacity.

Activity 3.2 includes one sub-activity:

Sub-activity 3.2a. Conduct bio-economic and fleet dynamics modelling to estimate changes in tuna catch and associated socio-economic benefits.

Activity 3.3: Provide AWS-related training to national institutions to negotiate in regional and international forums to address socio-economic losses due to the impacts of climate change on tuna.

99. This Activity will operationalise the AWS within national administrations and regional agencies. Operationalisation will focus on training national counterparts to use the AWS to develop and evaluate adaptation options that minimise the risk to Pacific Island economies from the impacts of climate-driven tuna redistribution (this includes attribution and adaptation evaluations). The activity will provide a professional development programme for 12 Pacific Island Fisheries Professionals[83] engaged by SPC and FFA (6 per organization) to use the AWS to develop scientific and economic adaptations to the impacts of climate change. In parallel, the AWS will be used to assemble the information necessary for PICs to contribute significantly to the ocean impacts section of the IPCC Assessment Report 7 cycle and negotiate for retention of historical fishing license revenue. This will include the preparation of annual report cards (endorsed by WCPFC, SPC-HoF, FFC, PNA, RFMM) on the bio-physical impacts of climate change and the economic impacts of climate change on tuna resources. FFA will support the implementation of this activity.

The sub-activities included in Activity 3.3 are:

Sub-activity 3.3a. Academic and vocational training to increase the number of Pacific Island fisheries and climate staff with enhanced capabilities to negotiate to retain the national benefits received from tuna.

Sub-activity 3.3.b. Assemble evidence for PICs to use in negotiations at regional and global scales to address the impacts of climate-driven tuna redistribution.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

100. The 15th Pacific Community Heads of Fisheries meeting (HoF15) stipulated that this GCF programme have strong country-led governance (SPC, the sole Executing Entity represents the interests and priorities of its Member states) and endorsed the high-level governance and administrative structure described in this section[84]. To ensure responsibilities are clearly understood during Programme implementation, HoF15 reiterated the need for clear separation of the AE responsibilities from those of the EE and partners implementing Programme activities. This is consistent with CI's practices of segregating the roles of AE and EE and aligns with GCF requirements. The requirement for member-led governance was also emphasised in the meeting, while it is acknowledged that the GCF requires the EE to retain final approval authority over any matter submitted to, or decision adopted by, the Programme's governance bodies.

Accredited Entity

101. Conservation International Foundation (CI), through its CI-GCF Agency, will serve as the Accredited Entity (AE) for the Programme. The CI-GCF Agency will be responsible for oversight of this Programme as defined in the Accredited Master Agreement between the GCF and CI, including technical, financial, and administrative monitoring and supervision (through reporting, audits, and periodic site visits) and review and approval of the Executing Entity's (EE) annual workplans and budgets. CI-GCF will also be responsible for providing support and guidance to the EE as mutually agreed; monitoring of the achievement of Programme results and Outputs; reporting to the GCF; and Programme closure and evaluation. CI-GCF will conduct

[83] Pacific Island Fisheries Professionals are short-term positions competitively selected to provide technical support to the Project while providing professional development. Nationals and residents of the Programme countries that are employed or contracted by government or NGO institutions in the field of fisheries science, management or monitoring in a Pacific Island country are eligible to participate.
[84] https://fame-archive.spc.int/en/meetings/262

these responsibilities, and disburse GCF funds to the EE, in line with CI's Accreditation Master Agreement (AMA) with the GCF. CI-GCF currently serves as AE for three approved GCF projects under implementation.

Executing Entity (EE)

102. The Pacific Community (SPC) will be the Executing Entity (EE) for all activities for the Programme. SPC will be responsible for execution of the Programme; management and monitoring of subsidiary agreements with sub-awardee partner organisations, Programme countries, and service providers receiving GCF funds and their related activities; reporting to the AE; and collaborating with Programme country governments to optimise alignment with national policies and contributions to achieve Programme outcomes, GCF-level goals, and sustainability of Programme interventions.

103. SPC is an international intergovernmental organisation established by treaty (Canberra Agreement) in 1947. SPC is governed by its 27 member countries to service the development, scientific and technical needs of the Pacific region. SPC's membership includes 22 Pacific Island countries (PICs) and territories, 14 of which will participate in this Programme. SPC's headquarters is located in Noumea, New Caledonia, and it has regional offices in four of the Programme countries. SPC implements a broad range of programmes spanning more than 20 sectors, addressing sustainable economic development, sustainable natural resource and environmental management, and human and social development. Recent large-scale programmes include:

- Pacific Women Lead programme through SPC's Human Rights and Social Development (HRSD) division. AUD$58 million; 2021- 2026; 10 PICTs
- Climate Change Flagship & Climate Science for ensuring Pacific Tuna Access. USD 35 million; 2022-2026; all 22 PICTs
- Pacific European Union Marine Partnership Programme. USD50 million; 2018-2025; for 14 PICTs

SPC's Fisheries, Aquaculture and Marine Ecosystems (FAME) division has been delivering fisheries, aquaculture, and marine ecosystems services to its members for more than 60 years. SPC's goal is that the fisheries and aquaculture resources of the Pacific region are resilient and managed sustainably for economic growth, food security, and cultural and environmental conservation. SPC is a GCF Accredited Entity with policies and procedures in place to meet GCF fiduciary standards.

104. SPC will execute the Programme in line with its policies, procedures and GCF requirements when using grant funds. It will recruit staff and contract experts required to support effective and efficient delivery of Programme outcomes, as appropriate. SPC will also enter into a variety of GCF-compliant legal arrangements for this Programme. SPC will enter into agreements that are compliant with SPC, CI, and GCF requirements with organisations serving as Implementing Partners (IPs) for the Programme (see details below).

105. SPC's execution of this Programme will be led by its FAME Division, which led design of the Programme. The Director's Office of FAME will have primary responsibility for Programme execution and will house full-time Programme staff and will be supported by technical expertise from FAME's Coastal Fisheries and Aquaculture Programme (CFAP) and Oceanic Fisheries Programme (OFP). FAME's current programming budget is approximately $25M/year.

106. SPC will establish and host a central Programme Management Unit (PMU) as well as other full-time Programme staff. The PMU and Programme team will consist of SPC staff and will be headed by a full-time Programme Director, who will be responsible for Programme delivery and coordination with all stakeholders. The Programme Director will be supported by two Programme Deputy Directors, who will each lead technical direction and management for one Component, as well as technical and operational staff dedicated full-time to the Programme.

107. The Programme staff will be responsible for executing the Programme and all of its components. PMU will be responsible for, *inter alia,*

- Coordinating, liaising to, and providing secretariat services to the Programme Steering Committee
- Liaising and collaborating with the AE to ensure efficient and effective Programme delivery
- Managing progress and financial reporting (including on co-financing) to the AE and GCF

- Managing and monitoring the Programme in line with Programme budget and workplans, and in accordance with GCF guidelines and policies
- Managing and monitoring subawards and legal agreements with Implementation Partners
- Coordination with, and ensuring the exchange of information, learning and knowledge across, the PICs and IPs
- Providing guidance, expertise, capacity-building, and monitoring as needed on Programme management, financial management, procurement, and technical implementation matters.

108. SPC will also be supported by part-time contributions from technical and operational experts across the organisation, including from Geoscience, Energy and Maritime Division (GEM); Climate Change and Environmental Sustainability (CCES); Human Rights and Social Development Division (HRSD); Public Health Division (PHD); and Statistics for Development (SDD) (see **Annex 04**).

109. The Programme Director will also liaise with members of the Programme Steering Committee, technical experts, government staff, civil society, other bilateral or multilateral programmes or projects with activities that may have relevance to the Programme, and other stakeholders to coordinate the implementation of proposed Programme activities with the aims of quality assurance and complementarity to other initiatives in the programme geographies.

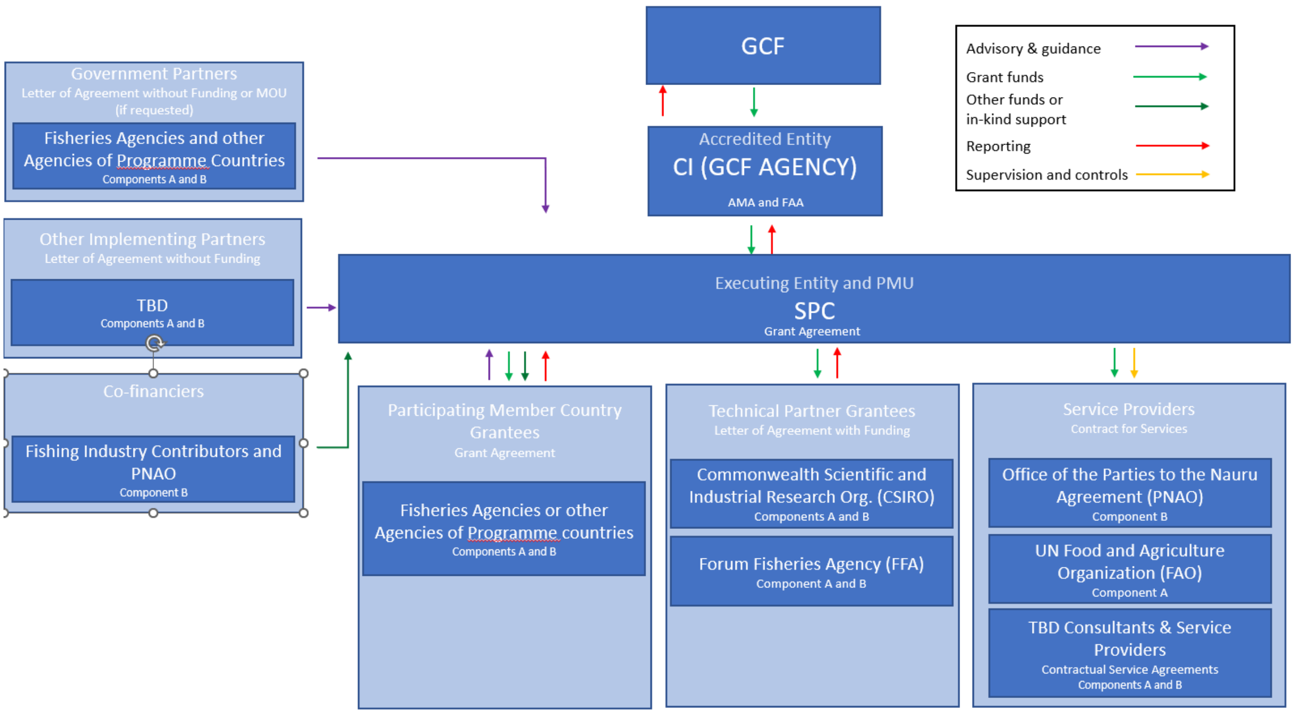


Figure 4. Financial and Contractual arrangements for the Regional Tuna Programme

Implementing Partners

110. As EE, SPC will engage with a wide array of Implementing Partners to implement or support implementation of Programme activities or sub-activities to execute a programme of this scope and scale. Of those IPs, SPC will provide GCF resources (funding and/or GCF-funded goods or services) to Technical Partner Grantees, Participating SPC Member Grantees, and Service Providers (collectively, "Sub-awardees"). SPC may also coordinate with Implementing Partners who do not receive GCF funds or resources.

Technical Partner Grantees

111. SPC will sub-grant GCF funding to Technical Partner organisations to support implementation of Programme Activities or sub-Activities. SPC will enter into a Letter of Agreement with funding (its sub-granting legal instrument that includes terms related to co-financing) with each of the Technical Partners, which will include contractual terms to ensure compliance with SPC, CI, and GCF requirements. SPC will manage and monitor

these subgrants including for ESS compliance, will approve grantees' workplans and budgets, and will have ultimate approval authority over their activities. Both partners will also provide in-kind staff time as co-financing to the project.

112. Australia's Commonwealth Scientific and Industrial Research Organisation (CSIRO) will support SPC implementation of certain sub-activities under Activities 2.1 and 3.1. CSIRO, Australia's national science agency, supports comprehensive research in the oceans and participates in regional and international science collaborations, particularly in relation to sustainable fisheries. Its research includes cutting-edge methods for stock assessment and management, including related to tuna. CSIRO will deliver updated projections on the gap in protein needs to be filled by bycatch and tuna, assist SPC establish baselines and indicators for quantifying the impacts of climate change on tuna through application of their genomic based fisheries management technologies, and develop advanced ocean monitoring tools.

113. The Pacific Islands Forum Fisheries Agency (FFA) will support implementation under Outputs 2 and 3. FFA is a trusted partner of SPC and contributed to the design of this Programme. FFA has been efficiently providing economic, legal, fisheries management and development assistance to its 15 member Pacific Island countries and a single territory since 1982. FFA works to strengthen national capacity and regional solidarity so its members can manage, control, and develop their tuna fisheries, including to address challenges posed by climate change. FFA will build MSME capacity to advance opportunities for value-adding to transhipped bycatch; and to develop, couple, and apply the economics component of the AWS.

114. SPC may also engage additional, TBD Technical Partners to support Programme activities during implementation as part of adaptive management for this complex multiyear programme, which will be selected in compliance with SPC's policies, which include but are not limited to SPC's Grants Policy or Procurement Policy and approved by the AE. Note that Technical Partners do not have discretion in implementing the Funded Activities; the EE (SPC) has final authority in decisions regarding the implementation of Activities and the use of GCF Proceeds.

Participating SPC Member Grantees

115. SPC will also provide GCF funding and/or GCF-funded goods or services to each of the 14 PICs to support each member country's participation in Programme activities. These resources will be provided for staff, consultants, service providers, goods and equipment, workshops, and travel under Activities 1.1, 1.2, 2.1, 2.2 for all PIC grantees as well as Activity 3.1 for some grantees (see details in **Annex 04** and **Annex 25**). GCF resources will be provided by SPC to Participating SPC Member grantees via cash and/or as goods or services procured by SPC using GCF funds, as negotiated between SPC and each member country and consistent with SPC grant risk assessments and policies and procedures. The EE will enter into a grant agreement with one entity in each member country with terms related to the provision of cash and in-kind goods and services; the agreement will include contractual terms to ensure compliance with SPC, CI, and GCF requirements, including ESS. SPC will manage and monitor these subgrants, will approve grantees' workplans and budgets, and will have ultimate approval authority over their activities conducted using GCF funds.

Service Providers

116. The UN Food and Agriculture Organization (FAO) will provide technical services to support the implementation of Activity 1.2. The FAO Subregional Office for the Pacific Islands (FAO SAP) supports 14 countries across the Pacific in addressing food availability, nutrition, agriculture, and rural development priorities and expertise for safety-at-sea initiatives. It has a track record of providing technical expertise and partnership to the region, a network of partners across the programme geography in the public and private sectors, comprehensive knowledge base on fishing vessel design, professional staff in fisheries, and ongoing technical programs supporting regional fisheries.

117. The Office of the Parties to the Nauru Agreement (PNAO) will provide technical services to support the implementation of Activity 2.1. The eight Pacific Island countries that are the Parties to the Nauru Agreement (together with Tokelau) manage the largest tuna fishery in the world. SPC will engage the PNA Office (PNAO) or its third-party provider to provide technical services to update activities of regional observer and port programmes for transhipment to estimate locally redirected tuna and bycatch.

118. SPC will engage additional consultants, vendors, and service providers to support the implementation of its Programme activities (please see **Annex 04** and **Annex 25** for more details). SPC will follow its procurement policy in procuring these goods and services and will enter into a contract for services, contract for goods, or purchase order with each contracted entity whereby SPC retains ultimate approval authority over activities conducted using GCF funds. FAO and PNAO have no discretion in implementing the Funded Activities and the usage of the GCF Proceeds; SPC, as the Executing Entity, has the final authority in decisions regarding the implementation of all activities as well as the use of the GCF Proceeds.

Co-financiers

119. PNAO and private-sector industrial fishing companies will provide in-kind co-financing support to the Advanced Warning System activities in Activity 3.1 by providing observer services, vessel time, and data. SPC will conduct first-level due diligence with each of these co-financing entities and enter into either an MOU or Letter of Agreement without funding from each entity. The EE and both Technical Partner Grantees will both provide co-financing (staff time and goods & equipment) as well.

Other Implementing Partners

120. SPC will also engage with relevant regional or technical organisations or entities for technical guidance or other collaboration, but which will not receive any GCF funds or resources. Example entities include Pacific Climate Change Centre (PCCC) at the Secretariat of the Pacific Regional Environment Programme (SPREP). SPC may enter into Letters of Agreement with such entities, with no transfer of funds, to define respective roles and responsibilities.

121. SPC and Participating SPC Member Grantees will work extensively with local beneficiary communities in project implementation, including by providing training, particularly in Output 1. As part of this training, equipment purchased by SPC and/or the Grantee and used for community trainings may then be transferred to the beneficiary communities. Transfer of any such goods will be documented by SPC's Assets External Transfer Form (or Grantee equivalent if deemed acceptable by SPC) that will include required CI and GCF terms on use of GCF-funded goods. Disposal and/or handover of all durable assets acquired with GCF proceeds will be undertaken through a disposition plan consistent with GCF policy and AMA clause 23.03.

Programme Governance

122. The 14 PICs will engage in the Programme through their designated national fisheries agencies (ministries, departments, or statutory authorities). GCF NDAs will ensure that activities implemented by the Programme align with strategic national objectives and priorities (see **Section D.5**). SPC will engage the NDAs throughout Programme implementation and will provide reporting on the status of Programme implementation and impacts. If requested by the PIC, SPC will enter into a Letter of Agreement or MoU to govern national-level Programme implementation with PICs during Programme inception.

Participating country	NDA	National Fisheries Agency / Participating SPC Member Grantee
Cook Islands	Climate Change, Cook Islands Division of the Office of the Prime Minister	Ministry of Marine Resources
Federated States of Micronesia	Department of Finance and Administration	Department of Resources and Development, and National Oceanic Resource Management Authority
Fiji	Office of the Prime Minister	Ministry of Fisheries
Kiribati	Department of Finance and Economic Development	Ministry of Fisheries Marine Resources and Development
Marshall Islands	Office of Environmental Planning and Policy Coordination	Marshall Islands Resources Management Authority
Nauru	Department of Foreign Affairs and Trade	Nauru Fisheries and Marine Resources Authority

Niue	Ministry of Finance and Planning	Ministry of Natural Resources
Palau	Office of the President	Ministry of Agriculture, Fisheries and Environment
Papua New Guinea	Office of Climate Change and Development Authority	National Fisheries Authority
Samoa	Ministry of Finance	Ministry of Agriculture and Fisheries
Solomon Islands	Ministry of Environment, Climate Change, Disaster Management and Meteorology	Ministry of Fisheries Marine Resources
Tonga	Ministry for Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications	Ministry of Fisheries
Tuvalu	Government of Tuvalu	Tuvalu Fisheries Department
Vanuatu	Ministry of Climate Change, Change Adaptation, Meteorology, Geo-Hazards, Environment, Energy and Disaster Management	Vanuatu Fisheries Department

Table 4. Primary Government agency contacts in each of the 14 participating countries

123. The national fisheries agencies may establish an inter-ministerial National Coordination Committee (NCC) that will, *inter alia*, review annual work plans and receive quarterly reports on progress and expenditure. In such cases, the national fisheries agencies will be accountable to the PMU and to the NCC for Programme progress and will submit scheduled progress reports to the PMU.

Programme Steering Committee

124. Programme governance will be headed by a Programme Steering Committee (PSC) that will meet at least once annually throughout Programme implementation and reach decisions by consensus. The PSC will consist of a senior official from each of the PICs. Meetings of the PSC will be organised and facilitated by the Programme Director. The principal functions of the PSC will be to provide strategic guidance and support adaptive management of Programme implementation, review progress and evaluation reports, discuss problems or strategic issues that might arise during implementation, and provide support for the necessary inter-governmental coordination of, and contributions to, Programme activities. The Programme's annual reports will be shared with PSC members and will be made freely available to other government departments. The PSC will provide guidance to the EE, which will have final decision-making responsibility for the Programme, per GCF requirements. The PMU will provide logistical and administrative support to the PSC. The PSC may adopt its own rules of procedure, if appropriate, and will follow a mutually agreed PSC Terms of Reference.

Regional oversight and coordination

125. Regional oversight will be achieved through annual reporting on Programme achievements and issues arising to SPC's Heads of Fisheries (HoF) meeting and to FFA's Forum Fisheries Committee (FFC). All 14 PICs are represented in these bodies at a head of agency/senior official's level. The PMU will coordinate national and regional reporting to these forums and report key outcomes to the AE.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

126. PICs collectively contribute <0.003% of global GHG emissions each year but are recognised as the most vulnerable to climate change, not only to sea-level rise but also to the effects of climate impacts on coral reefs and redistribution of tuna. At the same time, the 14 participating countries have relatively small Gross Domestic Products[85] and tax bases. GCF grant funding is essential for enabling Pacific Island countries to adapt due to the limited financial resources at their disposal.

[85] https://sdd.spc.int/dataset/df_national_accounts

127. Grant financing is justified for this Programme because of the acute vulnerability to climate change of PICs and their populations and the limited financial resources of these countries. Many PICs have a very small tax base and prioritise the allocation of government revenue, including the substantial proportion derived from tuna-fishing access fees, to meet their nation's basic needs, particularly health, and education. The overwhelming necessity to provide these services makes it difficult for the countries to make the investments required to adapt to the effects of climate change on the fish resources that sustain communities and national economies, particularly coral reef fisheries and the region's rich tuna resources.

128. The industrial fishing fleets that pay access fees to the governments will only be affected by climate change to a minor extent because they can follow the fish to the high seas as climate change forces redistribution of stocks. The negative impact of climate change is therefore, on the Pacific Island nations and not the fleets. There is little or no scope for the tuna-fishing industry to contribute to the costs of the proposed activities. However, the Programme will capitalise on the regular operations of industrial fishing fleets to cover the considerable costs of data collection across the vast area of the tropical/subtropical Pacific Ocean − costs that would otherwise need to be funded through the GCF grant. Industry is expected to collaborate in the collection of more than 300,000 tuna tissue samples needed for the close-kin mark-recapture assessment across an area of 28 million km^2 under Output 3.1. Industry actors have agreed to provide $22 million USD in in-kind vessel time for the collection of AWS data along with ~USD13.0 million USD provided by the regional observer programmes (PNAO) through observers for AWS data sampling while onboard vessels.

129. As regional fisheries organisations serving Pacific Island countries, SPC and FFA receive financial contributions from their members to assist them to manage the region's shared tuna resources. SPC also receives financial support from WCPFC as the 'Science Services Provider and Data Manager' for the Commission. However, these funding streams presently only enable SPC and FFA to monitor tuna catches, conduct regular stock assessments, and provide scientific and management advice. Both organisations depend on grant funding to engage in advising Pacific Island countries on how best to adapt to climate change and to assist countries to implement priority adaptations. None of SPC's and FFA's normal development partners can make the scale of investments needed to assist the 14 participating Pacific Island countries, including three LDCs, to adapt at scale to maintain nutritious food availability and the economic benefits they receive from tuna, their most valuable natural resource commodity, as the climate changes.

130. The budget for the Funding Proposal has been developed in a way that builds on other initiatives to support sustainable management of the region's small-scale and industrial tuna fisheries and will target the incremental costs of adapting tuna-dependent communities and economies to climate change. Grant finance will allow communities that benefit from increased food availability through the proposed activities to build their resilience without incurring more national debt. The substantial investment involved in developing an AWS for tuna-dependent economies that reduces the high levels of uncertainty in preliminary modelling used to estimate the likely impacts of climate change on the natural resource that underpins 4-70% of all (non-aid) government revenue for nine Pacific Island countries is justified because it will: i) enable these countries to identify adaptations that reduce risks to their economies with confidence, and ii) provide them with improved information to continue to manage tuna resources co-operatively, (iii) and will result in an estimated benefit of ~$40M/year in increased fishing access fees before negotiations in international fora. The AWS will empower these vulnerable countries to negotiate as a bloc to retain the rights to the levels of tuna catch traditionally made within their EEZs, regardless of climate-driven tuna redistribution. An example of an adaptation that could be informed by the AWS is the development of new transhipping facilities in Pacific Island countries where the information system predicts with considerable confidence that the abundance of tuna is expected to increase to levels that will support substantial industrial fishing activity.

B.6. Exit strategy (max. 500 words, approximately 1 page)

134. The objective of the RTP's Exit Strategy (see additional detail in **Annex 02, Chapter 5,** and **Annex 26, Technical Study 9**) is to provide the means to transition roles and responsibilities from the GCF Programme to participating countries and their regional agencies responsible for providing on-going logistical and technical support across the range of initiatives supported under the Programme. The Exit Strategy describes a plan for the phasing out of the technical and logistical support provided through the RTP to the 14 participating countries, the regional executing agency, SPC, and Technical Partners (FFA and CSIRO) over the 7-year Programme delivery timeframe. These institutions have the financial and technical capacity to undertake Programme activities and will continue to work with participating countries after the Programme

implementation period. This Programme's exit strategy will minimise the potential for Programme Outcomes to be jeopardised and maximise opportunities to sustain and enhance the achievements of the Programme.

Component A

135. During Programme design and consultation, transition arrangements that will apply to key elements of Component A were discussed with participating countries. The outcome was a commitment by all 14 participating countries to assume increasing responsibility for the employment of FAD management officers responsible for implementing all national activities, including data collection, within the fisheries departments as Programme implementation proceeds. The transition involves the Programme providing full support for such positions within the national fisheries agencies of each of the 14 participating countries for the first two years of the Programme, and a sliding scale for years 3-5 of implementation (75%, 50%, 25%) with the government assuming responsibility for the remaining salary in each of these years, and the government covering 100% of salary costs in year 6 and beyond. This commitment by participating countries to assume eventual full responsibility for the FAD-related aspects of the GCF Programme is a critical element of the Exit Strategy. The participating countries are aware of the need to include the salaries and other costs associated with maintaining their strengthened national FAD programmes in recurrent national budgets.

136. The Programme will facilitate a series of train-the-trainer capacity building exercises in FAD programme management, including construction, deployment, and maintenance. By building national and regional capacity in FAD programme management expertise, the sustainability of the RTP initiatives will be enhanced and a non-disruptive exit of the Programme will be facilitated as local experts assume full responsibility for national FAD programmes. In addition, training will be provided to local personnel in undertaking needs analysis, legislative reviews, and regulation drafting to strengthen the delivery and utilisation of bycatch and tuna transhipped or unloaded from commercial tuna fishing vessels for local consumption. Together, these trainings will identify a pool of local expertise that will realize the Exit Strategy (**Annex 02, Chapter 5).**

137. In nine of the participating countries, the Programme's FAD-related activities will be harmonised with the efforts that the World Bank PROPER project is also making to strengthen national FAD programmes (**See Annex 02, Chapter 4, Section 4.1.2**). In the larger countries, the PROPER World Bank project will enable access to tuna to be increased in more provinces. The positive impact of the combined efforts of the GCF Programme and World Bank project is expected to incentivise governments to continue to expand their national FAD programmes so that all provinces not yet reached by RTP or PROPER receive FADs.

138. For both national FAD programmes and bycatch interventions, the Programme will support government legislative review and policy updates that are expected to increase the effectiveness of activities during the Programme implementation period and sustain/grow these interventions once the Programme is completed.

Component B

139. For Component B, the Exit Strategy is institutionally firmly established. The Programme AWS will build on and improve the existing SEPODYM model which is an ecosystem and population dynamics model to inform decision-making relating to the management of tuna stocks in the context of climate and ecosystem variability, and to investigate potential changes due to anthropogenic activities including global warming and fisheries pressures and management scenarios. SEAPODYM is utilised by SPC as the primary tool to provide analytical advice to the Commission's Scientific Committee (SC) since 2008.[86]

140. The WCPFC currently provides financial support for the SEPODYM model and will continue to do so as the model is improved and delivered through the AWS. This support is captured in WCPFC's Resolution 2019-01. Although the WCPFC budgetary cycle does not support projections beyond three years, its history of support for Component B-type scientific activities augers well for the gradual integration of the work supported under Component B into the Commission's long-term work programme and budget. Current support includes:

[86] Lehodey, P., Senina, I., Sibert, J. and Hampton, J. 2008. SEAPODYM V2: A spatial ecosystem and population dynamics model with parameter optimisation providing a new tool for tuna management. Fourth Regular Session of the Scientific Committee, 11–22 August 2008 Port Moresby, Papua New Guinea. WCPFC-SC4-2008/EB-WP-10. Western and Central Pacific Fisheries Commission. 16 pages.

- Approved budgets for 2022-2025 for scientific work supported by WCPFC include strengthening of the Pacific tissue bank (US$321,000 in total for the period 2023-2025).
- Continuing regional tuna tagging operations (US$2.19 million for the period 2023-2025).
- Scientific work for close-kin mark-recapture (CKMR) method for tuna stock assessment (US$40,000).

141. The Exit Strategy also incorporates the ongoing contributions of industry to the development and operation of the AWS. Industry partners will collect AWS data during Programme implementation and are expected to continue data collection after the Programme period through the established Observer programme. This continued data collection will improve the resolution and accuracy of the AWS over time. As the Programme progressively produces advice that is both of improved resolution and predictive accuracy, industry will be encouraged to make increasingly significant contributions to support the ongoing development and maintenance of the AWS. In the medium to longer term, financial contributions by industry to initiatives under Component B are likely to become compulsory and a pre-condition of participation in the WCPO fishery. Although such a policy will potentially take more than 5 years to be debated and agreed to in the Commission, there is potential for such a commitment to be adopted before the conclusion of the RTP and to contribute significantly to the Programme’s Exit Strategy. ,

i. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	107.4	million USD ($)

GCF financial instrument	Amount	Tenor	Grace period	Pricing
(i) Senior loans				
(ii) Subordinated loans				
(iii) Equity				
(iv) Guarantees				
(v) Reimbursable grants				
(vi) Grants	107.4			
(vii) Results-based payments				

(b) Co-financing information	Total amount	Currency
	49.3	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
SPC	Grant	1.5	million USD ($)			
SPC	In kind	9.7	million USD ($)			
FFA	In kind	0.8	million USD ($)			
CSIRO	In kind	2.1	million USD ($)			
PNAO	In kind	12.9	million USD ($)			
Private Sector	In kind	22.0	million USD ($)			

(c) Total financing (c) = (a)+(b)	Amount	Currency
	156.8	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	
	The exit strategy (**see section B.6**) for the Programme includes transitioning costs related to national FAD programmes away from the Programme and to national budgets to ensure that national FAD programmes continue after the Programme ends. GCF funds will be used to pay for FAD-related staff positions within the national fisheries agencies of each of the 14 participating countries for the first two years of the Programme and a sliding scale for years 3-5 of implementation (75%, 50%, 25%) with the government assuming responsibility for the remaining salary in each of these years, and the government covering 100% of salary costs in year 6 and beyond. This contribution of in-kind staff time of FAD offers is estimated at 0.7 million USD. This commitment by participating countries to assume full responsibility for the FAD-related aspects of the GCF Programme is a critical element of the Exit Strategy. Furthermore, the participating countries have been alerted to the need to include the salaries, and other costs associated with maintaining their strengthened national FAD programmes and infrastructure, in recurrent national budgets. While these contributions are considered part of the exit strategy rather than co-financing, CI will report qualitatively on these contributions during implementation.

C.2. Financing by component[87]

Component	Output	Indicative cost million USD ($)	GCF financing		Co-financing		
			Amount million USD ($)	Financial Instrument	Amount million USD ($)	Financial Instrument	Name of Institutions
Component A: Adaptation to harness tuna to improve food availability in Pacific Island communities as coral reefs are degraded by climate change	Output 1: Increased access to tuna and other pelagic fish for coastal communities	**35.9**	34.8	Grants	1.1	In-kind	SPC
	Output 2: Increased supply of bycatch and tuna from industrial fishing operations for urban communities	**16.8**	16.5	Grants	0.3	In-kind	CSIRO FFA
Component B: Adaptation to reduce risks to Pacific Island economies from climate-driven tuna redistribution	Output 3: Science-based forecasts and projections that reduce uncertainty in climate change-driven tuna redistribution are available to all stakeholders	**92.1**	46.9	Grants	1.5 43.6	Grant In-Kind	SPC SPC CSIRO FFA Private Sector PNAO
Monitoring and Evaluation (M&E)		**4.4**	3.9	Grants	0.4	In-kind	SPC
Project Management Costs (PMC)		**7.4**	5.1	Grants	2.3	In-kind	SPC CSIRO
Indicative total cost (million USD)		156.8	107.4		49.3		

For this multi-country programme, the indicative GCF funding requested for each country is presented in **Annex 17**. In addition to GCF Funding, co-financing from multiple entities will be provided for Programme delivery (see **Annex 04** for details):

- The Executing Entity, SPC, will provide funding for goods and services to support Activity 3.1 and in-kind staff time to support Activities 1.1, 1.2, 1.3, 3.1, 3.2, 3.3, M&E, and PMC
- Technical Partner FFA will provide in-kind staff time to support Activities 2.1, 2.2, 3.1, 3.2, and 3.3
- CSIRO will provide in-kind staff time to support Activities 2.1 and 3.1
- Private-sector industrial fishing companies will provide in-kind vessel time and data to support the sampling needed to develop the AWS in Activity 3.1
- PNAO will provide the in-kind services of its Observer programme to support the AWS in Activity 3.1

During implementation, the Executing Entity will enter into agreements with each co-financing source that outline the amount of co-financing and co-financing reporting requirements (**see Annexes 14 and 25** for details).

87 The indicative budget figures presented are subject to further review by the AE and GCF. Columns may not add up to the total amounts due to rounding. Please see Annex 04 Detailed Budget Plan for exact figures.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)	
C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

142. Capacity building will be provided primarily through (i) promoting improved management skills for government personnel in charge of national FAD programmes (Sub activity 1.1d); (ii) training in effective FAD fishing methods (Sub activity 1.1g); (iii) training in the safe use of FAD fishing vessels and equipment (Sub activity 1.2c); (iv) training in improved post-harvest methods (Sub activities 1.3a and 2.2a);, and (vi) training to improve skills in AWS application (Sub-activity 3.1 and 3.2) and high-level international negotiations (Sub activity 3.3a).

143. The Programme will support technology development and transfer through (i) promoting the utilisation of advanced technology for disaster risk management for FAD fishers (Sub activity 1.1i); (ii) acquisition of data for more accurate forecasts and projections of Pacific tuna redistribution (Sub activities 3.1b and 3.1c); and (iii) improved monitoring of the status of tuna stocks at the scale of EEZs and tropical and subtropical Pacific Ocean ecosystems (Sub activity 3.1a).

144. The Programme will also support 18 Pacific Island Fishery Professional (PIFP) placements (12-month duration each) at SPC (8 to build capacity in Component A activity, and 6 to build capacity in Component B activity) and FFA (6 to build capacity in Component B activity). These placements are designed to build national capabilities for implementing key activities involved in strengthening national FAD programmes and developing and operating the AWS. PIFP's will work directly with the technical teams.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential (max. 500 words, approximately 1 page)

145. For millennia, coral reef fish have been a cornerstone of nutrition in the Pacific Island region. The effects of climate change on reefs and their resident fish populations are expected to have severe implications for the food availability of coastal communities in PICs.

146. To address this challenge, Component A of the RTP will assist growing Pacific Island populations to continue to have access to fish for good nutrition through the strengthening of national FAD programmes, directly improving the food availability for **560,000 individuals**. FADs enable coastal communities to fish for tuna and other large pelagic fish (hereafter 'tuna') in safe and effective ways – something that will be progressively more important in maintaining access to fish for good nutrition as coral reefs are degraded by climate change and populations continue to grow. For detailed information on the level of benefit (number of fish meals provided per beneficiary) please see **Annex 23.**

147. The Programme's activities related to improving the distribution of bycatch and tuna are expected to increase the food availability for **290,000 individuals** by providing up to four additional tuna meals per beneficiary per month in urban and peri-urban areas where transhipping of purse-seine catches occur (**Annex 23**). Unloading operations by commercial longliners in some ports (for example, Honiara, Majuro, Port Vila, Suva, and Nuku'alofa) also offer potential to supplement supplies of tuna and bycatch for local consumption, albeit at a lower level that the offloading of bycatch from purse-seine vessels.

148. Climate-driven redistribution of tuna threatens to undermine the economies of the PNA member countries and Cook Islands. By 2050 under RCP 8.5, preliminary modelling indicates that the climate-driven redistribution of tuna could cause an average of 20% (range 10-30%) reduction in purse-seine catch from the combined EEZs of the nine tuna-dependent countries, potentially reducing their total fishing access fees by an average of **~$90 million per year** (range $40−$140 million) at today's prices. Even under a more optimistic RCP 4.5 scenario, losses are projected to be ~$12M/yr. Such significant reduction in government finance will have direct impacts on vulnerable populations in these countries, with fewer government resources available for health, education, disaster preparedness and post-disaster recovery, and other related essential social services. Component B of the Programme will empower countries to use improved climate information in decision-making (investment options, national harvest strategies for fish resources) and to negotiate for the right to retain access to the tuna resources that have historically occurred within their combined EEZs. By delivering improved data through the AWS, the Programme will impact **2.5 M** indirect beneficiaries (**Annex 23**).

149. Programme alignment with the GCF's Integrated Results Management Framework and core indicators for adaptation is detailed in section E and will be demonstrated by attainment of the following targets in the 14 participating countries (or in subsets of countries where indicated) by the end of the Programme:

- Deployment of 333 new operational FADs, based on replacement of 195 of the FADs deployed during the Programme (noting that the working life of FADs is ~3 years, meaning that FADs deployed during years 1-3 will need to be replaced during years 4-7). Thus, a total of 528 FADs will be constructed and deployed.
- Training of approximately 800 people in safe and effective FAD fishing methods by the end of the programme (**Annex 26, Technical Study 3**).
- Provision of between ~9 and ~18 million nutritious tuna meals (150 g for adults and 75 g for children) are expected to be provided for 560,000 people across the region each year by strengthened national FAD programmes by the end of the programme (**see Annex 23**). Approximately 290,000 people will be provided with a total of ~13 million tuna meals from improved distribution of bycatch to urban and peri-urban communities where transhipping of purse-seine catches occurs at ports in the region (**see Annex 23**). The calculation of total direct beneficiaries **(790,000)** avoids double counting, as approximately 60,000 individuals will benefit from both FAD caught fish and improved distribution of bycatch.
- Improved livelihoods for 3,000 small-scale fishers using the 333 FADs deployed by the Programme, and the members of their households, totaling 15,000 people (co-benefit) (**Annex 23**).

- Training of 400 representatives of coastal communities in post-harvest methods for FAD-caught fish, resulting in improved food availability for their households, totaling 2,000 individuals, and establishing the basis of peer-to-peer learning within communities (**Annex 23**).
- Opportunities for an additional 130 people to sell the increased volume of bycatch in urban communities, improving the livelihoods of their households, totaling 650 individuals (co-benefit).
- Creation/strengthening of 10 MSMEs to apply post-harvest methods to bycatch in 10 of the countries where transhipping and unloading of these fish occurs (**Annex 23**).
- Providing **~670,000 people** with information on climate change impacts on coral reef fish and the need to consume tuna as an alternative to reef-associated fish resources (**Annex 23**; **Annex 26, Technical Study 6**).
- Development of improved meteorological and natural disaster forecasts for ~2.5 million people (**Annex 23**). Although forecasts will be improved to enable small-scale fishers to avoid hazardous sea conditions when planning fishing trips to FADs, timely warnings that are easy for communities to receive (**Annex 26, Technical Study 4**) will also benefit communities within 1 km of the coast in all 14 countries.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

150. The transition to making tuna a cornerstone of nutritious food availability for PICs across the region represents a fundamental transformational change in the development pathway. In the coming decades, under a “business as usual” scenario, the availability of nutritious animal protein (fish) per capita is projected to decline significantly in PICs due to (i) the effects of climate change on coral reef fish production; (ii) relocation of transhipping operations and population growth, and (iii) consumption of low-quality processed foods and the related increase in non-communicable (nutrition-linked) “lifestyle” diseases. Furthermore, in the absence of reliable climate information to accurately forecast the future distribution of tuna, economic development of tuna-dependent PICs will be significantly constrained.

151. The Programme paradigm shift will be achieved in part by strengthening two established mechanisms which will enable PICs to maintain their traditionally high levels of fish consumption under a changing climate: (i) National FAD programmes to increase access to tuna for coastal communities, and (ii) transhipping and unloading operations to supply tuna to urban and peri-urban communities by addressing the disruptions that are expected to occur as industrial fishing effort moves further to the east due to climate-induced ocean warming. These vehicles capitalise on the region’s rich tuna resources without threatening the status of tuna stocks, and will confer important health, livelihood, and employment benefits to communities. These interventions will be complemented by widespread campaigns to raise awareness of the need to rely more heavily on tuna for food availability and nutrition.

152. Paradigm shift will also be delivered through establishment of the Advanced Warning System (AWS) to inform tuna-dependent countries about the risks posed to their economies from climate-driven tuna redistribution, and the most appropriate adaptations to address this threat. The AWS will (i) empower island nations to secure equitable access to tuna resources as climate change alters the distribution of tuna; and (ii) provide the foundation for revisions to co-operative fisheries management arrangements within the region by producing the tuna ‘resource maps’ needed to document the changing spatial structure of tuna resources and identify what are expected to be new sets of stakeholders for some stocks. The AWS will provide reliable and timely information and risk-informed advisories on expected changes in the timing and extent of tuna catches in the EEZs of tuna-dependent Pacific Island countries. These data will empower countries to retain access rights to the shared fisheries resources that underpin their economies by increasing their capacity to negotiate for solutions founded on science and evidence-based decision making. Enabling governments to engage more effectively in negotiations to retain fishing revenue will help to ensure that more equitable outcomes are achieved for a vulnerable region and its populations.

Scale:

153. The Programme will ensure increased access to bycatch and tuna will be adopted and scaled up by:

- Embedding technical and financial capacity building, regulatory and institutional strengthening at local, national, and regional levels.
- Filling the gaps in support required to enable the participating countries to fulfil their adaptation and investment plans to implement national FAD programmes at the scale required (based on SPC guidelines) and to meet bycatch and tuna unloading requirements in local ports for commercial fishing vessels.

- Ensuring that support for FAD programmes is integrated into national recurrent budget planning for the operation and maintenance of existing FADs and for future upscaling beyond the GCF project. This will allow for FADs to be deployed at scale in the future to meet the needs of coastal communities.
- Leveraging private sector resources (industrial fishing vessels) to help deploy FADs.
- Raising awareness about impacts of climate change on nearshore coral reef fisheries and the need to catch tuna around FADs and use bycatch from transhipping and unloading operations to fill the gap in fish supply will be scaled up through public awareness campaigns.

154. In addition, the Programme will leverage the science needed to build the AWS to operate at the temporal and spatial scales needed to serve all Pacific Island countries.

Replicability

155. The Programme will support the replication of FAD technology and experience in improving the utilisation and distribution of bycatch and tuna from transhipping and unloading operations by:

- Enhancing technical and financial capacity and strengthening institutions at provincial, national, and regional levels.
- Converting lessons learned from the implementation of FAD activities and sourcing of bycatch and tuna from transhipping and unloading operations into knowledge and communication products and services and disseminating them widely, including cross-training, peer-to-peer training, communities of practice, and knowledge transfer as part of its awareness-raising activities.

156. Through these mechanisms, it is expected that FAD interventions, which will be carried out at a set of selected sites initially, will gradually be replicated at other sites within the target PICs (see Scale above), and possibly Pacific territories within the region and/or beyond the region.

157. As a region-wide system, the AWS will cover all 14 participating countries of the Programme. Given the large scale of the system, the mechanisms by which data generated through the AWS are applied in supporting multi-lateral agreements for access rights to tuna, will be rolled out in tranches, i.e., with some countries coming on-line earlier than others. Replication of lessons learned, and improvements made during the earlier tranches, can be applied to those countries coming on-line in the later stages. Eventually it will have region-wide application, including for Pacific overseas territories and the model may be replicated by small island developing states in other regions.

Sustainability

158. The Programme will sustain the use of FADs by ensuring that they are adopted by countries as part of their national infrastructure for nutritious food availability, and that the operation and maintenance of expanded national FAD programmes is included within ongoing national development plans and annual recurring budgets. The Programme will incorporate measures to ensure quick recovery and re-deployment of FADs following cyclones and other extreme weather events, thus improving sustainability of this initiative. Strong measures for awareness-raising and outreach which are built into the Programme will also help to improve overall sustainability.

159. The Programme will help sustain institutional technical capacity around FADs within Ministries of Fisheries in multiple ways. SPC will offer multiple regional and national level trainings related to safe and effective FAD fishing, deployment, and maintenance. These trainings will increase the technical knowledge and expertise within the fisheries ministries beyond the life of the Programme. The Programme will also allow SPC to fund FAD technical specialists to assist participating countries at the national scale, and the workplan and budget is designed so these key positions are made permanent and taken over fully by the countries by the end of the programme.

160. Depending on the context of the participating country, these measures can also be supplemented by (among others):

- Public private partnerships (PPPs) between commercial fishing companies and governments to provide technical and financial assistance with FAD deployment and maintenance as part of company Corporate Social Responsibility programmes.
- Creating legislation to prosecute actions that destroy / damage FADs or violate community-based FAD rules.
- Promoting models for community contributions to maintaining the national FAD infrastructure, e.g., through fisher association-based fees.

161. Greater distribution of bycatch and tuna to urban and peri-urban communities will be sustained through national regulations and improvements to transhipping and unloading operations and supply chains. These operations, in turn, will be sustained by providing long-term incentives for MSMEs and PPPs to participate in tuna transhipping and unloading operations to urban and peri-urban communities, and processing of surplus bycatch for export to neighbouring countries having shortages in fish supply. As an example, assistance could be provided to MSMEs and PPPs to access 'Performance-based climate resilience grants' administered by the United Nations Capital Development Fund (UNCDF).

162. Sustainability of the AWS will be ensured through the WCPFC and SPC member states' contributions. This sustainability will be strengthened through the promotion of strong economic and political alignment, brought about through the binding multilateral agreements which will govern the rights of access to tuna resources for the participating countries in the region. Operation of the AWS will also catalyse enhanced cooperation between WCPFC and the Inter-American Tropical Tuna Commission (IATTC) to ensure that tuna harvests can be sustained, even with a greater proportion of tuna eventually translocated to high-seas areas further to the east, outside the combined jurisdictions of PNA member countries.

D.3. Sustainable development (max. 500 words, approximately 1 page)

163. The interventions to be implemented under the RTP are focused on producing a range of positive economic, social, environmental, and gender-responsive development results, which are captured under the Programme's two outcomes and two defined co-benefits. Programme Outcomes 1 and 2 are expected to result in increased food availability, by promoting reliable access to tuna and other fish as a locally available source of nutritious dietary protein. This in turn will lead to improved public health outcomes.

For Co-benefit 1 (improved livelihoods), positive results will include the following:

- Increasing livelihood diversification and access to markets, thus providing opportunities to boost household incomes in rural and urban/peri-urban communities.
- Increasing gender equality and opportunity by encouraging greater participation of women in fisheries activities across the spectrum of value chain activities (e.g., harvesting, handling, processing, distribution, and marketing).

For Co-benefit 2, the expected results will include increasing the sustainability of ecosystem services by improving the management and sustainability of oceanic tuna and other pelagic fish stocks across the Pacific.

The Programme has been designed to promote greater gender equality and equity in fishing supply chains across programme countries. Three gender related outcomes have been identified in the Gender Action Plan (see **Annex 08**):

- Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing.
- Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs.
- Increased leadership and influence of women in FAD tuna fishery value chain decision-making.

164. The Monitoring and Evaluation (M&E) system of the Programme will ensure that the co-benefits will be monitored throughout the implementation of the Programme and integrated with monitoring systems of the relevant Ministries and other public authorities. These efforts will be coordinated through the Heads of Fisheries (HoFs) of the respective countries, and through SPC and other relevant regional entities. The M&E system, and the indicators to be monitored, are described in **Annex 11** and **Annex 23**.

The Programme is fully consistent with the UN Sustainable Development Goals (SDGs),[88] and will support the realization of the following goals and targets:

[88] Details regarding the SDGs and associated targets are articulated in: United Nations. 2015. Transforming our World: The 2030 Agenda for Sustainable Development.
https://sdgs.un.org/sites/default/files/publications/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf

- Goal 2 (End hunger, achieve food security and improved nutrition and promote sustainable agriculture): Target 2.1
- Goal 5 (Achieve gender equality and empower all women and girls): Targets 5.5, Target 5.a, 5.b
- Goal 8 (Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all): Target 8.2
- Goal 12 (Ensure sustainable consumption and production patterns): Targets 12.2, 12.a
- Goal 13 (Take urgent action to combat climate change and its impacts): Targets 13.1, 13.2, 13.3, 13.b
- Goal 14 (Conserve and sustainably use the oceans, seas, and marine resources for sustainable development): Targets 14.2, 14.4, 14.6, 14.7, 14.a, 14.b, 14.c

D.4. Needs of recipient (max. 500 words, approximately 1 page)

165. PICs are among the countries that are most vulnerable to the impacts of climate change. In a recent survey conducted as part of the Notre Dame University Global Adaptation Initiative (ND-GAIN),[89] nearly 200 countries worldwide were assessed in terms of their vulnerability to climate change. The ND-GAIN survey measured overall vulnerability by considering six life-supporting sectors – food, water, health, ecosystem services, human habitat, and infrastructure. It was determined that, for the countries surveyed, the majority of the 14 PICs participating in the Programme fall within the most-vulnerable quartile. The needs of this group of countries to combat the effects of climate change, and to strengthen resiliency, are great.

Absence of adequate financing

166. The participating PICs are generally small economies which lack adequate financial resources to develop robust systems to support well-coordinated national FAD programmes, as well as a regional AWS aimed at gathering critical data that would support the equitable sharing of tuna resources across the Pacific, as tuna stocks are redistributed due to climate change. The PICs are among a group of countries that are heavily reliant upon overseas development assistance (ODA) to fill gaps to meet their development needs.

167. The proposed Programme will help to provide the initial financial support needed to strengthen national FAD programmes, and to increase delivery of bycatch and tuna from transhipping and unloading operations, to improve access to tuna for coastal and urban/peri-urban communities, launch the AWS, and enable participating countries to identify costs for incorporation in national budgets to sustain activities over the long term.

Need for strengthening institutions and capacity.

168. Knowledge about climate change and the direct and indirect impacts on coastal and urban populations needs to be strengthened further in the PICs participating in the GCF Programme. The general public in these countries, while often heavily dependent on coral reef fish as their main source of dietary protein, may not be fully aware of the damaging impacts that climate change is having on coral reef ecosystems and their resident fish species. These countries also often struggle to obtain the financial resources needed to build the teams needed to manage the diversity of inter-related activities needed to adapt to climate change in the fisheries sector.

169. Prominent challenges are the need to (i) understand the potential adverse impacts of fishing activities on targeted fish stocks and other marine resources to enable users of these resources adapt to climate change; (ii) ensuing that the socio-economic benefits from sustainably managed fisheries are maximised and shared equitably; and (iii) create opportunities for greater participation of women and disadvantaged groups within the fisheries sector to improve the use of fish to increase food availability and improve livelihoods. The Programme will work to address these challenges.

Economic and logistical limitations (high dependency on limited resource base)

170. Nearly all the PICs participating in the RTP are categorised as Small Island Developing States (SIDs). The SIDs are characterised as limited in their economic development due in large part to the small size of their economies. In the case of the PICs, this condition is further exacerbated by their remoteness from major markets, and the non-

[89] Vulnerability assessment based on data published in 2021 at https://gain-new.crc.nd.edu/ranking/vulnerability

diversified nature of their economies. For example, 9 of the 14 participating PICs are considered tuna-dependent economies, reflecting their reliance on this single resource for their economic well-being.

171. The budgetary constraints of PICs make it difficult for these countries to adequately finance a range of vital social services and essential infrastructure, including education, health care, transportation infrastructure, and internet services. These systemic factors cannot be fully addressed through any single assistance programme. However, the RTP can make an important contribution towards addressing barriers to climate resilience. Through the development of a regional AWS, the Programme will support the main source of revenue generation for participating countries by creating the enabling countries to maintain or improve their potential to collect licensing fees from industrial tuna fishing operations.

D.5. Country ownership (max. 500 words, approximately 1 page)

172. Country ownership is fundamental to the design and implementation of the Pacific Island RTP. Considering the innovative approaches being undertaken to promote significant environmental, economic, and social improvements in response to the effects of climate change on coastal fisheries and the redistribution of tuna biomass in the waters of PICs, and to make countries' responses more climate resilient, engagement with a wide range of national and regional stakeholders is required to ensure the sustainability and success of the proposed interventions. The engagement process with national governments and a variety of other critical stakeholders was initiated during the conceptualisation and development of the Funding Proposal and will continue throughout the implementation of the Programme.

173. During development of this Funding Proposal, considerable efforts were made, on a frequent and regular basis, to engage and consult with the Heads of Fisheries (HoFs), the NDAs, and a wide range of stakeholders living and working within the target countries (**Annex 07**). These efforts enabled governments and other stakeholders to (i) understand the structure of the Programme; (ii) contribute to the identification and design of interventions to assist their tuna-dependent communities and economies adapt to climate change, and (iii) appreciate the nature of the funding that will be provided to support the key actions needed to enhance their capacity to adapt to climate change. The views and requirements of national stakeholders have been critical in shaping the framework and activities of the programme.

174. In-depth country engagement has focused attention specifically on how the Programme aligns with national NDCs, development plans, policies, strategies, and programmes. The Programme will assist many Pacific Island countries to implement their NDCs; 10 of the 14 participating countries have included the need for adaptation to the effects of climate change on the ocean and coastal marine habitats in their NDCs. **Appendix 2-M of Feasibility Study Chapter 2** details how the Programme will contribute to achieving climate goals identified in key climate change strategy documents in each participating country. **Annex 24** provides further detail on climate policies and strategies of each country.

175. It is important to note that national governments, through their heads of fisheries (HoF), determined the implementation arrangements for this program, selecting SPC to be the sole Executing Entity (EE) of the Programme. SPC has built strong, decades-long relationships and trust with its member states and their national stakeholders as their regional technical agency. SPC uses its extensive technical capacity to support sustainable development by applying a people-centred approach to science, research, and technology across all SDGs. SPC serves its members by interweaving and harnessing the nexus of climate, ocean, land, culture, rights, and good governance. As stewards of a vast Pacific Ocean domain, SPC is actively involved in mobilising its members to respond with urgent collective action to the threat of climate change.

176. Given the profound economic dependence of Pacific Island countries on tuna, and the close diplomatic, economic, and social ties between participating countries, the Programme is designed to be both regional and national in scope. The Programme is of regional significance because tuna are migratory fish species shared by PICs and interventions and management systems must be regional in scope. The Programme is also designed at the national level because adaptations to increase access to tuna for food availability must be customised for each country, due to varying contexts and the differences in population size and location of Pacific Island nations.

177. The framework for the Programme, together with the key components and activities, are designed to support countries' national efforts to embed climate resiliency at the core of their planning to maximise the sustainable socio-economic benefits they derive from their tuna resources. During Programme implementation, it is anticipated that strong country ownership will be further expressed through their co-financing of some activities, commitment of personnel, openness to public-private partnership arrangements, and mainstreaming of climate change adaptation into national policies, plans, and legislation--all of which will support successful outcomes of the RTP.

78. Pacific Island countries have also demonstrated their commitment to strengthening resiliency to climate change impacts, through their support of the following goals of the *Framework for Resilient Development in the Pacific: An Integrated Approach to Address Climate Change and Disaster Risk Management* (FRDP) [15]: i) 'Strengthened adaptation and risk reduction to enhance resilience to climate change, including managing risks caused by climate change within social and economic development planning processes'; and ii) 'Strengthened preparedness, response and recovery to natural disasters caused by climate change'.

79. Through extensive consultation and engagement, PICs have entered into a comprehensive series of agreements which reflect regional priorities for managing regional fisheries resources and are aligned with the Programme interventions and goals (**see Annex 02, Chapter 1, Section 4**). The relevant agreements include:

- Western and Central Pacific Fisheries Commission (WCPFC) *Resolution on Climate Change as it Relates to the Western and Central Pacific Fisheries Commission (Resolution 2019-01)*[90]
- Western and Central Pacific Fisheries Commission (WCPFC) *Resolution on Aspirations of Small Island Developing States and Territories (Resolution 2008-01)*[91]
- *Regional Roadmap for Sustainable Pacific Fisheries*[92] designed to improve sustainability of tuna resources, add value to tuna catches, increase employment, and provide better access to tuna for nutritious food availability, as well as building resilience of coastal habitats (by progressively shifting fishing effort from coral reefs to tuna)
- *A New Song for Coastal Fisheries – Pathways for Change*[93], an innovative regional approach to maintain the benefits of small-scale fisheries in the face of declining coastal ecosystems and associated fish stocks.
- The Parties to the Nauru Agreement (PNA) Vessel Day Scheme, which supplies 95% of tuna caught in the region.

180. The HoFs and NDAs in each of the 14 participating countries have acknowledged the high level of stakeholder engagement undertaken in the preparation of the GCF Funding Proposal. More than 133 institutions, including national agencies, NGOs, CSOs, private sector, academia, associations, and community groups, comprising over 1,070 individual stakeholders within the 14 partner countries were consulted and received an overview of the Programme and the associated presentations and documentation. In addition, efforts were made to ensure that a proportionate percentage of women were involved in the consultative process; of the total number persons consulted, ~38% were women. During over 50 consultative engagement events aimed at creating awareness about the Programme, stakeholders were provided with opportunities to share their views and stakeholder feedback has been used throughout the process to refine the Programme design. This process clarified the roles of key stakeholders in Programme development and implementation, ensuring that Programme components and activities incorporate country ownership and support from stakeholders and host communities. These efforts complement the regular dialogue and collaboration with HoFs and NDAs. Please see **Annex 07** for more information on the stakeholder engagement process.

181. The participating countries that have delivered their no-objection letters (NOLs) have duly followed their respective no-objection procedures as established by each country. The delivery of the NOLs represents the culmination of constructive engagement and collaboration with the countries, setting a foundation for effective implementation of the tuna adaptation programme.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

[90] https://cmm.wcpfc.int/resolution/resolution-2019-01
[91] https://www.wcpfc.int/doc/resolution-2008-01/resolution-aspirations-small-island-developing-states-and-territories
[92] https://ffa.int/system/files/Roadmap_web_0.pdf
[93] https://www.spc.int/DigitalLibrary/Doc/FAME/Reports/Anon_2015_New_song_for_coastal_fisheries.html

182. **Annex 03** of this Funding Proposal provides Economic Analyses (EA) and Financial Analyses (FA) of the Programme and alternative options under a variety of scenarios. **Annex 03** includes the Excel files for the EA and FA and the accompanying narrative is provided in **Annex 02, Chapter 3**. The economic analysis (EA) is conducted from the perspective of the national economies of the 14 Participating Countries.

183. The results of the economic analysis are summarized in Table 5 and indicate an overall Economic Internal Rate of Return (EIRR) of 18.4% with a Net Present Value (NPV) of USD58.81 million at a 9% discount rate and a benefit: cost ratio for the overall Programme of 4.26. The combined results for the Programme comfortably exceed a 9% EIRR. **Annex 03** also provides a variety of sensitivity analyses showing that the results are robust to a variety of assumptions (e.g. changes to fish catch around FADS, average fish price, changes to future investment in FADs, average FAD life, changes to Capital and Operational Expenditures of up to +/- 20%). It also provides analyses for the FAD component for each of the 14 countries in the Programme.

Output	PV of Costs @ 9% (USD million)	PV of Benefits @ 9% (USD million)	EIRR	NPV (USD million)	Benefit:cost Ratio
1. FADs	35.89	44.43	14.5%	8.55	1.24
2. Bycatch	11.62	13.41	11.1%	1.79	1.15
3. AWS	55.63	104.10	22.3%	48.47	1.87
Combined	103.14	161.95	18.4%	58.81	4.26

Table 5. Summary of results for the economic analysis of the combined FAD, bycatch and AWS activities.

184. The financial analysis (FA) considers the financial implications of the ongoing activities developed under the Programme on government costs and revenue for each of the 14 Participating Countries, and the increase in costs likely to occur to maintain the activities. The financial impact is considered for the three Outputs related to strengthening national FAD programmes, improving the distribution of bycatch, and developing and utilizing the AWS. The analysis shows that the total annual costs of replacing FADs, based on an average 3-year life for FADs, is $1.15 million at the end of the GFC Programme, increasing to $1.76 million per year in 2050. The government expenditure needed to improve the distribution of bycatch following the completion of the Programme are likely to be minimal because the role of government is limited to providing support for this initiative through its regulatory functions. The participating countries will contribute to the operation and maintenance of the AWS programme after the completion of the Programme through their membership of the regional organizations that develop and use the system. The ongoing annual costs associated with operating and maintaining the AWS are estimated to total $4.0 million, equating to ~$290,000 per country for each of the 14 PICs. The administration of the AWS will be the responsibility of the regional agencies acting on behalf of the PICs - FFA, WCPFC and SPC.

185. The activities proposed in this RTP represent a highly efficient way of building the resilience of 14 PICs with an extraordinary dependence on tuna for food availability and/or economic development. Component A (approximately 53% of the requested grant budget) will increase access to nutritious animal protein for an estimated 790,000 people by 2035. The estimated cost of this benefit is $85 per person over a seven-year period - approximately $12 per person per year. The proposed mechanisms for sustaining National FAD Programmes and distribution of tuna from transhipping and unloading operations will ensure that the food availability benefits of this investment will continue well beyond the life of the Programme.

186. The activities in Component B will empower nine countries to prevent estimated combined losses of government revenue of $90 million (range $40−$140 million) per year at today’s value by 2050. These estimates will be adjusted as uncertainty in the tuna modelling is reduced through development of the AWS. Even so, the benefits for tuna-dependent economies from the AWS are expected to be considerable (see Annex 03) and will exceed the total grant request for Component B within a relatively low number of years. **Annex 03** examines the payback period under a range of possible benefit scenarios and shows that payback is likely to be achieved within 5 to 14 years.

187. The regional Programme is also a sound investment because it will not only build the resilience of tuna-dependent communities and economies in 14 countries, it will also strengthen the capacity of SPC as the Science Service Provider to WCPFC, and to the other regional fisheries management agencies (FFA, PNA).

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max. 300 words)

The Programme will contribute to a paradigm shift from climate-induced food insecurity to improved resilience of communities by strengthening two established but under-performing vehicles to help Pacific Island countries maintain their traditionally high levels of fish consumption under a changing climate: (i) national FAD programmes to increase access to tuna for coastal communities as the supply of fish from coral reef ecosystems declines due to ocean warming and acidification, and (ii) transhipping and unloading operations to supply bycatch and tuna to urban and peri-urban communities by addressing the disruptions that are expected to occur as industrial fishing effort moves further to the east due to ocean warming and consequent redistribution of tuna stocks. These vehicles capitalise on the region's rich tuna resources without threatening the status of tuna stocks and will confer increased food availability and resilience to both coastal and urban communities, including enhanced livelihoods – benefits that would otherwise have been seriously diminished due to the effects of climate change on coral reefs and tuna distribution. Innovations for meeting per capita dietary animal protein requirements (**Annex 26, Technical Study 2)** of rapidly-growing Pacific Island communities will be complemented by widespread campaigns to raise awareness of the need to rely more heavily on tuna for food.

The Programme paradigm shift will also be delivered through establishment of an 'advanced warning system' (AWS) to inform tuna-dependent countries about the risks posed to their economies from climate-driven tuna redistribution, and the most appropriate adaptations to address this threat. The AWS will (i) empower island nations to secure equitable access to tuna resources as climate change alters the distribution of tuna; and (ii) provide the foundation for revisions to co-operative fisheries management arrangements within the region by producing the tuna 'resource maps' needed to document the changing spatial structure of tuna resources and identify what are expected to be new sets of stakeholders for some stocks. The AWS will provide reliable, timely and accurate information and advisories on expected changes in the timing and extent of tuna catches in the EEZs of tuna-dependent Pacific Island countries and pave the way for these countries to retain access rights to the shared fisheries resources that underpin their economies by increasing their capacity to negotiate for solutions founded on science and evidence-based decision making. Enabling governments to engage more effectively in negotiations for retention of historical fishing license revenue will help to ensure that more equitable outcomes are achieved for a vulnerable region and its populations.

Assessment Dimension	Current state (Baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		

Scale	(1) Investments have not been available to assist Pacific Island countries to adapt at scale to maintain the significant food availability benefits they receive from fish as the climate changes. Specifically, although scaling-up the use of nearshore FADs for capture of pelagic species has been recommended by SPC, Pacific Island countries have insufficient human and financial resources to do so. The policies governing the use of bycatch to feed urban populations have not taken the effects of climate change on the distribution of industrial fishing effort into consideration. In addition, awareness of causes of declines in stocks of coral reef fish (e.g., overfishing, climate change, pollution), as well as pathways for reversing this trend, are limited. (2) Modelling and monitoring of the anticipated climate-driven redistribution of tuna resources that underpin Pacific Island economies is still at a preliminary stage. The scale and resolution of modelling needs to improve so the impacts of climate change on local, sub-regional and regional tuna distribution can be explicitly included to provide a more reliable estimate of future tuna catches	<u>Low</u>	(1) FADs are adopted as part of the essential national infrastructure for food availability by Pacific Island countries and continued strengthening and maintenance of FADs is included in national budgets. Policies are adopted to improve access to bycatch in participating countries under the impacts of climate change. Public awareness of the need to supplement fish consumption with access to pelagic stocks is widely recognised. (2) The AWS provides tuna-dependent Pacific Island countries with state-of-the-art tools to forecast responses of tuna to ocean warming on time scales of 1-10 years and to project responses at time scales >15 years. The spatial scale of the AWS encompasses the exclusive economic zones of all Pacific Island countries and the entire tropical Pacific Ocean between 20° N and 20° S. The AWS enables the governments of tuna-dependent economies to understand	(1) The programme will ensure that FAD technology will be adopted and scaled by: • Providing technical and financial capacity building, regulatory and institutional strengthening at local, national, and regional levels • Filling the gaps in support required to enable the participating countries to fulfil their plans to implement national FAD programmes in an environmentally friendly way and at the scale required based on SPC guidelines. • Ensuring that support for FAD programmes is integrated into national budget planning for the operation and maintenance of existing FADs and for future upscaling beyond the GCF project. • Leveraging private sector resources (industrial fishing vessels) to help deploy FADs • Supporting policy development and planning to improve access to bycatch in urban/peri urban areas. • Raising awareness about impacts of climate change on nearshore coral reef fisheries and the need to supplement access to fish through use of nearshore FADs and the use of bycatch from transhipping operations through public awareness campaigns. (2) The programme will leverage the science needed to build the AWS to operate at the temporal and spatial scales needed to serve all tuna-dependent Pacific Island countries.

	in the waters of each Pacific Island country.		the impacts of climate-driven tuna redistribution with greater certainty and empowers them to negotiate to retain the present-day economic benefits they receive from tuna regardless of redistribution.	
Replicability	(1) Adoption of FADs for nearshore fishing for tuna and other pelagic fish is varied across the region, but generally well below the level needed to supply the amount of fish required for good nutrition of growing populations. (2) Preliminary modelling of the responses of tuna to the warming ocean has several weaknesses and sources of uncertainty. Current modelling assumes that each species of tuna forms one large mixed stock across the entire tropical Pacific Ocean and is at the coarse spatial resolution of 2 x 2 degrees. In addition, the ways in which the prey species on which tuna depend will respond to ocean warming is poorly known.	<u>Medium</u>	(1) Successful application of FAD technology results in targeted coastal communities catching more tuna and pelagic fish species, leading to improvements in nutrition and health, and increased household income. Successful models of FAD deployment and use are replicated for additional communities; lessons are learned to correct weaknesses as they are identified. Public awareness of declining coral reef fish stocks, and the benefits of FADs for increasing access to tuna and other pelagic fish species for local consumption spreads widely among coastal communities across the region (2) The AWS reduces uncertainty in the responses of tuna to ocean warming and is built as a 'living' system, enabling it to be continually improved as the modelling is progressively refined.	(1) The Programme will support the replication of FAD technology by: • Building technical and financial capacity, and strengthening institutions at provincial, national and regional levels • Converting lessons learnt from the implementation of FAD activities into knowledge and communication products and services and disseminating them widely including cross-training, peer-to-peer training, communities of practice and knowledge transfer as part of its awareness raising activities. (2) The programme reduces uncertainty in the modelling of the responses of tuna to ocean warming by ensuring that the AWS is based on identifying the stock structure of each tuna species, modelling the response of each stock, including better information on the impact of ocean warming on tuna prey, and increasing the

				spatial resolution of the modelling of the responses of each stock to 1 x 1 degrees.
Sustainability	(1) National FAD programs are currently supported in an ad hoc manner, without a source of initial investment to strengthen and develop these programmes and without ongoing financial support from government budgets or external donors. Transhipment regulations do not adequately consider the food availability needs of urban populations nor the redistribution of pelagic stocks due to climate change.	Medium	(1) The vital contributions of fish to the nutrition of Pacific Island people are sustained by i) strengthened national FAD programmes, supported by ongoing finance, and ii) regulations and mechanisms that provide for improved food availability of urban communities by improving transhipment from industrial fishing operations.	(1) The programme will sustain the use of FADs by ensuring that they are adopted as part of the national infrastructure for food availability, and that the maintenance of expanded national FAD programmes is included within ongoing national development plans and recurring budgets. Depending on the context of the participating country, these measures can also be supplemented by: • Public private partnerships (PPPs) between commercial fishing companies and governments to provide technical and financial assistance with FAD deployment and maintenance as part of company Corporate Social Responsibility programmes. • Creating legislation to prosecute actions that destroy or damage FADs, or violate community-based FAD rules; and • Promoting models for community contributions to maintaining the national FAD infrastructure, e.g., through fisher association-based fees. Improvements to transhipping operations and supply chains that result in greater distribution of tuna bycatch to urban communities will be sustained by regulatory reform and providing opportunities for MSMEs and PPPs to distribute tuna from transhipping operations to urban and peri-urban communities, and/or process surplus

	(2) The eight Parties to the Nauru Agreement (PNA) have an insufficient understanding the effects of ocean warming on the industrial tuna-fishing operations which currently underpin their economies to undertake adaptive planning. Countries need to know the timing and extent of tuna redistribution so that they can adapt to maintain access to their present-day share of the resource.		(2) Investments by GCF deliver an AWS that significantly reduces uncertainty in the timing and extent of tuna redistribution of tuna, and the effects of tuna redistribution on national economies. Continued operation of the AWS is sustained by SPC as the science service provider for the Western and Central Pacific Commission (WCPFC), empowering tuna-dependent Pacific Island countries to negotiate effectively to sustain the government revenue derived from tuna-fishing.	bycatch for export to neighbouring countries with shortages in fish supply. (2) The programme will ensure that investments by GCF result in an AWS that significantly reduces uncertainty in climate-driven tuna redistribution and the implications for Pacific Island economies by assembling the appropriate scientists to design the system, and mobilising industrial fishing fleets to contribute the data needed to build the AWS. The programme will also strengthen cooperation between WCPFC and the Inter-American Tropical Tuna Commission (IATTC) to ensure that tuna harvests remain biologically sustainable, and the current economic benefits derived by Pacific Island Countries are maintained into the future.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	**IRMF** **Core Indicators (1-4)**[94]	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Notes**
				Mid-term[95]	**Final**[96]	
Total number of beneficiaries	Core 2: Direct and indirect beneficiaries reached	See below for MoV for each category of beneficiary	**Direct:** **0**	**Direct:** **584,000** (292,000 female 292,000 male; 1/3 children)	**Direct:** **790,000** (395,000 female 395,000	Notes: Direct and indirect beneficiaries here are the sum of Core 2 beneficiaries for ARA1 and ARA2 noted below. Direct beneficiaries under Supplementary 2.2 = 850,000, but there is overlap between these populations – the Programme target

[94] The IRMF Indicators are set out in the Integrated Results Management Framework
[95] Mid-term targets are indicators to be achieved after 3.5 years of programme implementation. Mid-term evaluation will be conducted in year 4.
[96] The final target means the target at the end of project/programme implementation period.

			Indirect: **0**	Indirect: **350,000** (175,000 female 175,000 male)	male; 1/3 children) **Indirect:** **2.5 million** (1.25 million female 1.25 million male)	(avoiding double counting) is 790,000. Please see Annex 23 for additional detail. Mid-term and final beneficiaries based on projected population in 2030.
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	See ARA1/Supplementary 2.4 below for MoV	Indirect: 0	Indirect: **350,000** (175,000 female 175,000 male)	Indirect: 2.5 million (1.25 million female; 1.4 million male)	Notes: Beneficiaries here are the sum of beneficiaries noted below for Supplementary indicator 2.4. All indirect beneficiaries will be reached through the facilitation of improved meteorological forecasts There are no direct beneficiaries for this indicator. Indirect beneficiaries is the population living within 1km of the coast in all 14 participating countries[97]
ARA2 Health, well-being, food and water security	Core 2: Direct and indirect beneficiaries reached	See ARA2/Supplementary 2.2 below for MoV	Direct: 0	Direct: **584,000** (292,000 female 292,000 male; 1/3 children)	Direct: **790,000** (395,000 female 395,000 male; 1/3 children)	Notes: Beneficiaries here are the sum of beneficiaries noted below for Supplementary indicator 2.2. See footnote[98] on the use the term "Food availability" in this programme. Direct beneficiaries are individuals with improved food availability from increased access to tuna from FADs and those benefitting from improved distribution of bycatch and tuna from transhipping. **See Annex 23 for detail.** There are no indirect beneficiaries for this indicator.

[97] The full populations of all tuna-dependent economies except PNG are expected to benefit from an enhanced ability to negotiate for retention of fishing license revenue to support government funding and continued strong management of tuna stocks (AWS interventions). For PNG, the benefit is expected to apply only to the population living within 1 km of the coast. For the five subtropical countries benefits are also expected to be limited to people within 1 km of the coast. Populations living in coastal areas in all 14 countries will benefit from improved meteorological and disaster forecasts. **See Annex 23 for details.**

[98] The underlying issue is food availability because the aim is for fish to provide 50% of the dietary protein intake recommended by the World Health Organisation. This is the relevant dimension of food security for the Programme. Tuna and other large pelagic fish are available in the waters of the 14 Participating Countries, but access is currently difficult. Outputs 1 and 2 will improve access to this important source of dietary protein.

						~60,000 people are expected to benefit from FADs **and** improved distribution of bycatch. This overlap is included in the calculation of 790,000 direct beneficiaries of the GCF Programme (i.e. double counting is avoided).
ARA1 Most vulnerable people and communities	Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems	Indicator: The number of people in participating countries who benefit from deployment of the AWS and increased government capacity to negotiate for retention of tuna-fishing access fees; and/or the sustainable management of industrial tuna fishing fleets. Data sources: National census data for the programme countries	Indirect: 0	Indirect: **350,000** (175,000 female 175,000 male)	Indirect: **2.5 million** (1.25 million female 1.25 million male) Beneficiaries are assumed to be 50% female and 50% male.	Notes: These indirect beneficiaries are population living within 1 km of the coast in all participating countries in 2030.[99] There are no direct beneficiaries for this indicator. Assumptions: All specified populations in the 14 participating countries benefit from improved meteorological information to improve safety at sea for fishers and disaster preparedness / risk reduction for coastal populations.
ARA2 Health, well-being, food and water security	Supplementary 2.2: Beneficiaries (female/male) with improved	Please note that Supplementary 2.2 includes two sets of interventions: Indicator 1: The number of people with improved food availability from Programme deployed FADs Number of FADs deployed by the Programme per country and estimation of number of people likely to receive fish	Direct: 0[100]	**Direct:** **584,000** (292,000 female 292,000 male; 1/3 children)	**Direct:** **790,000** (395,000 female 395,000 male; 1/3 children)	Notes: This target (2.2) is a combination of targets under Output 1 (FADs) and Output 2 (Bycatch) – both contribute to improved availability of food for targeted populations. Please note that there is overlap between the two populations – double counting is avoided (please see Annex 23 for more information). FADs (Indicator 1): Mid-term target (354,000) is based on the percentage of new FADs expected to be installed after 3.5 years of implementation (65%) and the

[99] The coastal populations (within 1km) of all participating countries is included as the indirect beneficiaries of the Programme. The Programme will facilitate the delivery of improved meteorological information to this population during implementation which will improve both safety at sea for fishers and allow coastal communities to better prepare for cyclones and other extreme weather events.
[100] Although studies during the PPF showed there were ~220 FADs in the region in 2023 (see Study 3), they will all need to be replaced during the programme period. Therefore, this indicator is focused on the number of people supported by FADs put in place and maintained by this GCF programme. See **Annex 23 and Annex 17** for country specific calculations of the number of beneficiaries.

		from FADs is based on the method in **Annex 23**: Data sources: Programme records on FAD deployment; National census data; Analysis of number of people within 1km of the coast. Average annual catch per FADs will be monitored during years 1-3 to determine the yield from each FAD and the resulting number of additional tuna meals available (level of benefit) See **Annex 23**. Indicator 2: The number of beneficiaries (urban and peri-urban) receiving improved / additional distribution of bycatch and tuna from transhipping derived from: Data sources: Government census projection data (2030) for urban areas of ports where transhipping occur. Records of quantity of bycatch and tuna that comes ashore during transhipping operations (from national Fisheries Agencies / Regional Observer programmes)				corresponding coastal population with improved access to FAD caught fish. Final target (560,000) is based on the total number of Programme deployed FADs (333) and the corresponding coastal population with improved access to FAD caught fish. **See Annex 26, Technical Study 3.** There are no indirect beneficiaries for this indicator. Assumptions: The targeted coastal population has access to FAD caught fish; government agencies and communities support and participate in the programme activities; no differences in fish availability between genders and adults and children. Bycatch (indicator 2): Notes: Baseline is set at 0 as the Programme will increase the supply of transhipped fish. Final (290,000) and midterm targets (290,000) are the full urban populations where transhipping occurs. The entire population (which currently consumes offloaded fish) will benefit from the increase in supply.[101] Assumptions: The targeted urban population in each country has improved access to tuna throughout the implementation period and government agencies support and participate in the programme activities.

[101] The amount of increase will be measured to determine the level of benefit for this population. Programme target is to increase the amount of bycatch available in urban areas by 50% (level of benefit) See Annex 23 for more detail.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
IRMF Core Indicators (5-8)[102]	**Baseline context (Description)**	**Rating for current state (Baseline)**	**Target scenario (Description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Current institutional and regulatory framework inadequate to support increased access to tuna, in order to improve food availability for increased climate resilience	low	Strengthened institutions capable of implementing, monitoring, evaluating effective adaptations to increase access to tuna (Outcome 2)	(i) mainstreaming of national FAD programmes, with sustainable institutional and financial structures promoting climate resilience ii) policy strengthening in at least 10 PICs to secure increased availability and utilisation of tuna and bycatch from commercial fishing operations adapted to climate-induced redistribution of tuna to support national food availability threats, and (iii) institutionalization of AWS programme through multilateral cooperative agreements, which will enable vulnerable, climate-affected countries to negotiate for retention of the socio-economic benefits historically received from tuna.	Multi-countries
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	A. (i) Current investments in deployment of FADs are localised, fragmented, and not coordinated; (ii) minimal mechanism is in place for maintenance, repair, and replacement of damaged or lost FADs; (iii) coastal fishers often operate vessels without due consideration for maritime safety. B. PIC policy on utilising transhipped or unloaded tuna	low	A. (i) national FAD programmes are coordinated country-wide and operating effectively; coastal communities are well prepared to maintain, repair and replace FADs; (iii) maritime safety practices among FAD fishers are improved.	A. Programme will (i) assess current state of FAD deployment and provide support (policy reform, training, etc.) to scale-up national FAD programmes; (ii) provide training in construction and deployment of FADs, and facilitate procurement of materials needed for construction, maintenance, repair and replacement of FADs; (iii) customize meteorological data for delivery to small-scale fishers and coastal communities, (iv) facilitate procurement of safety equipment, and train fishers in use of safety equipment. B. Programme will review current practice, policy and legislation relating to the availability and	Multi-countries

[102] The IRMF Indicators are set out in the Integrated Results Management Framework

	and bycatch is a national prerogative and are varied / not regionally harmonised. Some countries restrict access to transhipped or unloaded tuna and bycatch to protect local fishers. There are concerns regarding the quality bycatch transhipped or unloaded and associated health risks. C. Climate-induced changes in tuna distribution in the Pacific are anticipated, but not known with sufficient accuracy to support the implementation of adaptations that ensure the economic benefits from tuna in tuna-dependent PICs' are maintained		B. Transhipped and/or unloaded tuna and bycatch from commercial fishing operations in at least 10 PIC ports is available and utilised by PIC communities to contribute to filling the increasing gap associated with a combination of decreased productivity of local coral reef fisheries and increasing populations. C. Increased generation and use of climate information and advisories as evidence to identify redistribution of tuna stocks with confidence for climate-informed planning, negotiations, and investments.	utilisation of transhipped and/or unloaded tuna and bycatch from commercial tuna fishing operations as a source of fish protein for urban/peri-urban communities. C. Programme will support the application of cutting-edge, innovative technology to identify biomass redistribution trends of major tuna stocks across the Pacific Ocean; this data will be necessary to enable tuna-dependent, climate-affected countries to negotiate to retain rights of access to tuna stocks.	
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	(i) Post-harvest handling of bycatch has not been optimised, from the standpoint of value-addition, food health and safety, nutritional value, and avoidance of waste (ii) potential relocation of transhipment hubs for industrial fishing operations in the Pacific is anticipated, due to projected climate-induced redistribution of tuna stocks	low	(i) Strengthened capacity to reduce vulnerability of economies to climate change-induced impacts, by retaining the benefits from fish through improved post-harvest handling (ii) Greater potential for offloading of transhipped or unloaded bycatch, by optimising the locations of transhipment hubs or unloading ports in response to climate-induced redistribution of Pacific tuna stocks	Programme will contribute to market transformation through (i) supporting improvements in post-harvest handling, processing, and preservation of fish, sourced both from FADs and fish transhipped or unloaded from commercial fishing operations. (ii) identification of future transhipment hubs or unloading ports, stimulating their development for offloading of bycatch from industrial fishing operations for urban and peri-urban consumers.	Multi-countries
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices,	Existing need to improve delivery of information to promote deeper understanding and appreciation, among resource users and key policymakers, of the threats posed by climate change to fisheries resources, and the	low	Increased awareness of threats to reef and coastal ecosystem fish supply from climate change and the potential of tuna as an alternative source of dietary protein	Programme will contribute to knowledge generation, information dissemination, and application of best practices by: (i) converting lessons learned into best practices, knowledge and communication products, services and platform for national and regional	Multi-countries

methodologies and standards	associated effects on the food availability of Pacific Island nations			peer to peer learning and training; (ii) raising awareness regarding the vulnerability of nearshore coral reef stocks to climate change impacts (iii) promoting adoption of safe and effective FAD fishing to help address gaps in local fish supply as a key source of dietary protein; (iv) replicating best practices from site to site across the region; (v) gathering critical new data on distribution of Pacific tuna stocks; (vi) applying tuna distribution data to equip tuna - dependent Pacific Island countries to negotiate for continued present-day benefits from commercial tuna fisheries; and (vii) conducting a range of awareness campaigns and training activities to disseminate critical climate change information to multiple stakeholder groups and end-users	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	**Project/programme specific Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Note**
				Mid-term	**Final**	
Outcome 1: Improved food availability of vulnerable communities in participating countries	Number of individuals with improved food availability	Programme and government records including: Number of FADs deployed Coastal population of target countries with access to FAD caught fish (census data) Urban population in target areas (census data)	0	Coastal communities: 354,000 (127,000 female and 127,000 male; 1/3 are children) Urban/peri-urban communities: 290,000 (145,000 females and 145,000 males; 1/3 are children)	Coastal communities: 560,000 (280,000 female and 280,000 male; 1/3 are children) Urban/peri-urban communities: 290,000 (145,000 females and 145,000 males; 1/3 are children)	Assumptions: See above ARA1 notes and assumptions as well as **Annex 23** Level of benefit (additional quantity of fish available) will be determined as described in **Annex 23**
Output 1: Increased national capacity to access tuna and other pelagic fish for coastal communities	a) Number of participating countries with strengthened national FAD programmes (from 2023 baseline)	a) National fisheries agencies' FAD management plans associated annual reports, and provisions within national fisheries legislation. Extent to which national FAD programmes have been strengthened will be measured against the	a) 0	a) 14 countries	a) 14 countries	a) National fisheries agencies and records from SPC's Coastal Fisheries and Aquaculture Programme (CFAP) support to national FAD activities.

	b) Number of dedicated staff provided to national FAD programmes under service agreements. c) Number of nearshore FADs installed and maintained. d) Number of individuals trained in effective and safe FAD fishing. e) Number of individuals trained in and applying post-harvest processing and value chain activities for FAD-caught fish. f) Number of individuals in coastal areas provided with information on climate change impacts on coral reef fish and the need to consume tuna and pelagic fish as an alternative to reef-	framework in SPC Policy Brief 31/2017 [103] b) Programme records / reporting from National FAD programmes c) Nearshore FAD deployment and maintenance records from national fisheries agencies, including provincial or district fisheries offices. d) Programme training records. e) Programme training records. f) Programme training / communications records	b) 0 c) 0 d) 0 e) 0 f) 0	b) 62 c) 215 FADs deployed and operational d) 512 fishers e) 200 individuals f) 200,000 individuals	b) 62 c) 333 FADs deployed and operational d) 848 fishers e) 400 individuals f) 450,000 individuals	b) One or two FAD officers per country + 2 FAD enumerators + one national coordinator per country - support for position transfer from Programme to national budgets. c) See **Technical Study 3**, noting that targets include regular maintenance of all deployed FADs. d) 512 trainees expected by the end of year 3 e) Community- representatives receive training in post-harvest methods for FAD-caught fish. f) Individuals in coastal areas across the region targeted with information relating to climate change impacts on coral reef ecosystems and the need to consume tuna as an alternative source of dietary protein.

[103] https://pacificdata.org/data/dataset/oai-www-spc-int-770d616e-1345-406f-bfce-6de7fc84c4b5"

	associated fish resources. g) Number of individuals provided access to improved meteorological and disaster forecasts for the fisheries sector. h) Level of benefit for coastal communities with improve access to FAD caught fish	g) Records and services from national authorities for meteorology, maritime safety, and disaster risk reduction. h) FAD yield surveys, extrapolated to all FADs deployed in each Participating Country	g) 0 h) 0	g) 350,000 individuals h) 8.9 million additional fish meals per year	g) 2.5 million individuals h) 17.8 million additional fish meals per year	g) Final target is total coastal population (within 1km of coast) of 14 participating countries in 2030 (indirect beneficiaries). h) Please see **Annex 23** for calculation methodologies and assumptions. These targets assume 10 tonnes of fish per year per FAD deployed. Midline assumes 65% of FADs deployed by midpoint of implementation. To be adjusted based on yield survey results.
Output 2: Increased supply of bycatch and tuna from industrial fishing operations for urban/peri-urban communities	a) Number of countries adopting and implementing policies promoting the increased availability of bycatch and tuna to urban and peri-urban communities. b) Number of countries provided with designs and potential financing mechanism options to develop new fish marketing facilities / outlets.	a) National fisheries management plans, port access/fisheries access agreements and associated annual reports, and national fisheries legislation. (b) Records maintained by national fisheries agencies, and national authorities for town planning and small business.	a) 0 b) 0	a) 10 Countries b) 1 country	a) 10 Countries b) 4 countries	a) Adopting and implementing defined as: commenced, or completed, reviews of national policy for the availability of bycatch and tuna transhipped or unloaded from commercial fishing to urban and peri-urban communities. a) and b) are limited to the 10 countries where there is potential to improve transhipping of bycatch from purse-seine vessels and unloading of bycatch and tuna from longline vessels.

	c) Number of people with improved access to bycatch and tuna in urban and peri-urban communities.	c) Reports by FFA Fisheries Development Division and SPC.	c) 0	c) 290,000 individuals	c) 290,000 individuals	c) See **Annex 23** for details. Note that PNG is not included because transhipping of purse-seine catch is well developed there, however, PNG will be included for offloading of longline bycatch.
	d) Number of individuals trained in post-harvest processing and value chain activities for brined bycatch.	(d) Programme training records.	d) 0	d) 20 individuals (10 MSMEs)	d) 20 individuals (10 MSMEs)	d) Training will be conducted for existing MSMEs to improve practices.
	e) Number of individuals benefitting from bycatch from purse-seine transhipping operations who are also provided with information on climate change impacts on coral reef fish (developed under Output 1) and informed about the need to consume bycatch and tuna as an alternative to reef fish.	e) Records of workshops, articles /broadcasts by media outlets, etc. related to outputs of Programme.	e) 0	e) 220,000 individuals	e) 220,000 individuals	e) The methods used to raise awareness of coastal populations in seven countries under Output 1 will also be used for the target urban populations in three of these seven countries where regional ports transhipping and unloading occurs.
	f) Increase in annual availability of transhipped and/or unloaded tuna and other large pelagic fish	f) Data from catch monitoring by national fisheries agencies / PNAO to identify average annual quantities of bycatch	f) 1,639 tonnes/year	f) 2,000 tonnes/year	f) 2,500 tonnes/year	f) . Baseline figure estimated – see **Annex 26, Technical Study 5** for details. These figures exclude PNG (where transhipment and offloading is well developed). See **Annex 23**

	for consumption by urban and peri-urban communities (level of benefit).	and tuna transhipped and unloaded				for details on number of additional fish meals per tonne of pelagic fish.
Outcome 2: Strengthened capacity of tuna-dependent Pacific Island nations to negotiate for benefits from climate redistributed tuna stocks	Pacific Island nations united and resourced to initiate and conduct long-term negotiations, in international fora to secure benefits from climate-driven redistribution of tuna stocks.	Number of project publications cited in the working and supporting documents used to inform negotiations at international forums, including IPCC AR7, UNFCC-COP, BBNJ, WCPFC, IATTC.	0	a) 8	a) 30	a) PIC Leaders endorse action in international fora to negotiate the retention of benefits for climate-induced re-distributed tuna stocks.
Output 3: Improved forecasts and projections for climate-driven tuna redistribution which facilitate effective adaptations for all stakeholders	a) An increasingly sophisticated integrated, Advanced Warning System (AWS) b) Published contributions by SPC utilizing AWS data endorsed by the Scientific Committee of WCPFC. c) Number of forecasts/ projections provided to national governments derived from the AWS. d) Number of national governments and regional agencies utilising improved data	a) Programme reports b) Programme reports c) Programme reports to national governments. d) Regional Report Cards	a) 0 b) 0 c) 0	a) 1 AWS b) 3 Reports C) 42 (14 x 3 reports)	a) 1 AWS b) 7 Reports c) 84 (14x6)	a) AWS will become more sophisticated over time as more data are collected and analysed. b) SPC provides at least one technical report annually to PICs describing the status of the AWS and its application in relation to forecasts and projections. These reports are endorsed by the SC of WCPFC. c) FFA and SPC collaborate to include economic forecasts and projections into annual AWS reports provided to PICs. Non-proprietary information is made freely available to other regional agencies. Information is utilised by PICs in regional and international fora to support negotiations relating to the retention of benefits from redistributed tuna stocks.

	on impacts of climate change on tuna biomass. e) Number of individuals from national governments trained in the use and interpretation of AWS data to enhance national negotiation capabilities within regional and international fora.	e) Senior representatives from the 14 participating countries (including nine countries with tuna-dependent economies as well as the other five countries), contributing to relevant negotiations within UNFCCC meetings and climate change-related WCPFC negotiations.	d) 0 e) 0	d) 14 Countries + 7 CROP agencies e) 25 individuals	d) 14 Countries + 7 CROP agencies e) 60 individuals	d) SPC Oceanic Fisheries Programme (OFP), Western and Central Pacific Fisheries Commission (WCPFC) technical and meeting documents / reports. e) Training and mentoring provided by FFA and SPC on use and interpretation of AWS-related information to enhance national negotiating positions in regional (WCPFC) and international fora (for example UNFCCC).
Project/programme co-benefit indicators						
Project/programme results **(outcomes/ outputs)**	**Project/programme specific Indicator**	**Means of Verification (MoV)**	**Baseline**	**Midterm**	**Final**	**Assumptions / Note**
Co-benefit 1: Improved livelihoods of vulnerable communities in participating countries	• Beneficiaries adopting improved and/or new climate-resilient livelihood options (**GCF Supplementary 2.1**)	Indicator: Sum of Supplementary 2.1 indicator targets from the three groups below. • See below for MoV of each element of the calculation.	0	**7,875 individuals**	**15,750 individuals**	Notes: Beneficiaries adopting improved and/or new climate resilient livelihood options benefit from livelihoods related to 1) 3,000 small-scale fishers around FADs and their households, 2) 130 new vendors of bycatch from transhipping/ unloading operations and their households; and 3) 10 MSMEs for post-harvest of bycatch.
Co-benefit 1: Improved livelihoods of vulnerable communities in participating countries	• Beneficiaries adopting improved and/or new climate-resilient livelihood options (note this is also **GCF Supplementary 2.1**)	Indicator: The number of people whose livelihoods benefit from fishing around FADs put in place by the Programme.	0	7,500 individuals	15,000 individuals	3,000 small-scale fishers expected to use the 333 new FADs, and their households (x5), totalling 15,000 people by endline.

		<u>Data sources</u>: Programme reports on number of FADs deployed; data collection on number of fishers reported by enumerators. Gender-disaggregated data from focused community social surveys on contributions of FADs to livelihoods in 14 PICs				Number of fishers to be determined during implementation as part of FAD yield surveys. uptake of improved practices secure buy-in from communities and long- term commitment to promoted climate resilient actions
Co-benefit 1: Improved livelihoods of vulnerable communities in participating countries	• Beneficiaries adopting improved and/or new climate-resilient livelihood options (note this is also **GCF Supplementary 2.1)**	<u>Indicator</u>: Number of people with livelihoods linked to transhipping operations who benefit from increases in the amount of bycatch available due to the Programme. <u>Data sources</u>: Data on number of people with livelihoods linked to transhipping operations from surveys; measures of bycatch available from transhipping operations from surveys.	0	325 individuals	650 individuals	<u>Assumptions</u>: Assumes a 50% increase in the availability of bycatch use due to the Programme activities, creating opportunities for an additional 130 bycatch vendors and their households (x 5), totalling 650 people by the end of the Programme.
Co-benefit 1: Improved livelihoods of vulnerable communities in participating countries	• Beneficiaries adopting improved and/or new climate-resilient livelihood options (**GCF Supplementary 2.1**)	<u>Indicator</u>: Number of people with livelihoods linked to the development of post-harvest bycatch products to the Programme. <u>Data sources</u>: Data on number of people with livelihoods linked to post-harvest operations from surveys; measures of bycatch available from transhipping operations from surveys.	0	50	100	Assumptions: There is active interest of MSMEs to explore possibilities of post-harvest operations. This is both for edible products and production of ingredients for animal/aquaculture feeds and plant fertilisers. Beneficiaries are the number of people in the households of 2 employees in each of 10 MSMEs.

Co-benefit 2: Strengthened management of industrial tuna fisheries by regional and national institutions	•Tonnes of fish stock brought under sustainable management practice (**GCF Supplementary 4.3**)	Data source: The AWS will be used to verify the amount of tuna brought under sustainable management by the Programme.[104]	0	0	**16M tonnes** (all four key tuna species in WCPO)	Assumes current predictions of general trends for west-to-east relocation of Pacific tuna stocks are reliable. The WCPFC will succeed in establishing conservation and management arrangements for tuna fishing in high seas areas.
Co-benefit 2: Strengthened management of industrial tuna fisheries by regional and national institutions	•Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under restoration and/or improved ecosystems (**GCF Supplementary 4.1**)	The data collection area for the AWS that will have the largest economic impact on countries – Please see **Annex 22, Appendix 1** for more information.	0	0 hectares	**75M hectares**	Notes: Please see **Annex 22 Appendix 1** Midterm is 0 because the full benefit of the AWS will be realised once the information it provides is used to guide WCPFC's CMMs / harvest strategies to adapt tuna fishing (after year 4).

E.6. Project/programme activities and deliverables			
Activities	**Description (see B.3 for details)**	**Sub-activities**	**Deliverables**
Component A – Adaptation to harness tuna for food availability of Pacific Island communities as coral reefs are degraded by climate change			
Output 1: Increased national capacity to access to tuna and other pelagic fish for coastal communities			
Activity 1.1: Provide technical and logistical support, including equipment,	The purpose of expanding the use of anchored, artisanal FADs is to provide the infrastructure needed to increase access to fish required for food availability of vulnerable coastal communities. Increasing the number of FADs in small island	1.1a. Audit progress towards requirements for scaling-up National FAD Programmes in the 14 participating countries.	1.1a-1. Audit reports against criteria specified in the SPC Policy Brief 31/2017

[104] At present, all tuna resources within the EEZs of Pacific Island countries are sustainably managed. Of particular relevance is the expectation that the AWS will confirm the proportion of the 4.5 million tonnes of tuna estimated to be within the combined EEZs of Pacific Island countries that is expected to move into the high seas. This is currently estimated to be 20% (900,000 tonnes) by 2050 under continued high GHG emissions. by preliminary modelling published in Nature Sustainability (Bell *et al*, 2021; https://www.nature.com/articles/s41893-021-00745-z) When the fish are in high-seas areas they are at greater risk of experiencing Illegal, Unreported and Unregulated (IUU) fishing because Monitoring, Control and Surveillance (MCS) can be less effective in high seas areas. The AWS will reduce uncertainty in the preliminary modelling on tuna biomass redistribution, enabling the Western and Central Pacific Fisheries Commission (WCPFC) to understand the risks more fully, an implement appropriate manage arrangements to sustain tuna biomass in high-seas areas.

to implement National FAD programmes	countries is a practical way of providing a greater number of small-scale fishers with the opportunity to catch tuna and other pelagic fish species (hereafter 'tuna') efficiently and safely. This Activity will commence by updating the assessment of the gaps to be filled to increase access to tuna in each country conducted - see **Annex 26, Technical Study 3**. National FAD programmes will then be strengthened to the level required to meet the standards promoted by SPC, through provision of the materials, equipment, training, and capacity-building needed to ensure that the required number of additional FADs are appropriately constructed, deployed, utilised, and maintained, and that small-scale fishers are equipped to harness the full potential of FADs safely. Across the 14 participating countries, 333 FADs will be operational, and routinely replaced when required, through this Activity. Furthermore, train-the-trainer sessions will be conducted with staff from national fisheries agencies in FAD rigging and deployment, and FAD-fishing methods. More than 200 staff from national fisheries agencies will be trained in each of these two areas. In turn, these trainers will pass on the skill sets to at least 800 small-scale fishers/community members across the region in each of the two areas. Regular monitoring of the performance of the FAD infrastructure will also be included in this Activity. Consideration will be given to the different roles played by men and women in increasing access to tuna for domestic food availability so that national FAD programmes are augmented in a gender-responsive manner (**Annex 08**).	1.1b. Develop workplans for scaling-up National FAD Programmes based on the audit. 1.1c. Review national policies and regulations to identify barriers to the equitable and sustainable use of FADs. 1.1d Design and implement capacity development activities to augment the skills of national staff to implement FAD programmes. 1.1e. Design and implement a gender-responsive consultative stakeholder engagement strategy for national fisheries agencies to utilise in consultation with communities to identify suitable FAD deployment sites. 1.1f. Procure materials to expand and maintain national FAD programmes. 1.1g. Enable small-scale fishers to catch tuna and other pelagic fish around FADs that integrate traditional knowledge. 1.1h. Improve the capacity of national FAD programmes to effectively collect and utilize catch data. 1.1i. Establish/strengthen national response mechanisms to natural disasters affecting small-scale fishers using FADs.	1.1b-1Work plans 1.1c-1 FAD Management Plans that reflect revised policies and regulations to remove barriers to the equitable and sustainable use of FADs in 14 Participating Countries 1.1d-1. Increased numbers of skilled staff designated to implement strengthened National FAD Programmes in 14 countries. 1.1e-1. Stakeholder engagement strategy applied during consultations for identifying FAD locations in 14 countries based on bathymetric analysis and past catch information from communities. 1.1f-1. Materials required to construct, deploy and maintain FADs are procured. 1.1g-1. Train-the-trainer courses in effective FAD-fishing methods for use in 14 countries 1.1h-1. Records of target fish catches from FADs. 1.1h-2. Records of non-target species interactions in FAD fisheries. 1.1h-3 FAD-related deployment and catch data for improving FAD design 1.1i-1. Improved response plans in 14 countries.

Activity 1.2: Augment national safety-at-sea initiatives	The Programme places a strong emphasis on safety at sea, given that i) some FADs will be up to 7 km offshore; and ii) the experience of many existing small-scale fishers encouraged to fish around FADs has been largely limited to targeting coral reef fish within lagoons and relatively sheltered inshore coastal waters. This activity will ensure that the risks for small-scale fishers operating around FADs are reduced in three important ways by augmenting national safety-at-sea initiatives. First, by evaluating the suitability of small vessels currently used for fishing around FADs and providing governments with improved vessel designs, and practical options for constructing/purchasing such vessels, where needed. Second, by upgrading forecasts of sea conditions by national meteorological agencies for dissemination to small-scale fishers and coastal communities by mobile phone for their use in planning fishing trips to FADs. Third, by providing training for small-scale fishers in the use of boating safety equipment recommended by SPC.	1.2a. Conduct a needs analysis for improved vessel safety for small-scale fishers using FADs. 1.2b. Establish the enabling conditions for national meteorological services to provide improved weather and extreme event warnings to small scale fishers. 1.2c. Support procurement and delivery of boating safety equipment, linked to vessel registration where practical. 1.2d. Increase national capacity to regulate and provide training in the use of the safety equipment.	1.2a-1. Needs analyses for improved vessel design for 14 countries; tested designs for safer and more efficient vessels; options for promoting improved vessel design uptake among coastal fishers. 1.2b-1. Improved systems for accessing natural disaster and meteorological forecasts via mobile phones and other methods for fishers in 14 countries. 1.2c-1. Materials required to support safer fishing around FADs from both canoes and motorised vessels procured. 1.2d-1 Train-the-trainer courses for small-scale fishers in the use of boating safety equipment in 14 countries. 1.2d-2 Review and needs analysis of national regulations on boat safety.
Activity 1.3: Strengthen post-harvest practices and improve market opportunities for FAD-caught fish	This Activity will be designed to provide training in post-harvest methods for tuna and associated species caught around FADs, and to develop supply chains and market opportunities for these fish. The main aims of improving the handling of tuna catches from FADs are to i) extend the shelf life of the fish, and (ii) add value, increasing the food availability for individuals and communities. Design of the training programme will commence with a needs assessment to determine the specific aspirations of the coastal communities receiving tuna from FADs. Training will incorporate information for a variety of processing methods. Training will then be tailored to meet the specific needs and conditions prevalent in the target communities/MSMEs. Processing methods may include, but are not limited to, increasing the supply of ice, freezing, drying, smoking, salting, pickling, and canning. Consideration will also be given to value-	1.3a. Provide training for subsistence households and MSMEs to improve preservation of FAD-caught fish using post-harvest methods, e.g., drying, smoking, and bottling, incorporating traditional knowledge. 1.3b. Provide coastal communities with basic equipment to apply post-harvest methods, including practical options for cold storage where appropriate.	1.3a-1. Training courses that result in utilisation of post-harvest methods to increase storage life and value of FAD-caught fish in 14 countries. 1.3b-1. Basic equipment for drying, smoking, or bottling FAD-caught fish utilised in selected countries, and operational cold storage for fish established at selected sites. 1.3c-1. Market opportunities for FAD-caught tuna for MSMEs identified and tested.

	adding through branding, certification, and similar mechanisms which may enhance the marketing of locally processed FAD-caught tuna products. Recognizing the different roles which may be played by men and women in post-harvest handling activities, the training will be developed and conducted in a transparent and gender-responsive manner.	1.3c. Identify and promote market opportunities for FAD-caught fish. 1.3d. Conduct communication campaigns to raise awareness of coastal communities about the impacts of climate change on coral reef fish and the need to consume more FAD-caught tuna for community food availability.	1.1.3d. Integrated, multi-media campaigns that raise awareness of climate-related impacts on coral reef ecosystems and promote FAD-caught fish as an alternative food source, in targeted coastal communities.
Output 2: Increased supply of bycatch and tuna from industrial fishing operations for urban/peri-urban communities			
Activity 2.1: Implement strategies to deliver more transhipped and unloaded bycatch and tuna to urban/peri-urban communities	This activity is designed to identify and address weaknesses in the entire supply chain for fish that come ashore from industrial tuna-fishing vessels while they are in port. This fish provides urban and peri-urban communities with a nutritious source of protein at low cost. It is comprised of bycatch caught by purse-seine fishing, i.e., fish that is not accepted by tuna canneries, which is separated from the high-quality tuna when these vessels tranship their catch to carrier vessels in regional ports, and lower-value fish from longline vessels that is also offloaded in port. The main transhipping ports are in Federated States of Micronesia, Kiribati, Marshall Islands, Solomon Islands and Tuvalu and PNG. Unloading of bycatch from longline vessels also occurs in some of these ports and is also common in Fiji, Tonga, and Vanuatu.	2.1a. Assess the supply of bycatch and tuna available for offloading at each transhipping and unloading port. 2.1b. Evaluate the projected shortfalls in the supply of fish needed for the food availability of urban and peri-urban communities by 2030 and in following decades. 2.1c. Use the Advanced Warning System (AWS) (see Activity 3.1.1 below) to assess the implications of tuna biomass redistribution for transhipping and unloading activities across the region. 2.1d. Build national capacity to conduct policy analysis on current and future transhipment and unloading of bycatch and tuna. 2.1e. Develop procedures and regulations to increase availability of transhipped and unloaded bycatch and tuna where needed to fill the gap in fish supply.	2.1a-1. Assessments of the availability of bycatch and tuna. 2.1b-1. Projected shortfalls in supply to meet future demand for fish by urban and peri-urban communities where transhipping and unloading occurs. 2.1c-1. Projected changes in location and frequency of transhipping and unloading operations identified for the countries where these activities occur. 2.1d-1. Policy analyses for current and future transhipment and unloading of bycatch and tuna for the countries where these activities occur, including increased capacity for local staff to undertake the required policy analysis. 2.1e-1. Procedures and regulations to increase the supply of transhipped and unloaded bycatch and tuna suitable for human consumption and/or production of animal feed and fertiliser.
Activity 2.2: Strengthen/develop post-harvest practices and improve market	This Activity is similar to Activity 1.3, but focusing on providing training in post-harvest methods, and expanding markets, for	2.2a. Provide training for urban communities to improve/develop post-	2.2a-1. Training courses in post-harvest methods to extend shelf life

opportunities to distribute bycatch and tuna from transhipping and unloading operations to urban/peri-urban communities	bycatch and tuna landed in transhipping and unloading operations. The main purposes of improving the handling of bycatch and tuna from these operations are to i) extend the shelf life of the fish, and (ii) add value to bycatch and tuna sold by MSMEs. The training will incorporate the processing methods requested by MSMEs and provide some of the equipment they need to process fish. The different roles played by men and women in post-harvest processing activities will be taken into consideration and the training will be developed and conducted in a transparent and gender-responsive manner. Activity 2.2 will also support three other sub-activities expected to increase demand for bycatch and tuna from transhipping and unloading activities: developing pilot marketing mechanisms; conduct communication campaigns about the benefits of bycatch and tuna; and providing governments with fish market outlet designs at various scales that they can use to improve the sanitary conditions under which the fish are sold.	harvest processing techniques for bycatch and tuna from transhipping and unloading operations. 2.2b. Pilot alternative marketing mechanisms to support increased trade in bycatch and tuna in urban areas. 2.2c. Conduct communication campaigns to raise awareness of urban/peri urban communities about the impacts of climate change on coral reef ecosystems and the need to consume more bycatch and tuna to meet future nutrition requirements. 2.2d. Provide fish market outlet designs at various scales for countries where transhipping and unloading occurs.	and add value to transhipped brined bycatch and unloaded frozen tuna. 2.2b-1. Novel post-harvest products and evaluations of the success of mechanisms to market them to support increased trade in bycatch and tuna. 2.2c-1. Integrated, multi-media awareness campaigns to raise awareness of climate-related impacts on coral reef ecosystems and promote bycatch and tuna as an alternative food source, in targeted urban/peri-urban communities. 2.2d-1. Fish market outlet designs
Component B – Adaptation to reduce risks to Pacific Island economies from climate-driven tuna redistribution			
Output 3: Improved forecasts and projections for climate-driven tuna redistribution which facilitate effective adaptations for all stakeholders			
Activity 3.1: Develop and deliver an Advanced Warning System (AWS) for climate-driven tuna redistribution	This Activity is designed to harness the latest in big data analyses, communications and climate and fisheries sciences to deliver an Advanced Warning System (AWS) that provides the decision-ready technical tools for PICs to develop adaptations that protect their economies from disruptions to income streams and oceanic food systems caused by the redistribution of tuna biomass. This activity will increase the capability of national administrations and their regional agencies to provide tailored information to 1) assist PICs (individually and collectively) to prepare for and respond to the impacts that climate change will impose on their oceanic ecosystem, and 2) negotiate mechanisms to reduce the disruptions to their economies and global mitigation options that minimise the severity of climate change. Significant co-benefits will be derived from the AWS, including the long-term	3.1a. Transition existing fisheries and ocean monitoring systems to produce higher-resolution forecasts and projections of tuna biomass redistribution. 3.1b. Establish baselines and indicators for quantification of climate change impacts on distribution of tuna biomass. 3.1c. Enhance collection and curation of physical oceanography and micronekton	3.1a-1. High-resolution version of the SEAPODYM tuna-climate model suitable for EEZ-scale analyses of changes in tuna biomass distribution. 3.1b-1. Geographical boundary and absolute population size baselines for each Pacific tuna stock, and population census information needed for optimisation of the SEAPODYM tuna-climate model. 3.1c-1. Baseline sub-surface ocean state observations to validate the SEAPODYM tuna-climate model.

	conservation of Pacific tuna stocks and oceanic biodiversity through improved stock assessments and CMMs adopted by the WCPFC and the IATTC.	data to inform the modelling of climate-driven redistribution of tuna biomass.	
Activity 3.2: Assess and socialise the impact of tuna biomass redistribution identified by the AWS on national economies at all levels	This Activity will couple tuna fisheries economics to the SEAPODYM tuna-climate model, providing the AWS with the capability to perform bio-economic analyses. This capability provides the AWS with all the component parts to simulate short-term and long-term changes in the productivity and locations of Pacific tuna fisheries under all future climate variability and climate change scenarios. Evaluation of simulations will allow national and regional fisheries management processes to develop options to secure national benefits derived from the tuna industry.	3.2a. Conduct bio-economic and fleet dynamics modelling to estimate changes in tuna catch, and the associated socio-economic benefits.	3.2a-1. Fully functional AWS, producing national and regional economic and resource outlooks.
Activity 3.3: Provide AWS-related training to national institutions to negotiate in regional and international forums to address socio-economic losses due to the impacts of climate change on tuna.	This Activity will operationalise the AWS within national administrations and regional agencies. Operationalisation will focus on training national counterparts with the capacity to use the AWS to develop and evaluate adaptation options that minimise the risk to Pacific Island economies from the impacts of climate-driven tuna redistribution (this includes attribution and adaptation evaluations), and the capability to prepare national forecasting and resource outlooks. In parallel, the AWS will be used to assemble the information necessary for PICs to contribute significantly to the ocean impacts section of the IPCC Assessment Report 7 cycle.	3.3a. Academic and vocational training to Increase the number of Pacific Island fisheries and climate staff with enhanced capabilities to negotiate to retain the national benefits received from tuna. 3.3b. Assemble evidence for Pacific Island countries to use in negotiations at regional and global scales to address the impacts of climate-driven tuna redistribution.	3.3a-1. Trained and fully competent national officers in AWS operations for sustainability of the AWS. 3.3b-1. Documented evidence for 14 countries and their regional agencies to use in negotiations.

E.7. Monitoring, reporting and evaluation arrangements (max. 300 words)

188. The Programme's Monitoring and Evaluation Plan is designed to measure the following (see **Annex 11** and **Annex 23** for more detailed information**)**:

- Progress towards accomplishment of Programme-level outcomes and co-benefits, as articulated in the Programme ToC and Logframe; including Programme contributions toward achieving Fund-level outcomes, as measured by IRMF core indicators (**see Sections E.3, E.4 and the co-benefits included in section E.5**); and
- Programme-level outcomes and outputs (**see Section E.5**).

Programme-level monitoring and evaluation will be undertaken in compliance with the monitoring, evaluation, accountability, and learning (MEAL) systems and processes utilised by SPC – referred to hereafter as M&E. Additionally, monitoring and evaluation for the Programme will be complementary to the applicable government monitoring and evaluation systems and policies of the various participant countries.

The Programme monitoring and evaluation staff will be responsible for designing studies to refine baselines and collect data for mid-term and final calculation of indicators for: GCF-level core indicators and outcomes (as defined in the IRMF) and Programme-level results and indicators.

Monitoring

189. The primary responsibility for day-to-day Programme monitoring and implementation rests with the Programme Director, supported by the monitoring and evaluation specialists to be located at SPC headquarters in Noumea. The Programme Director, in consultation with key stakeholders, will develop annual work plans based on the Inception Report to ensure the efficient implementation of the Programme. A Programme inception workshop, involving CI, SPC, FFA, CSIRO, senior representatives of each of the 14 participating countries, and other key stakeholders (e.g., PNAO) will be held within the first six months of Programme implementation. The overarching objectives of the inception workshop include:

- Assist the Programme team and stakeholders to understand and take ownership of the programme implementation approach, objectives and outcomes and discuss any changes in the overall context that influence programme implementation.
- Discuss the roles, support services and complementary responsibilities of the Programme team and the national government ministries including reporting and communication lines and conflict resolution mechanisms.
- Review the logical framework, re-assess baselines as needed, and discuss reporting, monitoring and evaluation roles and responsibilities and refine / elaborate on the Monitoring & Evaluation plan.

190. The Programme team will ensure that the indicators included in the Programme results logical framework are monitored and objectively report progress. Programme interventions including activities and outputs, will be monitored as well as the achievement of higher-level Programme results. The Programme M&E system will cover two levels of performance: GCF-level performance (expected performance against investment criteria) and Programme-level performance. Details of monitoring and evaluation implementation will be negotiated and included in the agreements between the CI-GCF Agency as AE and the Executing Entity (SPC). Agreements between SPC and Partners (FFA, PNAO and the CSIRO) will include similar obligations.

Evaluation

191. The Programme's mid-term evaluation process will include an internal impact evaluation and an independent process evaluation. Independent mid-term and final impact evaluations will take place to inform adaptive management of the Programme and capture overall impact, respectively. The evaluations will rely on key questions (to be initially developed during inception planning) to respond to the performance and impact of the Programme's completed activities and will include assessment against OECD-DAC and GCF evaluation criteria. These may include the following: relevance; effectiveness of the Programme and processes; the efficiency of processes; sustained impact and coherence in climate finance delivery; gender equity and inclusiveness; innovation and potential for paradigm shift; country ownership; coherence of climate finance; and potential for building scale and unexpected results (positive and negative). The Terms of Reference for the evaluations will be developed and agreed upon between the EE and AE.

192. Overall, evaluation will contribute to accountability and learning by reviewing emerging evidence on the performance and the impact and/or likelihood of impact of the Programme, and disseminating that evidence to Programme implementors, beneficiaries and stakeholders (including donors) to support evidence-based decision-making. The midterm evaluation will be instrumental in contributing – through operational and strategic recommendations – to improving implementation, setting out any necessary corrective measures for the remaining period of the Programme. The final evaluation will assess the relevance of the intervention, its overall performance, as well as

sustainability, scalability and replicability of results, differential impacts and lessons learned. The evaluation will also assess the extent to which the intervention has contributed to the Fund's higher-level goal of achieving a paradigm shift in adaptation to climate change with respect to the tuna resources of the Pacific.

Reporting

193. Annual monitoring will be based upon review and assessment of the indicators included in the programme results logframe and will be reported on through the Programme's annual performance reports (APRs). The results of independently produced mid-term and final impact evaluations will be reported first in draft reports, which will be reviewed and commented upon by the AE and other stakeholders. Final mid-term and final impact evaluation reports will be prepared considering the comments provided on drafts.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Note: Probability and impact ratings have been assessed based on inherent, pre-intervention risk (i.e. before mitigation measures).

Selected Risk Factor 1 Large-scale, climate-driven natural disasters that significantly disrupt tuna fisheries;

Category	Probability	Impact
Other	High	Low

Description

Natural disasters are a common occurrence in the Pacific Islands region and can disrupt plans to increase access to tuna and associated pelagic fish for food availability, as happened in Vanuatu in 2015 during Cyclone Pam.[105] Any area that is impacted by a cyclone, hurricane, tsunami, saltwater inundation or severe flooding or drought could be overwhelmed by associated recovery efforts, which in turn may decrease the ability of communities or governments to participate in Programme activities causing delays to implementation.

However, although natural disasters will have a high impact in the areas where they occur, but the overall risk to Programme implementation is considered to be low because most disasters are likely to be localised and have a relatively short duration relative to the implementation period of the Programme.

Mitigation Measure(s)

The EE will work with partners to assess potential risks of natural disasters, adaptively manage the risks of Programme activities, and change the impacted interventions as needed. Depending on the circumstances and the nature of the disaster, measures may include delaying some activities, providing adaptive management training to partners, or supporting post-disaster recovery efforts in coordination with other partners.

The Programme includes several activities that are specifically designed to mitigate risks from dangerous weather and natural disasters, especially storms and cyclones. **Activity 1.2** is focused on improving national safety-at-sea initiatives, including improvements to meteorological and natural disaster forecasts to inform small-scale fishers and coastal communities. In Fiji and Vanuatu, the two participating countries where cyclones impact the greatest number of coastal communities, SPC will assist the national fisheries agency to build resilience to these natural disasters. This will be done by establishing cyclone-proof storage for spare FAD materials in appropriate locations so that FADs lost due to extreme storm surge can be replaced quickly (**see Activity 1.1 described in section B.3**). Procurement of FAD infrastructure will also be phased over Programme implementation to reduce risks.

Selected Risk Factor 2 Large-scale ocean closures

Category	Probability	Impact
Legal	Low	Low

Description

There are multiple prominent entities supporting countries to designate more of their EEZs as no-take Marine Protected Areas and this can be expected to reduce fishing operations within these zones. For example, reductions in fishing effort by commercial fleets in the waters of some countries due to the 30x30 initiative[106] has the potential to adversely affect the plans to increase the availability of fish from transhipment and unloading operations for the food availability of some urban communities. It can also be expected to affect the full potential of the AWS to assist countries to optimise sustainable tuna catches, fishing access fees, and other socio-economic benefits derived from tuna caught in their

[105] https://www.adb.org/news/features/vanuatu-fisheries-and-food-security-after-cyclone-pam

[106] In December 2022, 30x30 was agreed at the COP15 meeting of the Convention on Biological Diversity, and became a target of the Kunming-Montreal Global Biodiversity Framework. The 30x30 concept refers to a target for at least 30 per cent globally of land areas and of sea areas, especially areas of particular importance for biodiversity and its contributions to people, to be conserved through effectively and equitably managed, ecologically representative and well-connected systems of protected areas and other effective area-based conservation measures, and integrated into the wider landscapes and seascapes.

EEZs. However, no-take zones are likely to be located in less productive fishing areas and the impact on Programme interventions is likely to be low.
Mitigation Measure(s)
The EE will work with participating countries to ensure consultation and overall communication, including all relevant government ministries and civil society stakeholders, including but not limited to the Ministries of Environment which oversee the planning of national-scale Marine Protected Areas and marine spatial planning processes to ensure sustainable protection and production in the fisheries sector. The EE will also oversee multiple activities related to an audit of national legislation that may need to be updated, created, or revised to effectively support strategic tuna fisheries goals and supply chains for FAD-caught fish and bycatch and tuna from transhipment and unloading operations. There are Programme activities specifically targeting this risk through communication campaigns and direct engagement with partner organisations and ministries. Further, there will be a Programme Steering Committee in place comprised of high-level individuals from participating countries who will have a sound understanding of national initiatives that may overlap with the Programme.

Selected Risk Factor 3 **Ineffective coordination between Regional Fisheries Management Organisations leads to increased uncertainties in managing redistributed tuna.**		
Category	**Probability**	**Impact**
Governance	Low	Low
Description		
Technical Study 8 prescribes "*Develop a preliminary framework for joint management of redistributed tuna stocks by WCPFC and IATTC, provides an overview of potential issues that will need to be considered by both RFMOs as the impacts of climate change on Pacific Ocean tuna populations continue to evolve*". There are existing legal frameworks governing the operation of the RFMOs and formal arrangements supporting their cooperation in relation to science, data, and compliance matters; however, the redistribution of tuna biomass due to climate change poses new management challenges that will require enhanced cooperation, particularly in the context of future negotiations for the retention of tuna fishing access fees by participating countries as tuna stocks are redistributed to the high seas and/or a different RFMO. The impact of ineffective coordination would be less effective management of redistributed tuna stocks in the long term.		
Mitigation Measure(s)		
The legal frameworks of the two RFMOs governing tropical tuna in the Pacific Ocean encourage and enable cooperation on various issues. There can be a specific conservation and management measure created that mandates the type of cooperation necessary. At the annual regular session of the WCPFC in 2022, the Commission adopted a Resolution that made climate change a permanent item on the Commission agenda (and the agenda of its subsidiary bodies) to ensure priority is given to the issue. The WCPFC and IATTC can act now to begin strategizing around a framework that will support strengthened, informed, and effective cooperation. SPC has existing MOUs with both RFMOs and will facilitate coordination and communication on the status and redistribution of tuna stocks as determined by the AWS.		

Selected Risk Factor 4 **Global or national economic or political situation becomes unstable, disrupting the Programme.**		
Category	**Probability**	**Impact**
Governance	High	Low
Description		
Political risk in the countries participating in the Programme is characterized by possible changes in government personnel in key positions or changing mandates of different government agencies or ministries. The main risk is that political disruption and changes in personnel and/or government structures could cause delays in Programme operations. While the probability of such disruption is high given the number of participating countries in this Programme, the overall impact of such disruption/delays over the lifetime of the Programme is considered to be low and geographically limited.		

Changing mandates of government agencies can impact how a country positions itself within global economic forums, for example, supporting maximizing benefits from tuna fisheries versus other national priorities such as deep-sea mining exploration and marine spatial planning to support MPAs. Other potential consequences include disruption or delays in participation in regional fora by relevant government officials; non-materialization of support for Programme staff through national budgets to sustain activities over the long term; and changes to the individuals assigned to engage with the Programme with potential negative consequences on effectiveness and / or efficiency.

Mitigation Measure(s)

SPC has a long history of working in the region and has been able to consistently collaborate with successive administrations and government structures in the Programme countries. Because the 14 participating countries are formal member nations of SPC as their regional technical agency, it is expected that they will continue to work closely with SPC despite changes in government personnel. The selection of SPC as single EE was an explicit instruction from the 14 Participating Countries.

Government ministries and agencies from all the participating countries have been involved in the Programme design, including during stakeholder consultations organized to provide input to the Funding Proposal development. The Programme activities have been designed to address government priorities related to climate change in the fisheries sector, and the Programme enjoys strong support from all participating countries.

Based on its 70 years of experience working in the region, SPC can swiftly respond to unexpected regional or national challenges. For example, SPC could quickly shift the timing and location of workshops and training events disrupted by political instability to virtual events.

Selected Risk Factor 5	**Prohibited Practices, including ML/FT**	
Category	**Probability**	**Impact**
Prohibited practices	Low	Medium

Description

This large-scale Programme involves a wide array of actors in 14 countries—including the EE, Implementing Partner organisations receiving GCF funds, participating SPC member country governments receiving GCF resources, and vendors and service providers—which may increase the likelihood of risk of prohibited practices (PP). However, both CI (the AE) and SPC (the EE) are accredited entities to the GCF and have policies and procedures in place to counter fraud, corruption, Money Laundering/Financing of Terrorism (ML/FT), and other prohibited practices. Both organisations and the Programme itself have grievance mechanisms through which prohibited practices violations, including those related to ML/FT, can and should be reported.

Concerning impact, serious breaches in prohibited practices, including ML/FT, would adversely affect the reputation of CI, SPC, and the Programme and relationships with participating countries.

Mitigation Measure(s)

The AE has several procedures and controls in place to prevent, mitigate, and identify the risks of prohibited practices. To comply with CI AML/CFT, U.S. law, and GCF fiduciary standards, CI applies rigorous CTF/AML screening to all CI funding recipients--including GCF EEs--that screen against global sanctions lists, including UN Security Council sanctions lists. CI also requires EE programme staff to undergo training on PP (including red flags and how to report violations), and all grant and services agreements (used by both CI and sub-grantees) must include language on compliance with the GCF Policy on Prohibited Practices, including Anti-Money Laundering / Countering the Financing of Terrorism (AML/CFT). Compliance with PP policies and mitigation measures is part of project monitoring by the AE, including site visits, throughout implementation.

As EE, SPC has a suite of policies related to ethics, fraud, and anti-corruption and a detailed Anti-Money Laundering and Counter-Terrorism Financing Policy consistent with international best practices to prevent, mitigate, and identify the risks of prohibited practices. SPC will screen all funding recipients against international sanctions lists to prevent ML/FT. In addition, SPC staff working on the Programme will be required to undergo CI's training on PP, including AML/CFT (including red flags and how to report violations).

In addition, all contractual agreements governing a transfer of GCF funds used by the AE or EE for the programme must include language on causing compliance with the GCF Policy on Prohibited Practices, including AML/CFT. It is also intended that ongoing AML/CFT compliance will be analysed, assessed, and monitored during program implementation. Appropriate contractual language will be added to all sub-agreements that will require notification about any alleged AML/CFT-related claims, and the EE and AE will take appropriate actions in such cases.

To report and remedy risks, any CI employee; employee of an EE, grantee, partner, or vendor; or beneficiary may report a grievance or violation of these policies anonymously online or by telephone through CI's Ethics Hotline. All reported grievances will be investigated and addressed using CI's Anti-Fraud Policy & Guidelines for Investigations. Full details can be found on CI's website at https://www.conservation.org/about/our-policies/reporting-illegal-or-unethical-conduct-statement. Grievances can also be submitted to SPC's complaints mechanism and will be investigated and addressed using SPC's policies and procedures.

Selected Risk Factor 6	**Unrealised or delayed co-financing**	
Category	**Probability**	**Impact**
Technical and operational	Low	Select High

Description

Component B relies on continued and ongoing oceanic data collection from a large area of the tropical Pacific Ocean. To collect the oceanographic data, successful collaboration with commercial fishing vessels that operate in EEZs and high seas areas is required. This data collection, and the associated value is considered in-kind co-financing for the Programme. This co-finance will occur in years 2-6 and there is a low probability this co-finance will not materialize in time and/or in amount.

Although SPC has pre-established relationships with many key industry players operating in the Western and Central Pacific region, the companies could consider this collaboration as an operational burden, given that commercial tuna fishing in the WCPO is an already demanding operation. Support from at least three fishing companies and their vessel captains is required to collect all the data needed to build the AWS; additionally, there may be legislative barriers in certain countries which limit the ability of fishing vessels to undertake this type of collaboration. No collaboration with commercial fishing vessels will result in the failure of Component B that relies on continued and ongoing oceanic data collection from a large area of the tropical Pacific Ocean.

Mitigation Measure(s)

CI and SPC conducted outreach to multiple companies operating commercial tuna vessels in the WCPO at side events of the Western and Central Pacific Fisheries Commission Meetings in December 2022 and 2023. Industry was briefed on the GCF Programme and requested to collaborate. Technical experts who have strong relationships with key industry leaders and captains have commenced a proactive engagement exercise to formalise collaborative arrangements with the industry. This exercise, which involves joint outreach and consultations to be undertaken by SPC with the support of the International Seafood Sustainability Foundation (ISSF)[107] aims to formally engage at least three fishing companies (one based in Solomon Islands, one based in Fiji, and one based in Pohnpei) in utilising their vessels as ''ships of opportunity'' to collect oceanographic data necessary to support Component B of the Programme. The collaborative arrangements for this exercise were discussed, and agreed, in the margins of the December 2023 WCPFC meeting in Cook Islands.

CI as AE will report to GCF 1. The amount of the Co-financing disbursed and applied to the implementation of the Funded Activity 2. Promptly inform GCF of any cancellation or reduction of the Co-financing that could hinder achievement of expected adaptation impact; per the Programme term sheet.

Selected Risk Factor 7:	**Risk of Inflation and Pricing Changes**	
Category	**Probability**	**Impact**
Technical and operational	Medium	Low

Description

[107] www.iss-foundation.org

The Pacific region, like the rest of the world, has recently experienced high inflation. Although the IMF forecasts that inflation is going to continue falling for the Asia-Pacific region during 2024 and 2025[108], the risk of inflation is considered a 'medium' risk to the Programme over the course of implementation because it could impact the budget, co-financing, and activity implementation. The Pacific region is particularly vulnerable due to dependence on external markets, the small economies of island states, limited foreign currency reserves, susceptibility to global supply chain disruptions, energy price fluctuations, and instability in major trading partners. Higher-than-expected inflation would increase costs across the whole Programme, making it harder to achieve Programme targets within the set Programme budget.
Mitigation Measure(s)
To mitigate the impacts of inflation, the budget has been prepared assuming an inflation rate of 2.5%, in line with SPC practice. The inflation rate used is in line with central bank targets for participating countries over the medium term (5-10 years). CI and SPC will regularly monitor inflation pressures on the Programme budget--including by SPC through its standard SPC financial management practices and by CI through the regular Programme financial reporting and annual budget updated. As needed, cost savings will be assessed and applied through adaptive management of the Programme by the EE in consultation with the AE.

Selected Risk Factor 8: **Health crises cause extended disruption to regular economic, societal, or institutional functions.**		
Category	**Probability**	**Impact**
Other	Low	High
Description		
Unanticipated situations like the COVID-19 pandemic could lead to countries participating in the Programme closing their borders in a complete lock-down manner, possibly for multiple years. COVID-19 made travel to the region impossible, and it disrupted many work streams, including the fisheries sector. Even now, some travel within the region is limited and subject to change, which could have an impact on certain national and regional workshops or Activities planned by the Programme. A significant increase in and spread of COVID-19 or other infectious diseases could lead to restrictions on movements at both regional and national scales, causing delays in Programme activities and other engagement work. Delays could also be caused because of limits on the implementation of in-person capacity-building activities/exchange experiences or limits on working directly with communities due to disease risks.		
Mitigation Measure(s)		
The EE and Implementing Partners will regularly revisit the workplan to apply adaptive management, including anticipation of different COVID-19 scenarios and appropriate sequencing of activities to permit continued progress. There will be strict adherence and compliance with the requirements of national health authorities and World Health Organization guidelines, in addition to the partner's own health and safety standards. As a result of COVID-19, all the partner organisations have procedures and guidelines related to highly infectious diseases, and recent experience in implementing measures such as social distancing, provision of personal protective equipment and project-related safety and security measures.		

Selected Risk Factor 9 **Participating governments contributions to Programme costs**		
Category	**Probability**	**Impact**
Technical and operational	Low	Medium
Description		

[108] https://www.imf.org/-/media/Files/Publications/REO/APD/2023/October/English/text.ashx; accessed 12 January 2024

As part of the Programme's exit strategy and sustainability beyond the implementation period, the governments of the 14 participating countries are required to pay for associated staff salaries from Years 3-5. There is a risk that the governments may not fulfil their commitment to the Programme due to changing priorities, instability, or other factors. Should the governments fail to fulfil their commitments, there may be adverse impacts on the implementation of Component A, resulting in delays in implementation and achievement of the Component's objective. It may also affect the sustainability of the Programme beyond the implementation period.		
Mitigation Measure(s)		
Failure to secure strategic partnerships or maintain collaborative agreements to maintain Programme initiatives beyond the conclusion of the Programme is a risk that has been associated with some initiatives across the Pacific Islands region. However, because, the 14 participating countries and their regional agencies are experienced in assessing and managing these risks, and in designing mitigation arrangements to minimize adverse implications, the risk is low (see also para 141 in the Exit Strategy section). The commitment by countries to provide indicated staff resources during implementation and beyond the Programme period of performance will be captured in the agreement to be developed between SPC and each participating country during Programme implementation.		
Selected Risk Factor 10 - EE Capacity to Implement the Programme		
Category	**Probability**	**Impact**
Technical and operational	Low	High
Description		
This multi-country Programme is large and complex by nature, and the EE has not managed or executed a single project of comparable size. Potential consequences of insufficient capacity to implement the Programme include cost overruns, delays in implementation, and ultimately failure to achieve adaptation impacts.		
Mitigation Measure(s)		
CI has conducted due diligence and capacity assessments of SPC as the EE, including financial management and procurement capacity, and SPC is implementing improvement/strengthening strategies including a risk registry for monitoring, improved procurement processes, and an appropriate staffing structure for implementation of the Programme. The EE has managed a variety of large-value projects concurrently across its varying divisions which cumulatively exceed USD$100 million. The AE will carefully monitor Programme implementation progress and suggest adaptive management measures if delays to implementation or other issues are encountered.		
Selected Risk Factor 11 - Partner Capacity		
Category	**Probability**	**Impact**
Technical and operational	Low	Medium
Description		
Full capacity assessments of Technical and SPC Member Grantees will take place during implementation of the Programme. If the technical and financial management capacity of partners is insufficient, consequences include delays to implementation of sub-activities due to the need to engage alternative grantees.		
Mitigation Measure(s)		
Technical partners were selected for their track record, expertise in specific sectors, and prior engagement with SPC. In addition, grantees' financial management capacity will be assessed during the implementation period prior to the award of the grants, and appropriate risk mitigating measures will be employed as necessary. In the case of SPC Member grantees, if financial management / procurement capacity is insufficient, SPC will procure goods and services on behalf of the participating country.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

191. During Concept Note development, the proposed activities were screened against the 10 Environmental and Social Standards (ESS) of the CI-GCF ESMF. The Screening was subsequently reviewed as part of the PPF grant application based on revised activity designs. The Screening was conducted again in March 2024 based on the final Programme design. No new risks were identified in the final screening. Below is a summary of the risks identified in the March 2024 screening, confirming the risk categorisation of C as the Programme is likely to have minimal adverse environmental or social impacts. The programme is expected to have minimal environmental and social impacts due to low-impact FAD designs that will be safely deployed at depths beyond those of coral reef ecosystems and informed by indigenous knowledge to avoid issues such as damaging coral reefs and marine mammal entanglement. In addition, there will be no infrastructure or construction activities implemented except for small-scale work for container pads and anchor blocks, and there will be no direct financing to industrial fishing vessels or fish processing plants. Identified impacts will be avoided through design or will be minimised through activity design and implementation management measures.

192. An Environmental and Social Management Plan (ESMP) was developed during the programme design to provide detailed guidance during implementation. The ESMP is provided in **Annex 06** of this Funding Proposal. The main risks anticipated are summarised against the CI/GCF ESMF standard below, along with mitigation measures to address them.

Summary of Risks	Mitigation Factors/Measures or Management Strategies
ESS 1: Environmental and Social Assessment, Management and Monitoring **(GCF ESS 1: Assessment and Management of Environmental and Social Risks and Impacts)**	Based on the detailed description of Programme activities, the results of the screening and identified inherent risks determine a Category C risk rating for this Programme. ESS screening finds that ‘there are minimal or no adverse environmental and/or social risks and/or impacts. An ESMP is determined to be an appropriate instrument for this Programme where there are still elements of activities which have not yet been determined (such as FAD deployment site selection), the ESMP and other safeguards instruments (Stakeholder Engagement Plan and Indigenous Peoples Plan) has been designed to provide coverage for avoidance of minimisation of potential risks and/or impacts without the need for additional site-specific screening.

ESS 4: Indigenous Peoples At the regional level, the 14 participating countries include communities with a diversity of customs, customary laws, norms, cultural practices, languages, and traditions meeting the broad GCF definition of Indigenous Peoples. However, at the Programme intervention level, communities across the PICs are, overall, considered to be homogenous in language, culture, and practices. This means that Programme benefits or impacts will not adversely or differently affect indigenous people under the GCF definition at the Programme site level.	**Low Risk** Affected People are considered at a community level encompassing marginalized and vulnerable groups and individuals. Best practice will be used and Free, Prior, and Informed Consent (FPIC) using traditional customer processes is integrated throughout Programme design and stakeholder engagement. An IPP was not required for this Programme as FPIC has been integrated into FAD site selection processes both in activity design and through the required processes of the ESMP. This is considered **low risk** given the inherent cultural norms across the PICs which prevent any type of Programme activities from taking place within communities without community support and/or approval.
ESS 5: Resource Efficiency and Pollution Prevention FADs which break loose from their mooring have the potential to become marine pollution.	The Programme is using a FAD system designed by SPC and will be securely anchored in a nearshore (or up to 7km offshore in limited circumstances in 3 participating countries) water for some FADs in three countries) setting which minimizes the usual risks to FAD loss (through damage from large shipping vessels). FADs are designed to minimize entanglement with marine mammals, turtles and sharks. FADs will also be installed with a GPS tracker to enable lost FADs to be easily recovered, repaired, and reinstalled. No specific management plan is required for this standard as SPC FAD design is specially contextualized to the nearshore environments of the PICs and recovery of lost FADs is built into activity design through GPS trackers. (Activity 1.1.6) This is considered **low risk** given the small artisanal nature of the proposed FADs and considering that SPC have specially targeted their FAD design to be suitable for these environments.

ESS 8: Community Health, Safety and Security	
1. Changing fishing methodology of some fishers from reef fishing to FAD fishing presents safety-at-sea risks. Safety-at-sea risks include: (i) operating new types or sizes of boat (mechanical and skills), (ii) fishing in new types of water conditions, (iii) inexperience or lack of data leading to exposure to weather events at sea, (iii) health and safety risks associated with using new type of fishing gear used for FAD fishing.	1. A CHSS Plan has not been developed as part of the PPF, instead an activity (with three sub-activities) has been developed in the Programme log frame to specifically include addressing the risks associated with safety-at-sea (Activity 1.2). Activity 1.2 will be implemented in alignment with the requirement of this ESMP to ensure all applicable aspects of ESS 8 will be captured during implementation. This is considered low risk as many fishers targeted by the Programme are already familiar with FAD fishing and are currently working under this risk. Through activity 1.2, the Programme is providing them with an increased level of safety compared to the status quo. Despite the specific comprehensive activities proposed under this Programme, any time a fisher enters the marine environment there is a risk to safety. That inherent risk of going to sea exists regardless of fishing method and complete avoidance of risk is impossible in any setting.
2. There is also a risk associated with the post-harvest handling and processing of tuna. These risks are associated with histamine poisoning from tuna left in the sun for too long and botulism risks from incorrect processing of harvested fish.	2. This is considered **low risk** as both risks are well known and understood within the local fishing communities and within the Executing Entity (EE), SPC. There are already existing materials for training in the PICs to minimise these risks and these training materials have been incorporated into the capacity building plans for activity implementation. Please see Annex 6a (ESMP) for more information.

193. Further details about the risks considered in the AE's risk screening process are provided in the ESMP in **Annex 06**. The ESMP shows how these risks were considered in the design of the Programme and any additional mitigation measures that may be required. In addition, given the processes to be carried out during Programme implementation, the ESMP requires that risks be managed through the measures listed below:

- Involving an experienced Gender Equity and Social Inclusion (GESI) officer throughout the activity planning, design, and implementation processes. This specialist will be part of the EE PMU.
- Ensuring the monitoring and evaluation (M&E) process includes adequate review of environmental and social risks. This will be overseen by the PMU's Monitoring, Evaluation and Learning (MEL) officer with specific experience in reviewing environmental and social considerations.
- Assigning formal roles and responsibilities within the ESMP for environmental and social management across the Programme team.
- Establishing a reporting procedure for environmental and social concerns related to the interventions, including through the Programme's Accountability and Grievance Redress Mechanism (AGRM).
- Incorporating environmental and social safeguard considerations into the training, capacity building and technical assistance delivered through the programme.
- Ensuring environmental and social risks and safeguards are considered in technical assistance activities and any documentation to be produced under the Programme, including in Terms of References and capacity-building materials.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

194. Across the Programme countries, apart from market sales, fisheries are generally considered a male domain given men's traditional role as fishers. Although FADs improve the availability of fish[109], they can only be directly used by those who have access to boats (mostly men). The Programme is designed to take a holistic view of the tuna value chain, not just the actual fishing of tuna, but the post-harvest processing, storing, and marketing which is where women feature more prominently, and which is also directly impacted by a transition to tuna FAD fishing. Women's roles in the value chain of fisheries are critical in sustaining the fishing families and community, protecting natural resources, and local food availability. While women and other vulnerable groups have an important stake in fishery programmes and outcomes, such as health, livelihoods, and quality of life, their needs can often be invisible and overlooked, limiting their access to the benefits and use of such programmes or projects. Such vulnerable groups can face persistent gender-based discrimination, which may impact their ability to gain benefits equitably.[110] They often have little to no say regarding decisions around the use of marine resources.

195. A Gender Assessment and Gender Action Plan have been compiled during Programme development to identify key gender concerns to be considered and addressed, as well as opportunities to drive gender-responsive Programme delivery (**Annex 08**). Data were collected from primary and secondary sources across the 14 Programme countries. The gender assessment identified gender risks pertaining to:

- Division of Labour, with men experiencing higher safety risks as active fishers using the FADs and women experiencing safety risks related to the processing and marketing of fish.
- Lack of acknowledgement or consideration of women's roles and contribution within the tuna fishery value chain leading to the focus primarily on men as the visible fishers. This can lead to restricted access to information or opportunities for women.
- Inadequate financial resources to address women's specific needs within the fishery value chain, particularly with respect to safe and hygienic marketplaces and processing/storage facilities.
- Decision-making roles and local governance of fisheries are often male dominated[111], making it difficult for women to advocate for safe working conditions.
- Traditional household roles and norms for women limit women's financial contribution to the family and restrict women from entering tuna fishing, perpetuating fishing as a male-dominated activity.

196. Given the results of the assessment, and noting that each country has different levels of infrastructure, resources, cultural norms, and levels of commitment to gender equality goals, the following broad strategies are proposed to address gender disparities in all participating countries:

- Improve awareness of gender and related gender dimensions in all levels of the Regional Tuna Programme to ensure that women and men benefit equitably from the Programme. This includes raising awareness on gender dimensions and women's roles among relevant Programme stakeholders and the broader coastal community in the region.
- Support and encourage women's participation in key local decision-making and implementation structures, processes and outcomes related to the Programme. This may include facilitating leadership skills development and peer mentoring for existing/emerging women leaders in the local tuna fishery sector.
- Collect sex-disaggregated data and information as a basis of planning and monitoring Programme outcomes, highlighting measures addressing gender equality objectives.
- Link gender equality outcomes to the broader RTP outcomes, indicating how these contribute to the overall Programme success. This may include producing communication materials that highlight lessons learned, good practices, and successes related to gender-responsive FAD deployment. This type of material is useful for the countries, the region, and the broader fishery/FAD user community to influence and support greater uptake of gender-responsive considerations in coastal FAD projects.

[109] See: Chapman L. *in prep*. Regional Report: Feasibility of scaling-up National Fish Aggregating Device (FAD) Programmes in all 14 participating countries. Technical Study prepared for the Pacific Community as a contribution to a funding proposal being prepared for submission to the Green Climate Fund (GCF). October 2023. Lindsay Chapman Consulting Pty Ltd, Brisbane, Australia. 92 p. https://purl.org/spc/digilib/doc/au6vm

[110] Pacific handbook for gender equity and social inclusion in coastal fisheries and aquaculture gender equity and social inclusion handbook (spc.int)

[111] Although note there's an increasing proportion of women taking on very senior roles in fisheries in both national and regional fisheries administrations

- Provide targeted capacity building on post-harvest techniques, including processing and marketing, with a focus on women given their primary role in this part of the value chain, linking awareness, resources, and support services to reduce GBV and SEAH programme-related risks.
- Actively implement inclusive approaches to engaging FAD fishers and others along the value chain. While women are not currently present as FAD fishers in the region, training and capacity building will be open and inviting for any women that may be interested in entering the profession. Communicating about and making programme activities and opportunities accessible and open to women is critical.
- The Programme's regional gender objectives must be flexible enough to be adapted to the individual country local realities, programme impacts, and related nuances. Each country will identify their priorities and opportunities to advance national gender equality goals in the context of FAD fishing. The Programme can provide an opportunity to change the narrative about gender inequality in the fishing sector and support women's economic empowerment.

Gender Action Plan (GAP)

197. A Programme-level Gender Action Plan has been developed to encourage women's participation, representation, and awareness in tuna fishing supply chains (**Annex 08**). This will seek to discourage gender biases among Programme participants and beneficiaries and/or ensure gender inequalities are not exacerbated by the Programme. These actions, in addition to those presented in paragraph 196 above, will encourage gender dimensions to be mainstreamed in each country's context, responding to each's varied risks & opportunities. This encourages achievable targets, sex-disaggregated data collection, gender-responsive programme design features, and quantifiable performance indicators. Some activities that the GAP recommends include:
    - Upskilling of programme stakeholders in Gender and GBV/SEAH dimensions to encourage greater awareness of men and women's roles and safety in the FAD tuna fishing supply chain.
    - Integrating gender considerations into important programme documents/outputs, including: workplans for country-level FAD scale-up and national natural disaster response mechanisms; inclusion of women in all programme components - consultations, monitoring activities, training, and workshops; and equitable stakeholder engagements to identify sites for FAD deployment.
    - Ensuring the Advanced Grievance Redress Mechanism (AGRM) (regional & national) are gender-responsive, with communications reaching women stakeholders and a mechanism to respond to GBV-related concerns/grievances.
    - Providing targeted, gender-responsive training courses for women in coastal communities and urban communities to improve the preservation and hygiene of FAD-caught fish, including marketing opportunities. This includes awareness & resources to respond to GBV/SEAH-type risks that women may face as fish processors/marketers.
    - Communications about the nutrition and benefits of FAD-caught tuna are gender-informed, recognising the roles that women play in household food preparation and decision-making.
    - Producing learning documents from the Programme that showcase lessons and successes concerning gender-responsive implementation.

198. The GAP includes indicators with baseline and targets for each activity to track progress, which are aligned with the GCF IRMF indicators:

    - Number of men & women who participate in Programme activities.
    - Number of women & men who receive direct benefits from the Programme.
    - Number of plans/strategies that integrate gender considerations.

199. The Regional PMU will have a GESI specialist to cover environmental and social safeguards and gender. Their primary focus will be social safeguards and gender elements as environmental risks are considered low **(section G.1)**. At the national level, each national implementation team/unit will have national officers who will have a range of technical areas to cover (**see section B.4**). The country level gender and safeguards responsibilities will be included in the terms of reference of the national technical specialists, and they will be supported by the regional GESI specialist. The regional specialist will provide training on safeguards and gender requirements including socializing the regional AGRM with the national staff.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

200. CI as Accredited Entity will follow financial management and procurement for this Project in line with relevant, established CI policies and procedures, as well as relevant provisions in the Accredited Master Agreement (AMA) signed between CI and the GCF.

201. SPC, the project's Executing Entity, is a GCF Accredited Entity and therefore, was deemed by the GCF to be compliant with its fiduciary standards. Furthermore, CI has assessed SPC and has determined that it has robust financial and procurement standards in place to meet CI and GCF fiduciary standards.

Financial management

202. The AE, CI, uses Unit 4 Business World software suite as its enterprise resource planning (ERP) system for financial accounting and monitoring in all its global operations. This system supports financial monitoring, reporting, human resources, procurement, and record-keeping in line with the requirements of CI's AMA. It supports a system of ledger accounts to separate and track income and expenditure by donor, project, and subaward. CI also provides a coding system and established procedures for monitoring and reporting of projects. The EE, SPC, uses the Dynamics Nav software suite as its enterprise resource planning (ERP) system for financial accounting and monitoring. This system supports a system of ledger accounts to separate and track income and expenditure by donor and project. SPC also has a coding system and established procedures for monitoring and reporting of projects.

203. SPC and its sub-grantees will be required to submit periodic financial reports that will be systematically reviewed by the AE and EE, respectively, according to their respective policies and procedures, which meet GCF standards.

204. An annual audit of the Programme will be conducted by an independent external audit firm in accordance with internationally recognized auditing and accounting standards and furnished to the GCF in accordance with CI's AMA.

205. Disbursements made to CI and SPC will be recorded in the CI ledger account for this GCF Programme, which can be tracked using the respective financial software systems. Disbursements from the EE to sub-grantees will be made in line with SPC financial policies.

206. CI and SPC have procedures and controls in place to prevent the risk of prohibited practices, including ML/FT (collectively, 'PP'). CI and SPC apply rigorous CFT/AML screening to all CI funding recipients. CI and SPC also assess grant recipients' PP risks and include mitigation measures in its grant agreements. All EE staff must also undergo training on PP (including red flags and how to report violations), and all grant and services agreements (used by the EE and sub-grantees) must include language on compliance with the GCF Policy on Prohibited Practices, including AML/CFT. Compliance with PP policies and mitigation measures is part of project monitoring, including site visits, throughout Programme implementation.

207. Violations of PPs, fraud, or financial misuse may be reported anonymously online or by telephone through the GCF's IIU, CI's Ethics Hotline, and/or SPC's complaints mechanism. All reported grievances reported to CI's Ethics Hotline will be investigated and addressed per CI's Anti-Fraud Policy & Guidelines for Investigations, and all grievances reported to SPC will follow its internal procedures.

208. CI grants management and subrecipient monitoring is supported by a grants management system that is integrated across its donor management, accounting, and budgeting systems. This system provides notifications for due dates of deliverables (from the grant recipient and to the donor), tracks disbursements, project expenses, milestones, audit findings (if applicable), and identified risks so that project supervision is informed and documented. SPC has a comprehensive Grants Policy governing subgrant management and subrecipient monitoring; the procedures and tools for SPC's application of these policies for the large-scale sub granting of this Programme will be further defined in a Programme granting manual during implementation and verified by the AE during implementation.

Procurement

209. All procurement for this project must follow the Procurement Process Standards as defined in the CI Procurement Policy, which requires that all procurements of goods or services ensure the best value and comply with all donor funding terms and conditions and that all goods be used only for their intended purposes and consistent with CI's Code of Ethics and suite of anti-fraud policies. CI's procurement standards are in line with GCF fiduciary standards.

210. SPC has detailed, written procurement policies consistent with CI and GCF procurement policies, and subsidiary agreements will include relevant flow-down terms to ensure full compliance (**see Annex 19**). All procurement done by SPC will follow its own policies and procedures and will be subject to the AE's procurement controls and risk-mitigation measures, including procurement planning, multi-level purchasing approvals (including AE approval over a set threshold), quarterly reporting on procurement & equipment, periodic inventories, and post-award reviews.

211. Consistent with CI's Procurement Policy, Programme sub-grantees that meet CI's Procurement Process Standards as defined in CI's Procurement Policy will use their own procurement policy and procedures for the Programme; otherwise, they will be required to use SPC's procurement policy and/or follow relevant contractual flow-downs. Detailed procurement requirements consistent with CI and GCF standards will be included in the subsidiary agreement signed between SPC and the grantee organisation.

212. CI will apply mitigation measures and monitor all procurements for this Programme to ensure that any procured goods are used only for their intended purposes. It will provide training on procurement and prohibited practices for SPC staff and will approve annual procurement plans and all SPC procurements over a set threshold. It will also conduct monitoring through quarterly reporting on SPC procurement and equipment and periodic site visits to SPC and subgrantees.

G.4. Disclosure of funding proposal

Indicate below whether or not the funding proposal includes confidential information.

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

H. ANNEXES

H.1. Mandatory annexes

☒	Annex 01	NDA no-objection letter(s) **(template provided)**
☒	Annex 02	Feasibility study - and a market study, if applicable
☒	Annex 03	Economic and/or financial analyses in spreadsheet format
☒	Annex 04	Detailed budget plan **(template provided)** (summary provided V.1)
☒	Annex 05	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 06	E&S document corresponding to the E&S category (A, B or **C**; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☒ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☒ Others - Accountability Grievance Redress Mechanism (AGRM)
☒	Annex 07	Summary of consultations and stakeholder engagement plan
☒	Annex 08	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 09	Legal due diligence (regulation, taxation, and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☒	Annex 13	Co-financing commitment letter, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval **(template provided)** (interim)
☒	Annex 16	Map(s) indicating the location of proposed interventions
☒	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence, or evaluation report for proposals based on up-scaling
☒	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☒	Annex 20	First-level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☒	Annex 22	GCF USP 2 target contributions
☒	Annex 23	Adaptation beneficiary calculation
☒	Annex 24	Summary of climate policies and strategies in participating countries
☒	Annex 25	Implementation arrangements detail
☒	Annex 26	Technical Studies
☒	Annex 27	Procedures for Granting
☒	Annex 28	Additional AWS Information

H

No-objection letter issued by the national designated authority(ies) or focal point(s)

CLIMATE CHANGE COOK ISLANDS

OFFICE OF THE PRIME MINISTER

GOVERNMENT OF THE COOK ISLANDS

Private Bag, Avarua, Rarotonga, Cook Islands, Phone: (682) 25 494, Fax: 20 856.
Email: wayne.king@cookislands.gov.ck **Website**: www.pmoffice.gov.ck,

To: The Green Climate Fund ("GCF")
"G" Tower
Songdo
South Korea

Rarotonga, Cook Islands
24 October 2024

Re: Funding proposal for the GCF by Conservation International regarding "Adapting Tuna-Dependent Pacific Island Communities and Economies to Climate Change"

Dear Madam, Sir,

We refer to the programme titled **"Adapting Tuna-Dependent Pacific Island Communities and Economies to Climate Change"** in the Cook Islands as included in the funding proposal submitted by Conservation International to us on 9 April 2024.

The undersigned is the duly authorized representative, Wayne King, the National Designated Authority of the Cook Islands.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of the Cook Islands has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Cook Islands;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

TAURANGA TAUI'ANGA REVA
CLIMATE CHANGE COOK ISLANDS
Office of the Prime Minister

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

____

Wayne King
Director
Climate Change Cook Islands, and NDA
Cook Islands

Government of the Federated States of Micronesia
Department of Finance & Administration
PO Box PS158 Palikir, Pohnpei FM 96941

Office of the Secretary

Date: August 26, 2024

To: The Green Climate Fund (GCF)
G-Tower, Songdo Business District
175 Art center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Re: Funding Proposal for the GCF by Conservation International regarding "Regional Program: Adapting tuna-dependent Pacific Island communities and economies to climate change"

Dear Executive Director Duarte,

We refer you to the program titled 'Adapting tuna-dependent Pacific Island communities and economies to climate change' in the Federated States of Micronesia as included in the funding proposal submitted by Conservation International to us on 26 August 2024.

The undersigned is the duly authorized representative, the Honourable Secretary Rose N. Nakanaga, the National Designated Authority for the Federated States of Micronesia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

a) The government of the Federated States of Micronesia has no-objection to the program as included in the funding proposal;
b) The program as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Federated States of Micronesia;
c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Rose N. Nakanaga
Secretary/National Designated Authority
Federated States of Micronesia

MINISTRY OF ENVIRONMENT AND CLIMATE CHANGE

Levels 1 & 2, Bali Tower, 318 Toorak Road
P. O. Box 2109
Government Buildings, Suva, Fiji

TELEPHONE NO: (679) 3311-699 **Email: info6@govnet.gov.fj**

To: The Green Climate Fund ("GCF")

Suva, Fiji, 9 April 2024

Re: Funding proposal for the GCF by Conservation International regarding '*Adapting tuna-dependent Pacific Island communities and economies to climate change*'

Dear Madam, Sir,

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities economies to climate change* in Fiji as included in the funding proposal submitted by Conservation International to us on 6 April 2024.

The undersigned is the duly authorized representative of the Climate Change Division, the National Designated Authority of Fiji.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Fiji has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Fiji;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

F. R. Tuivanualevu

Mr. Filimone Tuivanualevu Ralogaivau
Acting Director
Climate Change Division
Ministry of Environment and Climate Change
Fiji

GOVERNMENT OF KIRIBATI

MINISTRY OF FINANCE & ECONOMIC DEVELOPMENT

Phone: (686) 74021806, Address: PO Box 67, Tarawa, Kiribati

Date: 3rd July 2024

Mafalda Duarte
Executive Director
Green Climate Fund
Songdo International Business District
175 Art Center-daero
Yeonsu-gu, Incheon 22004
South Korea

Re: Funding proposal for the GCF by Conservation International regarding *'Adapting tuna-dependent Pacific Island communities and economies to climate change'.*

Dear Ms Duarte,

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change'* in the Cook Islands, Federated States of Micronesia, Fiji, Kiribati, Marshall Islands, Niue, Nauru, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu and Vanuatu, as included in the funding proposal submitted by Conservation International to us on 1 February 2024.

The undersigned is the duly authorized representative of the Ministry of Finance and Economic Development, the National Designated Authority of Kiribati.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Kiribati has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Kiribati;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,

Honourable Dr Teuea Toatu

Vice President and Minister of Finance and Economic Development

REPUBLIC OF THE MARSHALL ISLANDS
CLIMATE CHANGE DIRECTORATE
MINISTRY OF ENVIRONMENT

P.O. Box 975
Majuro, Marshall Islands 96960 Phone No: (692) 625-7944/7945
Skype: clarence.samuel E-mail: clarencesam@gmail.com

Ref: MI24-09

To: The Green Climate Fund ("GCF")

Majuro, Republic of the Marshall Islands, 19th March 2024

Re: Funding proposal for the GCF by Conservation International regarding Adapting tuna-dependent Pacific Island communities and economies to climate change.

Dear Madam, Sir,

We refer to the programme titled Adapting tuna-dependent Pacific Island communities and economies to climate change in the 14 Pacific Island Countries including the Republic of the Marshall Islands as included in the funding proposal submitted by Conservation International to us on 15th March 2024.

The undersigned is the duly authorized representative of the Climate Change Directorate, the National Designated Authority of the Republic of the Marshall Islands.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The Government of the Republic of the Marshall Islands has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the Republic of the Marshall Islands national priorities, strategies and plans;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Clarence Samuel
Director/NDA
Republic of the Marshall Islands

REPUBLIC OF NAURU

DEPARTMENT OF CLIMATE CHANGE AND NATIONAL RESILIENCE

To: The Green Climate Fund ("GCF")

Tuesday 9th April, 2024

Re: Funding proposal for the GCF by Conservation International regarding 'A*dapting tuna-dependent Pacific Island communities and economies to climate change'*

Dear Madam/Sir,

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change* in Nauru as included in the funding proposal submitted by Conservation International to us on 9 April, 2024.

The undersigned is the duly authorized representative of Reagan Moses, the National Designated Authority of Nauru.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Nauru has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Nauru;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

____

Reagan Moses
NDA
Nauru

PREMIER

8th April 2024

The Green Climate Fund ("GCF")
Mafalda Duarte
Executive Director

Dear Ms Duarte,

Re: Funding proposal for the GCF by Conservation International with Pacific Community (SPC) Adapting tuna-dependent Pacific Island communities and economies to climate change

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change* in Niue as included in the funding proposal submitted by Conservation International to us on the 1st of April 2024.

The undersigned is the duly authorized representative, Hon Premier Dalton Tagelagi, the National Designated Authority of Niue.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Niue has no-objection to the programme as included in the funding proposal.
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies, and plans of Niue;
(c) In accordance with the GCF's environmental and social safeguards, the programme included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Hon Dalton Emani Makamau Tagelagi
PREMIER OF NIUE
AND GREEN CLIMATE FUND NATIONAL DESIGNATED AUTHORITY

GOVERNMENT OF NIUE ★ OFFICE OF THE PREMIER ★

Office of the Premier, PO Box 27, Alofi, NIUE
Telephone: + (683) 4200, Email: niue.premier@mail.gov.nu

MINISTRY OF FINANCE
Bureau of Budget & Planning
P.O. Box 6011, Ngerulmud, Melekeok, Republic of Palau 96939
Telephone: (680) 767-1270 Fax: (680) 767-5642 Email: ropbudget@palaugov.org

July 4, 2024

The Green Climate Fund ("GCF")

Funding proposal for the GCF by Conservation International Foundation (CI) regarding Adapting Tuna-dependent Pacific Island communities and economies to climate change

Dear Madam, Sir,

We refer to the programme titled "Adapting Tuna-dependent Pacific Island communities and economies to climate change" in the Republic of Palau as included in the funding proposal submitted by Conservation International Foundation (CI) to us on 06 April 2024.

The undersigned is the duly authorized representative of the Ministry of Finance, the National Designated Authority/Focal Point of the Republic of Palau.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have been reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Republic of Palau has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Republic of Palau;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme. We further acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Casmir E. Remengesau
Director
and
National Designated Authority of the Republic of Palau

Climate Change & Development Authority

GREEN CLIMATE FUND

To: The Green Climate Fund ("GCF")

Port Moresby, 5th April, 2024

Re: **Funding proposal for the GCF by Conservation International regarding 'Adapting tuna - dependent Pacific Island communities and economies to climate change'**

Dear Madam, Sir,

We refer to the programme titled Adapting tuna dependent Pacific Island communities and economies to climate change in Papua New Guinea as included in the funding proposal submitted by Conservation International to us on the 4th April, 2024.

The undersigned is the duly authorised representative namely Mr Ruel Yamuna who is the National Designated Authority for the Green Climate Fund in Papua New Guinea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no objection to the programme as included in the funding proposal.

By communicating our no objection, it is implied that:

(a) The government of Papua New Guinea has no objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Papua New Guinea;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards

Mr Ruel Yamuna, LLB
Special Envoy for Climate and Environment
National Designated Authority (PNG)
Ministry of Environment, Conservation
and Climate Change

Please address all correspondence
To the Chief Executive Officer

In reply, please quote the file reference

File ref:

FA'AVAE ILE ATUA SAMOA

GOVERNMENT OF SAMOA

MINISTRY OF FINANCE

09 April 2024

Ms. Mafalda Duarte
Executive Director
Green Climate Fund
Songdo International Business District
175 Art Center-daero
Yeonsu-gu, Incheon 22004

Dear Madam,

Re: Funding proposal for the GCF by Conservation International regarding '*Adapting tuna-dependent Pacific Island communities and economies to climate change*'

With respect, we refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change* in Samoa as included in the funding proposal submitted by Conservation International to us on 8 April 2024.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The Government of Samoa has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Government of Samoa;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

Looking forward in working together in this important initiative of the Oceania region.

Best regards,

Tofilau Lae Tui Siliva
Deputy CEO - Policy Management
Ministry of Finance
Samoa

Private Bag
APIA, SAMOA
***Telephone**: (685) 34333*
***Fax**: (685) 21312, 24779*

***Website:** http://www.mof.gov.ws*
***E-Mail**: mof@mof.gov.ws*

1 Apia City Boulevard
Central Bank Building
SAMOA

Solomon Islands Government

Ministry of Environment, Climate Change, Disaster Management & Meteorology

P.O Box 21,
Honiara,
Solomon Islands.

Fax:
(677) 28054
Phone:
(677) 23031 / 23032

www.mecdm.gov.sb

To: The Green Climate Fund ("GCF")

Honiara, 27 August 2024

Re: Funding proposal for the GCF by Conservation International regarding Adapting Tuna-Dependent Pacific Islands Communities and Economies to Climate

Dear Madam/Sir,

We refer to the programme titled ***Adapting Tuna-Dependent Pacific Islands Communities and Economies to Climate Programme*** as included in the funding proposal submitted by **Conservation International** to us on 8 March 2024.

The undersigned is the duly authorized representative of **Ministry of Environment, Climate Change, Disaster Management and Meteorology (MECDM), the National Designated Authority of Solomon Islands.**

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Solomon Islands has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of **Solomon Islands**;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

MINISTRY OF ENVIRONMENT, CLIMATE CHANGE, DISASTER MANAGEMENT AND METEOROLOGY

Ms. Susan Sulu
Permanent Secretary, (MECDM)
National Designated Authority of the GCF
Solomon Islands

Solomon Is.

MINISTRY OF METEOROLOGY,
ENERGY, INFORMATION, DISASTER
MANAGEMENT, ENVIRONMENT,
CLIMATE CHANGE AND
COMMUNICATIONS (MEIDECC)
NUKU'ALOFA, TONGA

Ref: DCC/04/24
To: The Green Climate Fund ("GCF")
Date: 5 April, 2024.

Re: Funding proposal for the GCF by Conservation International regarding '*Adapting tuna-dependent Pacific Island communities and economies to climate change*'

Dear Sir/Madam,

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change* in Tonga as included in the funding proposal submitted by Conservation International to us on 2 April, 2024

The undersigned is the duly authorized representative of Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC), the National Designated Authority of Tonga

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Tonga has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Tonga;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

MINISTRY OF METEOROLOGY, ENERGY, INFORMATION DISASTER MANAGEMENT, ENVIRONMENT CLIMATE CHANGE & COMMUNICATIONS
MEIDECC TONGA

Mr. Sione Pulotu 'Akau'ola
Tonga NDA to GCF and
Chief Executive Officer for MEIDECC
Tonga.

GOVERNMENT OF TUVALU
Office of the Minister for Home Affairs, Climate Change, & Environment
GPO, Vaiaku, Funafuti, Tuvalu.

To: The Green Climate Fund ("GCF")

Government Building, 3 April 2024

Re: Funding proposal for the GCF by Conservation International regarding *'Adapting tuna-dependent Pacific Island communities and economies to climate change'*

Dear Sir,

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change* in Tuvalu as included in the funding proposal submitted by Conservation International to us on 27 March 2024.

The undersigned is the duly authorized representative of Honorable Maina Vakafua Talia, the National Designated Authority of Tuvalu.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Tuvalu has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Tuvalu;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Honorable Maina Vakafua Talia
Minister for Home Affairs, Climate Change &Environment
National Designated Authority of Tuvalu

GOVERNMENT OF THE
REPUBLIC OF VANUATU MINISTRY
OF CLIMATE CHANGE
ADAPTATION, METEOROLOGY,
GEO-HAZARDS, ENVIRONMENT &
ENERGY & NDMO
PMB 9074, PORT VILA
VANUATU

LONG GOD YUMI STANAP

GOUVERNEMENT DE LA
RÉPUBLIQUE DE VANUATU
MINISTÈRE DE L'ADAPTATION AU
CHANGEMENT CLIMATIQUE, LA
MÉTÉOROLOGIE, LES RISQUES
GÉOLOGIQUES, ENVIRONNEMENT &
ENERGIE & NDMO
SPR 9074, PORT-VILA, VANUATU

TEL : (678) 22068 FAX : (678) 22068

Ref: PV/MoCC/Fisheries/PROJ-GCF:2. B.4

To: The Green Climate Fund ("GCF")

Port Vila, Vanuatu

27 March 2024

Re: Funding proposal for the GCF by Conservation International regarding '*Adapting tuna-dependent Pacific Island communities and economies to climate change.*'

Dear Madam, Sir,

We refer to the programme titled *Adapting tuna-dependent Pacific Island communities and economies to climate change* in Vanuatu as included in the funding proposal submitted by Conservation International to us on 5 April 2024.

The undersigned is the duly authorized representative of Mr. Abraham Nasak, the National Designated Authority of Vanuatu.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Vanuatu has no-objection to the programme as included in the funding proposal;
(b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Vanuatu;
(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

REPUBLIC OF VANUATU
The Ministry of Climate Change Adaptation, Meteorolgy, Geohazards Environment, Energy & NDMO
Le Ministere De Climat Change Adaptation, Meteorologie Georisque, Environment
REPUBLIQUE DE VANUATU

Mr. Abraham Nasak
Acting Director General, Ministry of Climate Change and GCF National Designated Authority
Vanuatu

Secretariat's assessment of FP259

Proposal name:	Adapting tuna-dependent Pacific Island communities and economies to climate change
Accredited entity:	Conservation International Foundation
Country:	Cook Islands, Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Nauru, Niue, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu, Vanuatu
Project size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Tuna plays an important role in the Pacific, especially for countries that are highly dependent on selling licences for tuna fisheries for government revenue. To increase efficiency and effectiveness, this proposal chooses a programmatic approach. Instead of separate single-country projects, the programme covers 14 Pacific countries while addressing tuna as a source for food security and stabilizing the benefits from the sale of tuna fisheries' licences at the same time. This allows the programme to reach 790,000 direct beneficiaries and 2.8 million indirect beneficiaries, a significant number in the Pacific context.	The Pacific Community (SPC) is the sole executing entity (EE) for this large-volume programme. While it has never implemented stand-alone projects of the same volume, it has implemented programmes of a larger volume in sum. Conservation International as accredited entity (AE) has assessed the capacities of SPC and included mitigation measures, such as the hiring of sufficient staff, in the programme design. However, this remains a risk which needs to be closely monitored during implementation.
The proposal and its activities are embedded in existing regional cooperation structures of the Pacific, with a proven track record of successful cooperation, first and foremost SPC, a regional member/country-owned and governed organization that has strong ties with the 14 individual countries and a physical presence in many. This design increases the likelihood of success and sustainability of the measures even beyond the project lifetime.	
The programme will sustainably strengthen national programmes for the deployment of fish aggregating devices (FADs). To	

participate in the programme, countries will agree to increasingly include officers and the procurement of FADs into their government budgets. By the end of the programme the costs will be covered exclusively by the countries, ensuring continuation of the measures beyond implementation of the programme.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity (AE), and, if considered appropriate, subject to the conditions set out in annex II of document GCF/B.41/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. The GCF Regional Tuna Programme (RTP) supports 14 Pacific Small Island Developing States (3 of which are least developed countries - LDCs), which are among the most climate-vulnerable countries in the world. The programme will work with participating governments to build resilience and adapt to two major threats posed by climate change: the degradation of coral reef ecosystems and the redistribution of Pacific tuna resources due to ocean warming.

4. Ocean warming and acidification are causing coral reef ecosystems to degrade. Coastal communities heavily depend on coral reef fish for their protein intake. The projected decrease in fish stocks, combined with growing populations, will exacerbate the gap in fish supply required to provide the recommended protein intake required for nutrition security.

5. Of the 14 participating countries, nine are heavily dependent on the sale of tuna fishing licences in their exclusive economic zones (EEZs). Access fees contribute on average 32 per cent, and up to 70 per cent, to their total (non-grant) government revenue. Although Pacific tuna stocks are among the most sustainably managed in the world, ocean warming is predicted to cause tuna biomass to redistribute eastwards, out of these countries' EEZs. This shift could result in a loss of between 10 and 30 per cent of access fees by 2050.

6. The project aims to build resilience in the tuna-dependent communities through two main components:

(a) Enhancing food security by strengthening national government programmes to deploy nearshore, anchored FADs to increase access to tuna and other pelagic fish for coastal communities; and increasing access to by-catch and tuna from industrial fishing vessels at ports during trans-shipment to support the food security of urban populations; and

(b) Reducing economic risks to Pacific Island economies by delivering an advanced warning system (AWS) that will serve as an economic planning tool by providing improved forecasts of tuna redistribution that allow countries to adapt to climate-induced changes in the distribution of tuna biomass.

7. The total cost of the project is USD 156.2 million, with a GCF investment of USD 107.4 million in grants.

8. The project aims to benefit 790,000 people directly and 2,800,000 people indirectly. Together these beneficiaries will make up 20.5 per cent of the region's total projected population in 2030.

9. The environmental and social safeguards (ESS) category for this project is C.

2.2 Component-by-component analysis

10. The project is structured in two components and activities as follows:

Component A: Adaptations to harness tuna to improve food availability of Pacific Island communities as coral reefs are degraded by climate change (total cost: USD 52.7 million; GCF cost: USD 51.4 million in grants).

11. Component A aims to increase the availability of tuna to rural and urban communities to substitute protein from coral reef fish. This component contains sets of activities designed to (i) strengthen and invest at scale in national programmes through grants and technical assistance to support deployment and use of nearshore, anchored FADs to enable fishers in coastal communities to catch and process tuna and other pelagic species to improve access to dietary protein as the supply of fish from degraded coral reef ecosystems declines and human populations grow (Output 1); and (ii) to design and implement tuna trans-shipment mechanisms that will provide urban and peri-urban communities with greater access to by-catch from industrial fishing fleets for domestic food availability (Output 2).

12. Activities for Output 1 are to provide technical and logistical support to strengthen national FAD programmes (activity 1.1); to augment national safety-at-sea initiatives (activity 1.2); and to strengthen post-harvest practices and improve market opportunities for FAD-caught fish (activity 1.3).

13. Activities for Output 2 are to implement strategies to deliver more trans-shipped and unloaded by-catch and tuna to urban/peri-urban communities (activity 2.1); to strengthen and develop post-harvest practices; and improve market opportunities to distribute by-catch and tuna from trans-shipping and unloading operations to urban/peri-urban communities (activity 2.2).

Component B: Adaptations to reduce risks to Pacific Island economies from climate-driven tuna redistribution (total cost: USD 92.1 million; GCF cost: USD 47 million in grants)

14. Component B includes a set of activities to establish a region-wide AWS to gather and analyse the data needed to determine future abundance and distribution of tuna more accurately as stocks respond to ocean warming due to climate change. The evidence generated will support the preparation and negotiation of multilateral agreements, designed to enable tuna-dependent Pacific Island Countries (PICs) to maintain their access rights to shifting tuna stocks and the associated economic benefits regardless of the redistribution of the fish. The data from the AWS will also be used to inform management decisions (harvest strategies, conservation and management measures) made by the Western Central Pacific Fisheries Commission to ensure continued effective management of tuna stocks and industrial fishing under impacts of climate change.

15. Activities under this component are to develop and deliver an AWS for tuna redistribution (activity 3.1); to assess the impact of tuna biomass redistribution identified by the AWS on national economies at all levels (activity 3.2); and to provide AWS-related training to national institutions to equip them to engage in regional and international negotiations relating to impacts of climate change on the tuna resources historically caught in their EEZs; and provide the mechanisms needed to enable the tuna-dependent economies to maintain the socioeconomic benefits previously derived from their tuna resources, regardless of climate-driven redistribution of the fish (activity 3.3).

Project management (total cost: USD 6.9 million; GCF cost: USD 5.1 million in grants)

16. This activity refers to the support provided to a dedicated project management unit, (PMU) which is responsible for overseeing day-to-day execution activities under all components, including procurement and financial management. The PMU also conducts regular monitoring and evaluation of all activities implemented by the project.

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: High*

17. For millennia, coral reef fish have been a cornerstone of nutrition in the Pacific region. Climate change is expected to have severe implications on the availability of these fish, affecting the food security of coastal communities in the PICs.

18. To address this challenge, Component A of the RTP will help maintain fish availability for good nutrition by strengthening national fish aggregating device (FAD) programmes, benefiting 560,000 individuals by enabling safe and effective tuna fishing. The programme's activities to improve the distribution of by-catch and tuna are expected to increase food availability for 290,000 individuals, providing up to four additional tuna meals per month in urban and peri-urban areas where trans-shipping of purse-seine catches occurs.

19. Climate-driven redistribution of tuna threatens the economies of Parties to the Nauru Agreement member countries and the Cook Islands, potentially reducing purse-seine catches by an average of 20 per cent and fishing access fees by about USD 90 million annually by 2050, significantly affecting government resources for essential services. Component B of the programme will empower countries to use improved climate information in decision-making (investment options, national harvest strategies for fish resources) and to negotiate for the right to retain access to the tuna resources that have historically occurred within their combined EEZs. By delivering improved data through the AWS, the programme will impact 2.5 million indirect beneficiaries.

3.2 Paradigm shift potential *Scale: High*

20. The transition to making tuna a cornerstone of nutritious food availability for PICs marks a major shift in their development pathway. As climate change, relocation of trans-shipping operations, and population growth threaten fish availability and increase reliance on low-quality processed foods, the programme aims to maintain high fish consumption by enhancing national FAD programmes and improving trans-shipping operations to urban and coastal communities. This approach leverages tuna resources sustainably while improving health, livelihoods and employment. Additionally, the programme will establish an AWS to provide timely climate-related tuna distribution data, enabling PICs to adapt, secure equitable access and negotiate effectively to retain fishing revenues. This paradigm shift positions PICs to address climate-induced challenges, ensuring sustainable and nutritious food sources.

21. The programme aims to replicate FAD technology and improve the utilization and distribution of by-catch and tuna from trans-shipping and unloading operations by enhancing technical and financial capacity and strengthening institutions at various levels. It will convert lessons learned into knowledge and communication products, widely disseminate them through various training and awareness activities, and initially implement FAD interventions at selected sites, with gradual replication to other sites within target PICs and possibly beyond. The region-wide AWS system will cover all 14 programme countries, with data application mechanisms supporting multilateral tuna access agreements rolled out in stages, ensuring lessons learned and improvements are applied progressively. Ideally, it aims to have region-wide application and serve as a model for small island developing States in other regions.

3.3 Sustainable development potential *Scale: Medium to high*

22. The interventions planned under the RTP aim to generate positive economic, social, environmental and gender-responsive development outcomes, divided into two main outcomes

and two additional co-benefits. Outcomes 1 and 2 focus on boosting food availability by ensuring reliable access to tuna and other fish as nutritious dietary protein, thus improving public health. Co-benefit 1 aims to enhance livelihoods by diversifying income sources and market access for rural and urban communities and increasing gender equality by promoting women's participation in various fishery activities. Co-benefit 2 involves enhancing ecosystem service sustainability by improving the management of oceanic tuna and other fish stocks across the Pacific.

23. The programme emphasizes gender equality within fishing supply chains, with three specific gender outcomes defined in the gender action plan: increasing women's capacity in post-harvest tuna processing and marketing; raising awareness of gender roles in coastal fishery communities; and boosting women's leadership in decision-making related to the fishery value chain. The monitoring and evaluation system will track co-benefits throughout the programme's implementation, coordinated by the heads of fisheries in each country and supported by SPC and other regional entities.

24. The programme also contributes to the Sustainable Development Goals (SDGs), particularly SDG 2: zero hunger; SDG 5: gender equality; SDG 8: decent work and economic growth; SDG 12: responsible consumption and production; SDG 13; climate action; and SDG 14: life below water.

3.4 Needs of the recipient *Scale: High*

25. A recent Notre Dame University Global Adaptation Initiative survey assessed nearly 200 countries' vulnerability to climate change across six life-supporting sectors: food, water, health, ecosystem services, human habitat and infrastructure. The survey found that most of the 14 PICs fall within the most vulnerable quartile, indicating a high need for measures to combat climate change effects and strengthen resilience.

26. Many of these PICs are small economies lacking adequate financial resources to develop robust systems for well-coordinated national FAD programmes and a regional AWS for equitable tuna resource sharing. These countries heavily rely on overseas development assistance to meet their development needs. The proposed programme aims to provide initial financial support to strengthen FAD programmes, increase tuna access for communities, launch the AWS, and help participating countries incorporate costs into national budgets for sustained long-term activities.

27. There is a critical need to strengthen institutions and capacity within PICs to understand and mitigate the impacts of climate change on coastal and urban populations. Knowledge about climate change and its effects on coral reef ecosystems and fisheries needs enhancement. The programme will address challenges such as understanding adverse fishing impacts, maximizing and equitably sharing socioeconomic benefits from fisheries, and promoting participation of women and disadvantaged groups within the fisheries sector to improve food availability and livelihoods.

28. Nearly all participating PICs are small island developing States, with limited economic development due to their small economies, remoteness from major markets, and economic non-diversification. Of the 14 countries, 9 are heavily reliant on tuna for economic well-being. Budgetary constraints hinder their ability to finance essential services and infrastructures. Although a single programme cannot fully address these systemic issues, the RTP can contribute to overcoming barriers to climate resilience by developing a regional AWS to help PICs collect licencing fees from industrial tuna fishing operations.

3.5 Country ownership *Scale: Medium to high*

29. Country ownership is fundamental to the RTP design and implementation. The innovative approaches to promoting environmental, economic and social improvements in response to climate change effects on coastal fisheries require engagement with national and regional stakeholders. This ensures the sustainability and success of proposed interventions. The engagement process began during conceptualization of the programme and development of the funding proposal and continues through implementation of the programme.

30. Regular engagement with heads of fisheries, national designated authorities, and stakeholders living in the target countries has been crucial in shaping the RTP. This process allowed parties to understand the programme structure, contribute to intervention design, and appreciate the funding nature for climate change adaptation. National stakeholders' views and requirements have significantly influenced the programme framework and activities.

31. The programme aligns with nationally determined contributions, development plans, and policies of PICs, and is featured in several PIC GCF Country Programmes. Of the 14 participating countries 10 have included climate change adaptation for ocean and coastal habitats in their nationally determined contributions. SPC, chosen as the sole executing entity (EE), leverages its strong relationships and regional technical capacity to support sustainable development across the SDGs. SPC emphasizes a people-centred approach and mobilizes collective action against climate change threats.

32. The RTP design is both regional and national to address the migratory nature of tuna and specific country needs. Strong country ownership is expected through co-financing activities, personnel commitment, and mainstreaming climate change adaptation into national policies. Through extensive consultations, agreements aligned with the RTP goals have been developed, demonstrating the PICS's commitment to climate resilience. The programme’s stakeholder engagement process, involving over 1,070 stakeholders, including women (approximately 38 per cent), ensured comprehensive feedback and support for effective implementation of the tuna adaptation programme.

3.6 Efficiency and effectiveness *Scale: Medium to high*

33. The total cost for this project is USD 156.8 million, with a GCF contribution of USD 107.4 million in grant and co-finance amounting to USD 49.3 million.

34. The economic analysis is conducted from the perspective of the national economies of the 14 participating countries. The analysis reveals that the FAD, by-catch, and AWS activities support a sustainable investment with an overall positive economic impact, demonstrating significant benefits relative to costs. The financial analysis examines the financial implications for each of the 14 participating countries, focusing on government costs and revenue. Key aspects include maintaining national FAD programmes, improving by-catch distribution, and developing AWS. The analysis indicates minimal government expenditure will be needed post-programme to continue by-catch initiatives and shared operational costs for AWS among the countries, totalling approximately USD 4.0 million annually.

35. Component A aims to increase access to nutritious animal protein for around 790,000 people by 2035 at a cost of approximately USD 12 per person annually. Component B enables nine countries to prevent significant revenue losses estimated at USD 90 million per year by 2050. The AWS component and its expected benefits will likely recoup the grant investment within 5 to 14 years. Moreover, the programme strengthens SPC’s capacity as a key scientific service provider to regional fisheries’ management organizations.

36. Overall, the proposed regional programme is efficient in building resilience among the 14 countries that heavily depend on tuna.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

37. The proposed programme is categorized as Category C for environmental and social impacts, in accordance with the GCF revised Environmental and Social Policy and the AE's accreditation level. Minimal environmental and social risks and impacts are expected from the deployment of low-impact FADs, and the conduct of capacity-building activities. Except for small-scale works for FAD anchor blocks and concrete pads for the 20-foot/6-metre containers supporting the cyclone-proof storage for FAD materials in two of the target countries, the programme will not support infrastructures, construction activities, or any direct financing to industrial fishing vessels and/or processing plants. The AE has developed an environmental and social management plan (ESMP) that includes the screening procedures for selecting sites for FADs and for managing the potential minimal environmental and social risks and impacts from the programme activities.

38. The AE also developed a stakeholder engagement plan to ensure continuous stakeholder engagement during implementation and the Accountability Grievance Redress Mechanism to serve as the channel for addressing any issues and complaints from communities and other stakeholders. The PMU will be established to implement the ESMP, stakeholder engagement plan and Accountability Grievance Redress Mechanism.

39. **GCF Indigenous Peoples Policy and ESS7 (Indigenous Peoples).** The populations of the project countries – the Cook Islands, the Federated States of Micronesia, Fiji, Kiribati, the Marshall Islands, Nauru, Niue, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu and Vanuatu – are largely comprised of Indigenous Peoples. Affected People are considered at a community level and encompass marginalized and vulnerable groups and individuals. Best practice will be used and free, prior, and informed consent using traditional customary processes is integrated throughout programme design and stakeholder engagement. An Indigenous Peoples Policy was not required for this programme as free, prior, and informed consent has been integrated into FAD site selection processes both in activity design and through the required processes of the ESMP. This process will ensure the project is low risk given the inherent cultural norms across the PICs which prevent any type of programme activity from taking place within communities without community support and/or approval. Additionally, the AE will meet the requirements of the GCF Indigenous Peoples Policy by applying its relevant policies (in particular its ESS 4 Indigenous Peoples) in a manner that meets the GCF requirements at all times. As a recommendation, the AE and EEs may benefit further from the guidance contained in the GCF Indigenous Peoples Policy and operational guidelines regarding development of the programme. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE who is also encouraged to share emerging good practices and success stories with the Advisory Group.

40. **Sexual exploitation, sexual abuse, and sexual harassment (SEAH)**. The AE has identified SEAH risks associated with the programme's activities that involve targeted training, especially those focused on women, who play a primary role in the fishery value chain, and the potentially remote locations where some activities may be carried out. The screening that was undertaken assessed these risks to be at a low level. To address these risks, the AE has proposed the following measures: designing and delivering capacity-building and awareness-raising on SEAH for personnel, such as national and PMU staff, working in the programme, and contractual requirements with partners to comply with relevant SEAH requirements. Additional SEAH risk analyses will be undertaken as part of the national programmes' investigation and documentation of gender roles. The AE will ensure that reported SEAH cases are investigated, and survivors are provided with support at the most local level. Examples of some country-level referral services are listed in the ESMP. The programme-level grievance redress mechanism will

be available for receiving and responding to any SEAH complaints as well as the GCF independent redress mechanism. In addition, national-level grievance redress mechanisms will be designed and implemented. Grievances, including SEAH cases, will be monitored by the safeguards focal point.

4.2 Gender policy

41. The AE provided a gender assessment and action plan with the funding proposal and therefore complies with the requirements of the GCF Updated Gender Policy.

42. The AE combined a desk review of regional and national gender policy and gender strategy documents, including interviews and consultations with a selection of fishing supply chain stakeholders, to develop the gender assessment. The gender assessment findings reveal that many (PICs) have adopted policies aimed at fostering an enabling environment that promotes gender equality, encouraged by several international agreements and commitments. Despite the gender equality efforts put in place by the PICs, studies indicate gender disparities compounded by social cultural norms and slow implementation of policy and study recommendations. Some of the barriers to the advancement of women include poor working conditions and environments, a lack of employment and personal security, and gender-based violence which is prevalent in all the countries.

43. The gender assessment includes findings on the roles, barriers and opportunities for women and men in Pacific fish value chains. The fishing sector in PICs is largely dominated by men, while women play supportive roles such as post-harvesting activities. These supportive roles keep women's contribution invisible yet there is great potential in expanding their activities including protecting natural resources and enhancing food security. There are immense opportunities to enhance gender equality and safer working spaces within the fish value chains by promoting more equitable gender roles, recognition and rewarding of women's contributions, and enhancing collaboration between women and men fishers in the sector. There is also an opportunity to improve awareness of gender-based violence among the value chain actors and develop mechanisms to identify and mitigate related risks in the programme and value chain.

44. The AE has developed a gender action plan (GAP) consisting of activities, performance indicators, baseline, targets, budget, timelines and responsibilities. The GAP includes the following activities that will be undertaken as part of the programme: capacity-building and awareness-raising on gender dimensions of the FAD fishery value chain and women's roles within the fishery; consulting women fishers on needs analysis for improved vessel safety; training women fishers on boat safety equipment; providing support for post-harvest practices; and improving market opportunities. The GAP foresees training courses for women in coastal and urban communities to improve preservation and hygiene of FAD-caught fish, for example the bottling and canning of fish, and to improve hygienic and safe working conditions for sellers of fish, especially women. Two gender and environmental and social safeguards officers will oversee the implementation of the GAP in coordination with a national-level officer who will be responsible for the country-level implementation of the GAP.

45. Each country will identify its priorities and opportunities to advance the national gender equality goals in the context of this programme. The AE is advised to ensure that additional and more detailed gender analyses are undertaken at the country level prior to implementation to assess specific gender contexts and identify actions that are relevant to addressing the needs and priorities of women and men in the fishery value chain in the context of national projects.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

46. As sole EE, SPC has not managed or executed a single project/programme of comparable funding size in the region. We acknowledge SPC is an AE and has an existing track record with two approved GCF projects, FP191 and FP169, within its accreditation scope (small size), both of which are single-country projects, and has managed large-value projects concurrently across its varying divisions which cumulatively exceed USD 100 million.

4.3.3. Project-specific execution risks (low risk)

47. Considering the broad geographical footprint of the project (14 countries), the shift in governmental political focus and priorities of the participating countries may pose risks to the implementation and long-term sustainability of the project, e.g. implementation delays, budgetary changes, etc. However, we derive comfort from SPC's experience in managing projects across the region and collaborating with the respective governments.

4.3.4. GCF portfolio concentration risk (low risk)

48. In case of approval, the impact of this proposal on the GCF concentration risk remains within the monitoring thresholds of the Risk Appetite Statement in terms of results areas, single proposal or AE concentration.

4.3.5. Compliance risk (medium risk)

49. The proposed project, spanning 14 countries and involving a multitude of stakeholders—including CI as AE, SPC as EE, as well as government agencies and vendors—presents inherent risks in managing prohibited practices (PP). With a wide array of actors and geographical diversity, there exists an elevated risk of potential non-compliance with GCF's AML/CFT and PP policies. While the probability of these risks is assessed by CI as low, their impact on the program's reputation and the GCF's relationships with partner countries is deemed medium.

50. To address these risks, CI as AE and SPC as EE have implemented a variety of controls to mitigate PP and ML/FT risks, aligned with GCF fiduciary requirements. CI ensures all program staff undergo AML/CFT training, covering PP red flags and reporting protocols, and mandates that grant and service agreements embed compliance with GCF's AML/CFT and PP policies. Additionally, SPC employs an AML/CFT policy, to screen all fund recipients against international sanctions lists. and its agreements mandate compliance with GCF's PP policies. CI's Ethics Hotline and SPC's complaint mechanisms allow all stakeholders to anonymously report concerns, reinforcing transparency and enabling effective response measures. Regular project monitoring, including site visits and in-depth due diligence processes, further strengthens these controls.

51. Considering the established mitigation strategies and the project's compliance framework, the residual risk level for this initiative is assessed as Medium.

4.4 Fiduciary

52. The Regional Tuna Programme is spearheaded by CI as the AE and executed by SPC as the EE. CI is responsible for programme oversight, monitoring, and fund disbursement, while SPC manages programme execution, including financial reporting, subgrant administration, and

stakeholder coordination. The programme will establish a PMU within the SPC Fisheries, Aquaculture and Marine Ecosystems Division to oversee all operational aspects, led by a Programme Director and supported by dedicated and part-time technical and operational staff from SPC internal divisions. This PMU will coordinate with a diverse network of implementing partners to ensure technical and operational support across programme components. Additionally, a Programme Steering Committee will provide governance oversight and strategic guidance to align activities with GCF and regional priorities, reinforcing transparency and accountability throughout implementation.

53. CI and SPC utilize separate enterprise resource planning systems for financial accounting, allowing both entities to monitor income and expenditure meticulously by project and donor. CI employs Unit 4 Business World, while SPC uses Dynamics Nav. These systems support rigorous financial tracking, segregation of funds, and systematic reporting. Financial reports will be prepared by SPC and undergo review by CI to ensure compliance with established protocols. Furthermore, SPC and its subgrantees are required to provide periodic financial updates, reinforcing financial transparency at all stages of programme execution. Annual audits will be conducted by independent firms in line with international standards, and findings will be shared with GCF, providing an additional layer of fiduciary oversight.

4.5 Results monitoring and reporting

54. This is an adaptation project that will contribute to GCF adaptation results areas 1 and 2. The project is expected to reach 790,000 direct beneficiaries, and around 2.5 million indirect beneficiaries across 14 countries in the Pacific region.

55. The results management arrangements proposed under the project were refined over four rounds of review and engagement with the AE. Specifically, changes were made to the project theory of change to ensure that it was clear, logically coherent, and linked to the climate rationale and activities proposed under the project.

56. The project logical framework was also refined over multiple rounds to ensure that the proposed indicators and targets were realistic and linked to the proposed theory of change. In particular, the AE was provided feedback to ensure that beneficiary targets at the level of the adaptation results areas were accurately articulated, and there was low risk of inaccurate or double counting of beneficiaries. It is positive to note that the AE has defined project-level outcome and output indicators, in addition to the indicators against different GCF results areas.

57. The AE was also provided advice for refining the supporting annexes, including the Implementation Timetable, Monitoring and Evaluation Plan, and beneficiary calculations. Guidance was provided to the AE to ensure that the beneficiary calculations, especially for indirect beneficiaries, are realistic and aligned with the proposed project rationale and activities.

58. The AE has responded positively to guidance provided and refined the funding proposal accordingly.

4.6 Legal assessment

59. The Accreditation Master Agreement was signed with the Accredited Entity on 13 July 2017, and became effective on 17 August 2017, which was amended and restated pursuant to a first amendment and restatement agreement dated 15 August 2022, and which became effective on 17 August 2022 (the “AMA”).

60. The Accredited Entity has provided a legal certificate confirming that it has obtained all internal approvals, and it has the capacity and authority to implement the programme.

61. The proposed programme will be implemented in fourteen host countries. The GCF has signed bilateral agreements on privileges and immunities with nine of these host countries, namely Cook Islands, Kiribati, Federated States of Micronesia, Niue, Papua New Guinea, Samoa, Tonga, Vanuatu and Solomon Islands. The bilateral agreement on privileges and immunities with the Solomon Islands is not effective yet. In the case of the six host countries, namely Fiji, Marshall Islands, Nauru, Palau, Solomon Islands, and Tuvalu, GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

62. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the country or has been provided with appropriate privileges and immunities for GCF and its personnel.

63. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP259

Proposal name:	Adapting tuna-dependent Pacific Island communities and economies to climate change
Accredited entity:	Conservation International Foundation
Countries:	Cook Islands, Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Nauru, Niue, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu and Vanuatu
Project size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. The proposed project, the Adapting Tuna-Dependent Pacific Island Communities and Economies to Climate Change, or Regional Tuna Program (RTP), has a total cost of USD 156.8 million, with USD 107.4 million requested from GCF entirely as a grant. The total co-financing amounts to USD 49.38 million. This co-financing reflects a mix of cash and in-kind contributions, combining technical expertise, equipment and operational support. It is broken down as follows[1]:

(a) USD 22,080,000 (in-kind contributions) from the Fiji Fishing Industry Association (FFIA), representing access to vessels equipped with suitable instruments for recording tuna data and physical oceanography parameters in support activity 3.1.

(b) USD 11,295,405 from the Secretariat of the Pacific Community (SPC), including USD 9,710,655 for staff costs (for activities 1.1, 1.2, 1.3, 3.1, 3.2, 3.3. monitoring and evaluation, project management costs), USD 1,244,750 for equipment and supplies for Activity 3.1, and USD 340,000 for professional and contractual costs under Activity 3.1;

(c) USD 2,162,493 from the Commonwealth Scientific and Industrial Research Organisation, (CSIRO) including USD 1,490,191 for staff costs supporting the implementation of Activities 2.1 and 3.1, and USD 672,213 for project management costs; and

(d) USD 885,976 (in-kind contributions) from the Pacific Islands Forum Fisheries Agency (FFA) for staff costs for Activities 2.1, 2.2, 3.1, 3.2 and 3.3.

(e) USD 12,960,000 from the Office of the Parties to the Nauru Agreement (PNAO) and private-sector industrial fishing companies will provide in-kind co-financing support to the Advanced Warning System activities in Activity 3.1 by providing observer services, vessel time, and data.

2. The proposed project has an implementation period of seven years, while its total lifespan is expected to be 20 years, ensuring long-term sustainability and impact.

3. Assuming the project is successful, it will have 790,000[2] direct beneficiaries (50 per cent male, 50 per cent female). It will directly benefit 5.8 per cent of the region's projected total

[1] See funding proposal annex 4 CI Tuna detailed budget
[2] See funding proposal annex 23 for details

population in 2030 (31.9 per cent of the population within 1 km of the coast) and indirectly benefit 18.3 per cent of the region's total projected population in 2030.

4. **Theory of change.** The RTP theory of change is that "IF Pacific Island governments are supported to implement effective programmes for assisting coastal and urban communities to obtain and utilise more tuna, and are provided with improved information on climate change-driven redistribution of tuna; THEN Pacific Island nations will be transformed to become more resilient to key climate change threats facing the fisheries sector; BECAUSE communities and governments will be better informed and equipped to make optimal use of the fisheries resources on which they depend for food, livelihoods, and economic development."[3]

5. The funding proposal has two major outcomes. The first is improved food availability for vulnerable communities in participating countries (Outcome 1) and the second is strengthened capacity of tuna-dependent Pacific Island countries (PICs) to negotiate for benefits from climate- redistributed tuna stocks (Outcome 2). Outcome 1 will be achieved via two strategic interventions: increasing capacity to access tuna and other pelagic fish for coastal communities (Output 1) and increasing supply of by-catch (undersized fish unfit for canning) and tuna from industrial fishing operations for urban communities (Output 2). Outcome 2 will be achieved via a strategic intervention focused on improving forecasts, and projections on climate-driven tuna redistribution that will facilitate effective adaptations for all stakeholders. The co-benefits identified relate to improved livelihoods of vulnerable communities in participating countries for Outcome 1, and strengthened management of industrial tuna fisheries by regional and national institutions.[4]

6. The programme of activities under Outcome 1 (also called Component A in "Adaptations to harness tuna to improve food availability of Pacific Island communities … as coral reefs are degraded by climate change"[5]), are as follows:

(a) Output 1 seeks to (a) provide technical and logistical support, including equipment, to implement national fish aggregating device (FAD) programmes, (b) augment national safety-at-sea initiatives and (c) strengthen post-harvest practices and improve market opportunities for FAD-caught fish;

(b) Output 2 seeks to (a) implement strategies to deliver more trans-shipped and unloaded by-catch and tuna to urban and peri-urban communities and (b) strengthen and develop post-harvest practices and improve market opportunities to distribute by-catch and tuna from trans-shipping and unloading operations to urban and peri-urban communities; and

(c) Under Outcome 2 (also called Component B in "Adaptations to reduce risks to Pacific Island Economies from climate-driven tuna redistribution"[6]), the activities under Output 3 seek to (a) develop and deliver an advance warning system (AWS) for climate-driven tuna redistribution, (b) assess and share useful and actionable information on the impact of tuna redistribution as identified by the AWS among national economies at all levels and (c) provide AWS-related training on how to negotiate regional and international forums to address the socioeconomic losses resulting from the impacts of climate change on tuna.

1.2 Impact potential *Scale: Medium to High*

7. For many generations, coral reef fish have been a cornerstone of nutrition in the Pacific Island region. However, the combined impacts of growing populations and climate change on

[3] See funding proposal section B.2, para. 54.
[4] See funding proposal section B.2.
[5] See funding proposal Outcome 1 statement.
[6] See funding proposal Outcome 2 statement

reefs and their resident fish populations are expected to have severe implications for the food availability of coastal communities in PICs. Climate change is causing warming and acidification of the ocean, changing the suitability of areas for specific fish species, and is also interacting with direct human pressures to cause the degradation of coastal ecosystems such as coral reefs, which are important fish breeding grounds. Ocean warming is causing tuna to migrate towards cooler waters outside PIC economic zones. Overall, PIC governments, businesses, people and ecosystems are becoming more vulnerable to the cascading future effects of climate change.

8. To address this challenge, Component A of the RTP will assist growing Pacific Island populations to continue to have access to fish for good nutrition by strengthening national FAD programmes, directly improving food availability for 560,000[7] individuals. FADs enable coastal communities to fish for tuna and other large pelagic fish in safe and effective ways – something that will be progressively more important in maintaining access to fish for good nutrition as coral reefs are degraded by climate change and populations continue to grow.[8]

9. Those of the programme's activities related to improving the distribution of by-catch and tuna are expected to increase food availability for 290,000[9] individuals by providing up to four additional tuna meals per beneficiary per month in urban and peri-urban areas where trans-shipping of purse-seine catches occurs.[10] Unloading operations by commercial longliners in some ports (e.g. Honiara, Majuro, Nuku'alofa, Port Vila and Suva) also offer the potential to supplement supplies of tuna and by-catch for local consumption, albeit at a lower level than the offloading of by-catch from purse-seine vessels.

10. Climate-driven redistribution of tuna threatens to undermine the economies of the Parties to the Nauru Agreement member countries and the Cook Islands. By 2050 under Representative Concentration Pathway (RCP) 8.5, preliminary modelling indicates that the climate-driven redistribution of tuna could cause an average of a 20 per cent (range 10–30 per cent) reduction in purse-seine catch from the combined exclusive economic zones (EEZs) of these nine tuna-dependent countries, potentially reducing their total fishing access fees by an average of about USD 90 million per year (range USD 40–140 million) at today's prices. Even under the more optimistic RCP 4.5 scenario, the losses are projected to be about USD 12 million per year. Such significant reduction in government finance will have direct impacts on vulnerable populations in these countries; governments will have less budgetary resources for priority sectors (i.e. health, education, disaster preparedness and post-disaster recovery, and other related essential social services). Component B of the programme will empower countries to use improved climate information in decision-making (e.g. investment options and national harvest strategies for fish resources) and to negotiate for the right to retain access to the tuna resources that have historically occurred within their combined EEZs. By delivering improved data through the AWS, the programme will impact 2.5 million indirect beneficiaries.[11]

11. It is hoped that the RTP will deliver on the following targets in the 14 participating countries (or in a subset of countries where indicated) by the end of the programme:

(a) Deployment of 333 new operational FADs, with replacement of 195 of the FADs deployed during the programme (noting that the working life of FADs is about three years, meaning that FADs deployed during years one to three will need to be replaced during years four to seven). Thus, a total of 528 FADs will be constructed and deployed;

(b) Training of approximately 800 people in safe and effective FAD fishing methods;[12]

[7] See funding proposal annex 23 for details
[8] See funding proposal annex 23 on the level of benefit (i.e. number of fish meals) provided per beneficiary.
[9] See funding proposal annex 23 for details
[10] As footnote 3 above.
[11] See funding proposal annex 23.
[12] See funding proposal technical study, annex 26.

(c) Provision of between 9 million and 18 million nutritious tuna meals (150 grams for adults and 75 grams for children) for 560,000 people across the region each year through strengthened national FAD programmes.[13] Approximately 290,000 people will be provided with a total of around 13 million tuna meals from improved distribution of by-catch to urban and peri-urban communities where trans-shipping of purse-seine catches occurs at ports in the region.[14] Approximately 60,000 individuals will benefit from both FAD-caught fish and improved distribution of by-catch but the calculation of total direct beneficiaries (790,000) avoids double counting.

(d) Improved livelihoods for 3,000 small-scale fishers using the 333 FADs deployed, and the members of their households, totalling 15,000 people (co-benefit);[15]

(e) Training of 400 representatives of coastal communities in post-harvest preservation methods for FAD-caught fish, resulting in improved food availability for their households, totalling 2,000 individuals, and establishing a basis for peer-to-peer learning within communities;[16]

(f) Opportunities for an additional 130 people to sell the increased volume of by-catch in urban communities, improving the livelihoods of their households, totalling 650 individuals (co-benefit);

(g) Creation or strengthening of 10 micro, small or medium enterprises (MSMEs) to apply post-harvest preservation and processing methods to by-catch in 10 of the countries where trans-shipping and unloading of these fish occurs;[17]

(h) Provision – for about 670,000 people – of information on climate change impacts on coral reef fish and the need to consume tuna as an alternative to reef-associated fish resources;[18] and

(i) Development of improved meteorological and natural disaster forecasts for about 2.5 million people.[19] Forecasts will be improved to enable small-scale fishers to avoid hazardous sea conditions when planning fishing trips to FADs, and timely warnings (that are easy for communities to receive[20]) will benefit communities within 1 km of the coast in all 14 countries.

12. The AWS-linked interventions will provide short-term coping strategies, such as inputs for (i) detailed short-term forecasts and identification of immediate impacts on tuna fisheries at the EEZ level to support quick responses and (ii) operational planning and facilitation of near-term decision-making for fishing activities and resource management (e.g. harvest strategies).

13. The AWS will also provide inputs to long-term coping strategies, such as (i) projections of climate change impacts, where it will inform strategic adaptations to long-term changes in tuna distribution and abundance, and (ii) economic planning, where it will provide countries with data to help them plan for shifts in fishing revenues and licence structures over time.

14. The RTP will greatly enhance the capacity of PIC government-mandated entities (e.g. fisheries departments and ministries) to integrate climate considerations into their budgeting systems and broader operations. The planned allocation of government budget to support FADs and the AWS potentially demonstrates how lessons learned about climate-related initiatives can be replicated across the whole of government.

[13] See funding proposal annex 23.
[14] As footnote 7 above.
[15] As footnote 7 above.
[16] As footnote 7 above.
[17] As footnote 7 above.
[18] See funding proposal annex 23 and technical study 6 annex 26.
[19] See funding proposal annex 23.
[20] See funding proposal technical study 4, annex 26.

15. One risk to impact has been identified. While the RTP design provides valuable short- and long-term coping strategies, one notable weakness lies in its limited emphasis on preparing PIC stakeholders (i.e. government representatives, local communities and MSMEs) to effectively manage the risk of climate change impacts on their natural and ecological systems that are likely to occur within their lifetimes. For example, PIC stakeholders' potential lack of awareness, resources, and technical capacity to process and understand the useful information from the AWS may stymie their capacity to use these real-time data and information to effectively craft timely adaptive policies, strategies, programmes and other initiatives that will address the impacts of climate change on tuna resources. In addition to a number of risks identified by the accredited entity (AE), the iTAP has identified the risk of occurrence of a large-scale climate event or natural disaster (e.g. category 4/5 cyclone or tsunami) that may significantly disrupt the overall impacts of the RTP; thus, there is a need to climate-proof relevant project components.

16. The funding proposal's impact potential is assessed as medium to high.

1.3 Paradigm shift potential *Scale: Medium to High*

17. The scale-up of the FADs across the 14 countries included in this project, and the potential for inclusion in the national budget (for maintenance) of each of the PICs, would bring about a paradigm shift in the way sustainable fisheries programmes are run to deal with climate change. Previously reliant on donor funding, the deployment and maintenance of FADs as part of national programmes would make an important contribution to mainstreaming climate resilience among governments. This potential inclusion in national budgets (based on the view of FADs as part of the national infrastructure for food security) could trigger further domestic resource mobilization and, most importantly, attract much-needed climate co-funding from other donors and/or private sector organizations in support of similar climate-related initiatives.

18. The optimization of tuna and by-catch as a food and protein source, coupled with the development of a more organized business value chain for tuna from trans-shipments and other by-catch, points to a potential paradigm shift. Through the RTP, local communities could begin to diversify and secure their food and protein options. Most importantly, the RTP would empower PIC stakeholders to develop innovative business and eventually economic models along the fishing value chain, fostering diversified livelihoods and income sources in response to climate impacts on their traditional tuna-linked business value chains.

19. The implementation of the AWS has the potential to strengthen the fisheries sector both within PICs and on a global scale. By providing detailed data on the redistribution of tuna resources due to climate change, the AWS would decrease uncertainty in the use of existing models (e.g. Spatial Ecosystem And Population Dynamics, or SEAPODYM) and enable PIC stakeholders to craft effective adaptation policies (e.g. sustainable harvest strategies),[21] resilient commercial strategies and fishing fleet management approaches to these evolving adverse climate conditions. This integration of climate resilience into tuna management plans and conservation and management measures by PICs and the Western and Central Pacific Fisheries Commission (WCPFC) would set a precedent for sustainable fisheries management worldwide. The potential major contributions of the AWS[22] would be as follows:

(a) **Tuna resource mapping.** Utilizing genetic population analyses and close-kin mark–recapture techniques, the AWS would accurately identify and track tuna stocks, facilitating precise monitoring of climate-induced shifts in biomass;

[21] See funding proposal annex 26 CI Tuna Technical Studies V3.
[22] See funding proposal annex 26 CI Tuna Technical Studies V3

(b) **Enhanced tuna-related climate models.** Improvements to the SEAPODYM model would increase spatial resolution and incorporate climate variables such as ocean warming and prey dynamics, offering more accurate predictions of tuna distribution under changing environmental conditions;

(c) **Industry collaboration.** Partnering with industrial fishing companies for data collection would validate global climate models and deepen understanding of how tuna prey respond to oceanic changes, enhancing the reliability of predictions;

(d) **Operationalizing the AWS.** Conducting regular assessments to forecast sustainable tuna catch levels within PIC EEZs and the high seas would inform the refinement of WCPFC harvest strategies, ensuring they account for climate impacts; and

(e) **Knowledge-sharing about the AWS.** This would foster global knowledge-sharing and collaborative efforts in sustainable fisheries management. The insights gained through the AWS would be shared during WCPFC deliberations, contributing to a collective understanding and responsibility among all members to manage tuna resources effectively within the WCPFC Convention Area.

20. This comprehensive approach would not only transform fisheries management in the Pacific but also serve as a model for integrating climate resilience into fisheries governance globally.

21. The AWS would be a transformative tool that enabled PICs to negotiate effectively for equitable resource sharing. By providing reliable, climate-informed data and enhancing negotiation capacities, it would ensure PICs are better positioned to retain economic benefits from tuna fisheries amid climate change.

22. There are two risks to the paradigm shift potential, as follows.

23. As a multi-country initiative, the programme's success hinges on strong collaboration among PICs. Weak or problematic cooperation would pose a significant risk to achieving the programme's intended outcomes. This highlights the need for robust mechanisms to maintain unity and alignment among stakeholders.

24. The iTAP believes that there is also a perceived risk that the RTP initiatives will be perceived as insufficiently transformational in their approach to climate adaptation. Transformational adaptation would extend beyond merely maximizing tuna access to bolster short-term resilience in existing fisheries; the RTP should address the deeper systemic changes required to ensure long-term food security amid the challenges of climate change. To mitigate this risk, the iTAP recommends strengthening the project by supporting national dialogues in each country. These dialogues could explore additional transformational adaptation strategies that involve systemic economic shifts, such as developing mariculture or aquaculture sectors.

25. Overall, the paradigm shift potential is assessed as medium to high.

1.4 Sustainable development potential *Scale: Medium*

26. **Sustainable Development Goals (SDGs).** The funding proposal cuts across several of the SDGs. The RTP will contribute to the achievement of SDG 13 (Climate Action), which is "intrinsically linked to all 16 of the other Goals of the 2030 Agenda for Sustainable Development".[23] The RTP will also contribute to SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being), SDG 4 (Quality Education), SDG 5 (Gender Equality), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequalities), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible

[23] *Climate Action.* Available at www.un.org/sustainabledevelopment/climate-action.

Consumption and Production), SDG 14 (Life Below Water) and SDG 17 (Partnerships for the Goals).

27. **Socioeconomic co-benefits.** The potential for the development of business value chains for the tuna and by-catch from FADs and trans-shipments, including its processing and distribution to new markets, offers diversified economic possibilities, primarily for local communities. The food processing of this by-catch and tuna will enable increased resilience of local livelihoods; this will promote livelihood diversification and provide market access, boosting household incomes and creating economic opportunities for both rural and urban populations.

28. **Environmental co-benefits.** While the funding proposal mentions the impact of global "high GHG [greenhouse gas] emissions" on the ocean and coral reefs, it does not consider GHG mitigation within this project. At the very least, baselining of GHG emissions by source and by type of fishing activity, or according to the fuel used for each type of fishing operation, could have been integrated into the project design. The funding proposal could have explored unmitigated GHG emissions from fishing (longline, purse seine and coastal), processing (cold chain logistics), and tuna and by-catch fish waste to understand whether the industry itself also needs to address GHG emissions to prevent unwanted consequences, or maladaptation. All of these areas could have offered opportunities to attract additional carbon or mitigation finance in support of the tuna fishing industry.

29. **Gender-sensitive development impact.** The funding proposal promotes gender-responsive development as one of the key aspects of the project's sustainable development strategy by fostering greater gender equality across fisheries value chains. It also proposes opportunities for women to participate in and benefit from fisheries-related activities, such as enhancing women's capacity in the FAD business value chains, particularly in post-harvest processing and marketing, and raising awareness of gender dimensions in coastal communities managing FADs. It furthermore emphasizes increasing women's leadership and influence in decision-making processes within the value chain. By integrating gender considerations into fisheries activities and value chains, the RTP would contribute to livelihood diversification, increased household incomes and more inclusive economic opportunities, ensuring that women have equitable access to the benefits of sustainable fisheries development.

30. Overall, the sustainable development potential is assessed as medium.

1.5 Needs of the recipient *Scale: Medium to High*

31. **Need for latest evidence-based data among decision makers.** The AWS will provide evidence-based data to support tuna fisheries management by reducing uncertainty in near-term predictions (1–10 years) of tuna distribution, compared to the current long-term models (30–50 years). The AWS will provide a scientific foundation for evidence-based, adaptive tuna management strategies in the western and central Pacific Ocean (WCPO), ensuring sustainable fisheries amid climate change. The AWS will also provide information for forecasting and evaluating fishing fleet dynamics, including within the individual EEZs, and the projected economic impacts for the 14 participating countries. The AWS scientific data will enhance the capacity of decision makers to sit down and discuss collaboration on tuna sharing, should the AWS demonstrate a shift of WCPFC tuna stocks into the jurisdiction of the Inter-American Tropical Tuna Commission (IATTC). The AE states that the WCPFC and the regional fisheries management organization for the eastern Pacific Ocean, the IATTC, are mandated to collaborate.

32. **Needs of the PICs and WCPFC tuna stock management.** Although the WCPFC has implemented robust binding conservation and management measures that have successfully sustained tuna harvests in the WCPO large marine ecosystem, the progressive, climate-driven redistribution of tuna biomass from the EEZs of PICs to the high seas presents a significant risk

to the current management regime. This shift could undermine efforts to sustain tuna stocks within the EEZs of PICs. The AWS will mitigate this risk, serving as a critical tool for guiding adaptive management decisions. It will help the WCPFC and its member governments to better protect 75 million hectares of the tropical WCPO within the EEZs of the participating countries, where industrial purse-seine tuna fleets are predominantly active. By reducing uncertainty around the impacts of ocean warming on tuna distribution, the AWS will support the adaptation of conservation and management measures and harvest strategies, ensuring they remain effective in sustaining productive fisheries for key tuna species despite changing environmental conditions.

33. **Needs of coastal communities.** Warming of water temperatures will continue to degrade the health of coral reefs and impact the food security and livelihoods of PIC coastal communities. Programme interventions will directly address these needs. These interventions include deployment of 333 FADs across community fishing grounds, optimization of tuna by-catch from trans-shipping and unloading operations by industrial fishing fleets, and training and awareness campaigns emphasizing the importance of shifting consumption from fish caught from coral reefs and other coastal ecosystems (which will continue to be degraded by climate change) to tuna and other pelagic fish species.

34. Overall, the potential for the project to serve the needs of the recipients is assessed as medium to high.

1.6 Country ownership *Scale: Medium to High*

35. The RTP is a multi-country initiative. For it to be successful, there must be strong country ownership and capacity of the delivery partners to bring the intended outcomes to fruition; failure to fully achieve this is the biggest risk. The RTP is supported by a robust team of AE, executing entity (EE) and implementing partners with extensive experience in PICs.

36. **Conservation International (CI).** Serving as the international AE, CI has been active in Pacific Ocean conservation and sustainability for over 30 years. Notable initiatives include the Climate Change Flagship and Climate Science for Ensuring Pacific Tuna Access (2022–2026) and the Pacific-European Union Marine Partnership programme (2018–2025). CI also leads projects aimed at strengthening fisheries management in the Eastern Tropical Pacific Seascape and enhancing longline tuna fisheries in Fiji and New Caledonia.

37. **Secretariat of the Pacific Community (SPC).** As the sole EE and a direct access entity, SPC is an international intergovernmental organization, governed by 27 member countries, established in 1947 to address the development and the scientific and technical needs of the Pacific region. Overall, 14 of the 22 Pacific Island countries and territories will participate in the RTP. SPC has a history of assisting national offices with statistics, planning and economic development, including conducting household income and expenditure surveys on tuna consumption. Its existing memorandums of understanding with regional fisheries management organizations facilitate effective coordination and communication regarding tuna stock status and redistribution.

38. **Pacific Islands Forum Fisheries Agency (FFA).** As a programme implementing partner, FFA will enhance overall project management, particularly for Outputs 2 and 3. Since 1982, FFA has provided economic, legal, fisheries management and development assistance to its 15 member Pacific Island countries and a single territory. Its mandate includes assisting members in utilizing the best available science for management decision-making within and beyond PIC EEZs. The 14 PICs involved in this programme are also FFA members.

39. **Commonwealth Scientific and Industrial Research Organisation (CSIRO).** The national science agency of Australia, CSIRO, will assist SPC in executing Activities 2.1 and 3.1. Renowned for extensive ocean research and participation in regional and international

scientific collaborations, particularly in sustainable fisheries, CSIRO employs advanced methods for stock assessment and management, focusing on tuna species. In this collaboration, CSIRO will provide updated projections on how to address protein needs through by-catch and tuna resources, assist SPC in establishing baselines and indicators to quantify climate change impacts on tuna populations using genomic-based fisheries management technologies, and develop advanced ocean monitoring tools to enhance data collection and analysis.

40. **Food and Agriculture Organization of the United Nations (FAO).** Through its Subregional Office for the Pacific Islands (FAO SAP), FAO will provide technical support for Activity 1.2. FAO SAP assists 14 Pacific countries in areas such as food security, nutrition, agriculture, rural development and safety-at-sea initiatives. It offers extensive technical expertise, a robust network of public and private sector partners, and a comprehensive knowledge base on fishing vessel design, supported by professional fisheries staff and ongoing regional fisheries programmes. This collaboration aims to enhance the implementation of safety-at-sea measures and related activities within the project.

41. **Office of the Parties to the Nauru Agreement (PNAO).** PNAO will provide technical services for Activity 2.1. Representing eight PICs that manage the world's largest sustainable tuna purse-seine fishery, PNAO will assist in updating regional observer and port programmes to estimate locally redirected tuna and by-catch. Additionally, PNAO will utilize the AWS to support its members in developing adaptive policies and strategies. Although the PNAO constitution limits the body's ability to receive external funds, it will contribute as a co-financier of the programme.

42. These organizations, with their extensive experience and demonstrated collaborative efforts in the Pacific region, form a credible and reliable team to implement the RTP effectively.

43. One risk to country ownership has been identified. There is a risk of limited participation or consultation with local communities, governments, and businesses and MSMEs. This may lead to limited direct involvement in higher-level decision-making and policy formulation for tuna fisheries management, insufficient integration of local and traditional knowledge into resource management and adaptation strategies, and limited participation of MSMEs in value-added activities such as fish processing or aquaculture development.

44. Country ownership is assessed as medium to high.

1.7 Efficiency and effectiveness *Scale: Medium to High*

45. Given the co-financing commitments of SPC, FFA, FFIA and CSIRO, the total leverage factor is 0.34. The GCF grant of USD 107.4 million will be matched by FFIA with USD 22.08 million (1:0.21), SPC with USD 11.21 million (1:0.10), CSIRO with USD 2.2 million (1:0.02) and FFA with USD 0.89 million (1:0.008). Even if these contributions are mostly in kind, they show that the EEs have an active share in the risk.

46. **Continued financing of the AWS, FAD and by-catch programmes.** The nine tuna-dependent countries and five subtropical PICs benefiting from the AWS will ensure its continued operation and maintenance through their membership in regional organizations that use the system. SPC estimates the annual cost of maintaining the AWS at USD 4.045 million, averaging about USD 290,000 per country across the 14 PICs. The total ongoing costs for all programme components (FADs, by-catch and AWS) amount to about USD 6.0 million annually, with the AWS representing the largest share.

47. **Strong partnerships.** The expected close collaboration between and among the PICs will enhance the success of the programme components. The success of the programme will rely mostly on the governments of the PICs, which will need to carefully enforce rules and

regulations, and standards on optimizing the efficiency of the fishing industry (e.g. processing of tuna and by-catch, and waste management).

48. The FAD component – when viewed in complementarity with the World Bank Pacific Islands Regional Oceanscape Program (PROP)[24] – will enhance the overall effectiveness of the project in addressing the food, health (nutrients) and livelihood needs of the PIC communities. The World Bank PROPER (the second phase of the PROP) is under way and involves many of the 14 countries participating in the RTP. PROPER is expected to be active throughout much of the implementation phase of the RTP, and preliminary talks with the World Bank PROPER team on a collaborative approach have been promising. A collaboration between the two programmes to promote synergies and avoid duplication will enable more provinces in the larger countries to receive FADs, and the number of FADs for some provinces proposed under the RTP may be increased through investments by the World Bank where this is a national priority.

49. **Good exit strategy.** The RTP has established an exit strategy to ensure that participating countries and regional agencies can sustain and enhance RTP initiatives beyond its seven-year implementation period. In **Component A**, the RTP will initially fund FAD management officer positions within national fisheries agencies, with a gradual reduction in support over five years, leading to full governmental responsibility by year six. Additionally, the RTP will conduct capacity-building exercises in FAD programme management and legislative processes to strengthen national capabilities. Collaboration with projects such as the World Bank PROPER will further harmonize efforts to expand national FAD programmes. For **Component B**, the exit strategy focuses on enhancing the existing SEAPODYM model, utilized by SPC since 2008, to improve decision-making related to tuna stock management in the context of climate and ecosystem variability. The WCPFC currently funds the SEAPODYM model and is expected to continue its support as the model is enhanced and delivered through the AWS. Industry partners will also contribute by collecting data during programme implementation and are anticipated to continue data collection post-programme through established observer programmes, enhancing AWS resolution and accuracy over time. This structured exit strategy ensures that the participating countries and regional agencies are well prepared to sustain and advance the initiatives introduced by the RTP, thereby minimizing risks to programme outcomes and maximizing long-term benefits.

50. **Risk to efficiency and effectiveness.** The financial numbers presented on the recurring costs of FADs and FAD-related activities after the project implementation period do not factor in the potential impacts of extreme climate events (e.g. category 4/5 storms), which are known to have frequented this region in the past five years. Moreover, there is a risk of there being a lower number of beneficiaries in the medium to long term due to these climate impacts, but the funding proposal assumes that the number of individuals benefiting is proportional to the number of FADs deployed – for example, if 50 per cent of the planned FADs for a given country are operational, 50 per cent of the beneficiary population in that country will be reached.

51. Given the above, efficiency and effectiveness are assessed as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

52. The iTAP recommends this funding proposal for approval by the Board.

53. The proposed project is well prepared and documented. Its interventions cover the short-, medium- and long-term needs of the PICs in relation to the impacts of climate change on the countries' primary economic resource – tuna. The AE, the EE and the implementing partners

[24] *Forum Fisheries Agency: Pacific Islands Regional Oceanscape Program – Second Phase for Economic Resilience (P177661).* Available at https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fdocuments1.worldbank.org%2Fcurated%2Fen%2F099925206282215250%2FP177661082ba4d000854e01689da4c6a79.docx.

show good working relationships with the PICs, and it is hoped that the PICs will maintain their collaboration with the AE, EE and implementing partners.

54. To enhance the impact potential, paradigm shift potential and effectiveness of the proposed project's interventions, the iTAP recommends the following to the AE:

55. **Climate-proofing FADs.** Given the increasing intensity and frequency of cyclones in PICs, FADs should be designed for resilience. Options include dynamic mooring techniques, elastic or adjustable anchoring systems, redundant anchoring mechanisms, and careful placement to avoid shallow waters.

56. **Facilitating transformative adaptation dialogues.** Include budgeted national dialogues with governments and stakeholders to identify practical, context-specific transformative adaptation strategies. Where feasible, integrate these strategies into project implementation or support alternative funding efforts for transformative adaptation actions.

57. **Capacity-building and stakeholder engagement.** Conduct tailored workshops and training programmes to address specific stakeholder needs. Improve preparedness with real-time, accessible data from the AWS. Strengthen community engagement by increasing opportunities for participation in policy dialogue, decision-making, and training on sustainable practices and climate adaptation. Enhance governance and enforcement by providing resources and training to local governments and agencies. Improve coordination between national and regional agencies.

58. **Promoting business involvement.** Develop incentives for businesses to adopt sustainable practices. Support MSMEs in diversifying income streams through mariculture, aquaculture and/or value-added processing.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP259)

Proposal name:	Adapting tuna-dependent Pacific Island communities and economies to climate change
Accredited entity:	Conservation International Foundation
Countries:	Cook Islands, Fiji, Kiribati, Marshall Islands, Micronesia (Federated States of), Nauru, Niue, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu and Vanuatu
Project size:	Medium

Impact potential - *Medium to High*
CI notes iTAP's positive assessment with thanks. CI acknowledges the risk of limited PIC stakeholder capacity to effectively manage climate change impacts. The Programme will address this risk through strengthening capacities of national FAD programmes (Sub-activity 1.1d), improving local skills to make better use of tuna and bycatch transshipped/unloaded by industrial tuna fleets (Activities 2.1 and 2.2), conducting communication campaigns on the impacts of climate change on availability of coastal fish for food security (Sub Activity 1.3d), and building capacity to understand/utilize AWS data effectively under Activities 3.2 and 3.3. The Programme seeks to address the risk of major cyclones by incorporating rapid FAD replacement in the most cyclone-prone Participating Countries.
Paradigm shift potential - *Medium to High*
CI notes iTAP's positive assessment with thanks. The identified risks to achievement of paradigm shift are well noted. Collaboration between PICs is historically strong, and will be reinforced through the Programme's governance structure, transparent information sharing, and by implementation led by SPC – the Participating Countries' technical organisation with extensive experience working on fisheries-related issues. The iTAP recommendation to support national dialogues in each country to support additional adaptation strategies is also appreciated; these strategies may include interventions like those currently led by SPC's FAME Division to develop small pond aquaculture to increase access to fish in rural communities.
Sustainable development potential - *Medium*
CI notes iTAP's positive assessment with thanks. iTAP's recognition of the strong development potential across SDGs is appreciated, as well as the acknowledgement of the Programme's emphasis on socio-economic benefits and focus on gender-responsive development.

The comments on environmental co-benefits are well received. CI notes that this Programme responds to national priorities by focusing on climate-adaptation support to vulnerable communities and national economies - reducing the GHG emissions profile of industrial tuna fleets is outside the Programme's mandate. However, the Programme will develop small vessel designs that improve fuel-efficiency and reduce emissions.
Needs of the recipient - *Medium to High*
CI notes iTAP's positive assessment and appreciates the accurate summary of how the Programme is designed to explicitly address the identified needs of participating governments, local stakeholders, and vulnerable communities.
Country ownership - *Medium to High*
CI notes iTAP's positive assessment with thanks. CI notes that Programme design was genuinely country-led and that the organizations involved in Programme implementation are well placed to effectively interact with and serve the Participating Countries. Ongoing engagement with and guidance from countries during implementation will take place through meetings of the Programme's government-led Steering Committee, SPC's Heads of Fisheries meetings, and FFA's Forum Fisheries Committee meetings. The iTAP identified risk of limited participation / consultation with various other stakeholders during implementation is noted. CI is confident that this risk will be minimized through the extensive consultations planned by Programme proponents and captured in the Stakeholder Engagement Plan (Annex 07).
Efficiency and effectiveness - *Medium to High*
CI notes iTAP's positive assessment with thanks. CI recognizes the importance of collaboration between Participating Countries, and with ongoing and planned interventions in the region. CI will support these efforts during implementation to avoid duplication and build complementarity. CI thanks iTAP for its recognition of the Programme's strong exit strategy, which was a significant focus of Programme design. CI also notes the risk of cyclones to FAD infrastructure and potential impacts to beneficiary support. This risk will be addressed by rapid re-deployment of damaged/lost FADs, and through capacity building of national FAD programs to effectively respond to climate-driven natural disasters.
Overall remarks from the independent Technical Advisory Panel: CI thanks iTAP for its thorough review of the Funding Proposal package and the overall positive assessment of the Programme. The assessment document provides an excellent summary of the Programme against the GCF's six investment criteria. CI sincerely appreciates iTAP's recommendation to the GCF Board to approve this Funding Proposal.

CI also notes and welcomes iTAP's recommendations on Programme design and is confident that the points raised will be effectively addressed during implementation.

Pacific Community
Communauté du Pacifique

CONSERVATION INTERNATIONAL

Adapting Tuna-dependent Pacific Island Communities and Economies to Climate Change

Annex 8: Gender Action Plan

Introduction

Conservation International (CI) as the Accredited Entity and Pacific Community (SPC) as Executing Entity, are supporting the governments of fourteen Pacific Islands Countries (PICs), namely Cook Islands, Federated States of Micronesia, Fiji, Kiribati, Marshall Islands, Niue, Nauru, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu and Vanuatu, to develop a GCF Funding Proposal (FP) for the regional Programme "Adapting Tuna-dependent Pacific Island Communities and Economies to Climate Change". This Programme is based on the Concept Note that has been endorsed by GCF and supported through a Project Preparation Facility (PPF) grant**.** This Programme is structured around two Components that aim to improve the climate resilience of the 14 participating countries:

- Component A is aimed at strengthening national programmes to employ fish aggregating devices (FADs) for the sustainable harvest of tuna and associated pelagic fish species. Principal beneficiaries for this set of activities are residents of coastal communities across the participating Pacific Island Countries (PICs). Through the implementation of the activities included here, the programme will:
    - provide technical and logistical support to strengthen National FAD programmes (Activity 1.1);
    - augment national safety-at-sea initiatives (Activity 1.2); and
    - strengthen post-harvest practices and improve market opportunities for FAD-caught fish (Activity 1.3).

Successful implementation of this set of activities leads logically to programme Output 1: the increased access to tuna and other pelagic fish for coastal communities and vulnerable households.

- Component B – Adaptation to reduce risks to Pacific Island economies from climate-driven tuna redistribution: This component includes a set of activities to establish a region-wide Advanced Warning System (AWS) to gather and analyse the data needed to more accurately determine future abundance and distribution of tuna as the fish respond to ocean warming due to climate change. The evidence generated will support the preparation and negotiation of multi-lateral agreements, designed to enable tuna-dependent PICs to maintain their access rights to shifting tuna stocks and the associated economic benefits regardless of the redistribution of the fish.

This Gender Action Plan (GAP) is informed by the Gender Assessment and is designed to support women and men to share in Programme benefits and decision-making equitably and meaningfully. Programme activities will be gender responsive, defined as: "identifying and understanding gender gaps and biases, and then acting on them, developing and implementing actions to overcome challenges and barriers toward improving and achieving gender equality. A stronger term than 'gender sensitive', gender responsive has come to mean more than "doing no harm"; it means "to do better."[1] As described in the Gender Assessment, key strategies for

[1] International Union for the Conservation of Nature

avoiding gender-related risks and promoting gender equality in the Programme will be through capacity building and institutional development, facilitating gender analysis of issues and women's participation, and capturing and reporting on gender outcomes through the Monitoring and Evaluation (M&E) Plan. Noting that each country has different levels of infrastructure, resources, cultural norms and levels of commitment to gender equality goals, the following broad strategies are proposed to address gender disparities and advance equity & equality in the Programme's activities:

- **Awareness raising**: Improve awareness of gender and related gender dimensions on all levels of the Programme to ensure that women and men engage in, and benefit equitably and meaningfully, from the Programme's outputs, outcomes, and activities. This includes raising awareness on gender dimensions in the coastal communities who manage FADs and women's roles in FAD fisheries among national Programme stakeholders.
- **Capacity and skills strengthening:** Support and encourage women's participation in practical decision-making and implementation of programme activities that are meaningful to them. This includes technical capacity building on post-harvest techniques, e.g., processing and marketing, with a focus on women given their primary role in this part of the value chain, as well as capacity building on leadership and management skills and peer mentoring for women in the local fishery sector.
- **Contribute to knowledge sharing**: Link gender equality outcomes to the broader Programme outcomes, indicating how these contribute to the overall Programme success. This may include producing communication materials that highlight lessons learned, good practices, and successes related to gender-responsive FAD deployment. This type of material is useful for the countries, the region, and the broader fishery/FAD community to influence and support greater uptake of gender-responsive considerations in coastal FAD projects.
- **Track and report on efforts**: Mainstream gender in the Monitoring and Evaluation Plan of the Programme through the collection of sex-disaggregated data and information as a basis for evaluating and monitoring gender-related Programme outcomes and its effectiveness and highlighting measures that address gender equality objectives.
- **Address potential gender risks**: Link resources and support services at the national level to reduce and respond to gender-based violence (GBV) and Sexual Exploitation, Abuse, and Harassment (SEAH) project-related risks. Please also note both the EE (SPC) and AE (CI) have The AE (Conservation International) senior staff members trained on investigation of SEAH incidents (training provide by Investigator Qualification Training Scheme), and the AE is prepared to lead the investigation if a SEAH incident escalates to the AE level.
- **Design inclusive activities**: Actively implement gender inclusive approaches to engage FAD fishers and others along the value chain. While women are not well represented as FAD fishers in the region, training and capacity building should still be open and inviting for any women that may be interested in entering the profession. Support awareness on project activities and opportunities and ensure it is accessible and open to women, is therefore important.

Scope of the Gender Action Plan: This Gender Action Plan is designed to set out regional priorities and actions to which the national programmes will align and contribute, recognizing the need for national-level tailoring and variety based on context. It is recommended that national-level workplans be designed in a gender responsive manner to guide country-specific activities and ensure alignment with and reporting to, the regional GAP. For example, once the exact locations for FAD deployments are selected and adjacent communities identifies, more detailed and appropriate timeline for consultations and development of site-specific qualitative indicators will be considered for that specific community. The regional GAP is managed and reported on by the PMU (SPC).

Roles and Responsibilities: At the EE level (within the PMU), this GAP is overseen by two SPC Gender & Environmental and Social Safeguards(Inclusion) officers (GESI Officers) who will provide backstopping, training, support, and coordination for National Fisheries Agencies by training one programme staff in Gender & Environmental and Social Safeguards to act as the GESI focal points to the programme (National GESI FPs). The national GESI focal point role will be assigned to one of three full-time programme staff appointed by the Fisheries Ministries/Departments at the start of implementation included in the overall programme budget (1 programme coordinator, 2 FAD Officers). The GESI Officers are a full-time position and considered core programme and PMU staff. While the PMU GESI Officers are ultimately responsible for implementing the GAP, national GESI FPs are responsible for the national-level tailoring and supporting implementation of the GAP, including the implementation of specific gender considerations for the country-level Programme activities and reporting to the EE GESI Officers on GAP activities and results.

Timeline: The GAP lists "ongoing and in alignment with timelines of project activities to which these contribute" in the timeline column below, given the highly integrated nature of the activities and recognizing that timelines will likely adjust. It is expected that the GESI officers will be well integrated on the PMU and be able to ensure specific GAP activities are implemented at the right time.

Impact Statement:	**Support climate adaptation through promoting greater gender equality and equity in FAD Fishing supply chains across programme countries.**				
Outcomes:	• Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing. • Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs. • Increased leadership and influence of women in FAD tuna fishery value chain decision-making.				
Programme Activity	**Gender Activities**	**Performance Indicators with Targets and Baselines**	**Timelines**	**Responsibility**	**Budget – estimate for each country's GAP activities $USD**
Output 1: Increased national capacity to access tuna and other pelagic fish for coastal communities					
1.1: Provide technical and logistical support, including equipment, to strengthen National FAD programmes	1. Investigation and documentation of gender roles (including risk of, and response to, GBV/SEAH) within localized FAD fishery value chains is integrated into the workplans for scaling up of national FAD programmes.	1. # of workplans that include tailored gender considerations in the national FAD programmes. **Baseline = 0** **Target = 14**	Ongoing and in alignment with timelines of project activities to which these	SPC GESI officers to coordinate and support; national GESI officers to support workplan design, delivery training,	Refer to Budget Summary Table below or full programme budget

Impact Statement:	Support climate adaptation through promoting greater gender equality and equity in FAD Fishing supply chains across programme countries.				
Outcomes:	• Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing. • Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs. • Increased leadership and influence of women in FAD tuna fishery value chain decision-making.				
Programme Activity	**Gender Activities**	**Performance Indicators with Targets and Baselines**	**Timelines**	**Responsibility**	**Budget – estimate for each country's GAP activities $USD**
	2. Design and deliver capacity building and awareness raising on gender dimensions (including GBV & SEAH) of FAD fishery value chain, and women's roles within the fishery, to augment the skills of national staff who implement FAD programmes and PMU staff. 3. Ensure the consultative stakeholder engagement strategy for national fisheries agencies and communities to identify FAD sites is gender responsive and includes relevant qualitative indicators. 4. Design skills strengthening for small-scale FAD fishers in a gender-inclusive way that engages and encourages women's participation. 5. Integrate gender considerations into national natural disaster response mechanisms.	2. # of staff (national FAD programmes & PMU) who have received training. **Baseline = 0** **Target = 50% women** 3. # of FAD consultative stakeholder engagement strategies that are gender responsive. **Baseline = 0** **Target = 14** 4. Percentage of women participants attending FAD skills strengthening training sessions. (regional aggregate) **Baseline = 0** **Target = (>30% women)[2]** 5. # of disaster response plans that are gender responsive. **Baseline = 0** **Target = 14**	contribute.	and input to National FAD Programme consultation strategy.	

[2] Target to be revised after first year of implementation based on experience; target should be realistic yet ambitious.

Impact Statement:	**Support climate adaptation through promoting greater gender equality and equity in FAD Fishing supply chains across programme countries.**				
Outcomes:	• Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing. • Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs. • Increased leadership and influence of women in FAD tuna fishery value chain decision-making.				
Programme Activity	**Gender Activities**	**Performance Indicators with Targets and Baselines**	**Timelines**	**Responsibility**	**Budget – estimate for each country's GAP activities $USD**
1.2: Augment national safety-at-sea initiatives.	1. Ensure needs analysis for improved vessel safety for SS FAD fishers includes input from women fishers (as available). 2. Ensure training on boat safety equipment is inclusive and inviting of women fishers.	1. # of needs analyses that have sought/included feedback from women fishers. **Baseline = 0** **Target = 14** 2. % of women participants attending training on boat safety. **Baseline = 0** **Target = (>5% women)**[3]	Ongoing and in alignment with timelines of project activities to which these contribute.	SPC GESI officers to coordinate and support national GESI FPs.	Refer to Budget Summary Table
1.3: Strengthen post-harvest practices and improve market opportunities for FAD-caught fish in countries with active transhipping.	1. Targeted training courses for women in coastal communities to improve preservation and hygiene of FAD-caught fish, including bottling and canning of fish, and marketing opportunities. 2. Provide support to improve hygienic and safe working conditions for sellers of fish, especially women. This includes GBV/SEAH awareness training, including referral services.	1. % of participants in training courses that are women. **Baseline = 0** **Target = (60% women)**[4] 2. (same as 1) 3. # of national FAD projects include provision of equipment for FAD caught fish preservation that is gender equitable. **Baseline = 0**	Ongoing and in alignment with timelines of project activities to which these contribute.	SPC GESI officers to coordinate and support national GESI FPs.	Refer to Budget Summary Table

[3] Target based on fact that pelagic fishing is very heavily male dominated. Target to be revised after first year of implementation based on experience; target should be realistic yet ambitious.

[4] Target to be revised after first year of implementation based on experience; target should be realistic yet ambitious.

Impact Statement:	Support climate adaptation through promoting greater gender equality and equity in FAD Fishing supply chains across programme countries.				
Outcomes:	• Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing. • Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs. • Increased leadership and influence of women in FAD tuna fishery value chain decision-making.				
Programme Activity	**Gender Activities**	**Performance Indicators with Targets and Baselines**	**Timelines**	**Responsibility**	**Budget – estimate for each country's GAP activities $USD**
	3. Facilities are gender equitable for FAD caught fish preservation. 4. Communication campaigns are designed to be gender informed, recognizing the roles that women and men play in HH food preparation and decision-making.	**Target = 100%** 4. # of communications campaigns that raise awareness of gender roles in small scale tuna fishery. **Baseline = 0** **Target = 14**			
Output 2: Increased supply of bycatch and tuna from industrial fishing operations for urban / peri-urban communities					
2.1 Implement strategies to deliver more transhipped and unloaded bycatch and tuna to urban/peri-urban communities.	Not gender specific, no activities and indicators needed for Output 2.1. Please refer to M&E table.				
2.2: Strengthen/develop post-harvest practices and improve market opportunities to distribute bycatch and tuna from transhipping and unloading operations to urban/peri-urban communities.	1. Develop or update post-harvest handling national regulations for fish sellers that provide a quality standard equal to FAD fish selling, e.g., support improved hygienic and safe working conditions for sellers of fish, especially women and fish market outlet designs consider specific needs of women fish sellers. 2. Targeted training courses for women in urban communities to	1. Post-harvest handling workplans and policies include gender responsive facilities, e.g, fresh water, shade, access to bathrooms. **Baseline = 0** **Target = 100%**[5] 2. # of participants (participant sex disaggregated) in post-	Ongoing and in alignment with timelines of project activities to which these contribute.	SPC GESI officers to coordinate and support national GESI FPs.	Refer to Budget Summary Table

[5] Applicable to countries that do not have existing post-harvest handling regulations.

Impact Statement:	Support climate adaptation through promoting greater gender equality and equity in FAD Fishing supply chains across programme countries.				
Outcomes:	• Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing. • Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs. • Increased leadership and influence of women in FAD tuna fishery value chain decision-making.				
Programme Activity	**Gender Activities**	**Performance Indicators with Targets and Baselines**	**Timelines**	**Responsibility**	**Budget – estimate for each country's GAP activities $USD**
	improve preservation and hygiene of FAD-caught fish, including marketing opportunities.	harvest handling training and GBV/SEAH awareness training. **Baseline = 0** **Target = 100%**			
Output 3: Improved forecasts and projections for climate-driven tuna redistribution which facilitate effective adaptations for all stakeholders					
3.1: Develop and deliver an Advanced Warning System (AWS) for climate-driven tuna redistribution.	Not gender specific, no activities and indicators needed for Output 3.1. Please refer to M&E table.				
3.2: Assess and socialise the impact of tuna biomass redistribution identified by the AWS on national economies at all levels.	Not gender specific, no activities and indicators needed for Output 3.2. Please refer to M&E table.				
3.3: Provide AWS-related training to national institutions to negotiate in regional and international forums to address socio-economic losses due to the impacts of climate change on tuna.	1. Training for staff on negotiation capabilities is designed/delivered with gender considerations in mind.	1. % of participants (sex-disaggregated) skilled in high-level negotiation procedures. **Baseline = 0** **Target = (>40% women)**[6]	Ongoing and in alignment with timelines of project activities to which these contribute.	SPC GESI officers to coordinate and support national GESI FPs.	none

[6] Target to be revised after first year of implementation based on experience; target should be realistic yet ambitious.

Impact Statement:	Support climate adaptation through promoting greater gender equality and equity in FAD Fishing supply chains across programme countries.				
Outcomes:	• Increased capacity of women in the FAD tuna fishery value chain focusing on post-harvest tuna processing and marketing. • Increased awareness of gender roles and gender dimensions in the coastal fishery communities managing and operating FADs. • Increased leadership and influence of women in FAD tuna fishery value chain decision-making.				
Programme Activity	**Gender Activities**	**Performance Indicators with Targets and Baselines**	**Timelines**	**Responsibility**	**Budget – estimate for each country's GAP activities $USD**
GAP specific operational activities					
4. Programme is delivered in a gender-responsive manner.	1. Annual assessment of Gender Action Plan implementation, i.e., activities, indicators, and targets, whereby the Programme has contributed to gender equality. 1a. PMU GESI Officers supported by GESI focal points conduct a representative # of gender-disaggregated focus groups[7] with women and men project participants to assess changes in empowerment which contribute to the annual assessment of GAP implementation. 2. Ensure regional and national-level AGRMs are designed in a gender sensitive manner and able to process gender-related grievances. 3. Skills/experience are gender equitable in salary in selection of new hires across the Programme.	1. Report/communication material annually that describes the project's gender approach, outcomes, and lessons learned. 1a. Baseline will be collected in first year[8]; target is increase of at least 10% in each domain of power per year.	Annually, in line with programmatic reporting.	SPC GESI officers to coordinate and support national GESI FPs.	Captured within budget for salary, trainings and travel of SPC GESI officers.

[7] Methodology for the focus groups will be co-designed by the SPC GESI specialist & national GESI focal points, tailored to the context of each country project, but following similar structure and topics. The methodology will be informed by the Abbreviated Women's Empowerment in Fisheries Index which measures the empowerment, agency and inclusion of women in fisheries context.

[8] Specific indicators to be tracked annually will be identified during the baseline year.

Budget Summary for GAP

Budget	Actions	Costs USD
Staff		
Full time salary for 2 SPC Gender and Social Inclusion (GESI) Officers Travel budget for GESI Officers • 28 trips (14 countries, 2 visits each over 5 years)	GESI Officers annual salary & travel	Staff time: $1.56 million Consultants: $340,000 Travel: $145,751
Trainings		
Quarterly training for first year, i.e., 4 trainings (different target audiences) Annual training in each country for 4 years	Delivery of gender sensitisation and awareness training (Corresponds with GAP Activity 1.1.2 above)	Included in the programme travel/training budget
Annual training for each country for 5 years	Post harvest handling of fish training, with a gender-sensitive approach that accounts for issues of childcare, location/transportation, time of day, and other specific challenges that may prevent women from attending.	Included in the budget for programme meetings & workshops
Annual training for each country for 5 years	Hygienic and safe working conditions for women and men inclusive of GBV/SEAH awareness training (including referral services), with a gender-sensitive approach that accounts for issues of childcare, location/transportation, time of day, and other specific challenges that may prevent women from attending.	Included in the budget for programme meetings & workshops
Annual training for each country	Boat safety training including first aid (>15% female attendees), with a gender-sensitive approach that accounts for issues of childcare, location/transportation, time of day, and other specific challenges that may prevent women from attending.	Included in the budget for programme meetings & workshops

Budget	Actions	Costs USD
GAP specific operational activities		
Training with Fisheries Ministries	AGRM meetings are held annually for any grievances brought up by the community.	Included in the budget for programme travel, meetings & workshops

Pacific Community
Communauté du Pacifique

CONSERVATION INTERNATIONAL

Adapting Tuna-dependent Pacific Island Communities and Economies to Climate Change

ANNEX 08.a: Gender Assessment

Abbreviations

ADB	Asian Development Bank
BPA	Beijing Platform of Action
CEDAW	Convention on the Elimination of all forms of Discrimination against Women
CI	Conservation International
FAD	Fish Aggregating Device
FHH	Female Headed Households
GAP	Gender Action Plan
GBV	Gender Based Violence
GCF	Green Climate Fund
GRM	Grievance Redress Mechanism
HIV	Human Immunodeficiency Virus
HOH	Head of Household
MDGs	Millenium Development Goals
MMR	Maternal Mortality Rates
PICs	Pacific Island Countries
PIFWLM	Pacific Islands Forum Women Leaders Meeting
SDG	Sustainable Development Goals
SEAH	Sexual Exploitation Abuse and Harassment
SPC	Secretariat of the Pacific Community
STIs	Sexually Transmitted Infections
UN	United Nations
UNCLOS	United Nations Convention on the Law of the Sea 1982
UNDP	United Nations Development Program

Table of Contents

Table of Figures

1 Executive Summary

The fisheries sector is largely considered to be a male domain, with the role of women in the sector generally confined to support roles (e.g., post-harvest). Women's roles are often invisible, yet they can be quite significant and contribute to sustaining the fishing families and community, protecting natural resources and local food security.

This assessment provides a high-level overview of gendered roles, opportunities, and barriers within the context of coastal FAD fisheries in the Pacific region. This assessment provides context and analysis of the power dynamics, relative vulnerabilities, and varying roles and responsibilities of men and women in the communities that will be involved in, and impacted by, Programme activities in each participating Pacific Island country.

The assessment highlights five targeted gender equality regional objectives for the Programme's gender approach, noting that these will become more specific at country level.

- Provide a pathway for greater recognition of the roles both women and men play in the coastal FAD fishing context through training and equal remuneration with men doing the same roles.
- Measure benefits and empowerment (personal viability) that men and women receive working in the Programme encouraging gender inclusion to be a 'norm'. For example, fostering working partnerships between men and women fishers.
- Provide opportunities to improve the working conditions of women selling fish in unhealthy and unsanitary conditions, including investments that support institutional strengthening and improving women's access to productive assets related to the sector. These include several options for support, for example: shade, cold storage, lighting, handwashing facilities.
- Provide basic equipment to maintain shelf life of fish and provide training to women on the safe and effective processing and marketing of tuna.
- Provide an increased supply of FAD fish at an affordable price for urban and peri-urban communities. The market will determine the price.

These regional objectives provide enough flexibility to be adapted to the individual country local realities and nuances. Each country will identify what their priorities and opportunities are to advance national gender equality goals in the context of this programme.

2 Background and Introduction

2.1 Background and objectives

The Gender Assessment and Gender Action Plan is a necessary safeguard requirement by the Green Climate Fund (GCF). Under the GCF Regional Tuna Programme: Adapting Tuna-dependent Pacific Island Communities and Economies to Climate Change, the gender assessment explores how women and men are currently engaged in the value chain of the fisheries sector in the Pacific and the barriers and the foreseeable impacts to gender roles in transferring fishing efforts from coral reefs to offshore tuna. The shift in fishing efforts from coral reefs to tuna fishing will require the deployment and monitoring of Fish Aggregating Devices (FADs) by national/provincial fishery officers in coastal and relatively deep water, and small-scale fishers going further out to sea to fish. This change has come about due to a decline in the availability of reef fish for human consumption due to the degradation of coral reefs coupled with increasing human population in the Pacific Islands region, and all of this is exacerbated by subsequent climatic changes and storm events.

Over the last few decades there has been a large shift towards using FADs to attract tuna and other pelagic fish, and this has become one of the major fishing modalities worldwide. FADs are artificial structures placed at sea to attract fish and make them easier to catch. It is expected that the use of FADs can help increase the supply of fish for food security to local fishers and growing coastal communities and reduce fishing pressure on reefs. The Programme seeks to increase the sustainability of healthy fish supply for local consumption to decrease imports of unhealthy and inexpensive alternatives such as high-fat meats (e.g., lamb and mutton flaps, turkey tails etc.) and is also environmentally supportive.

While there are many benefits to the use of FADs, they do not come without challenges and FADs will have impacts on fishing, marketing, and consumption of the catch. Because of this, and the gendered roles and responsibilities of women and men across the domestic tuna fishery supply chains, particularly the value chain of small-scale tuna fisheries, implementation of this Programme may also impact men and women differently across the full regional geography. The increased risks from this programme will be reduced through risk avoidance or mitigating/minimizing activities. The programme also provides an opportunity to change the narrative about gender inequality in the fishing sector and support women's economic empowerment in this space by providing greater recognition of women's roles in fisheries in the Pacific Islands.

Gender equality is about equal opportunities, rights and responsibilities for women and men, girls and boys. It does not suggest that women and men are the same. Gender inequality is a result of unequal power distribution between women and men, exacerbated by ongoing discrimination, weaknesses in laws, policies and institutions, and social relations that normalise inequality. Experience over recent decades shows that gender equality, economic growth and development are mutually reinforcing and significantly correlated. This acknowledges that one cannot occur effectively without the other.

While fisheries are generally considered a male domain, the Pacific Islands region has made significant progress in this arena in both regional (currently the leading regional fisheries management agencies in the Pacific such as the Forum Fisheries Agency and the WCPFC are led by women) and national scales (https://fame.spc.int/publications/bulletins/womeninfisheries/39). This Gender Assessment demonstrates that both women and men are involved in the sector at different levels of the product value chain and identifies areas to promote gender equality.

This report provides a high-level overview of gendered roles, opportunities, and barriers within the fisheries sector in the Pacific Islands region. This will provide context and regional-level analysis of the power dynamics, relative vulnerabilities, and varying roles and responsibilities of men and women in each participating Pacific Island country. Understanding the social nuances in each context is important, and while this assessment includes short country-specific overviews, it could not go into too much detail at the national level given that specific sites (where FADs will be placed) have not yet been identified. The social and cultural contexts of each country and each community are unique, and we did not want to homogenize at the national level. Therefore, we have built into the GAP (Activity 1.1.1) a process for collecting and documenting gender roles at the community level, where the FADs will be deployed, and where support across the value chain (including targeted post-harvest support to women) will be prioritized. This assessment also contains an annex describing country-level details which are meant to help the national gender and safeguard focal points ensure a tailored approach to national-level activities.

It is expected that national-level implementation will be further developed and tailored to each context. The Programme will strive to be gender responsive by addressing identified gender issues through a well-defined Gender Action Plan (GAP) that will be fully integrated into Programme design and implementation. The Gender Action Plan also aligns with the safeguard plans for the Stakeholder Engagement Plan, Accountability and Grievance Redress Mechanism, as well as the Environmental & Social Management Plan.

4.1 Methodology and scope of the assessment

To develop this gender assessment and highlight local realities and characteristics of the Pacific Island Region, a mixed method approach was undertaken. Note that this assessment is focussed on gender impacts specific to the proposed Programme, rather than a broader consideration of gender dimensions in general.

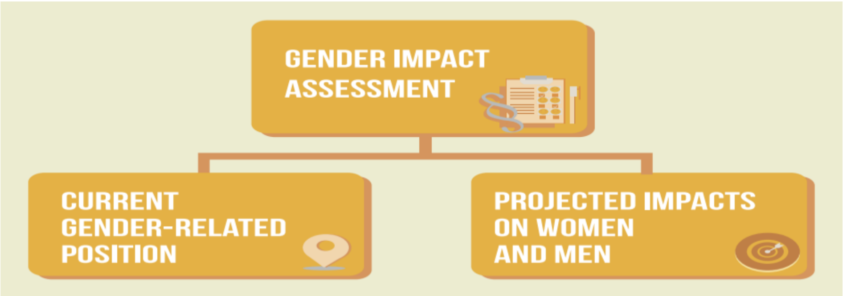
Further, given the specific data limitations in terms of gender characteristics in some Pacific Island Countries (PICs), (no territories are participating in the Programme) an assessment based on available gender related information from each targeted Pacific Island has been considered in the context of FAD fishing. Note that it is important to recognise that each PIC have their own realities and nuances, and this is only captured broadly in the current gender assessment. A short summary highlighting each country's context forms part of this assessment to provide a glimpse into the major gender related nuances of each Pacific Island country. When country projects are identified at the next phase of the GCF FAD Programme, the gender dimensions will be integrated into the project document from the start of the design phase onwards. These will be tailored according to the country nuances and current gender mainstreaming efforts.pro

The methods used for this gender assessment included:

- Literature review of available statistics and information including national censuses, International Labor Organization documents, Household Income and Expenditure surveys, Health and Education Statistics and other available data;
- Review of regional and national gender policy and strategy documents;
- Interviews and consultations at the national level with a selection of fishing supply chain stakeholders. That is: fisher folk, fish sellers in stalls and markets; consumers of fish who purchase from local markets and impacted local stakeholders from affected communities. This included fishing communities from 13 PICs (the RMI chose not to participate in Component A) with a total of approximately 190 + interviewees; and
- Consideration of other recent studies and projects which may offer a glimpse into local realities (referenced in annex).

National consultants were recruited from each of the PICs to conduct community consultations using an approved survey to understand the gender dimensions in the fishery sector of their country. A line of questioning with a combination of qualitative and quantitative questions was developed to provide a glimpse into national gender dimensions for the consideration of FADs. The information from the questionnaires and interviews have been analysed and reported herein in line with above project objectives in Chapter 6 Country Gender Profiles.

Figure 1: Gender Impact Assessment Structure



This Gender Assessment provides information on general gender dimensions inherent to the Pacific Islands Region and where available the local islands proximate to the proposed FAD project sites. The Assessment then details the potential risks and impacts of the Fish Aggregating Device in the Programme. Finally, an overview of primary gender-related gaps and opportunities, as well as project-specific recommendations to manage and/or prevent the negative impacts (and build on positive ones) are provided.

3 Gender and Development Issues in the Fisheries Sector

It is pertinent to commence this section highlighting the keynote speech by the recently elected Prime Minister of Samoa, (Fiame Mata'afa) at the first Pacific Islands Forum Women Leaders Meeting (PIFWLM) in 2022. She noted that to pursue gender equality across the Pacific, it's time for leaders "to move beyond rhetoric", it is important to focus on the process required to strengthen outcomes for women and girls, leaving no one behind.

4.2 International policies

There are several international agreements, commitments and overarching principles that seek to improve gender equity and equality across all facets of development in the Pacific. Momentum is occurring to recognize women's role in the fishing industry, but more needs to be done. These include:

- Convention for the Elimination of All Forms of Discrimination against Women (CEDAW),
- Universal Declaration of Human Rights 1948-Gender Equality was made part of international human rights recognized that,

> "*All human beings are born free and equal in dignity and rights*" and that "*everyone is entitled to all the rights and freedoms set forth in this Declaration, without distinction of any kind, such as race, colour, sex, language, religion, … birth or other status*."

- Sustainable Development Goals outlines 17 global objectives to be achieved by 2030.

- Millennium Development Goals (MDGs),
- Revised Pacific Platform for Action on Advancement of Women and Gender Equality (2005 to 2015);
- Pacific Plan;

- 42nd Pacific Island Forum commitment

4.3 Current stance on gender equality for the Pacific Island Region

Many Pacific Island countries have adopted policies aimed at working towards greater gender equality, encouraged by several international agreements and commitments. These include: the Convention for the Elimination of All Forms of Discrimination against Women (CEDAW), the Millennium Development Goals (MDGs), the Revised Pacific Platform for Action on Advancement of Women and Gender Equality (2005 to 2015); the Pacific Plan; the 42nd Pacific Island Forum commitment to increase the representation of women in legislatures and decision making; and the 40th Pacific Island Forum commitment to eradicate sexual and gender based violence.[1]

In 2012, PICs governments committed to the Pacific Leaders Gender Equality Declaration that stands to lift the status of women in the Pacific and empower women and girls as active participants in economic, political, and social life. These include commitments towards responsive programmes and policies to advocate for:

I. Improved production and use of sex disaggregated data and gender analysis to inform government policies and programmes.
II. Increased representation of women in decision making as part of private and local and national governance (e.g., advocacy groups, political parties, school boards and market committees).
III. Economic empowerment through improved access and equal opportunity to employment opportunities and a reduction in discriminatory pay conditions for women and improved facilities for local produce markets to increase profitability and efficiency and support for women entrepreneurs.
IV. Ending violence against women through implementation of essential services (protection, health, counselling, legal) for women and girls who are survivors of violence, enact legislation regarding sexual and gender-based violence and impose appropriate penalties for perpetrators of violence.
V. Encourage education on reproductive health and awareness of programmes and gender parity in informal, primary, secondary, and tertiary education and training.

Despite these commitments, overall progress in PICs towards gender equality is slow. Many studies and reports have been undertaken about women's participation in the tuna industry highlighting impacts, costs and benefits, constraints, and opportunities. However, it appears that little has been done to implement the recommendations of these studies and reports[2] given that in all relevant PICs, women's role in the fishing industry remains obscure, even invisible. There is some progress towards improved education for girls and some positive initiatives to address violence against women,

[1] Pacific Leaders Gender Equality Declaration 30 August 2012, Rarotonga, Cook Islands – Forum Sec

[2] Gender Issues in the Pacific Islands Tuna Industry 2006 Gender issues in P. I. Tuna Industries 1 0.pdf (ffa.int)

however there is considerable room for improving outcomes in such areas. Gender inequality poses a high cost on personal, social, and economic outcomes for Pacific peoples and there are significant benefits if greater gender equality is acquired. Actions to encourage gender equality as part of this Programme, contribute to cultivating a more prosperous and secure environment for communities in the Pacific.

4.4 Characteristics of Pacific Island Communities in Gender Equality

Gender equality in Pacific Island countries is reportedly limited by several factors including representation in parliament and public decision making, legislation that protects women and girls against violence, lack of women's rights over their sexual and reproductive health (e.g., birth control and abortion) and fewer opportunities for economic independence (e.g., access to public and private finance) and customary delineation of domestic roles and responsibilities.

3.3.1 Literacy and Education

Education Levels and Access

In general, most PICs are on track to achieve targets for universal primary education and gender equality in enrolments to primary and secondary school *(see Figure 2: Gross* **Enrolment Rates and Gender Parity Indexes in Primary and Secondary Education for PICs***.* Interestingly, boys are more likely to drop out of school compared to girls, and enrolments to university are also higher in females compared to males. There is a still a strong inclination for girls to choose courses that are more female dominant, and girls are often under-represented in scientific, vocational and technical fields.[3] Although advances in educational opportunities have been demonstrated in PICs, this has not yet resulted in a marked increase in women's economic empowerment through employment or entrepreneurship.[4]

In the context of this Programme, an **opportunity** lies in developing or increasing focused support to women in areas such as adding value to fish through cooking, barbecuing or smoking fish which can also be an entry point for women into formal roles in the fishing industry. This would require training, mentorship through small businesses incubators, and entry funding.

Women remain largely tied to traditional roles, and the majority of those who have professional careers carry these out in addition to their traditional home and community roles. This can place a considerable time constraint on those with professional roles, limiting the opportunity to advance into senior roles.

In the context of this Programme, a **risk** therefore stems from women not being able to engage in project activities actively or meaningfully due the great burden of other priorities. At the national level, gender approaches should take this into account and find times, locations, strategies that work best for women in the fish processing & marketing sphere.

Figure 2: Gross Enrolment Rates and Gender Parity Indexes in Primary and Secondary Education for PICs

[3] Review of the Revised Pacific Platform for Action on the Advancement of Women and Gender Equality 2005–2015 (spc.int)

[4] PIFS and SPC, Addressing Inequalities: the Case of Small Island Developing States in the Pacific, Issues Paper for the Small Islands Developing States Conference 2013

Country	Year	Primary Gross Enrolment Rate Male	Primary Gross Enrolment Rate Female	GPI-Primary	Year	Secondary Gross Enrolment Rate Male	Secondary Gross Enrolment Rate Female	GPI-Secondary
Cooks	2014	103.0	103.0	100.0	2014	JS 99.0	JS 100.0	JS 101.0
						SS 64.0	SS 78.0	SS 122.0
FSM	2010	96.0	98.1	102.0	2010	72.5	79.9	110.0
Fiji	2013	109.0	110.0	101.0	2013	–	–	111.0
Kiribati	2011	89.0	90.0	101.0	2011	JS 79.0	JS 93.0	JS 118
						SS 38.0	SS 52.0	SS 137.0
RMI	2010	95.0	93.9	99.0	2010	57.1	63.5	111.0
Nauru	2013	106.6	100.4	94.0	2013	77.0	81.2	106.0
Niue	2010	100.0	100.0	100.0	2013	113.4	97.5	86.0
Palau	2011	120.2	117.5	98.0	2011	67.9	73.4	108.0
PNG	2009	89.9	80.8	90.0	2013	23.1	16.5	71.0
Samoa	2012	102.0	105.0	103.0	2012	67.0	76.0	113.0
Tuvalu	2011	101.0	101.0	100.0	2011	41.0	63.0	154.0
Vanuatu	2011	120.4	116.7	97.0	2011	39.1	42.8	110.0

Source: National Minimum Development Indicators database, SPC; Cook Islands: Cook Islands 2014 Statistics Report from http://www.education.gov.ck/attachments/article/46/2014%20Education%20Statistics%20Report.pdf ; FSM 2010 Census Summary; RMI Ministry of Education; Nauru and Palau derived from Ministry of Education Annual Statistics Reports and SDD population estimates.
Note: JS = Junior Secondary; SS = Senior Secondary.

3.3.2 Women's Health and Nutrition

A few PICs have made substantial advances in women's health, yet in other countries there remains a lot to be done. Many of the health challenges are avoidable. These challenges can vary because of both biological and gender-related differences. In many societies in the Pacific, women are disadvantaged by discrimination rooted in sociocultural factors and subsequently, women's health suffers. Non-communicable diseases are a significant threat to women's health due to higher susceptibility and because women are more likely to undertake carer roles for others with non-communicable diseases.[5] Heart-disease and stroke, diabetes, cancer, and chronic lung disease continue to be the most common causes of death among Pacific Island women. Subsequently, many PICs have not achieved the health Millennium Development Goals (MDG) targets.

Furthermore, underlying socio-cultural circumstances are major contributors to maternal and newborn deaths, low antenatal coverage, high rates of home birth, low contraceptive use (<33% in some PICs), and high rates of unplanned pregnancy. And yet, improved maternal health; including antenatal care coverage and attendance at births by trained personnel, can result in a decrease in maternal mortality in PICs. However, accessibility to these services is often difficult in those countries with dispersed populations and those living on remote islands or in mountainous areas.[6] In 2020, it was estimated that 75% of nurses and 48% of doctors are women (SPC 2022), offering the opportunity for improved health outcomes for women's health issues in the future. Women health workers generally provide a more comfortable and credible environment to support women's and family health issues, due to both cultural norms as well as through having a sound understanding on what other women experience.

In the context of this Programme, there is good **opportunity** to influence diet towards tuna and other fish taken in association with FADs and away from imported or other less healthy foods. Given that women are usually responsible for household meals, women would be a key audience for these messages.

---

[5] Ibid.
[6] Review of the Revised Pacific Platform for Action on the Advancement of Women and Gender Equality 2005–2015 (spc.int)

3.3.3 Vulnerability and Poverty

The Pacific Island countries face significant challenges which contribute to their growing vulnerability and the increased poverty experienced across the region. The following are some of the key challenges relating to vulnerability and poverty:

- Geographical location:
- Climate Change:
- Economic concerns:

Women and children are at a greater risk of poverty in PICs, particularly in urban areas where women's subsistence agriculture is no longer viable as a survival measure. Social protection programmes have been introduced in some countries; however, slow economic growth limits the effect of such measures.[7] Socially disadvantaged groups and those who are discriminated against, often have limited rights to resources. As this inequality and vulnerability is normalised, these populations often overlook their rights and do not seek support from institutions and services that are available.[8]

3.3.4 Gender Based Violence (GBV) and Related Dimensions

Domestic Violence and Sexual Abuse

Violence against women and girls in PICs is amongst the highest in the world. Global evidence demonstrates that gender inequality is a root cause of violence against women and girls. In countries where prevalence studies have been undertaken, up to 68% of women in the Pacific have reported experiencing physical or sexual violence by a partner in their life (*see Figure 3: Prevalence* **and Patterns of Violence Against Women in PIC***.*[9] Men's power controls over women and girls, traditional gender roles and social norms as well as male sexual entitlement are drivers of violence in PICs. In many cases, women in PICs who experience domestic violence believe it is 'normal' and that it is their responsibility to 'behave' in a way that does not anger their partner.[10] In some communities' women may accept this way of thinking more than their male partners.

A high level of alcohol consumption in males and childhood experience of, and exposure to violence are also factors that reinforce violence against women and girls in PICs.[11] While some countries and/or communities are matrilineal, this does not deter the level of violence against women.

As a result of domestic violence, sexual harassment and trafficking women and girls, females are often more likely to be exposed to sexually transmitted infections (STIs) and HIV is increasing in some PICs. HIV is considered a low-level epidemic in countries such as Kiribati according to the 2015 Global AIDS report.[12]

---

[7] Ibid

[8] Pacific handbook for gender equity and social inclusion in coastal fisheries and aquaculture gender equity and social inclusion handbook (spc.int)

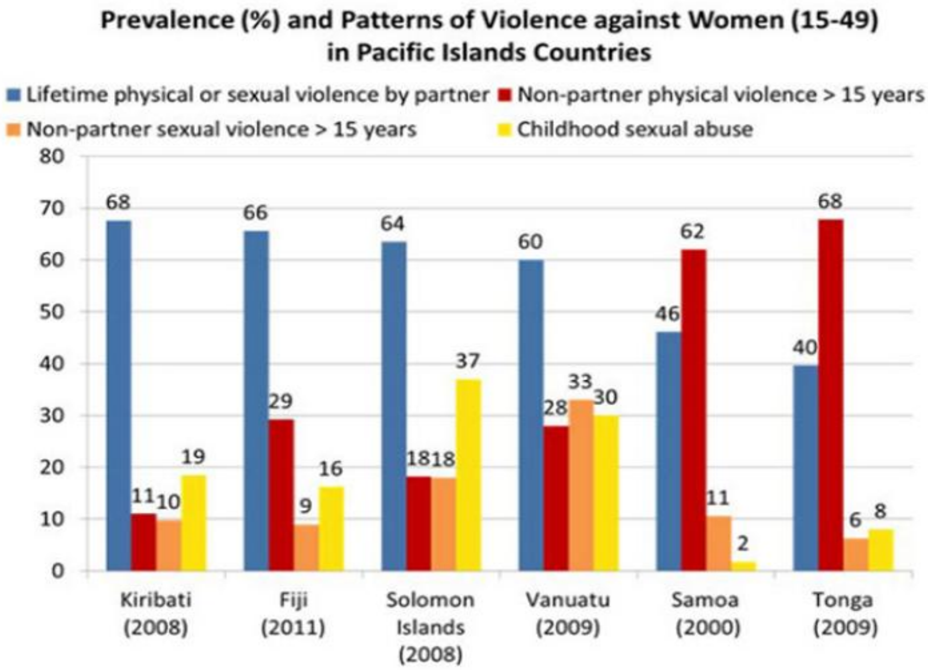
[9] Ending Violence Against Women and Girls | UN Women – Asia-Pacific

[10] Pacific handbook for gender equity and social inclusion in coastal fisheries and aquaculture gender equity and social inclusion handbook (spc.int)

[11] UN Women Asia-Pacific, Violence against women and girls in South Tarawa Kiribati, Findings from a 2019 baseline study.

[12] MoH, 'Global AIDS Response Progress: Kiribati Country Progress Report 2015', viewed on 15 February 2023, <http://www.unaids.org/sites/ default/files/country/documents/KIR_narrative_report_2015.pdf>

Figure 3: Prevalence and Patterns of Violence Against Women in PIC[13]



In the context of this Programme, there is a **risk** that activities could contribute to, or exacerbate GBV during the course of implementation. Not necessarily because FAD fishing is known to be a high risk GBV activity, but because of the contextual/background/existing high rates of GBV in the fisheries sector more broadly. At the same time, there is an **opportunity** within this programme to build awareness about GBV, women's/girl's rights, and resources that exist to help support survivors.

3.3.5 Employment opportunities and labour force participation

Overview of Employment and Gender Constraints

As described earlier, advances in educational opportunities have not resulted in a marked increase in women's economic empowerment through employment or entrepreneurship.[14] Gender-based segregation and occupational streaming in vocational training and occupations tends to be gender stereotypical, which can limit both women's and men's access to certain jobs. Women are often siloed into gendered positions of employment and household roles such as preparing food and caring for family and children which limits their participation in the labour force.

Employment Status and Labour Force Participation

Aside from reports from Solomon Islands and Fiji, there is very little information on employment opportunities specifically for women in the tuna industry. Overall, there is a lack of job opportunities for women in paid employment, including within the fisheries sector. Most women occupy lower-level positions in the public sector and the number of women who hold senior public service positions is very gradually increasing.[15] The majority of women participate in the informal economy which limits them to low earnings and no protection. Women are most frequently home-based and reliant on subsistence agriculture, marketing agricultural products and petty trading.[16] Whilst this is the case, the fishing industry remains a predominantly male dominated area. Section 4 discusses the gender

[13] Ending Violence Against Women and Girls | UN Women – Asia-Pacific
[14] Review of the Revised Pacific Platform for Action on the Advancement of Women and Gender Equality 2005–2015 (spc.int)
[15] PIFS and SPC, Addressing Inequalities: the Case of Small Island Developing States in the Pacific, Issues Paper for the Small Islands Developing States Conference 2013
[16] International Finance Corporation, Economic Opportunities for Women in the Pacific, 2012

dimensions related to tuna fishing and how utilising FADs can increase the potential for greater food security for communities. Roles of women in FADs tend to be overlooked, however while women rarely go out on the fishing boats, they have a significant role in post-harvest handling and marketing of the catch.

While some efforts are made to improve the safety and facilities for women vendors in marketplaces, further efforts are needed to improve access to productive resources, financial literacy and safe working conditions in post-harvest handling and sales of fish for women in rural and remote areas and for young women and women with disabilities.

In the context of this programme, and as described in more detail in the following section, the highly disaggregated labour force in FAD fisheries presents a **risk** that women will not be in a position to directly benefit from the FADs. At the same time, the programme could present an **opportunity** to support and encourage women to enter into the FAD fishery if they choose.

4 Gender and Fish Aggregating Devices in the Pacific Region

4.5 Brief overview of fish aggregating devices in the Pacific Region

FADs have the capability of improving food security through increased availability and accessibility to localized aquatic animal protein, increasing the reliability of income from fishing and creating employment in coastal areas through fish trading and processing. Over the last 10 years, FADs for both artisanal and commercial/industrial fisheries have proliferated in the Pacific region.

While FADs are already located across Pacific Island seas, these are in varying conditions ranging from effective to washed up on reefs. The proposed Programme seeks to disseminate further FADs across various strategic locations to provide fishing locations and food to communities within PICs including the Cook Islands, Federated States of Micronesia, Fiji, Kiribati, Marshall Islands, Niue, Nauru, Palau, Papua New Guinea, Samoa, Solomon Islands, Tonga, Tuvalu, and Vanuatu.

The proposed Programme seeks to develop more effective FADs that are secure, functional, and more accessible to local fishermen. The project also seeks to facilitate the catch of quality non-reef fish to supplement the current nutritional sources in each location and reduce fishing pressure on reefs.

Primary stakeholders and target beneficiaries include:

1. Coastal fishers, processors, traders and their families, and communities
2. Government institutions responsible for the administration, management, and development of coastal fisheries at local, district and national levels.

These stakeholders are important to consider with the implementation of the regional tuna programme in and around the participating Pacific Island Countries.

4.6 Gender in the fisheries sector in the Pacific

Women play a crucial role in value chains of fisheries sectors. This is particularly true in industrial fishing particularly in processing factories, in the field of seafood sales as well as management of the fisheries sector. These and the relationship between sustainable development, fisheries, and the crucial role of women in different areas of fishing activities are reflected in various international conventions, agreements, and policies. Examples of these international agreements can include:

- United Nations Convention on the Law of the Sea 1982 (UNCLOS) – considered the Constitution of the Sea.
- International Health Regulations (2005).
- Pacific Islands Regional Ocean Policy.

Gender dimensions shapes the differential identities of women and men, their norms, roles and responsibilities. It influences people's (unequal) access to resources and decision making and people's agency. Fisheries are generally considered a male domain.

Although FADs are thought to greatly improve the availability of fish to the entire community, FADs can only be directly used by those who have access to boats (mostly men), therefore the benefits are not always equitably shared. Women's roles in fisheries—along the value chain, are in sustaining the fishing families and community through shore-based activities such as processing, coastal harvesting, selling, and trading. In a Gender Mainstreaming study[17] done by WWF in Fiji's offshore tuna industry, gender participation of women in processing and post-harvesting at the Pacific Fishing Company (PAFCO) in Levuka, Fiji was 33% more than the participation of men. While this is commercial fishing it is also reflected in other countries such as Kiribati, Solomon Islands and Tuvalu whereby women have a key role in shore-based activities and coastal harvesting[18].

Women face persistent gender-based discrimination and marginalization in the tuna value chain of the fisheries sector that is often defined by diverse social context. Women, youth, and other marginalised groups often have little to no say regarding decisions around the use of natural resources and which may impact their ability to gain benefits from marine resources equitably.[19]

In commercial fisheries, women are most commonly overrepresented in vulnerable categories of employment. They generally lack tenure security, access to productive assets and market opportunities and decent work conditions as shown by the women selling fish along the roadside often in dusty unhealthy conditions. They also have limited access to services like healthcare, childcare, credits, insurance, legal aid, and capacity building. Women are also often disproportionately impacted by sexual violence, prejudices, and other forms of harassment.

Across the Pacific, women are less represented in fisheries' associations, cooperatives, and unions. They rarely have a say in the decisions that govern their fisheries and other matters that affect their lives and livelihoods. Lack of quantitative data further undermines women's role in fisheries.

This Programme thus presents opportunities to create sustained change within the fisheries sectors of these countries through the gender-sensitive (perhaps beyond tuna). For example, the designation of a national GESI focal point in each country, housed within the Fisheries Ministry and supported by 2 full time GESI officers from the PMU, could significantly contribute to institutional strengthening on gender, including policy implementation.

Further, a primary approach of this programme (and the project level activities most directly impacting women) is to focus on capacity building & training in post-harvest methods for processing & marketing fish caught around FADs and in MSMEs. This will include basic supplies and equipment to apply post-harvest methods (e.g., freezers, smokers/dryers, bottling) and safe/clean post-harvest workplaces (such as shade). Additional productive assets are outside the scope of this project.

---

[17] WWF, Gender Mainstreaming in Fiji's Offshore Tuna Industry, 2021

[18] FAO Factsheet, Sustainable Fish Value Chains for Small Island Developing States (SVC4SIDS), 2017

[19] Pacific handbook for gender equity and social inclusion in coastal fisheries and aquaculture [gender equity and social inclusion handbook (spc.int)](spc.int)

4.2.1 Fish Aggregating Device investments can overlook women's needs and priorities.

Gendered Roles in Fisheries in The Pacific

"Fishing" tends to be narrowly viewed as only catching fish at sea, which often leaves women out of sight in the industry. While offshore fishing activities are predominantly carried out by men in the Pacific, women are involved in supportive activities in some forms of fishing such as the post-harvest processing of fish and some smaller-scale styles of fishing i.e. 'kai' fishing in Fiji and shoreline net fishing in Tonga, and it is estimated that women carry out almost 50 percent of fishing activities that are crucial to Pacific livelihoods. Their role in the sector in the Pacific is much higher than the international average and yet women's contributions to the sector have not been well documented. A reason is that women mainly work in the less visible fishing roles, such as: shoreline fishing and picking shellfish; mending nets and processing and selling fish, doing the accounting; and roles in processing, and marketing[1].

Women working in the fishery sector can face poor working conditions and environments, lacking job and even personal security. Some roles are either under or unpaid making them invisible within the industry. This oversight not only creates a gender gap in wages but excludes them from opportunities to advance in the industry, having less access than men to resources such as technology, loans, insurance, industry networks and information. Women's traditional household and childcare responsibilities further limits what they can do in the industry both timewise and knowledge related.

Historically women's needs and priorities and women in the fishing sector have had limited access to services such as healthcare, childcare, credits, insurance and legal aid.

Gendered Roles in Market-Stalls for The Sale of Fish from FADS

There are safety and security issues related to selling fish in many markets across the project countries. The health of women, who are the primary sellers of fish, may be compromised while selling tuna at the market, and while travelling to and from the market. Selling locations may be in less than hygienic locations at the roadside and the women are subjected to dust, fumes and other risks such as harassment. Women may be at increased risk of infection (waterborne and food-borne illnesses) while handling the fish or have their safety compromised at markets and while travelling. Women and girls in PICs are often disproportionately impacted by sexual violence, abuse and harassment and prejudice in the context of this project, there is not a risk that women will be exposed to increased safety and security issues when selling tuna (vs. other fish), however there is an **opportunity** to improve the safety and security of women fish sellers through capacity building and awareness about support services.

Gendered Governance of Local Fisheries Resources and Fishing Grounds

Governance controls access to natural resources, local tenure arrangements, levels of education, wealth, and cultural values. Policies and procedures need to be in place to ensure the entire population benefits from fisheries and aquaculture. Women have limited roles in governance for local fisheries and fishing grounds, making it difficult for women to advocate for safe working conditions, health protection and programmes and policies to prevent sexual abuse and harassment.

In the context of this programme, there is a **risk** that women's specific needs and priorities may not be well represented in project governance, while there is also an **opportunity** within this programme to increase women's representation and interests in local fisheries governance.

4.2.2 Implementing Fish Aggregating Devices can promote gender equality.

Projects that seek to strengthen the role of women in skilled roles in the fishing industry help to address the gender inequality gap in the industry, and as highlighted above, there are both risks and opportunities in this programme. FAD programmes and projects need to integrate gender dimensions to maximise the opportunities available and enable greater sustainability of efforts to improve community food security and nutrition.

The following Table summarises the Risks & Opportunities that are likely from the Regional Tuna Programme as noted above. Note that these impacts are largely positive and can encourage greater gender equality in the fishing industry.

Table 1: Gender-related risks and opportunities to support gender equality from the Outcomes of GCF FAD Tuna Fishing Projects

Gender Impact	Recommended actions/approaches at the national level	Likelihood	Impact Level on project
Potential gender-related Risks			
Women not being able to actively or meaningfully engage in project activities due the great burden of other (gendered) priorities.	At the national level, gender approaches should take this into account and find times, locations, strategies that work best for women in the fish processing & marketing sphere. This particularly important in capacity building/training activities of the project.	High	High
Project activities could contribute to, or exacerbate, GBV during the course of implementation.	At the national level, GESI focal points should assess each project activity for potential to increase or exacerbate GBV and provide for mitigation where needed. Additionally, GBV support services should be mapped and provided to communities as a resource for care. The programme and national level AGMs should also be designed to be prepared for GBV-related grievances (see AGRM plan for more details).	Low; while GBV is prevalent in the fishery sector, project specific activities do not pose a significant likelihood of increasing GBV.	Low
Women will not be in a position to directly benefit (as fishers) from the FADs due to the highly disaggregated labour force in FAD fisheries.	Project activities should be designed to be open and accepting to anyone who wants to attend; for example, if a woman fisher wants to attend a FAD training, she should be made to feel welcome.	High	Low; while women will likely not benefit directly (as fishers) from the FADs, other activities have been incorporat

			ed to support their benefit in processing & marketing.
Women's specific needs and priorities may not be well represented in project governance	Targets have been set to help encourage women's representation at key governance spaces and incorporating of gender-specific considerations will be incorporated into workplans, communications, and other project-related activities.	Medium	Low
Positive impacts/opportunities to advance gender equity			
Opportunity for fish sellers to increase the hygiene and safety of fish, and/or through adding value (e.g., BBQ or smoking fish)	Fish sellers trained in post-harvest handling to maintain shelf life of tuna.		
Opportunity to influence diet towards tuna and other FAD fish and away from imported or other less healthy foods.	Given that women are usually responsible for household meals, women would be a key audience for these messages.		
Opportunity within this programme to build awareness about GBV, women's/girl's rights, and resources that exist to help support survivors	Community engagement should include links to GBV resources and clarity on women/girls' rights.		
Opportunity to support and encourage women to enter into the FAD fishery if they choose.	Ensure that FAD training is open and welcoming to all. Futher it is primarily women who handle post-harvest processing so trainings with this focus will be geared towards women in the fishery.		
Opportunity to improve the safety and security of women fish sellers	Tailored capacity building and awareness about safety and security for fish sellers (e.g., GBV support services)		

From this standpoint each country can prioritise specific activities or approaches that consider local realities, resources and mainstream gender dimensions into these to enhance gender equality. The following table indicates negative impacts of the programme are few and which can be managed through enforcement and communication.

5 Recommended Strategies for Regional and Country Level Gender Equality Approach

There are several agendas behind the impetus to encourage Gender Equality in Fisheries. For example, fisheries organisation's view gender equality as a pathway to increase healthy fish stocks and improve community nutrition (instrumental); development agencies view gender equality as a core value or principle (inherent); fishers see gender equality either as a greater source of reliable labour and skills (if gender sensitised) or as a threat to men's roles and masculinity (if not gender sensitised). Obviously increasing gender equality needs to be managed and may vary by country and even location. These different viewpoints and agendas will most effectively be applied through a partnership and multidiscipline approach. This requires working collaboratively to bring together the business case and the altruistic case (Mangubhai et al 2022).

5.1 Recommended processes and activities to advance Gender Equality in fisheries

Several recommendations have been highlighted throughout this Assessment to encourage greater access to healthy tuna stocks for men and women in communities. It is important to emphasise the approach and related processes required to strengthen gender equality in the fishing sector. Furthermore, Gender and climate change also need to be interwoven into all activities and changes and benefits measured intermittently through country defined indicators. Some cross-cutting activities are as follows:

Upskilling women and men to improve fish products and services

- Opportunities for training provided to both men and women in various aspects of fishing including *gender dimensions; fisheries extension activities; safety dimensions* in all phases of the fish supply chain and conservation measures.
- Provide basic supplies and training to maintain shelf life of fish and provide training to women on the processing and marketing of tuna. This includes targeted capacity building on post-harvest techniques for women and men, with an emphasis on trainings for women as this post-harvest dimension of the fishery is mainly done by women
- Provide an increased supply of FAD tuna fish at an affordable price for all households.
- Provide training for communities to better understand and thus manage effects of climate change on coral reefs, reef fish and in communities. For example, consuming more tuna instead of reef fish.
- Identify and encourage activities that women and men and collective communities undertake to manage and conserve reef fish resources, particularly the health of the reef.

Gender Sensitisation – building an awareness

- Upskilling of project stakeholders in Gender and GBV dimensions to encourage greater awareness of men and women's roles in the industry, behavioural changes towards better family nutrition, and encourage participation of women in non-traditional work opportunities.
- Identify the gender related issues that can be faced by men and women in the industry and develop training to minimise risks of these issues. This can include safety risks, health concerns, GBV and SEAH issues and other behavioural risks. These are included in activity 1.1.1 in the GAP.
- Build greater awareness of women's contribution to family livelihoods and food security through roles in the fishing industry including in the artisanal tuna fishery sector.

Improve communication and collaboration

- Link project AGRM process to other national service support providers through greater collaboration between relevant agencies. This can help prevent as well as provide greater support to victims of GBV and SEAH project-related risks. A more detailed mechanism for reporting and responding to these types of incidents is outlined in the AGRM. Further both SPC and CI have Senior level staff who are trained in how to handle cases involving GBV and SEAH.

Understanding and encouraging Role diversity

- Men and women do not have to do the same roles in fishing to promote gender equality. However, it is important that they have a choice.
- Inclusion of a training for greater recognition of the roles both women and men play in the fishing industry, both coastal and offshore fishing, through training, accreditation and commensurate remuneration between women and men doing the same roles.
- Encourage women to pursue careers in the fisheries sector through scholarships, recruitment encouragement and improved workplace conditions – for all stages of the supply chain.
- Actively implement inclusive approaches to engaging FAD fishers and those along the value chain.
- Provide resources and training for both men and women in fisheries development and management such as leadership training, extension services and conservation activities.
- Peer mentoring for relevant emerging women leaders in the tuna fishery sector will encourage their participation in non-traditional roles.

Monitor improvements in gender awareness and equality

- Improve sex disaggregated industry data which provides greater recognition of women's roles in the industry both directly (fishing and post-harvest processing) and indirectly (nutrition aspects).
- Monitor developing shifts in knowledge and skills developed by women and men relating to fish catches and how they are using these to strengthen nutrition and food security outcomes for families within communities. This involves three core indicators including,
    a. # of men & women who participate in project activities,
    b. # of women & men who receive direct benefits,
    c. # of plans/strategies/plans that integrate gender considerations
- Measure benefits and empowerment (personal viability) that men and women receive working in the project encouraging gender inclusion to be a 'norm'. For example, fostering working partnerships between men and women fishers. This can include leadership training and peer mentoring for emerging women leaders in the tuna fishery sector.

These priorities, strategies and recommendations will be tailored in each country, as resources and socio-cultural characteristics at the country level will vary. Yet these offer some practical and viable insights into practical means to enhance gender equality in the RTP.

An intersectional analysis of gender and fishing

Intersectionality offers an analytical framework which can consider how societies treat people based on inherent social and political identities, such as their gender, ethnicity, and sexuality. Depending on those identities, a person may be privileged or oppressed. (*Women's changing productive practices, gender relations and identities in fishing through a critical feminisation perspective - ScienceDirect* Intersectionality: An Underutilized but Essential Theoretical Framework for Social Psychology | SpringerLink)

However, what is subsequently important is how this framework can be implemented at a program and project level to deliver longer term sustainable change. While the approach offers various ways to analyse complex relationships and activities between men and women in different cultural environments, the outcomes of this approach can benefit both men and women and the fishing industry itself.

How this is important to the Regional Tuna Programme

The intersectional framework helps to systematically clarify the largely 'invisible' but significant contributions of women in sustaining fishing families, communities and industries in all Pacific countries (Frangoudes and Gerrard, 2018; Zhao et al., 2013). In recent years there has been a greater consideration of the contributions of women in fishing families, communities and industries globally, regionally and nationally in the Pacific. Yet whilst it is important to recognise these contributions, there has been little attention to how women's changing roles and practices are managed within unchanged gender relations shaping, and being shaped by, women's (fishing) identities in different ways. The basic premise requires women to consider their roles and activities from their own perspectives and the subsequent value of their fishing roles to be reflected. Simply put: there has been little attention given to the identities and activities of women's lives in fishing and women's knowledge is often ignored in research and policy. This can serve to cement the invisibility of women's roles in the industry.

While women's roles and responsibilities in fisheries are often different from men's, they still have an active and important role. For example, women harvest and sell seafood and seaweed in shallow waters, sometimes on foot and using simple gear. They also engage in fish trading and fish processing in large numbers.

There was little included in the country questionnaires and responses regarding the benefits of the intersectionality approach of gender and fishing for several reasons:

- Little to no information available on the intersections between gender and fishing especially in the Pacific whereby information covered largely practical benefits and outcomes of fishing, yet there is limited understanding on the longer-term strategic benefits. This was an important aspect which was unable to be covered in such a short time and with available resources.
- An unwieldy number of responses were collected from a Pacific perspective which provided a baseline of practical roles of women in the fishing industry. This can be broken down at a national project level and investigated further to consider this strategic approach in terms of project activities and outcomes.

An example of responses which could be further broken down to build on an intersectional approach country specific

Country	**Activities which can further intersectional gender outcomes**
Tonga	Communities need focused training on managing parts of the FAD fishing supply chain – eg fixing ice machines Communities working together
PNG	Investigate mechanisms to drive community ownership of FADs Strengthen governance and justice systems in transhipping ports to reduce human trafficking and sexually transmitted diseases from crews of fishing vessels
Fiji	Need for an analysis of the number of active small and medium-scale enterprises, and the affiliated needs, costs, etc.
Samoa	Provide fish storage coolers for operators Ice making machines in key districts

Tuvalu	Build supply chains for distributing fish caught from FADs – include considering where the elderly and disabled can assist
Vanuatu	Some FADS put closer to shore than others so women may also be involved

How the Regional Tuna Programme will seek to integrate an intersectional framework

Consequently, this program will acknowledge the intersectionality of gender dimensions and women's roles in FAD fishing to develop a more systematic understanding of these roles, opportunities and benefits to understand women's varying and changing roles and practices and how this can (re)shape gender relations and identities. Primarily this will be guided through the GAP Activity 1.1.1 when a much more detailed investigation of gender roles within localized FAD fishery value chains is integrated into the workplans for scaling up of national FAD programmes. For example, the measure of women's many tasks can be supported through such elements of projects that both support women's reproductive roles while they are employed in productive roles in the FAD fishing industry. Key considerations can include:

- subsidised childcare support at market locations so children do not need to sit in the dust and face safety issues while their mothers sell fish.
- Training in maintenance of equipment's such as ice machines can include women to develop their more technical skills.

6 Country Gender Profiles

As aligned with the SDG gender indicators and targets, the regional and country gender equality briefs cover several thematic areas including Women's Human Rights, Women's Representation and Leadership, Women's Economic Empowerment, Education, Health/Sexual and Reproductive Health, Ending Violence against Women and Girls, and Gender and Protection in Humanitarian Action. Women undertake considerable roles in the fishing industry; however, their roles are not well articulated, hence many do not consider themselves fishery professionals.

Key information that can strengthen the use of FADs and benefit local communities can include:

- Country specific approaches that can deliver a baseline of sex disaggregated information as it applies to FAD fishing and to further inform proposed activities to ensure that both women and men benefit from the RTP.
- Improved disaggregated fishing industry data which provides greater recognition of women's roles in the industry both directly (fishing) and indirectly (food preparation aspects).
- Opportunities for training provided to both men and women in various aspects of fisheries including *gender dimensions; safe and effective FAD fishing activities; safety dimensions* in all phases of the fish supply chain.

A brief assessment of key gender dimensions has been prepared for each respective country in the proposed Programme. While some of the data is not consistent between countries, it is important to identify any gaps in information.

6.1 Cook Islands

International Gender Policies

Cook Islands has shown commitment to strengthening gender equality in the fisheries sector through various international commitments and national priorities and policies, including but not limited to the following:

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – acceded to in 2006.
- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs, and oceans.
- Pacific Leaders Gender Equality Declaration signed 2012 – recent report (2016) noted that while women actively participate in village councils they are still under-represented in senior management and decision-making roles.

National Gender Policies

- 'National Policy on Gender Equality and Women's Empowerment' together with a 5-year strategic plan of action 2011-2016. Recognises women and men are equal partners in the development of the Cook Islands. Also established the Gender and Development Division under the Ministry of Internal Affairs as the official national women's machinery.
- CEDAW Law Reform programme - began in 2008 with revisions noted to a few acts recommended, notably the Marriage Act and the Crimes Act and recently the Family Protection and Support Law Bill in 2017 - ensure children and families are protected from violence.
- Enactment of the Employment Relations Act 2012, which provides maternity leave benefits for women in the private sector and provisions for the prevention of sexual harassment.

Outcomes

- Women's Participation in Decision Making - progress in increasing women's representation and political participation, for example, in 2020 four of the 55 Island Government Councillors (7.3%) were women (Public Service Commissioner's Annual Report).
- The Cook Islands Women Parliamentarians Caucus, launched 2018, - bipartisan group that advocates for gender equality through law-making, budgeting, oversight, and representation.
- The percentage of women in parliament is 17%, women in management roles at 48% with managerial positions including senior government officials, corporate and general manager positions.
- At the local level, numerous initiatives provide capacity building and support for women candidates in island council elections:
    - Cook Islands National Council of Women, Cook Islands Gender Equality Policy SPC
    - 2012 Stocktake of the Gender Mainstreaming Capacity of Pacific Island Governments

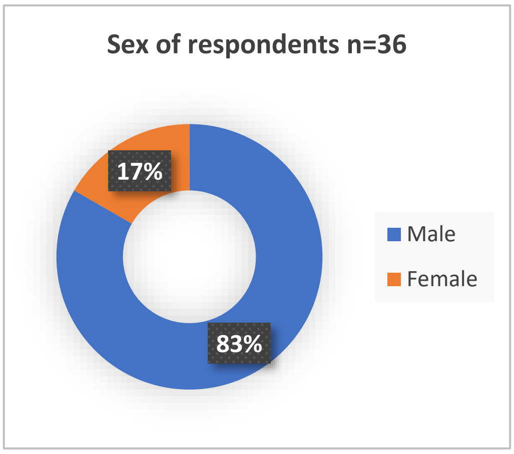
Some Bottlenecks

Sexual offences are defined in the 1969 Crimes Act. While updates have been under discussion since 2017 a new Crimes Bill has yet to be accepted and passed into law. This updated Bill,

- seeks to remove exemptions related to marital rape.
- criminalise sexual connection with a person under the age of 16 if married to the child in question.
- decriminalise same-sex sexual contact between adult men upon mutual consent.

Key input from national consultations

30 men and 6 women respondents were interviewed in focus group discussions, aged between 30 to 70 years of age. This was broken down into 23 from Mangaia (5 female, 18 male) and 13 from Rarotonga (12 male, 1 female) villages in July 2023.



Of these respondents, 13 respondents from Rarotonga and 15 from Mangaia were fishermen. Note that several people had multiple work roles. Most head of households (HOH) were men – 17 in Mangaia compared to 1 female-headed household (FHH). In Rarotonga, there was a different response with 5 FHH compared to 8 male HOH.

Roles and Responsibilities

The main roles for men and women in fishing as reported in the survey are as follow,

- Offshore fishing – male dominated – no female respondent is a fisher in this type of offshore fishing including FAD fishing. While many respondents did not consider there were constraints to women carrying out FAD fishing, traditional roles and time commitments make up the main reasons women do not participate in offshore fishing.
- Coastal fishing – 100% men carry out coastal fishing.
- Boat management and repair – only men are involved in these tasks – apart from one woman in Mangaia.
- Post-harvest handling – predominantly men, however a significant number of women are also involved in post-harvest handling – 28.5% respondents in Mangaia and 46% respondents in Rarotonga noted that women are involved in post-harvest handling.
- Fish cooking for home or selling use is cited to be the role of both men and women fairly equally.
- Cleaning fish is carried out by both men and women.
- Selling fish is the role of both men and women.

6.2 Federated States of Micronesia

Given that no FAD fisheries supply chain surveys were able to be carried out in FSM, this section has drawn from a few comprehensive gender equality briefs carried out in FSM in recent years, while some are becoming dated, they still provide a coverage of the main gender dimensions in the different States and basically reflect several key local realities.

Context

Approximately 22% of residents live in urban areas of the FSM island States, this means that ~78% of the population live in rural areas. Different levels of Governance exist in the nation with the overall national government forming one level of governance and each of the four states Chuuk, Yap, Kosrae and Pohnpei manage each States governance and affairs. This multi system governance can both offer support and yet constraints to actions geared to strengthen gender equality in the country. This which is summarised in this country brief.

While there is some political will at the national level towards encouraging greater gender equality, there is limited action to actively address issues pertaining to gender equality. Gender sensitisation has been undertaken but more is required with adequate allocation of resources.

International Commitments

FSM has shown commitment to strengthening gender equality through several international commitments and national priorities and policies, including the following:

- The 1995 Beijing Platform for Action (BPA).
- The 2000 Millennium Development Goals (MDGs).
- Revised Pacific Platform for Action on the Advancement of Women and Gender Equality 2005–2015 (RPPA).
- Pacific Plan (2005, revised in 2007).
- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – acceded to in 2006. FSM maintains a few reservations to CEDAW, including in respect of Article 11(1) (d) on equal remuneration in employment; Article 11(2) (b) on maternity benefits; and Articles 2(f), 5, and 16 on the elimination of discriminatory cultural stereotypes.
- Convention on the Rights of the Child 2004.
- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs, and oceans.
- Pacific Leaders Gender Equality Declaration signed 2012 – recent report (2016) noted that while women actively participate in village councils they are still underrepresented in senior management and decision-making roles.

National Gender Policies

Legislation in FSM is shared between the National Congress and the four state parliaments.

- National Constitution and the constitutions of the four states protect the legal rights of women, all prohibiting discrimination on the grounds of sex ().
  - Article IV of the FSM Constitution (GoFSM 1978) provides for a 'Declaration of Rights'. Section 4 of this Article confirms that equal protection of the laws may not be denied or impaired on account of, among other things, sex. However, the Constitution contains no definition of discrimination against women.
  - States' constitutions (GoFSM 1978) have provisions that can protect women's human rights, including:
    - Article III of the Chuuk State Constitution, section 2, - rights to be enjoyed by a person irrespective of, among other things, sex.
    - Article II of the Kosrae Constitution, sub-sections 1(b) and (c), set out provisions for protection under the laws based on equality.
    - Article 4, section 3 of the Pohnpei State Constitution and Article 4 of Yap State Constitution set out parallel provisions on equality.
      - All constitutions, apart from Kosrae, contain provisions through which a person who has experienced discrimination can seek redress.
- National Gender Policy for FSM (endorsed 2018) which guides training in gender mainstreaming in fisheries management, development, and enforcement at national and state levels, in collaboration with the Women's Interest Office and women's associations.
- The FSM Strategic Development Plan (SDP) 2004–2023 (GoFSM 2004: 525) includes a strategic goal to mainstream gender issues into decision-making, policies, and strategic development plans.
- Women's Participation in Decision-Making - progress in increasing women's representation and political participation, for example, in 2020 four of the 55 Island Government Councillors (7.3%) were women (Public Service Commissioner's Annual Report).

Some bottlenecks

Institutional challenges include:

- Many government agencies with overlapping mandates and responsibilities in coastal fisheries management, as defined in their respective legislation.
- Coastal fisheries management is carried out at national, state and local/municipal government levels, as well as at community level. The four different levels of governance adding complexity to fisheries management and development and requires strong coordination.
- However, from a gender mainstreaming perspective, the management of coastal fisheries at a community level can also provide an entry point for stronger engagement and visibility of organised women's groups in each state.
- Gender analysis of the fisheries sector in FSM coastal waters (this includes territorial seas and internal waters) are under the responsibility of states, and the national government has the role of coordination.
- There is a lack of support for women wanting to move into roles that have traditionally been dominated by men. Leadership at the national, state, and municipal levels is based strongly on traditional forms of local leadership, in which women are not highly represented.
- There is insufficient collection, compilation and analysis of sex disaggregated data and gender indicators across sectors to provide the basis of positive gender mainstreaming.

6.2.1 Literature Summary of the Federated States of Micronesia

Gender relations in FSM are shaped by cultural norms and practices which vary across the four states. While dated information is available, it is pertinent to include some cross cutting gender related dimensions as summarised:

Education

- There is close parity between male and female student numbers at schools,
- However, at tertiary level there tends to be stereotypical fields of study (2008 FSM Statistical Yearbook). For example, 60 boys compared to 14 girls in marine science. The lack of women in technical and professional positions in employment reflects this situation.
- Traditional gender roles can limit girls' and women's choices in education and careers options.

Employment

- The number of female wage and salary earners was on average less than half that of males in 2019 (4,514 vs. 9,286) (UN 2022).
- Furthermore, women in the paid workforce, are concentrated at the lower levels of the hierarchy, with comparatively lower pay.
- Formal employment in the FSM fisheries sector consists of only about 250 people working for wages, the majority of these being men. Overall, less than 2% of all wages earned comes from formal fisheries-related employment. Both men and women, however, stand to benefit from proposals by the government to support the development of tuna-loining plants in Pohnpei and Kosrae States.
- As far as women's roles in the National Fisheries Corporation (NFC), they are employed in the office as administration and finance staff, with the crew of the fishing vessels being men.
- Twenty-five per cent of male employees come from FSM Maritime Training College, which offers an entry point for training on prevention of sexual harassment and domestic violence.

Health

- Although maternal mortality rates (MMR) have fallen significantly, FSM still has one of the highest in the Pacific region (206 in 2003) (GoFSM and UNDP 2009).
- All four states report high teenage pregnancy rates (SPC Stocktake 2012). Teenage pregnancy is a major reason for girls dropping out of high school and college.

Access to water

In 2010 data suggested that only 41 per cent of the population has access to an improved water source and about 45 per cent to improved sanitation (ADB 2010). This is of particular concern for women, both in terms of their role in food preparation and their hygiene requirements.

Gender Based Violence

GBV in all its forms including sexual harassment is a common cause of physical and mental harm to women and children and a major contributor to social problems (2004 Situational Analysis Report). Some concerns inherent to FSM include:

- Until recently, the age of consent in the four states has been 13–16 years of age, with Chuuk recently increasing the age of consent to 18 and Yap State planning to do the same (UPR 2015).
- While there is mandatory prosecution of sexual offences, bail can be gained for defendants even where there is a risk of violence to survivors. Furthermore, customary practices of 'forgiveness' are given due recognition by the court.

Actions need to integrate both legal and cultural practices that strengthen the protection of vulnerable women and children. Some actions undertaken have included:

- As a pilot project in 2018, the Education Department conducted a domestic and sexual violence prevention and education training in 94 per cent of the elementary schools in Pohnpei.
- Women's groups in all four states have lobbied extensively for protective legislation. Pohnpei passed the Family Protection Act in 2017, addressing some gaps in protection of women and children. Kosrae passed similar legislation in 2014. over the years for legislation that protects women from domestic violence and marital rape. Yap and Chuuk are yet to institute specific legislation that provides for the protection of women against domestic violence and related issues.

Land ownership

While FSM is largely a matrilineal nation, most decision making related to land ownership and land use is managed by male members of the family.

Governance and Decision Making

Women rarely become traditional leaders, deferring to men in community affairs. No woman has held one of the 14 positions in the FSM National Congress. However, there has been some representation of women at national cabinet and State Congress levels.

6.3 Fiji

In Fiji, women reportedly provide around 80% of the seafood catch for their communities' annual subsistence needs (Mangubhai 2022). However, the role of women in fisheries is often unrecognized, not recorded or poorly understood, leading to their overall limited engagement in fishing activities and especially decision-making in fisheries management.

International Gender commitments

Fiji has shown commitment to strengthening gender equality in the fisheries sector through several international commitments and national priorities and policies, including but not limited to the following:

[20] Progressing gender equality in fisheries by building strategic partnerships with development organisations Sangeeta Mangubhai a,b,⇑ , Sarah Lawless c , Anna Cowley d , Jayshree P. Mangubhai e,f , Meryl J. Williams

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – ratified in 1995.
- the 2030 Sustainable Development Agenda (SDG 5) on gender equality and women empowerment.
- Pacific Leaders Gender Equality Declaration signed 2012.

National Gender Policies and Strategies

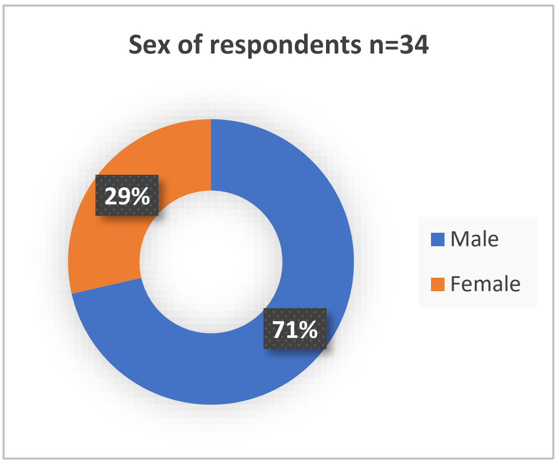
- Fiji National Gender Policy which should be reflected in both coastal and offshore fishing.
- Fiji's Women in Fisheries Network (WiFN) an active group of women in the sector.

A few studies have been carried out in Fiji which sadly reflect that despite policies aimed at creating employment, women's contribution to the offshore fisheries sector labour force is impacted by cultural beliefs, traditional norms, and gender stereotypes. Many local fishing vessels are over 40 years old and do not have separate crew cabins, washroom facilities etc. Furthermore, boat owners are reluctant to take females on their vessels since they do not want to be accountable to any sexual harassment, abuse or worse cases. Subsequently, women are traditionally confined in their fishing activities to coastal areas, mainly harvesting (e.g., reef gleaning), seafood processing, and food preparation for home consumption.

However, women are increasingly participating in economic opportunities such as marketing or small businesses that involve the harvesting capture and sale of marine resources. For example, there is greater participation of women in the Management/Administrative areas and in processing.

Key inputs from national consultations

34 Participants composed of 10 women and 24 men were interviewed. Ages of men ranged from 25 to 60 and women's ages ranged from 30 to 55. This was broken down into 18 men and 8 women from Natarawau and some Nakavika as well as Lami villages in July 2023.



Several interesting insights came out of the surveys in terms of women's roles and opportunities.

- Most head of households (HOH) were men – 3 FHH were interviewed.
- People live in extended families, with about 6-12 people in a household.
- Most in the community own their own home. This will be the same in all other traditional fishing communities where FADs are deployed.

Natarawau village & Nakavika in the Ba Province in Western Viti Levu are settlements that comprise people who are originally from other Provinces. People spoken to were from Vanua Levu, Kadavu, Lau and other parts of Fiji who had moved to these settlements. Settlements are those sites that have no traditional mechanism of governance, no traditional clans (mataqalis) with their different mataqalis. For example, in Traditional Fijian villages there would be a chief, his spokes people clan, the carpenter clan and the traditional fisher's clan. This is not the case in this settlement.

Fishers Cooperatives/Fishers Association

Some of those interviewed are part of the Natawarau Fisherman Association (NFA) set up in the Community. This Cooperative is a collaboration between the Ministry of Fisheries and the

Department of Cooperative Business (DCB), within the Ministry of Industry, Trade and Tourism (MITT), which is the regulatory and promotional authority on cooperatives.

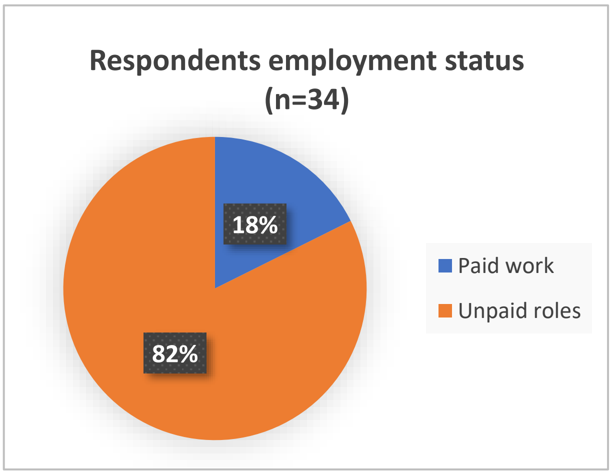
In March 2023, the Deputy Director-General of the Food and Agriculture Organization of the United Nations (FAO), visited the Natawarau fishing community in Ba, Fiji to present fishing families with Fish Aggregating Devices (known as FADs) used to enhance nearshore fisheries.[21]

- Foods consumed are almost the same with most households relying dominantly on fish and marine products. This will be the same for all other locations where FADs had been deployed.
- **Gender roles** - Both men and women fish in the two communities, with men fishing in deeper waters and offshore reefs while women fish in mangroves, sand flats and in inshore reefs. This will be the same case for communities in Lau, Kadau, Vatulele, Ra Yasawa and other locations where FADs had been deployed.
- **House designs** - There is not much difference in house designs in this community and those in neighbouring land. This will be the same in outer islands and other locations where FADs had been deployed.
- **Markets** - Markets for fish are at the Ba and Lautoka municipal markets, Ba and Roadside stalls or middle sellers buy from the communities and sell independently.

Employment

Main livelihoods among community residents

There are a few people employed in these communities as most are dependent on fisheries and farming for subsistence and income sources. Two only work for the government, and one is a trades person. This will be the same in the outer islands where most FADs were deployed- Only 6 respondents had paid employment. Most women are engaged in unpaid work and selling of fish is the main occupation. This means “unpaid work” in this context and defines women in care giving roles and carrying out domestic chores. However, all these women are involved in post processing and marketing and distribution of men’s catches.



Both men and women carry out fishing, women are engaged in coastal fishing as well as run a small-scale fishing trading /business. Roadside stalls are communally owned as people sell together- and now with the cooperative selling are in central positions where everyone sells.

When women are carrying out shoreline fishing or are selling fish at the market, the men then assist with these domestic roles. Hence the family can operate as a unit undertaking complementary roles.

Most fishers work full time unless there is bad weather when there is no fishing. Seasonal fishing takes place for certain fish species along coastal areas and reefs, which women often target.

Income Levels

[21] Fiji Sun, March 2023. FAO Deputy Director-General visits fishing families in Fiji to view successes in sustainable fishing practices.

The average weekly wage from fishing is $150- $300. When the weather is good, a 6-day fishing week can earn them up to a thousand dollars. If it is bad weather, there is no fishing and thus no income. Many male fishers also market their fish by the roadside.

Expenditure

On average weekly expenditure is approximately $200, spent mainly on food items.

Impacts of climatic events on fishers

The effects from climate change and increasing changes in weather and increasing number of storms, hinder fishing. Key impacts are as follows:

- There is also a lot of coral dying (bleaching) so it is harder to catch fish when diving.
- Safety issues- people are aware of the changes in climate and safety becomes an issue when fishing when storm warnings are in place. There are sometimes unusually bigger waves that catch fishers by surprise and safety is becoming more an issue for fishers.
- Higher Costs – 'Closer reefs are dying- we have to go out further to fish thus costs us more'.
- Health concerns - longer hours fishing for men and women- and can affect health especially during cold weather.
- Many men are divers, and they are affected when the weather is cold.
- Food availability - Because of depletion of resources from reefs dying- there are less fish.
- Others- Because of climate change, there has been changes to fishing seasons. Certain fish were in certain areas, but now they do not appear or may come months later. Hence, traditional knowledge of fishing that they know and when to fish for certain species may in the future not be practical or useful.

Fishing Practices

Reef fishing (onshore) - 90 % of fisherfolk fish on reefs and free dive.

Reef fishing (offshore) - 20% of men- diving.

Bottom trawling - Rarely mentioned, although may be used. But diving is preferred.

FAD fishing. When a FAD was installed- it was used for about 2 months, then the floater was cut and then the FAD was no longer used. The men are traditional divers, thus using fishing lines at the FADs was new. The FAD was deemed too far- and the community only has a few boats which are used for inshore fishing- It was pricier (fuel) and too time consuming to go out to the FADs.

Gleaning-Eighty percent of women gleaned and used cast nets and fishing line. Some women also dive and fish day and night, although only 2 women have previously dived with men.

Crabbing and collecting sea cucumbers are among the fishing activities that women undertake**.**

Seasonality and location of fish. Women also know seasonality of fish – that is where to fish for certain species and when to fish (seasons). There are times in the year when there is not enough fish for sale due to bad weather. At such times the household mainly eats tinned fish and vegetables.

Roles of men and women in the fishing industry

Fishing is for subsistence livelihoods and for income. All family member fish, yet roles may differ between women and men as highlighted below:

- Boat operation – men
- Offshore fishing including FADs – men
- Boat repair/ maintenance – men

- Post-harvest handling – women
- Cleaning of fish products – women
- Use of fish for food or sale – both men and women
- Selling of fish, all kinds: shoreline, reef, and offshore – both men and women
- Catch coastal fish and other sea products – both men and women
- Sell fish (all types of fish) – both men and women
- Cook fish for home use – Women and men sometimes
- Cook fish for sale, e.g., restaurants, stalls, etc. – women.

Health issues

The main health issues suffered in the household are related to the colder weather including flus and COVID. Some people also suffer from NCDs.

COVID posed a considerable impact due to all markets and towns closing. People could not travel around to buy produce and fish. This was because all markets and towns closed.

Nutrition

Households prefer fish over meat, which is considered to take more effort to prepare. Fish are sourced from their fishing activities. Meat (mainly chicken) is eaten about 2 times a week, whereas fish is eaten daily. In this context fish includes fish, crabs, shellfish, and seaweed.

6.4 Kiribati

General information[22]

Kiribati has a population of population is over 119,000 as of 2020 census and comprises 33 islands. It is located in one of the world's largest exclusive economic zones (EEZ) of 3.5 million km^2 and has a coastline of 1296km. There are 3 main island groups and more water area than land, therefore the communities of Kiribati rely heavily on fishing activities for subsistence and commercial purposes.

Fishing and seaweeds contribution to GDP was an estimated 13.6 million USD in 2015. Estimated exports of fish/fishery products were valued at 121.4 million USD, tuna being the major fish exported. In 2016, 5000 people engaged in marine fisheries full time or part time.

The total, with more than half living on Tarawa atoll[23].The capital, South Tarawa, is the most populated area. Kiribati is at the forefront of islands impacted by threats associated with climate change. Saltwater intrusion, coastal erosion, and food insecurity is a current reality for many I-Kiribati people[24]. All survey respondents were involved in some part of the fishing supply and demand chain as fisher people, sellers or consumers.

Gender policies

Kiribati is a member of a number of international agreements

- 1979 United Nations Convention on the Elimination of all forms of Discrimination against Women (CEDAW) and includes 1999 Optional Protocol
- 1992 Rio Conference 1992: woman as environmental managers
- 1995 Beijing Platform for Action of 4th UN World Conference on Women
- Millennium Development Goals (MDG 3, 5) - Gender equality and WASH

---

[22] Fisheries and Aquaculture Kiribati. Food and Agriculture Organization of the United Nations [viewed 2023 July 20] https://www.fao.org/fishery/en/facp/kir

[23] https://en.wikipedia.org/wiki/Kiribati

[24] https://thecommonwealth.org/news

- MDG 3 seeks to promote gender equality and empower women.
- MDG 5 (reducing child mortality). A healthy and productive population and natural environment are critical for achieving economic growth and poverty reduction.
  - Infant mortality rate - 46 per 1,000 live births - among the highest in the world and partially attributed to infantile diarrhea.
  - Average of two to three outbreaks of acute diarrhoeal disease in South Tarawa every year which is directly linked to inadequate sanitation and poor hygiene.

Key input from national consultations

The following information was collected through a mixture of desktop information and complemented by a community engagement survey in August 2023. Key considerations focussed on fish consumption, fishing activities, health and nutrition and FADs in Kiribati. Thirty-one respondents were surveyed from Betio and Teaorereke in August 2023. This included: 14 male and 17 females. Traditionally, fish is the main dish in Kiribati, showing the 'true cultural identity of the Kiribati people'. Fish is also cheaper than meat.

Women are traditionally not involved in the offshore fishing, however, are active participants in the shoreline fishing and gleaning. They also participate in the selling of fish in the roadside or main market areas. Boys above the age of 10 can go fishing with the fishers.

Fish sellers indicated that they earned AUD$300-450 a month, whereas fishers noted that income depended on catch which was usually under $300 a month.

Fish consumption – health, nutrition and food security[25]

There is a significant reliance on marine resources for the community's livelihood, government revenue and nutrition. Kiribati has one of the highest per capita fish consumptions in the world at 76.3kg (2013) per person annually.

The 2015 census demonstrated that 12 196 households (67% of households) had at least one member who fished regularly. Most households fished for consumption purposes, especially those on outer islands. The census also demonstrated that 75% of households fished for consumption only, 19% for consumption and sale and 4% for sale only.

- Many islands have refrigeration for storage of local sale and shipment to Tarawa. However, catches taken by small-scale fishers in South Tarawa are mainly sold on the roadside by women vendors from insulated ice boxes. It is likely that the limited cold-chain facilities may contribute to the sale of fish that are not fresh which may compromise the health and safety of the community, contributing to communicable infectious diseases.
- Much of the catches are transhipped to other islands and overseas. As such, the purchase of reef fish from outer islands for frozen export may jeopardise long-term and future food security.

FAD Fishing

Kiribati's fisheries sector has two main categories; coastal fisheries (subsistence and small-scale commercial – nearshore areas) and offshore fisheries – predominantly industrial scale fisheries.[26]

[25] Fisheries and Aquaculture Kiribati. Food and Agriculture Organization of the United Nations [viewed 2023 July 20] https://www.fao.org/fishery/en/facp/kir

[26] Ibid

Coastal fishing is primarily carried out for subsistence purposes and for sales in local markets and only some coastal fisheries export.[27]

In 2021 the Fish FAD project deployed 4 FADs in the waters of Tarawa. It was expected that the FADs support the livelihoods of Tarawa's large urban fisher population, increasing catches of larger fish, improving domestic food security and revenue generation.[28] Three respondent groups from Betio and Teaoraeke indicated that the FADs were no longer operational.

Safety in the fisheries industry

Fishing boats[29]

The total number of artisanal fishing vessels in 2015 was estimated to be 4,766. Respondents indicated that the boats available are reliable for near shore fishing and there are skilled boat operators.

Subsistence and small-scale commercial fishing is conducted using traditional canoes, plywood canoes powered by outboard motors and larger outboard-powered craft.

From 2005 to 2010 there was an overall increase in the number of boat-owning families. To a certain extent catch size is dependent upon the number of fishing vessels licensed by Kiribati.

Fishing Infrastructure[30]

The majority of Kiribati's offshore catches are exported and therefore transhipped to other ports. Catches from small scale fisheries are mostly landed in South Tarawa and smaller catches are landed at villages throughout Kiribati.

Climate Change

Fishers are aware of the changes in climate and safety can become an issue when there is a storm, hence safety and health are key challenges. Furthermore, if they cannot fish there may not be enough food to feed their families.

Training

Fishers have reportedly had a few trainings including:

- PPE
- First Aid
- Flares and appropriate safety equipment
- Boat carriage limits
- Radio Communication systems – eg GPS.

6.5 Republic of Marshall Islands

Comprehensive studies into the fishing sector in RMI were carried out in 2019 by RMI Government and some of the questions duplicate those in the current survey. Hence, it was deemed adequate to review the responses in this survey.

[27] Ibid
[28] FAD deployment in Kiribati a success. Food and Agriculture Organization of the United Nations [viewed 2023 July 20] https://www.fao.org/in-action/sustainable-nearshore-fisheries-improves-livelihoods-pacific/news/details/en/c/1419206/
[29] Fisheries and Aquaculture Kiribati. Food and Agriculture Organization of the United Nations [viewed 2023 July 20] https://www.fao.org/fishery/en/facp/kir
[30] Ibid

A number of measures to address gender equality in RMI have been adopted over the last 10 years. Some of these International and others National are summarised below. For example, two significant Agreements have included:

- Ratification of CEDAW
- Ratification of CRC

National Gender policies

- The National Gender Mainstreaming Policy, a critical policy that seeks to "engage adolescents and youth as strategic groups for ending the cycle of domestic violence." According to the policy, the RMI government is taking additional actions to remove all obstacles to gender equality and empower women in areas where they have been disadvantaged. Under the National Gender Mainstreaming Policy, the RMI government have adopted strategic measures for addressing gender issues (Figure XX). These measures seek to eliminate discrimination against women and girls and promote gender equal rights in civil, political, economic, social, health, and cultural areas.
- Comprehensive National Study on Family Health and Safety 2014. This study (FHSS) was initiated through the RMI Ministry of Internal Affairs and conducted by Women United Together Marshall Islands (WUTMI).
- Dedicated Office in the Ministry of Internal Affairs with an operational budget.

Civil society organisations

- **Women United Together Marshall Islands (WUTMI)** is an organization representing the voices of women's clubs in RMI. WUTMI's focus includes but not limited to - women empowerment (especially for young women), life skills development, raising awareness on gender-based violence and child abuse, while protecting and strengthening Marshallese values and culture.

National Gender targeted legislation in RMI

Key gender equality legislative efforts include:

- Domestic Violence Prevention and Protection Act of 2011 to address gender equality and GBV
- Child Rights Protection Act of 2015
- Human Rights Committee Act of 2015
- Rights of Persons with Disability Act of 2015
- Birth, Death and Marriage Registration Act of 2016
- Prohibition of Trafficking in Persons Act of 2017 (ADB 2020)
- National Climate Change Policy Framework – covers goals and outcomes on gender – gender sensitive strategies for climate change responses

An issue relates to implementation and enforcement of the law which have been challenging, and addressing political, economic, social, and cultural barriers pose significant barriers to progress.

6.6 Niue

Gender Policies

Niue has committed to strengthening gender equality in the fisheries sector through international commitments and national priorities and policies, including but not limited to the following:

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – efforts to ensure women benefit from resource conservation and development especially in rural areas. Ratified under the auspices of Niue's relationship with NZ prior to 1994. However, the country has not independently ratified CEDAW or incorporated it into domestic law. Niue does not have legislation to prevent discrimination based on sex (non-compliant in this area).

- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs, and oceans.
- Pacific Leaders Gender Equality Declaration signed 2012 – recent report (2016) noted that while women actively participate in village councils they still under-represented in senior management and decision-making roles.
- Beijing Declaration and Platform for Action signed 1995 – Niue has maintained good maternal healthcare provisions and outcomes.
- Niue Treaty on Cooperation in Fisheries Surveillance and Law Enforcement in the South Pacific Region 2015 - an agreement on cooperation between Forum Fisheries Agency members about monitoring, control, and surveillance of fishing.
- 2015 Draft National Policy on Gender Equality – need for greater awareness and increased budget to address gender equality challenges. Less than one per cent of the national budget allocation goes to the Department of Women. Four priority areas that include,
    - Healthy, safe, and harmonious families.
    - Developing full potential of women and men for economic development and food security.
    - Women and men participate equitably in decision making bodies and leadership positions in all sectors.
    - Gender-responsive government policies and programmes are implemented in all sectors.
- Draft Family Law Bill – process defined for dealing with GBV through legal system – still in draft after 10 years.

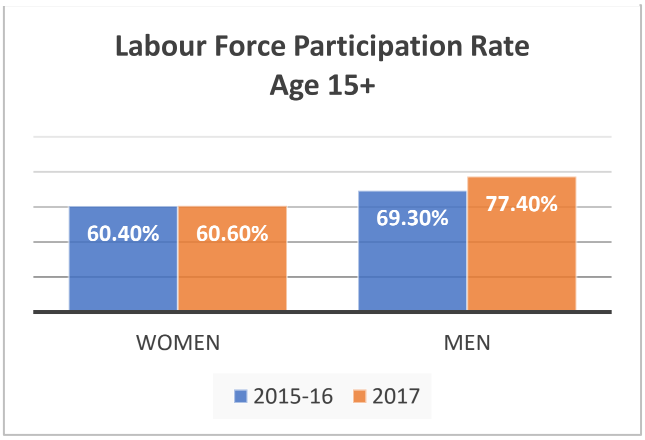
Overview of Gender Dimensions in Niue

- There is a good representation of women in the National Parliament (25 per cent) – the highest in the Pacific (apart from Australia and New Zealand).
- Women are under-represented in senior management positions in the public sector.
- Strong gender stereotypes as to what constitutes women's and men's roles have created an inequitable participation in the labour force and food production.
- Police and health services indicate that both physical and psychological abuse are present in Niue. Local response continues to be largely managed within the village or extended family network as violence against women and girls is a taboo subject and can be considered a private family matter. This can discourage victims seeking help, with a corresponding lack of services available for victims, e.g., safehouse or counselling.
- Men and women of Niuean descent have equal rights to family land under the Land Act 1969. Sixty seven percent of households own the house they live in.
- In 2017, 27.7% of households were headed by women.
- The Niue National Disaster Plan of 2010 does not mention gender or the vulnerability of any group.

Changing gender stereotypical subjects such carpentry, mechanics, cooking, and sewing are increasingly being undertaken by both boys and girls in Niue (SPC 2015).

However as noted in Figure on the right, labour force participation rate is far more gradual for women than men. More men are in the private sector than women (20.0% of men and 11.8% of women), and women are more likely to be homemakers (2.9% of men and 14.2% of women).

Figure 4: Labour Force Participation Rate Age 15+



Source: 2015-16 HIES

Key Input from National Consultations

Characteristics of Respondents

Fifteen participants in total were interviewed which consisted of the following:

The median age of participants was 40 and they were selected from the following villages: Avatele, Vaiea, Liku, Lakepa, Mutalau, Tuapa, Makefu, Alofi Nth, and Alofi South.

Other characteristics include,

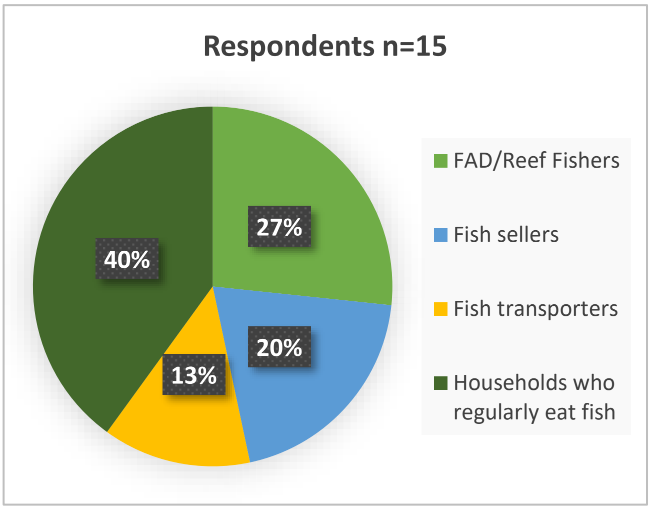
- 7 males and eight females were interviewed.
- 14 interviewees indicated they had one or more children and one interviewee also cared for an elderly person in their household.
- On average three people lived in each household.
- All participants were of Niue ethnicity.
- 14 interviewees indicated that an ethnically diverse group of people lived in their community. The main groups highlighted were of Tongan or Samoan descent. There were no significant differences highlighted among the groups as indigenous groups present tended to
    - speak vagahau Niue,
    - belonged to a Christian church,
    - held similar cultural beliefs,
    - consumed similar foods and
    - are generally involved in Niue community events and activities.
- Interviewees were either running a small business or working for the Niue government. All interviewees were engaged in farming work.
- Of the 15 interviewees, 9 were in full time work.
- All interviewees earned over NZD 600 per month.

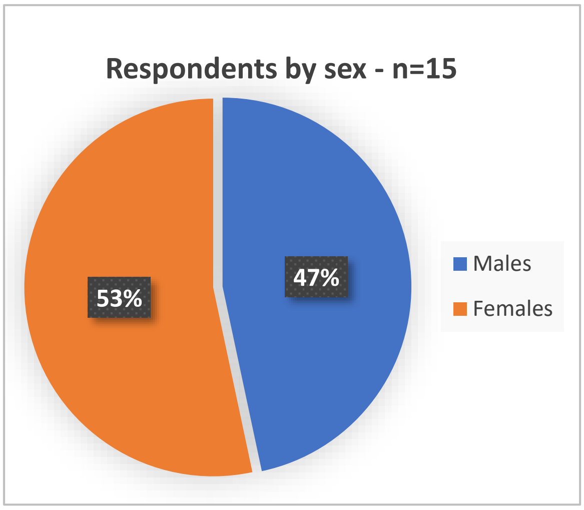
Effects of Climatic Events and Fishing Activities

Most participants indicated that climatic events affected their ability to carry out their work. The following characteristics of FAD and reef fishing were mentioned,

- FADs were accessible via Niue canoes however, weather events prevented regular fishing at FADs.
- Fishermen were more likely to reef fish onshore.
- Tools used included fishing rods, lines, hooks, sinkers, soft bate and traditional fishing rods. Bottom trawling was used by all fishermen.
- Women were more likely to engage in gleaning compared with men. One interviewee indicated she did not appreciate FADs being used as a model for attracting fish as she felt that the FADs drew the fish away from the reef. One fisherman however indicated that FADs attracted different fish that were not normally present.
- Cast netting was not a method used by any of the interviewees nor was it considered an approach commonly used in the community.
- Interviewees who owned boats and canoes noted they had access to reliable boats suitable for nearshore fishing and they were in good working order. Operators of boats were also highly experienced.
- There has been significant investment in marine education in recent years which has helped

FAD fishermen and/or reef fisherfolk	4
Fish sellers	3
Fish transporters	2
Households who eat fish regularly	6





to improve community awareness in marine safety.

6.7 Nauru

Gender policies

Nauru has committed to strengthening gender equality in the fisheries sector through international commitments and national priorities and policies, including but not limited to the following:

International

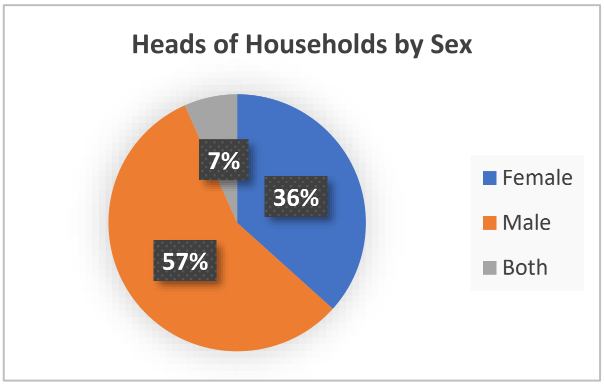
- Convention on the Rights of the Child (CRC) (Accession 1994) •

- Optional Protocol to the CRC on the involvement of children in armed conflict (Signed 2000). Optional Protocol to the CRC on the sale of children, child prostitution and child pornography (Signed 2000
- International Convention on the Elimination of all forms of Racial Discrimination (Signed 2001) • Convention on the Elimination of All forms of Discrimination against Women (CEDAW) (Accession 2011). Nauru has not signed the Optional Protocol to CEDAW.
- Convention on the Rights of Persons with Disabilities (Accession 2012)
- Convention against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment (CAT) (Signed 2001, ratified 2012; Optional Protocol on CAT (Accession 2013)
- Nauru is a signatory to the 2011 Joint Statement on Ending Acts of Violence Related Human Rights Violations Based on Sexual Orientation and Gender Identity at the UN Human Rights Council.
- Nauru is not included within the World Risk Index on disaster risk or the Global Climate Risk Index due to its small size and lack of relevant data.

National legislation

- Nauru National Women's Policy 2014-2019 2014. Policy reviewed in 2021
- 2016 Crimes Act criminalises rape and sexual offences within marriages and de facto relationships, and states that evidence of physical resistance is not necessary to prove non-consent
- Under the Public Service Act 2016, women in the public sector who have been in service for six months are entitled to 12 weeks' paid maternity leave, while men are entitled to two weeks' paid paternity leave.
- Nauru's 2015 Framework for Climate Change Adaption and Disaster Risk Reduction (RONAdapt) recognises the gender-differentiated impact of disasters on the vulnerability of households and individuals, and states that the empowerment of both men and women is necessary in order to build resilience and future capacities
- The definition of domestic violence in the Domestic Violence and Family Protection Act 2017 includes not just physical and sexual violence but also coercive control, economic and financial abuse, stalking, and cruel and degrading treatment.

Key input from national consultations

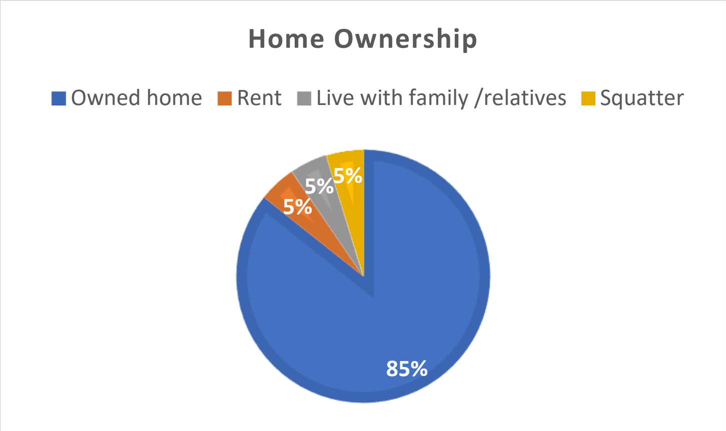


Thirty respondents from Nauru were surveyed in October 2023 regarding tuna fishing and the fish supply chain. Respondents included 10 women and 20 men. As shown in the following diagram, eleven households or 36% of respondents were FHH. Twelve households (40%) also had elderly family members residing in the home. All respondents earned over $600 a month in income.

In 2013, 34% of households were headed by women which reinforces the results of this study.

The respondents came from the following villages:

Yaren, Boe, Aiwo, Buada, Denig, Nibok, Uaboe, Batisi, Ewa, Anetan, Anabar, Ijuw, Anibare, and Meneng.



Eighteen respondents (85%) owned their own home while the remainder either rented, lived with relatives or was a squatter.

6.8 Palau

Gender policies

An approximate 70% of Palauans rely on ocean resources for food, livelihoods, and recreation. On average, Palauans prepare about 50% of their own seafoods within the household, indicating a high reliance on local marine resources for food security. This indicates the importance of sustainably managed and accessible marine resources.

Palau is a matriarchal and matrilineal society whereby lineage and titles are inherited from the mother's side of the family. Women traditionally hold positions of power and respect and hold central positions in Palau society.

International Gender policies

Palau is committed to strengthening gender equality in the fisheries sector through several international commitments and national priorities and policies, including but not limited to the following:

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – signed in 2011, but they are yet to sign the Optional Protocol.
- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs and oceans.
- Beijing Declaration and Platform for Action signed 1995 – Palau has introduced a National Gender Mainstreaming Policy in 2018 and established a Bureau of Aging, Disability and Gender.

National Gender policies

- Palau's Constitution prohibits discrimination based on sex, race, place of origin, language, religion, or belief. Discrimination based on sexual orientation or gender identity is not explicitly prohibited.
- Family Protection Act 2012 provides protection for victims of GBV.
  - National Gender Mainstreaming Policy, 2012:
  - Provides support to victims of GBV.
- All women and men have the same opportunities to earn incomes and fulfill needs.
- Voluntary National Review 2019 - five key areas that needed improvement in Palau:
  - Universal mandated maternity leave.
  - Legislation to address sexual harassment in the workplace.
  - Expanded childcare facilities.
  - Improved support services for victims of GBV.
  - Removal of gender discriminatory marriage and inheritance laws.

Violence against women is considered by many in Palau to be a private family matter that should be dealt with within the family, not through the courts, which prevents women's ability to access services.

Health and nutrition

Many Palauans currently have a diet rich in seafoods, but also eat a range of unhealthy tinned meats such as Spam. There is the local interest to consume even more seafood and reduce this intake of unhealthy canned meat (kansume) (Ferguson and Singeo, unpublished data).

Largely because of poor diets there are high levels of Non-Communicable Diseases (NCDs) in Palau. For example, in 2017 and 2018, 36% of school age girls and 40% of school age boys were overweight or obese. In 2018, 36.3% of men and 11.2% of women reported that they currently smoked tobacco. In 2019, it was reported that 64% of women aged 25-64 chew betel nut daily, compared to 57% of men. Additionally, women are more likely than men to add tobacco to betel nut.

Gender Roles

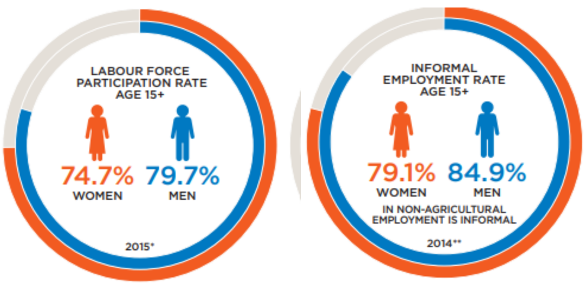
As recorded by the 2015 Census, 30.7% of households were headed by women. Eighty one percent of the population reside in urban areas (not disaggregated by sex). Twenty two percent of women and 5.6% of men were unpaid family workers helping with basic household duties in 2015 (Census 2015). This included full time students (40.8% of men and 34.7% of women outside the labour force).

The 2015 Census identified that women undertaking tertiary education were more likely to specialise in business, administration, and law (34.0% of women and 15.7% of men); education (15.2% of women and 6.7% of men); and health and welfare (11.2% of women and 3.0% of men). Men comprised 77% of students undertaking Science, Technology, Engineering, and Mathematics (STEM) majors were men (Census 2015).

Employment

Data from the census 2014/2015 shows that 79.1% of women and 84.9% of men were engaged in informal employment in the non-agricultural sector. Informally employed workers do not benefit from national labour legislation or social protections (UN Women, 2022).

Figure 5: Palau-Employed Men and Women in Labour Force and Informal Sector



Source: Census 2015, cited UN Women 2022

Key input from national consultations

Fifteen respondents from the villages of Ngatpang and Aimeliik were interviewed. Of these respondents 8 were female and 7 males. Ages were between the ages of 29 to 60.

Gender roles in fisheries

Fishing is a male dominated in Palau, while gleaning is dominated by women. Different fishing modes largely are carried out by men and women as follows:

- Fishers – 67% men.
- Spearfishers – 86% of spearfishers in Palau are men.
- Gleaners – 85% of gleaners are women, with 72% of sea cucumber gleaners being women (Ferguson and Singeo, unpublished data).

Fishing related Role	Men	Women
Boat operation	Men	
Offshore fishing including FADs	Men	
Boat repair/ maintenance	Men	
Coastal fishing and other sea products	Men	Women
Post-harvest handling	Men	Women
Cleaning of fish products	Men	Women
Selling of fish	Men	Women
Cleaning of fish	Men	Women
Cook	Men when there is no female around to cook.	Women
Cook fish for sale, e.g., restaurants, stalls, etc.		Women

Source: 2023 survey for GCF proposal

Gleaners depend heavily on their catch for income and food and rely on nearshore coastal resources that can be accessed without a boat. These women are vulnerable to food insecurity and loss of income from unsustainable coastal developments such as dredging, sand mining, and land reclamation, including polluted, and unproductive fishing grounds (Gender and Natural Resources Report 2020). Many gleaners (89%) and fishers (86%) have personally and directly observed the effects of climate change on their marine resources (Gender and Natural Resources Report 2020). Gleaners and fishers reported the same four leading threats to their resources:

- climate change,
- overfishing,
- sedimentation, and
- plastics pollution.

Solutions to climate change requires global action, overfishing must be addressed locally.

6.9 Papua New Guinea

Considerable efforts have been made to mainstream gender equality across all sectors in PNG. Sadly, GBV is widely accepted and “culturally condoned”. While challenging this remains a commitment of the Government. Key international and national policies are summarised:

International Gender policies

PNG is committed to strengthening gender equality across sectors including the fisheries sector through a number of international commitments:

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – ratified on 12th January 1995.
- Beijing Declaration and Platform for Action signed 1995.

National Gender policies

- PNGs Constitution provides for gender equity and equality, but customary law, recognized by the constitution, discriminates against women in relation to rights and property.
- The National Policy for Women and Gender Equality, 2011–2015 – the government's mission to promote improved equality, participation, and empowerment of women.
- The government's Vision 2050; the Development Strategic Plan (DSP), 2010–2030 and the Medium-Term Development Plan (MTDP), 2011–2015. These outline broad aspirations about gender equality, providing 'what should be done' but this requires implementation instructions.

Key input from national consultations

Undertaking the community surveys posed some challenges in PNG related to the size of the country and processes to organise meetings in each location, and selection of a representative sample that provides an example of local reality differences. However, four locations were selected, and surveys were undertaken in July 2023. Locations surveyed were as follows:

- Karasau Island
- Vekeo Island
- Koil Island
- Koil Island

Twenty seven respondents were interviewed from different stages of the fishing supply chain from fishers, market vendors and consumers. Of these respondents 21 were male and 6 were female. Interviewees were aged between 27 and 53.

Employment and Income generation

Only one respondent had a full-time job while the remainder 26 carried out fishing and home duties. Fishing was carried out for subsistence and excess fish catches were sold. Earnings reportedly per month are ST$55 - $300 a month. Local Government earnings reportedly were ST$200 a month.

Consumption of fish

All respondents consume fish every day and indeed favour fish over meat. Respondents indicated that they eat meat when there are fish shortages such as bad weather when the fishers cannot go out in their boats. The fish are caught on the reef and deep waters around each island.

Roles in Fishing

Both men and women carry out fishing activities. All respondents indicated that there were no constraints to women fishing. However, women tend to fish in the coastal waters and do not go out to the deep waters. Reasons provided were sea safety and personal security. Thus, women do participate in the coastal FAD fishing.

Fishing related Role	Men	Women
Offshore fishing including FADs	Men	
Boat repair/ maintenance	Men	
Coastal fishing and other sea products	Men	Women

Post-harvest handling	Men	Women
Cleaning of fish products	Men	Women
Selling of fish	Men	Women
Cleaning of fish	Men	Women
Cook	Men.	Women
Cook fish for sale, e.g., restaurants, stalls, etc.		Women

Source: 2023 survey for GCF proposal

FAD Fishing

All respondents have heard about FAD fishing and understand that it is 'safer' and reduces fishing costs. However, concerns regarding perceptions are as follows:

- that volume of fish caught will be reduced.
- If FADs are placed in areas with strong currents, fishers safety can be compromised.

6.10 Samoa

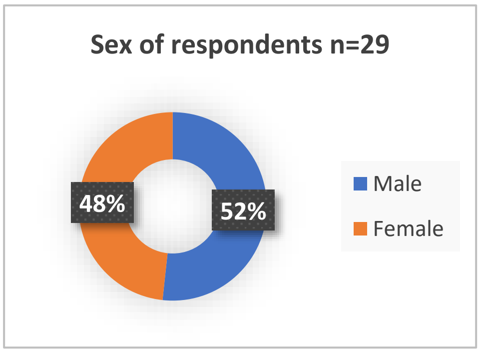
Gender Policies

Samoa has committed to strengthening gender equality in the fisheries sector through several internal commitments and national priorities and policies, including but not limited to the following,

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – efforts to ensure women benefit from resource conservation and development especially in rural areas.
- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs and oceans.
- The Agriculture Sector Plan 2016–2020, Ministry of Agriculture and Fisheries - opportunities for women in the fisheries sector.
- Ministry of Agriculture and Fisheries, Corporate Plan 2016–2020 - specific outputs to increase women's capacity and access to opportunities. **Key input from national consultations.**

Key input from national consultations

15 men and 14 women respondents were interviewed and were aged between 25 to 83 years of age. This was broken down into 15 from Manono (7 female, 8 male) and 14 from Siumu (7 female, 7 male) villages in July 2023. Of these respondents 9 buy fish and 22 get fish from fishing and two sell fish. Note some roles overlap.



While the majority of HOH were men, there were 2 FHH in both Manono and Siumu respondent – 14% of respondents were FHHs.

Roles and Responsibilities

It was interesting to see the results from the survey whereby both men and women self-identified as fishers. While the fishing carried out by women predominantly covers reef fishing, aquaculture, and gleaning activities, at least 9 women surveyed do consider themselves fishers. Some of these female fishers also carry out other roles in the household such as care giver. This is different from a 2018 SPC Survey[31] the 'aquaculture study' that found that women seldom self-identify as aquaculture farmers or fishers, even though they are actively engaged in fish farm maintenance and post-harvest processing and marketing. This recognition can enable women fishers to also receive necessary training and supports that they may have previously missed out.

Despite this gradual change in thinking regarding the roles of women in fishing, when this is broken down further the roles remain much as they have always been with women having extra home roles on their fisher roles.

The main roles for men and women in fishing as reported in the survey are as follows,

- Offshore fishing – male dominated – no female respondent is a fisher in this type of offshore fishing including FAD fishing. While most respondents consider the constraints to women carrying out FAD fishing, home commitments and lack of interest are the main reasons women do not participate.
- Coastal fishing – 100% men and 45% of women reportedly carry out coastal fishing.
- Boat management and repair – only men are involved in these tasks.
- Post-harvest handling – predominantly men, however a significant number of women are also involved in post-harvest handling.
- Fish cooking for home or selling use is predominantly the role of women, however a number of men also undertake this role.
- Cleaning fish is carried out by both men and women.
- Selling fish is the role of both men and women.

6.11 Solomon Islands

International Commitments

Solomon Islands has committed to strengthening gender equality in the fisheries sector through several international and national commitments, priorities and policies, including but not limited to the following:

- CEDAW – acceded 2002 – efforts to ensure women benefit from resource conservation and development especially in rural areas.
- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs, and oceans.

National Commitments

- Solomon Islands National Development Strategy – Objective 2 – highlights 'the need for the benefits of development to be more equitably distributed for the whole of Solomon Islands which includes improved gender equality and targeted actions to empower and include everyone'.
- National gender equality and women's development policy, enabling policies and initiatives that facilitate gender mainstreaming - enabling policies and initiatives.

Key input from national consultations

[31] Gender and fisheries in Samoa Summary of key issues June 2018. Online at: Gender and fisheries in Samoa - Summary of key issues. (windows.net)

Three respondents from a fishing village in Honiara and 5 respondents from the Honiara Central Market were interviewed in July 2023. Some key demographics of the respondents are summarised in the following table.

Location	Respondents		Household residents	HOH	Indigenous residents	Home ownership
	Male	Female				
Fishing village	2	1	6 to 7	All male	All local Solomon Islanders	All owned home.
Honiara central market	3	3	6 to 7	All male	Mixed clans from across SI	5 owned home and 3 rented houses.

Women participants mentioned they are part of household decision making.

Health

Many respondents and their families have been afflicted by the flu and colds and even COVID which has meant that respondents cannot earn while they are sick. Some market venders were also affected by Dengue and Malaria and even knee problems.

During COVID the market was closed and there was no income for families. Even after COVID respondents noted that fish selling is not as good as previously – reportedly the only good days to sell fish are Thursday to Saturday, and on Sunday for the SDA market.

Nutrition

Respondents all noted that they prefer fish over meat. Respondent families each meat once a week or less which is mainly chicken wings. However, families eat fresh fish 2-3 times a week and tin fish every 5-6 days. The main fish consumed are reef fish, bonita and mackerel.

Most respondents indicated that it is faster to cook fish than meat. They also noted that fish prices are higher now than 5 years ago. With a family of 5/6 people it costs more than SBD 50 to feed the family fish. Furthermore, the size of the fish sold is smaller compared to 5 years ago.

Respondents indicated that the main reasons for not eating more fish relate to the following:

- Cost
- Cost of fishing
- Quality / freshness of fish concerns
- Access to quality fish
- Fear of food poisoning. Many fish sellers at the market sell salted fish from fish that are not fresh.

Employment and earnings

All respondents' interviewed work in the informal sector as fishermen or fish sellers. When there are no fish due to bad weather and other issues, they sell vegetables, clothing, cooked food, and other products in the market. While bad weather can affect sales, all respondents work full time.

Each respondent earns over SBD 600 a month.

Expenditure

Fish market respondents reportedly spend around SBD 250 a month whereas the Fishing village report around SBD 550 a month.

Climate change effects

All respondents noted that climate change affects their work. They noted that bad weather and strong winds affect safety, cost to attain and sell their fish, personal health concerns as well as broader access to services and commodities.

Fishing gender roles

Fishing roles tend to be dictated by cultural taboos, especially in communities. Only men allowed in some communities. Respondents indicated that at times women know how to fish better than men, but cultural norms limit them to specific roles. See the table below indicating main fishing industry roles of men and women in Solomon Islands.

Fishing related Role	Men	Women
Boat operation	Men	
Offshore fishing including FADs	Men	
Boat repair/ maintenance	Men	
Coastal fishing and other sea products	Men	Women
Post-harvest handling	Men	
Cleaning of fish products	Men	Women
Selling of fish	Men	Women
Cleaning of fish belle	Men	Women
Cook	Men when there is no female around to cook.	Women
Cook fish for sale, e.g., restaurants, stalls, etc.		Women

6.12 Tonga

Gender policies

Tonga has committed to strengthening gender equality in the fisheries sector through several international commitments and national priorities and policies, including but not limited to the following:

- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs, and oceans.

Tonga is one of xx countries in the Pacific that have not ratified CEDAW.

National Commitments to Gender Equality

- The Constitution of Tonga states, "No laws shall be enacted for one class and not for another class, but the law shall be the same for all the people of the Land". Laws not always applied consistently to men and women.
- Tonga's Strategic Development Framework –'Strong inclusive communities, by engaging districts/villages/communities in meeting their prioritised service needs and ensuring equitable distribution of development benefits'
- National Women's Empowerment and Gender Equality Tonga Policy and Strategic Plan of Action 2019–2025.

Key input from national consultations

Eighteen respondents from Ha'atafu, Tongatapu and 20 respondents from 'Eua were interviewed in July 2023. Ages of respondents varied between 32-72 in Ha'atafu and 35-85 in 'Eua.

Some key demographics of the respondents are summarised in the following table.

Table 1.

Location	Respondents		Household residents	HOH	Indigenous residents	Home ownership
	Male	Female				
Ha'atafu	13	5	4 to 5	All male	All respondents were Tongan.	All owned home and looked after homes of overseas relatives.
'Eua	17	3	5	All male	All respondents were Tongan	

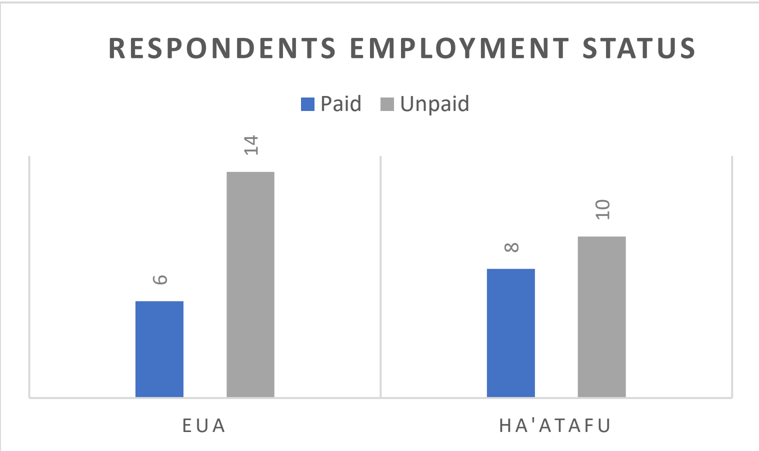


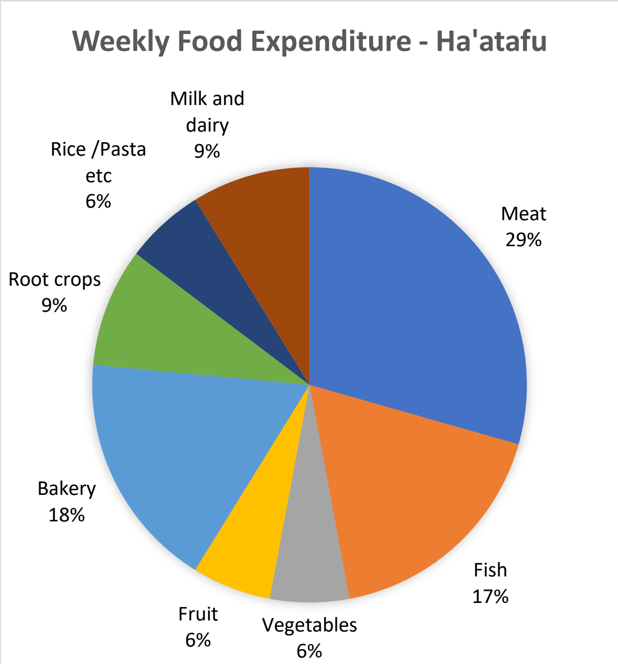
Figure 1.

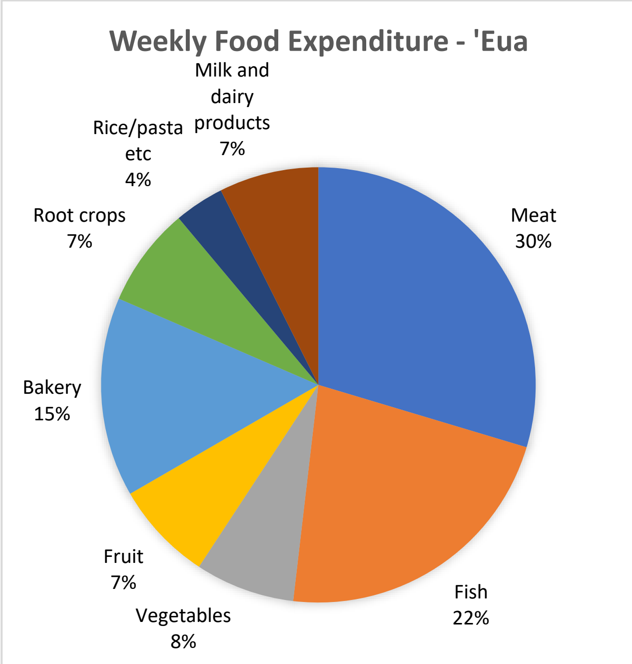
Employment and earnings

Most of the respondents did not receive an income as noted in Figure 1 above. Those who earned an income included 7 respondents from 'Eua and 5 from Ha'atafu who were in full time employment. The remainder were either in part time employment or self-employed. Income earned each month reportedly was between $150 -300 in Ha'atafu and 'Eua.

Food Expenditure

Respondents reportedly spend TOP 170 a month on food. Of this approximately 18% (Ha'atafu) and 20.7% ('Eua) is spent on fish and 29% on meat in Ha'atafu and 29.5% ('Eua). See below the expenditure pattern of respondents from Ha'atafu and 'Eua.





Climate change effects

All respondents noted that climate change affects their work. They noted that bad weather and strong winds affect safety, cost to attain and sell their fish, personal health concerns, food availability as well as broader access to services and commodities.

Health

Many respondents and their families have been afflicted by the flu and colds and even COVID which has meant that respondents cannot earn while they are sick.

During COVID the market was closed and there was no income for families. Even after COVID respondents noted that COVID affected the way the marketing operates, and it has been slow to re-establish normal operations and generate incomes.

Nutrition

Respondents all noted that they prefer fish over meat. Respondent families reportedly eat meat daily, and this includes mainly lamb flaps or chicken thighs or wings. However, families eat fresh fish 2-3 times a week in ‘Eua and weekly in Ha’atafu. The main fish consumed are **reef fish and tuna from longline fishing vessels**. Most respondents indicated that it took longer preparatory time cooking fish than meat.

Respondents from both Ha’atafu and ‘Eua noted that cost and limited access to quality fish were the main reasons that they did not eat more fish.

Fishing gender roles

Fishing roles continue to follow traditional patterns whereby offshore fishing and more laborious roles are carried out by men. Women fish and glean in coastal areas and carry out some of the less physically demanding roles such as cleaning and selling the fish. See the table below indicating main fishing industry roles of men and women in Tonga.

Fishing related Role	Men	Women

Boat operation	Men	
Offshore fishing	Men	
Boat repair/ maintenance	Men	
Post-harvest handling	Men	Women
Cleaning of fish products	Men	Women
Use of fish for food or sale	Men	Women
Selling of fish	Men	Women

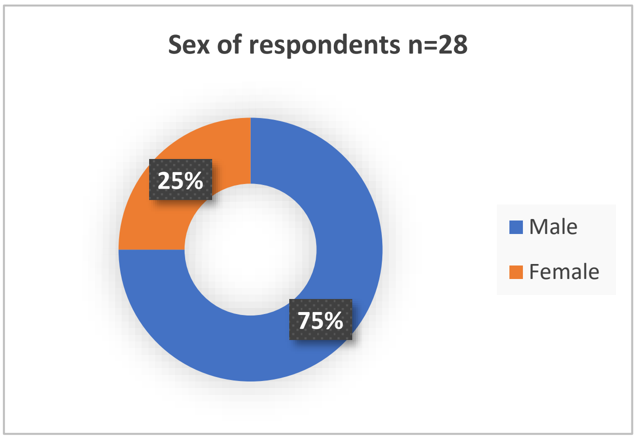
6.13 Tuvalu

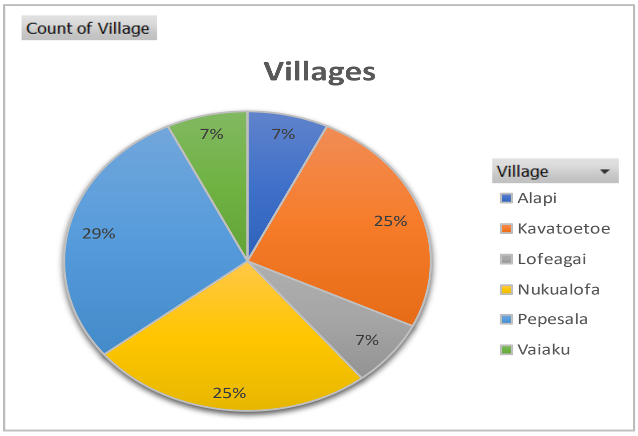
Gender Policies

Tuvalu has committed to strengthening gender equality in the fisheries sector through a number of internal commitments and national priorities and policies, including but not limited to the following:

- The Convention on Elimination of all Forms of Discrimination Against Women (CEDAW) – ratified in 2000 - efforts to ensure women benefit from resource conservation and development especially in outer islands.
- The Convention on the Rights of the Child ratified in 2000.
- The UN Framework Convention on Climate Change and the Convention on Biodiversity. This considers men's and women's roles in resource management, including management of rivers, reefs and oceans.
- Tuvalu National Gender Policy - opportunities for women in the fisheries sector.

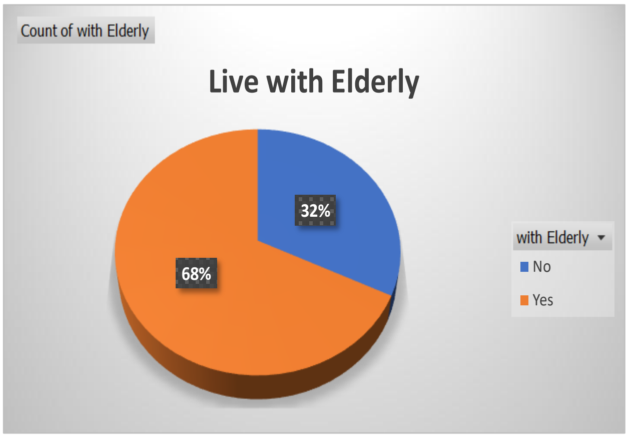
Key Input from National Consultations





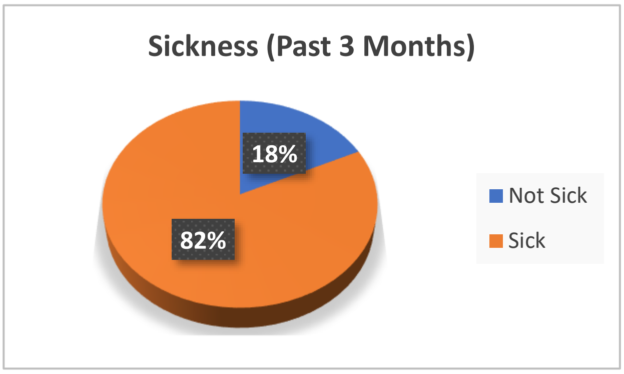
7 women and 21 men respondents were interviewed and were aged between 23 to 66 years of age. This was broken down into 13 from Funafuti (5 women and eight men) and 15 from Nukulaelae (13 men and two women) villages in July 2023.

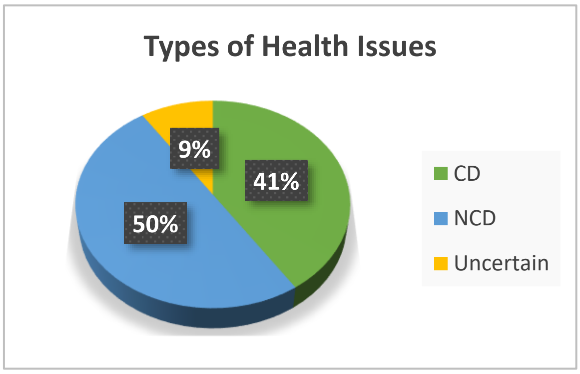
Of these respondents 27 were fishers who also farmed and undertook several other tasks including house duties. Figure XX highlights respondents from villages surveyed in both Funafuti and Nukulaelae.



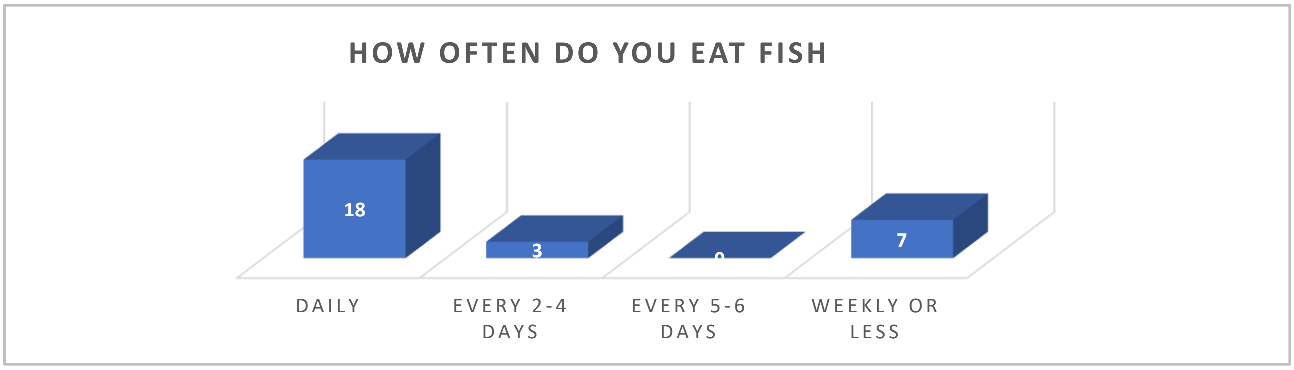
A significant number of respondents reside with elderly family members, who are likely to face increased vulnerability to climatic events.

The high level of sickness in respondent households and the further fact that the illnesses suffered are predominantly non-communicable in nature, can suggest the vulnerability of respondents' households as determined by health-related causes.

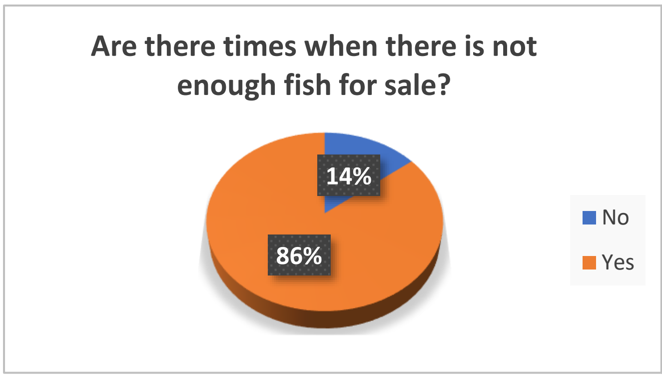


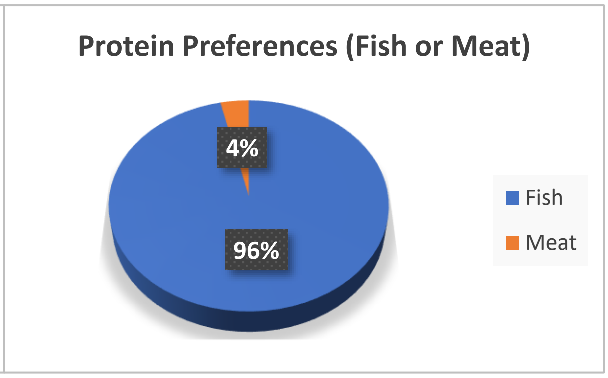


Consumption of Fish

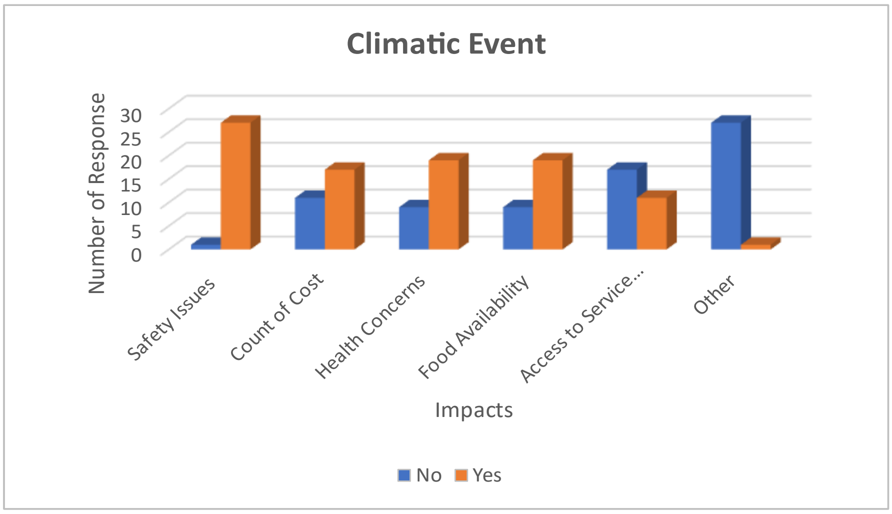


64% of respondent eat fish daily. Furthermore, fish is the favoured protein source in respondents' diets. Most respondents indicated that there were times that no fish were available for sale in the community.





Climatic Events

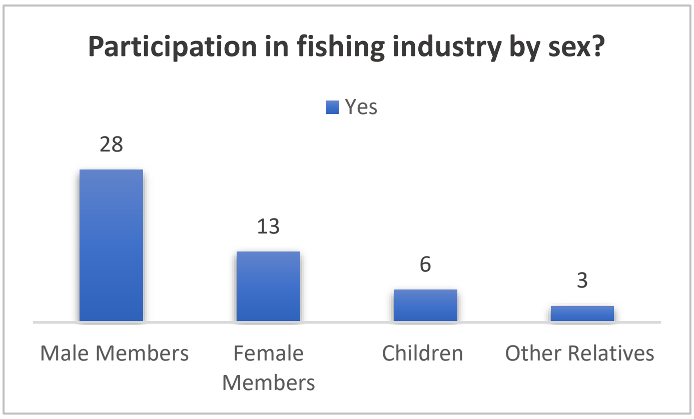


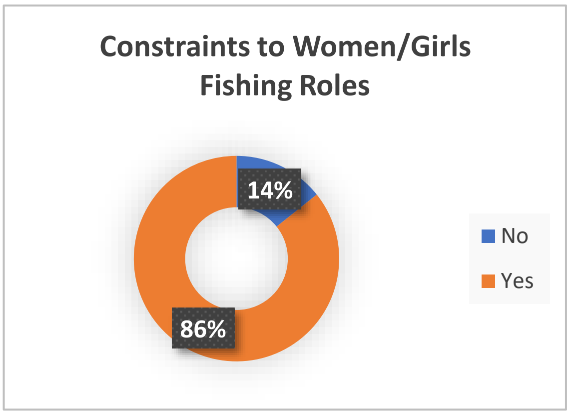
Reflected herewith are the key concerns of respondents in a climatic event.

Understandably, especially with many vulnerable household members, 'safety' is the dominant concern. This is subsequently followed by 'health, costs of climate induced impacts, access to services and food availability'.

Roles and Responsibilities

Constraints of traditional working roles highlighted fishing sector roles and responsibilities. Women do not carry out offshore fishing but are responsible for selling the fish. This is the same situation in both Funafuti and Nukulaelae. Respondents (96%) indicated that there are constraints to women being involved in offshore fishing, the main ones being 'traditional roles and lack of interest' are the main reasons women do not participate in offshore fishing. Both men and women are involved in sale of fish in Funafuti, whereas it is largely the role of women in Nukulaelae.





6.14 Vanuatu

Gender policies

'The Vanuatu Government recognises that gender equality is part of the fundamental right and duty enshrined in the National Constitution of Vanuatu' (National Gender Equality Policy 2015-2019).

'The future of Vanuatu is shaped by active participation and meaningful contribution of all citizens'.

International commitments

Vanuatu is a signatory to several international and regional agreements on gender equality and the advancement of women. These include:

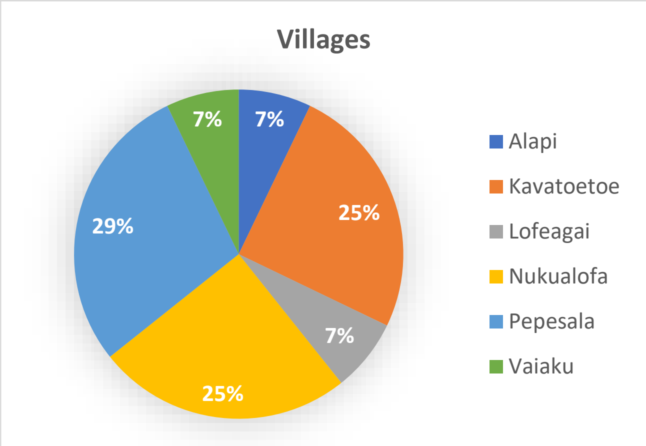
- Convention on the Elimination of all forms of Discrimination Against Women (CEDAW) and Optional Protocol;
- Millennium Development Goals;
- Beijing Platform for Action (BPA);
- Accra Agenda for Action and the Busan Partnership for Effective Development Cooperation;
- International Conference on Population and Development (ICPD) Programme of Action;
- Pacific Plan;
- Revised Pacific Platform for Action on the Advancement of Women and Gender Equality;
- Pacific Leaders Gender Equality Declaration

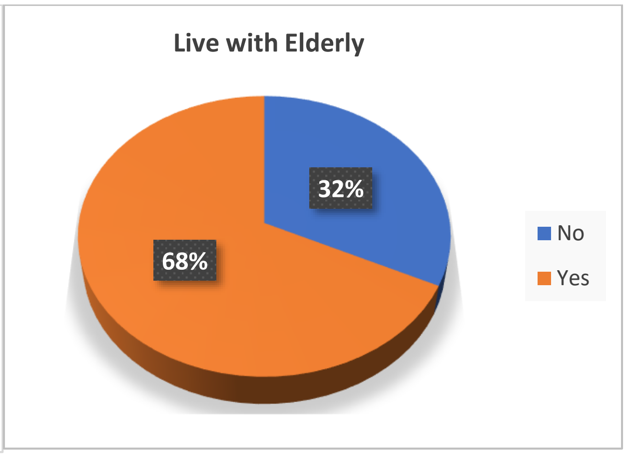
National Commitments

- Constitution recognises the rights and freedoms of all individuals without discrimination on the grounds of sex, race, place of origin, religious or traditional beliefs, political opinions or language.
- The first National Policy on Gender Equality affirms the Vanuatu Government's commitment towards gender equality across all sectors and at all levels of society and the elimination of discrimination and violence against women and girls.
- the 2006 National Women's Forum;
- the National Plan of Action for Women 2007-2011;
- Policy development consultations undertaken in Torba, Sanma, Penama, Malampa, Shefa and Tafea provinces, 2012-2013; and
- the 2015 national and provincial policy validation. towards men and women.
- the Comprehensive Reform Programme (CRP) and later in the Priorities and Action Agenda (PAA) 2006 – 2015.

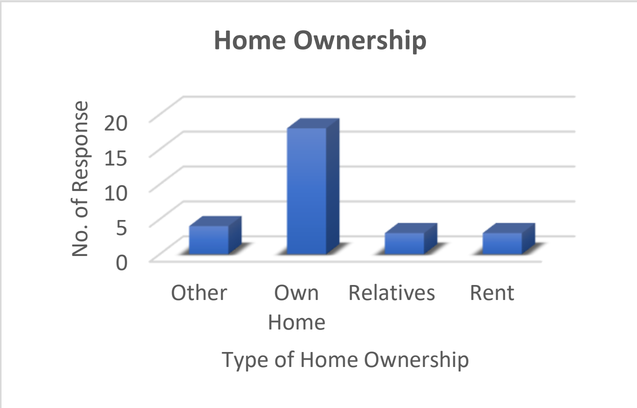
Key input from national consultations

28 respondents from Vanuatu were surveyed from islands of Efate, Pele, Nguna, Emao, Moso and Lelepa. Ages of respondents were between 23 and 66. There were seven females and 21 male respondents.

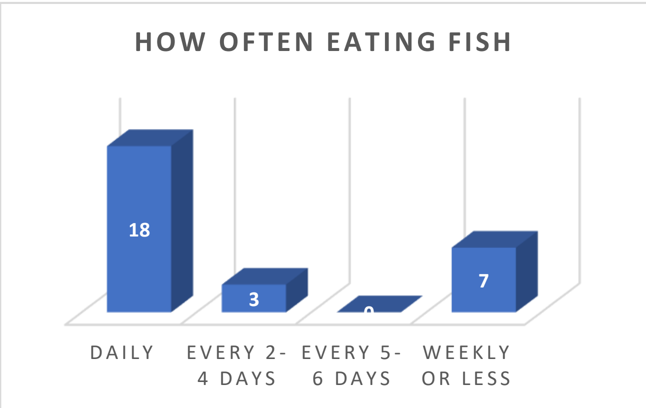




What is shown clearly here is that most (68%) of respondents live with elderly family members. This indicates a high dependency rate which can increase household vulnerability.



16 respondents owned their own home, while 12 either lived with relatives or rented.



Fish is important to both the culture as well as the diet as shown in Figure xx. Eighteen respondents (64%) of respondents eat fish daily. In comparison to 28% or 7 respondents ate fish weekly or less.

Only one respondent preferred meat over fish.

85% percent of respondents indicated that women are constrained from fishing. The main constraints to women actively fishing is traditional roles (93% of respondents) and lack of interest (96.5% of respondents). However, 75% of respondents indicated that women sell fish. Eighty six percent of respondents noted that there were times that there was not enough fish to sell.

7 References

ADB Country Gender Assessments. Online at: gender-assessment-fp036-adb-cook-islands-marshall-islands-micronesia-nauru-papua-new-guinea-samoa.pdf (greenclimate.fund)

Collins, J 2022, 'Gender equality in the Pacific: less talk and more action'. Lowry Institute.

Online at: Gender equality in the Pacific: less talk and more action | Lowy Institute

FFA, 2010. 'Gender issues in the Pacific Islands Tuna Industries'. Online at:Gender issues in P. I. Tuna Industries 1_0.pdf (ffa.int)

Mangubhai, S., Lawless, S., Cowley,A. Mangubhai,J., Williams M., 2022, 'Progressing gender equality in fisheries by building strategic partnerships with development organisations'. Online at: Progressing gender equality in fisheries by building strategic partnerships with development organisations - ScienceDirect

Pacific Leaders Gender Equality Declaration 30 August 2012, Rarotonga, Cook Islands – Forum Sec
[1] Gender Issues in the Pacific Islands Tuna Industry 2006 Gender issues in P. I. Tuna Industries 1_0.pdf (ffa.int)
[1] Review of the Revised Pacific Platform for Action on the Advancement of Women and Gender Equality 2005–2015 (spc.int)
PIFS and SPC, Addressing Inequalities: the Case of Small Island Developing States in the Pacific, Issues Paper for the Small Islands Developing States Conference 2013
[1] Ibid

Sharples, CC. 2021, Online at: Fisheries need to make gender inclusion a norm, not just 'reach' women, says Pacific study (mongabay.com)